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**Department of War
Fiscal Year (FY) 2027 Budget Estimates**

April 2026



Defense Advanced Research Projects Agency

Defense-Wide Justification Book Volume 1 of 5

Research, Development, Test & Evaluation, Defense-Wide

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Defense Advanced Research Projects Agency • Budget Estimates FY 2027 • RDT&E Program

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Department of War
 FY 2027 President's Budget
 Exhibit R-1
 Total Obligational Authority
 (Dollars in Thousands)

Apr 2026

Appropriation

	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
Research, Development, Test and Evaluation, Defense-Wide	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
Total Research, Development, Test, & Evaluation	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370

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Department of War
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Total Obligational Authority
(Dollars in Thousands)

Apr 2026

	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
<u>Summary Recap of Budget Activities</u>							
Basic research	366,226	345,053	12,500	357,553	387,633		387,633
Applied research	1,519,756	1,738,137	344,500	2,082,637	2,098,760		2,098,760
Advanced technology development	2,254,291	2,109,021	110,000	2,219,021	2,439,493		2,439,493
Management support	268,718	129,296		129,296	113,484		113,484
Total Research, Development, Test, & Evaluation	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
<u>Summary Recap of FYDP Programs</u>							
Research and Development	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
Total Research, Development, Test, & Evaluation	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370

*Budget activities unique to the Golden Dome for America Fund Acct (3007D) for efforts associated with Opn & Maint, Procurement, and Research, Dev, Test & Eval activities

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Apr 2026

	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
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Total Research, Development, Test, & Evaluation	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
<u>Summary Recap of FYDP Programs</u>							
Research and Development	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
Total Research, Development, Test, & Evaluation	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370

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(Dollars in Thousands)

Apr 2026

Appropriation

	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
Defense Advanced Research Projects Agency	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370
Total Research, Development, Test and Evaluation, Defense- Wide	4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370

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Defense-Wide
FY 2027 President's Budget
Exhibit R-1
Total Obligational Authority
(Dollars in Thousands)

Apr 2026

Appropriation: 0400D Research, Development, Test and Evaluation, Defense-Wide

Line No	Program Element Number	Item	Act	Sec	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
2	0601101E	Defense Research Sciences	01	U	279,971						
5	0601117E	Basic Operational Medical Research Science	01	U	86,255						
7	0601122E	Emerging Opportunities	01	U		345,053	12,500	357,553	387,633		387,633
	Basic research				366,226	345,053	12,500	357,553	387,633		387,633
12	0602023E	Access and Awareness	02	U		98,269	7,000	105,269	110,096		110,096
13	0602024E	Warfighting Performance	02	U		307,727	87,500	395,227	364,141		364,141
14	0602025E	Making, Maintaining, Supply Chain and Logistics	02	U		1,318,411	250,000	1,568,411	1,624,523		1,624,523
15	0602026E	Effects	02	U		13,730		13,730			
16	0602115E	Biomedical Technology	02	U	128,307						
21	0602303E	Information & Communications Technology	02	U	352,961						
26	0602702E	Tactical Technology	02	U	112,715						
27	0602715E	Materials and Biological Technology	02	U	304,669						
28	0602716E	Electronics Technology	02	U	621,104						
	Applied research				1,519,756	1,738,137	344,500	2,082,637	2,098,760		2,098,760
46	0603286E	Advanced Aerospace Systems	03	U	241,495						
47	0603287E	Space Programs and Technology	03	U	175,105						
56	0603467E	DARPA Advanced Technology Development	03	U		1,528,100	90,000	1,618,100	1,568,124		1,568,124
57	0603468E	Advanced Complex Systems	03	U		263,461		263,461	540,362		540,362
58	0603469E	Advanced Enabling Technologies	03	U		317,460	20,000	337,460	331,007		331,007
68	0603739E	Advanced Electronics Technologies	03	U	148,382						

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Total Obligational Authority
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Appropriation: 0400D Research, Development, Test and Evaluation, Defense-Wide

Line No	Program Element Number	Item	Act	Sec	FY 2025 Actuals	FY 2026 Discretionary Enacted	FY 2026 PL 119-21 Spend Plan	FY 2026 Total	FY 2027 Discretionary Request	FY 2027 Mandatory Request	FY 2027 Total
69	0603760E	Command, Control and Communications Systems	03	U	591,042						
70	0603766E	Network-Centric Warfare Technology	03	U	838,984						
71	0603767E	Sensor Technology	03	U	259,283						
	Advanced technology development				2,254,291	2,109,021	110,000	2,219,021	2,439,493		2,439,493
175	0605001E	Mission Support	06	U	122,998	115,673		115,673	108,101		108,101
190	0605502E	Small Business Innovative Research	06	U	130,133						
199	0605898E	Management HQ - R&D	06	U	15,587	13,623		13,623	5,383		5,383
	Management support				268,718	129,296		129,296	113,484		113,484
Total Research, Development, Test and Evaluation, Defense-Wide					4,408,991	4,321,507	467,000	4,788,507	5,039,370		5,039,370

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2	01	0601101E	DEFENSE RESEARCH SCIENCES.....	Volume 1 - 1
5	01	0601117E	BASIC OPERATIONAL MEDICAL SCIENCE.....	Volume 1 - 15
7	01	0601122E	EMERGING OPPORTUNITIES.....	Volume 1 - 19

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Line #	Budget Activity	Program Element Number	Program Element Title	Page
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13	02	0602024E	WARFIGHTING PERFORMANCE.....	Volume 1 - 63
14	02	0602025E	MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS.....	Volume 1 - 89
15	02	0602026E	EFFECTS.....	Volume 1 - 159
16	02	0602115E	BIOMEDICAL TECHNOLOGY.....	Volume 1 - 165
21	02	0602303E	INFORMATION & COMMUNICATIONS TECHNOLOGY.....	Volume 1 - 171
26	02	0602702E	TACTICAL TECHNOLOGY.....	Volume 1 - 187

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27	02	0602715E	MATERIALS AND BIOLOGICAL TECHNOLOGY.....	Volume 1 - 197
28	02	0602716E	ELECTRONICS TECHNOLOGY.....	Volume 1 - 209

Appropriation 0400: Research, Development, Test & Evaluation, Defense-Wide

Line #	Budget Activity	Program Element Number	Program Element Title	Page
46	03	0603286E	ADVANCED AEROSPACE SYSTEMS.....	Volume 1 - 227
47	03	0603287E	SPACE PROGRAMS AND TECHNOLOGY.....	Volume 1 - 231
56	03	0603467E	DARPA ADVANCED TECHNOLOGY DEVELOPMENT.....	Volume 1 - 235
57	03	0603468E	ADVANCED COMPLEX SYSTEMS.....	Volume 1 - 257
58	03	0603469E	ADVANCED ENABLING TECHNOLOGIES.....	Volume 1 - 273
68	03	0603739E	ADVANCED ELECTRONICS TECHNOLOGIES.....	Volume 1 - 291
69	03	0603760E	COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS.....	Volume 1 - 297
70	03	0603766E	NETWORK-CENTRIC WARFARE TECHNOLOGY.....	Volume 1 - 307
71	03	0603767E	SENSOR TECHNOLOGY.....	Volume 1 - 317

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Appropriation 0400: Research, Development, Test & Evaluation, Defense-Wide

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190	06	0605502E	SMALL BUSINESS INNOVATION RESEARCH.....	Volume 1 - 329
199	06	0605898E	MANAGEMENT HQ - R&D.....	Volume 1 - 333

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Program Element Table of Contents (Alphabetically by Program Element Title)

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ADVANCED AEROSPACE SYSTEMS	0603286E	46	03.....	Volume 1 - 227
ADVANCED COMPLEX SYSTEMS	0603468E	57	03.....	Volume 1 - 257
ADVANCED ELECTRONICS TECHNOLOGIES	0603739E	68	03.....	Volume 1 - 291
ADVANCED ENABLING TECHNOLOGIES	0603469E	58	03.....	Volume 1 - 273
BASIC OPERATIONAL MEDICAL SCIENCE	0601117E	5	01.....	Volume 1 - 15
BIOMEDICAL TECHNOLOGY	0602115E	16	02.....	Volume 1 - 165
COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS	0603760E	69	03.....	Volume 1 - 297
DARPA ADVANCED TECHNOLOGY DEVELOPMENT	0603467E	56	03.....	Volume 1 - 235
DEFENSE RESEARCH SCIENCES	0601101E	2	01.....	Volume 1 - 1
EFFECTS	0602026E	15	02.....	Volume 1 - 159
ELECTRONICS TECHNOLOGY	0602716E	28	02.....	Volume 1 - 209
EMERGING OPPORTUNITIES	0601122E	7	01.....	Volume 1 - 19
INFORMATION & COMMUNICATIONS TECHNOLOGY	0602303E	21	02.....	Volume 1 - 171
MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	0602025E	14	02.....	Volume 1 - 89
MANAGEMENT HQ - R&D	0605898E	199	06.....	Volume 1 - 333
MATERIALS AND BIOLOGICAL TECHNOLOGY	0602715E	27	02.....	Volume 1 - 197

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Program Element Title	Program Element Number	Line #	BA	Page
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NETWORK-CENTRIC WARFARE TECHNOLOGY	0603766E	70	03.....	Volume 1 - 307
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SMALL BUSINESS INNOVATION RESEARCH	0605502E	190	06.....	Volume 1 - 329
SPACE PROGRAMS AND TECHNOLOGY	0603287E	47	03.....	Volume 1 - 231
TACTICAL TECHNOLOGY	0602702E	26	02.....	Volume 1 - 187
WARFIGHTING PERFORMANCE	0602024E	13	02.....	Volume 1 - 63

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Executive Summary for Research, Development, Test and Evaluation, Defense-Wide Defense Advanced Research Projects Agency (DARPA)

Table 1: Funding Overview

	<u>FY 2025 (\$K)</u>	<u>FY 2026 (\$K)</u>	<u>FY 2027 (\$K)</u>
Discretionary	\$4,408,991	\$4,321,507	\$5,039,370
Mandatory	\$0	\$467,000	\$0
Total	\$4,408,991	\$4,788,507	\$5,039,370

Table 2: Detailed Mandatory Breakout

<u>BLI</u>	<u>Reconciliation Bin Title</u>	<u>FY 2025 (\$K)</u>	<u>FY 2026 (\$K)</u>	<u>FY 2027 (\$K)</u>
0601122E	Quality of Life	\$0	\$12,500	\$0
0602023E	Low-Cost Weapons	\$0	\$7,000	\$0
0602024E	Quality of Life	\$0	\$87,500	\$0
0602025E	Low-Cost Weapons	\$0	\$250,000	\$0
0603467E	Low-Cost Weapons	\$0	\$90,000	\$0
0603469E	Cyber Security	\$0	\$20,000	\$0
Mandatory Total		\$0	\$467,000	\$0

MANDATORY FUNDING JUSTIFICATION: The FY 2027 request for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$5,039,370 thousand of discretionary and \$0 thousand of mandatory for a total of \$5,039,370.

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$4,321,507 thousand of discretionary and \$467,000 thousand of mandatory (reconciliation) for a total of \$4,788,507.

The FY 2025 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$4,408,991 thousand of discretionary and \$0 thousand of mandatory (reconciliation) for a total of \$4,408,991.

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Defense Advanced Research Projects Agency

FY 2027 President's Budget

Approved PE Crosswalk¹

Current Program Element (FY 2025)				New Program Element (FY 2026 and FY 2027)		
Act	Line #	Program Element Number	Program Element Title	Line #	Program Element Number	Program Element Title
01	2	0601101E	DEFENSE RESEARCH SCIENCES	7	0601122E	EMERGING OPPORTUNITIES
	5	0601117E	BASIC OPERATIONAL MEDICAL SCIENCE			
02	26	0602702E	TACTICAL TECHNOLOGY	12	0602023E	ACCESS AND AWARENESS
	16	0602115E	BIOMEDICAL TECHNOLOGY	13	0602024E	WARFIGHTING PERFORMANCE
	27	0602715E	MATERIALS AND BIOLOGICAL TECHNOLOGY			
			MATERIALS AND BIOLOGICAL TECHNOLOGY	14	0602025E	MAKING, MAINTAINING, SUPPLY CHAIN and LOGISTICS
	28	0602716E	ELECTRONICS TECHNOLOGY			
	21	0602303E	INFORMATION & COMMUNICATIONS TECHNOLOGY			
	28	0602716E	ELECTRONICS TECHNOLOGY	15	0602026E	EFFECTS
03	69	0603760E	COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS	56	0603467E	DARPA ADVANCED TECHNOLOGY DEVELOPMENT
	70	0603766E	NETWORK-CENTRIC WARFARE TECHNOLOGY			
	71	0603767E	SENSOR TECHNOLOGY			
	46	0603286E	ADVANCED AEROSPACE SYSTEMS	57	0603468E	ADVANCED COMPLEX SYSTEMS
	47	0603287E	SPACE PROGRAMS AND TECHNOLOGY			
	68	0603739E	ADVANCED ELECTRONICS TECHNOLOGIES	58	0603469E	ADVANCED ENABLING TECHNOLOGIES
	69	0603760E	COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS			
06	175	0605001E	MISSION SUPPORT	175	0605001E	MISSION SUPPORT AND MGMT HQ
	199	0605898E	MANAGEMENT HQ - R&D	199	0605898E	MANAGEMENT HQ - R&D
	190	0605502E	SMALL BUSINESS INNOVATION RESEARCH	190	0605502E	SMALL BUSINESS INNOVATION RESEARCH

Notes:

1. One of the key recommendations of the Commission on Planning, Programming, Budgeting, and Execution (PPBE) Reform, established by section 1004 of the National Defense Authorization Act for Fiscal Year 2022, is to review and consolidate budget line items (BLI). The consolidation would streamline the current resourcing and execution processes, retain and increase transparency for Congress, end unnecessary duplication in the existing budget structures, and eliminate redundancies. The consolidation would also increase the Department of War's ability to adjust rapidly to changing circumstances in the year of execution and inject innovation or adopt new technology to address changing threats. This crosswalk details DARPA's implementation of the Commission's recommendation, and subsequent review and implementation guidance issued by the Department of War (DoW), and reflects initial consolidation of the Agency's Program Element (PE) structure from 16 to 11 PEs.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>	R-1 Program Element (Number/Name) PE 0601101E / <i>DEFENSE RESEARCH SCIENCES</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	279.971	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
CCS-02: <i>MATH AND COMPUTER SCIENCES</i>	0.000	180.734	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
ES-01: <i>ELECTRONIC SCIENCES</i>	0.000	2.824	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
ES-02: <i>BEYOND SCALING SCIENCES</i>	0.000	42.970	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MS-01: <i>MATERIALS SCIENCES</i>	0.000	53.443	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Basic Research associated with the Defense Research Sciences Program that provides the technical foundation for long-term National Security enhancement through the discovery of new phenomena and the exploration of the potential of such phenomena for Department of War applications. This PE supports the scientific study and experimentation that is the basis for more advanced knowledge and understanding in electronic, mathematical, computer, and materials sciences. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Math and Computer Sciences project supports scientific study and experimentation on new mathematical and computational algorithms, models, and mechanisms in support of long-term national security objectives. Modern analytic and information technologies enable important new military capabilities and drive the productivity gains essential to U.S. economic competitiveness. Conversely, new classes of threats, in particular threats that operate in or through the cyber and information domain, put military systems, critical infrastructure, and the civilian economy at risk. This project aims to magnify these opportunities and mitigate these threats by leveraging emerging mathematical and computational capabilities including artificial intelligence (AI), computational social science, machine learning and reasoning, data science, quantum science, complex systems modeling and simulation, and theories of computation and programming. The basic research conducted under the Math and Computer Sciences project will produce breakthroughs that enable new capabilities for national and homeland security.

The Electronic Sciences project was for basic exploration of electronic and optoelectronic devices, circuits, and processing concepts to meet the military's need for near real-time information gathering, transmission, and processing. In seeking to continue the phenomenal advancement in microelectronics innovation that has characterized the last few decades, the project sought to provide the DoW with new, improved, or potentially revolutionary device options for accomplishing these critical functions. The resulting technologies will help maintain knowledge of the adversary, communicate decisions based on that knowledge, and substantially improve the cost and performance of military systems. Research areas included analog, mixed signal, and photonic circuitry for communications and other applications; alternative computer architectures; and reducing the size of Electromagnetic (EM) and sensing systems. Other research supported field-portable electronics with reduced power requirements, and new approaches to nanometer-scale structures, molecules, and devices.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide I</i> BA 1: <i>Basic Research</i>	R-1 Program Element (Number/Name) PE 0601101E / <i>DEFENSE RESEARCH SCIENCES</i>
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The Beyond Scaling Sciences project supports investigations into materials, devices, and architectures to provide disruptive improvements in electronics performance that can be realized by techniques other than transistor scaling. Examples include circuit specialization, non-volatile memory devices that combine computation and memory, and new automated design tools using machine learning. Additionally, new design and manufacturing advances for three-dimensional microelectronics integration will underpin continued performance improvements as silicon transistor scaling plateaus.

The Materials Sciences project provides the fundamental research that underpins the design, development, assembly, and optimization of advanced materials, devices, and systems for DoW applications in areas such as robust diagnostics and therapeutics, novel energetic materials, and complex hybrid systems.

Beginning in FY 2026, efforts in this PE will be funded in PE 0601122E, Emerging Opportunities.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	293.145	0.000	0.000	-	0.000
Current President's Budget	279.971	0.000	0.000	-	0.000
Total Adjustments	-13.174	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	-3.680	0.000			
• SBIR/STTR Transfer	-9.494	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES				Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
CCS-02: MATH AND COMPUTER SCIENCES	0.000	180.734	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Math and Computer Sciences project supports scientific study and experimentation on new mathematical and computational algorithms, models, and mechanisms in support of long-term national security objectives. Modern analytic and information technologies enable important new military capabilities and drive the productivity gains essential to U.S. economic competitiveness. Conversely, new classes of threats, particularly threats that operate in or through the cyber and information domain, put military systems, critical infrastructure, and the civilian economy at risk. This project aims to magnify these opportunities and mitigate these threats by leveraging emerging mathematical and computational capabilities, including artificial intelligence, computational social science, machine learning and reasoning, data science, quantum science, complex systems modeling and simulation, and theories of computation and programming. The basic research conducted under the Math and Computer Sciences project will produce breakthroughs that enable new capabilities for national and homeland security.

Beginning in FY 2026, efforts in this Project will be funded in PE 0601122E, Project EMR-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Advanced Research Concepts (ARC) Description: The Advanced Research Concepts (ARC) program will rapidly explore revolutionary, nascent high-risk/high-reward ideas to accelerate the discovery of paradigm-shifting technologies. The program will focus on answering fundamental research questions to investigate the boundaries of what is possible and push the state of the art of science and technology. By exploring a high volume of ideas through targeted and limited-scope investments, this program will identify promising ideas early and assess the impact of further investment on problems of importance to the Department of War. The program will cover a broad spectrum of science and engineering disciplines that have applications for national security research. Individual efforts will explore numerous emerging technologies in various thrusts including, but not limited to, artificial intelligence and machine learning, mathematics, biology, chemistry, quantum technologies, social science, material science, electronics, sensing and measurement, and modeling. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.	30.000	-	-
Title: Advanced Tools for Modeling and Simulation Description: The Advanced Tools for Modeling and Simulation thrust is developing foundational mathematical, computational, and multi-physics theories, approaches, and tools to better represent, quantify, and model complex Department of War (DoW) systems from multimodal data analysis through part/system design and fabrication. One focus area of this thrust is developing a unified mathematical framework to enable better visualization and analysis of massive, complex data sets. Rigorous mathematical	22.924	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
theories are also being developed to address uncertainty in the modeling and design of complex multi-scale physical and engineering systems, incorporating capabilities to handle noisy data and model uncertainty beyond the scope of previously-existing capabilities. Other work in this thrust focuses on developing the mathematical and computational tools required to generate and better manage the enormous complexity of design, ultimately allowing designers to more easily discover non-intuitive (yet realizable) designs that fully leverage new materials and advanced manufacturing approaches now available. Outcomes from this thrust seek to improve the speed and accuracy of modeling and simulation, as well as enable management of complexity across DoW devices, parts, and systems. Another focus area of this thrust is the development of quantum-assisted approaches for predicting behavior and non-intuitive failure pathways for complex, dynamic physical systems.				
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Intrinsic Cognitive Security (ICS) Description: The Intrinsic Cognitive Security (ICS) program extends computational formal methods with cognitive guarantees and models to protect mixed reality (MR) users from cognitive attack. MR systems integrate virtual and real worlds in real time, and, given their increasing prevalence in military missions, they must be resilient against a wide variety of adversary attacks that may seek to exploit the intimate connection between users and MR equipment. Formal methods are rigorous, mathematics-based approaches that provide guarantees about computer-based systems, such as the absence of exploitable weaknesses. Cognitive models represent aspects of human perception, action, memory, and reasoning. The ICS program extends formal methods by explicitly creating and analyzing cognitive models as part of MR system development to protect the user from cognitive attacks. ICS will create cognitive guarantees that address MR vulnerabilities and are expressed in languages suitable for proofs from models; build cognitive models for reasoning about users of MR systems with sufficient fidelity relative to human behaviors; and evaluate model, proof, and guarantee validity using automated reasoning tools and prototype implementations of proved guarantees. The cognitive protections being developed under ICS will prevent exploitation of MR systems by adversaries.		22.000	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Enhanced SBOM for Optimized Software Sustainment (E-BOSS) Description: The Enhanced SBOM for Optimized Software Sustainment (E-BOSS) program is creating enhanced software bill of materials (eSBOM) technologies with new types of rich metadata and is developing cyber reasoning algorithms and tools that leverage eSBOMs to defend against potential flaws during the software development process as well as triage and remediate flaws found in operation. The global impacts of flawed software deployed at scale, such as the Log4Shell vulnerability found in Log4j cloud and web app deployments, where mitigations took from one week to months, motivated new SBOM requirements in Executive Order 14028. E-BOSS is developing software technologies integrated with modern software build chains to enable		19.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
rapid triage and remediation of vulnerabilities at enterprise scale. The enhanced metadata incorporated in the eSBOMs will enable trace back of discovered flaw evidence to derive the vulnerability triggers, starting from a crash and tracing back through complex inter-component interactions, transfers, and transformations. Successful E-BOSS technologies will enable cyber-reasoning for improved remediation and sustainment of large-scale software systems.				
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Young Faculty Award (YFA) Description: The goal of the Young Faculty Award (YFA) program is to encourage junior faculty at universities and their equivalent at non-profit science and technology research institutions to participate in sponsored research programs that will augment capabilities for future defense systems. This program focuses on cutting-edge technologies for greatly enhancing microsystems technologies, biological technologies, and defense sciences. The long-term goal for this program is to develop the next generation of scientists, engineers, and mathematicians in key disciplines who will focus a significant portion of their careers on Department of War (DoW) and national security issues. The aim is for YFA recipients to receive deep interactions with DARPA program managers, programs, performers, and the user community. Current activities include research in fifteen topic areas, including Machine Learning and Many Body Physics, as well as Wideband Transmitter-Antenna Interfaces and Multi-Scale Models of Infectious Disease Dynamics.		18.015	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: In The Moment (ITM) Description: In the Moment (ITM) supports the development of algorithms that are trusted to make decisions independently in difficult domains. Military operations, such as mass casualty triage and disaster relief, require complex and rapid decision-making in dynamic situations where there is often no human consensus and no ground truth. The ITM program generates a quantitative alignment framework for trusted human decision-makers and algorithms. ITM investigates these decision-making problems in the context of medical triage in military environments. If the ITM program is successful, the alignment framework will support the development of algorithms that can be tuned to align with specific, trusted humans.		15.000	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Translating All C To Rust (TRACTOR) Description: The Translating All C To Rust (TRACTOR) program aims to automate the process of transitioning legacy C to Rust, with the same quality and style that a skilled Rust developer would employ. At present, buffer overflow vulnerabilities and		10.700	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES		Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
other related memory safety software flaws allow an attacker to inject messages that hijack control of a computer. TRACTOR combines programming language and machine learning techniques, including large language models, to eliminate memory safety vulnerabilities. TRACTOR researchers are building C to Rust translation systems for single-threaded application translation, and, subsequently, multi-threaded application translation. TRACTOR is intended to enable the wholesale translation of C programs to Rust, thereby strengthening security in mission-critical and other high-priority software-reliant systems.					
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.					
Title: Scientific Feasibility (SciFy) Description: The Scientific Feasibility (SciFy) program is developing computational methods to measure the feasibility of claims to enable accurate assessments of scientific content. Automated scientific content generation, via rapidly improving large pre-trained models, has the potential to disrupt the U.S. technology base in times of crisis and to distort the global race for technological dominance in key areas. Similarly, false capability claims can have significant negative implications for national security and international relations. To address these threats, SciFy focuses on methods for assessing the scientific feasibility of claims by using automated reasoning to decompose claims into constituent, verifiable parts. Assessing each component involves referencing existing technological advancements, foundational scientific principles, data, software, models, simulation results, and industry standards or benchmarks. SciFy will create methods that go beyond automated fact-checking by addressing complex component interactions and operational constraints, and evaluating logical consistency, system integration, and compatibility considerations. Success with the SciFy program will enable the U.S. to reliably determine whether claimed scientific and technological capabilities are practical and realistic when considered as a whole, even when theoretically possible in parts. The SciFy program is also funded in PE 0602303E, Project IT-04.			9.000	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01 and PE 0602025E, Project MSL-05.					
Title: Foundational Artificial Intelligence (AI) Science Description: The Foundational Artificial Intelligence (AI) Science thrust is developing a fundamental scientific basis for understanding and quantifying performance expectations and limits of AI technologies. Current AI technologies are challenged in handling uncertainty and incompleteness of training protocols and data. This has inhibited integration of AI technology into a number of important Department of War (DoW) applications. To address these limitations, the Foundational AI Science thrust focuses on the development of new learning architectures that enhance AI systems' ability to handle uncertainty, reduce vulnerabilities, and improve robustness for Department of War AI systems. One focus area of this thrust is the ability to detect and accommodate novelty - i.e., violations of implicit or explicit assumptions - in AI applications. Another focus area is the development of a model framework for quantifying performance expectations and limits of AI systems as trusted human partners and			7.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
collaborators. A third focus area is the development of new tools and methodologies that enable AI approaches for accelerated scientific discovery. The technology advances achieved under the Foundational AI Science thrust will ultimately remove technical barriers to exploiting AI technologies for scientific discovery, human-AI collaboration, accommodating novelty, and other DoW relevant applications.				
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Exponentiating Mathematics (expMath) Description: The Exponentiating Mathematics (expMath) program is developing and applying artificial intelligence (AI) techniques to enable breakthrough advances in national security-related mathematical fields. At present, important problems in mathematics take decades or centuries to resolve, but recent advances in generative AI (genAI) and large language models (LLMs) offer the promise of accelerating advances in mathematics. expMath aims to use AI to automate the discovery of new mathematical knowledge. Creating an AI with these mathematical capabilities will require advances along two fronts: decomposition/abstraction and formalization/informalization. expMath focuses on mathematical domains of particular importance for national security. If successful, expMath will provide the basis for potential leap-ahead mathematical capabilities for the U.S.		3.510	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01.				
Title: Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) Description: The Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) project aims to capture and solve selected artificial intelligence (AI) research challenges to accelerate the development and fielding of frontier AI for national security. At present, a relatively small number of frontier AI companies drive most innovation in AI with a strong focus on commercialization and return-on-investment. AI FORGE seeks to make national security considerations more prominent in the generative AI research and development (R&D) community. These considerations primarily involve the life- and mission-critical nature of national security generative AI use cases, which span information-centric intelligence analysis to grounded planning of logistics and tactical operations to embodiment in autonomous systems, and the highly adversarial nature of battlefield operational environments. For such use cases and environments, matters of assurance, predictability, reliability, resilience, and transparency are of heightened importance. AI FORGE is conducting high-risk, high-reward efforts that address cross-cutting challenges that arise when using frontier AI for national security applications, systems, and missions. AI FORGE combines the talents of universities with resources from frontier AI companies to create an R&D ecosystem focused on national security.		3.500	-	-
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-01 and PE 0602025E, Project MSL-05.				
Title: Knowledge Management at Scale		11.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) CCS-02 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Description: The Knowledge Management at Scale thrust focused on the development of knowledge management tools that could efficiently capture, analyze and reason with expertise, experience and data. The technology development under this thrust helped to address a critical need for assimilating and preserving critical national security knowledge and expertise that is currently being lost due to attrition and other factors. Specific objectives included the following: 1) effective, trustworthy, and easily accepted approaches for domain agnostic knowledge acquisition at scale; 2) capabilities to identify correlations or hidden factors relating to knowledge acquired from different sources; and 3) techniques for incorporating domain models and other data sources for more extensive reasoning-based applications. Example approaches towards achieving these objectives included identifying and demonstrating robust knowledge acquisition tools, exploiting artificial intelligence techniques to establish a framework for knowledge analysis and causal reasoning, and developing automation tools that effectively elicit and impart acquired knowledge via user friendly interfaces.			
Title: Environment-driven Conceptual Learning (ECOLE) Description: The Environment-driven Conceptual Learning (ECOLE) program created artificial intelligence (AI) agents capable of continually learning from linguistic and visual input to enable human-machine collaborative analysis of image, video, and multimedia documents during time-sensitive, mission-critical Department of War analytic tasks, where reliability and robustness are essential. ECOLE transformed current machine learning approaches by developing algorithms that can identify, represent, and ground the attributes that form the symbolic and contextual model for a particular object or activity through interactive learning with a human analyst. Knowledge of attributes and affordances, learned dynamically from data encountered within an analytic workflow, enables joint reasoning with a human partner. The acquired knowledge also enables the machine to recognize when an observed object or activity is novel rather than misclassifying the newly observed object or action as a member of a previously learned class and to readily learn a new symbolic representation through interaction with its human partner.		9.085	-
Accomplishments/Planned Programs Subtotals		180.734	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES				Project (Number/Name) ES-01 / ELECTRONIC SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ES-01: ELECTRONIC SCIENCES	0.000	2.824	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Electronic Sciences project is for basic exploration of electronic and optoelectronic devices, circuits, and processing concepts to meet the military's need for near real-time information gathering, transmission, and processing. In seeking to continue the phenomenal advancement in microelectronics innovation that has characterized the last few decades, the project will provide DoW with new, improved, or potentially revolutionary device options for accomplishing these critical functions. The resulting technologies will help maintain knowledge of the enemy, communicate decisions based on that knowledge, and substantially improve the cost and performance of military systems. Research areas include analog, mixed signal, and photonic circuitry for communications and other applications; alternative computer architectures; and reducing the size of Electromagnetic (EM) and sensing systems. Other research could support field-portable electronics with reduced power requirements, and new approaches to nanometer-scale structures, molecules, and devices.

B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Atomic-Photonic Integration (A-PhI) Description: The Atomic-Photonic Integration (A-PhI) program showed a path to reducing the size, weight, and power of atomic clocks and gyroscopes for position, navigation, and timing (PNT) applications through the development of integrated photonics. Specifically, A-PhI demonstrated that a compact photonic integrated chip could replace the optical assembly for trapped atomic gyroscopes and clocks without degrading the performance of the device. PNT is a critical resource for all DoW missions such as communications, navigation, reconnaissance, and electronic warfare. While PNT needs are usually met by using the global positioning system (GPS), GPS signals are vulnerable to disruption and a fallback from GPS is essential. In the absence of GPS, tactical-grade clocks and tactical/navigation grade inertial measurement units (IMUs) currently can provide GPS-like accuracy only for the short term, and longer-term GPS-independent strategies are highly desirable.	2.824	-	-
Accomplishments/Planned Programs Subtotals	2.824	-	-

C. Other Program Funding Summary (\$ in Millions)
N/A

Remarks

D. Acquisition Strategy
N/A

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES				Project (Number/Name) ES-02 / BEYOND SCALING SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ES-02: BEYOND SCALING SCIENCES	0.000	42.970	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Beyond Scaling Sciences project supports investigations into materials, devices, and architectures to provide disruptive improvements in electronics performance that can be realized by techniques other than transistor scaling. Examples include circuit specialization, non-volatile memory devices that combine computation and memory, and new automated design tools using machine learning. Additionally, new design and manufacturing advances for three-dimensional microelectronics integration will underpin continued performance improvements as silicon transistor scaling plateaus.

Beginning in FY 2026, efforts in this Project will be funded in PE 0601122E, Project EMR-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Joint University Microelectronics Program 2.0 (JUMP 2.0) Description: The Joint University Microelectronics Program 2.0 (JUMP 2.0) program is developing and demonstrating innovative next-generation microelectronics technologies through a public-private consortium with universities, the defense industrial base, and the semiconductor industry. The JUMP 2.0 program addresses the grand technical challenges of our increasingly connected world that must be overcome including: the need for innovation in analog hardware, increasing demand for more memory and data storage, the imbalance between data generation and communication capacity, the emerging security vulnerabilities in highly interconnected Artificial Intelligence systems, and the unsustainable growth in energy demands for computing. Therefore, the JUMP 2.0 program sponsors academic research teams focused on related key technology areas that will not only impact future defense and national security capabilities but also strengthen U.S. leadership in information and communication technology. The JUMP 2.0 program will push fundamental technology research themes in cognition, communications, sensing to action, computing and processing, memory and storage, integration and packaging, and high-performance energy efficient devices to enable key disruptive advances in microelectronic technology. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-02.	28.000	-	-
Title: Compartmentalization and Privilege Management (CPM) Description: The Compartmentalization and Privilege Management (CPM) program is developing new system frameworks, architectures, and tooling to provide fine-grained, least-privileged compartmentalization that enables prevention and containment of cyber attacks. Today's information systems are structured around a monolithic core, the kernel, that operates within a single protection domain at a single high privilege-level. A monolithic kernel contains many separate components without protection	12.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) ES-02 / BEYOND SCALING SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
boundaries between them. A single compromise anywhere in the system allows attackers effectively unlimited access through an extended sequence of exploits, privilege escalation, and lateral motion. CPM is developing technologies and tools to compartmentalize large, legacy software systems automatically and designing processor architectures and system software to enforce a compartment and privilege-level regime. CPM tools and architectures will prevent initial penetrations from propagating into successful cyber attacks. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-02.			
Title: Advanced Microsystems Manufacturing Technologies Studies and Concepts Description: The Advanced Microsystems Manufacturing Technologies Studies and Concepts examined the infrastructure needed to bring sustainability to advanced microsystems. Studies examined technologies that could reduce the cost and time-to-market footprint of microsystems manufacturing as well as foundational ecosystem enhancements that could provide far-reaching impact. This program also examined ways to take advantage of commercial scaling while building in superior capabilities for defense systems through design and integration. These studies led to the development of new programs in these areas.		2.970	-
Accomplishments/Planned Programs Subtotals		42.970	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES				Project (Number/Name) MS-01 / MATERIALS SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MS-01: MATERIALS SCIENCES	0.000	53.443	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Materials Sciences project provides the fundamental research that underpins the design, development, assembly, and optimization of advanced materials, devices, and systems for DoW applications in areas such as robust diagnostics and therapeutics, novel energetic materials, and complex hybrid systems.

Beginning in FY 2026, efforts in this Project will be funded in PE 0601122E, Project EMR-03.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Fundamental Limits Description: Understanding the Fundamental Limits (i.e., achievable boundaries) of scientific principles, processes and technologies is critical to better anticipate technological surprise for our adversaries and ourselves. This thrust explores boundaries across fields such as physics, chemistry, mathematics, biology, and engineering to address critical questions for national security, addressing foundational theory and approaches that include, for example, the fundamental limitations of optical technologies, potential implications for basic biology on national security, and the ability for modeling and simulation to provide a better understanding of complex systems. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-03.	37.033	-	-
Title: Molecular Systems and Materials Assembly Description: The Molecular Systems and Materials Assembly thrust is exploring new approaches for the synthesis, assembly, characterization and application of molecules and materials for a variety of DoW applications from the atomic to the product scale. Ultimately, materials and methods developed in this thrust will support a wide range of DoW applications that will leverage novel materials to extend the range, duration, and capabilities of DoW systems and the warfighter. Through control of the arrangement, interactions, and assembly of atoms and molecules, new materials and manufacturing processes are being developed to address long-standing challenges in supply chains, logistics, and sustainment while simultaneously enhancing the warfighter's capabilities on the battlefield. Efforts in this thrust range from fundamental science to better understand the chemistry and physics related to each application, to developing means to utilize such capabilities in future test systems and prototype devices. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-03.	16.410	-	-
Accomplishments/Planned Programs Subtotals	53.443	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601101E / DEFENSE RESEARCH SCI ENCES	Project (Number/Name) MS-01 / MATERIALS SCIENCES
C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity					R-1 Program Element (Number/Name)							
0400: Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research					PE 0601117E / BASIC OPERATIONAL MEDICAL SCIENCE							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	86.255	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MED-01: BASIC OPERATIONAL MEDICAL SCIENCE	0.000	86.255	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Basic Operational Medical Science Program Element (PE) will explore and develop basic research in medical-related information and technology leading to fundamental discoveries, tools, and applications critical to overcoming DoW challenges. This PE will address the Department's identified need for warfighter medical care related to prevention and treatment of infectious disease, real-time healthcare interventions of acute and chronic illness and injury, and interventions for improved warfighter resilience and performance against operational stressors. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA-funded technologies take root in the U.S. and provide new capabilities for national defense.												
Beginning in FY 2026, efforts in this PE will be funded in PE 0601122E, Emerging Opportunities.												
B. Program Change Summary (\$ in Millions)				FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total				
Previous President's Budget				89.143	0.000	0.000	-	0.000				
Current President's Budget				86.255	0.000	0.000	-	0.000				
Total Adjustments				-2.888	0.000	0.000	-	0.000				
• Congressional General Reductions				0.000	0.000							
• Congressional Directed Reductions				0.000	0.000							
• Congressional Rescissions				0.000	0.000							
• Congressional Adds				0.000	0.000							
• Congressional Directed Transfers				0.000	0.000							
• Reprogrammings				0.000	0.000							
• SBIR/STTR Transfer				-2.888	0.000							
Change Summary Explanation												
FY 2025: Decrease reflects SBIR/STTR transfer.												
FY 2026: N/A												
FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.												
C. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Synthetic Hemo-technologies to Locate & Disinfect (SHIELD)									16.052	-	-	

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>		R-1 Program Element (Number/Name) PE 0601117E / <i>BASIC OPERATIONAL MEDICAL SCIENCE</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: Bloodstream infections (BSI) are a significant source of morbidity in service members who sustain combat-related injuries. Trauma temporarily degrades the efficacy of the host immune system, thereby increasing the risk of life-threatening opportunistic infections from fungi and bacteria that enter into the blood. If unchecked, bloodborne fungi and bacteria lead to debilitating conditions such as invasive fungal infections, sepsis, and shock. The Synthetic Hemo-technologies to Locate and Disinfect (SHIELD) program is developing a systems-level approach to prevent BSI in warfighters who suffer trauma from burn or blast. Preventative medical countermeasures circulating in the blood will be developed to bind infectious particles in the blood early and label pathogens for clearance and deliver drugs to destroy pathogens and/or restore healthy physiology. Ultimately, this program will develop novel technologies that will protect service members from morbidity and mortality associated with BSI. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.				
Title: Anesthetics for Battlefield Care (ABC) Description: The Anesthetics for Battlefield Care (ABC) program aims to produce safe, battlefield-ready anesthetics to reduce the trauma associated with injury and improve combat casualty outcomes. Current therapeutics that enable life-saving interventions and wound stabilization must be used in hospitals or highly monitored settings due to their lack of safety. Prolonged peer or near-peer conflict could severely impact medical evacuation times, resulting in extended time before patients reach a hospital. The ABC program seeks to uncover mechanisms of anesthesia at multiple biological levels ranging from the molecular to the organismal. Novel treatments developed under the program will exhibit the desirable properties of anesthetics, including calming effects, loss of sensation and loss of consciousness but will have vastly improved safety profiles, making them usable in the field by warfighters with minimal medical training. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.		15.706	-	-
Title: CoasterChase Description: DoW warfighters are often required to perform complex tasks in a high-stress situation. Warfighters under stress can make poor decisions with immediate and/or lasting consequences that may also contribute to long-term trauma. The CoasterChase program seeks to evaluate host response to stimulation of the digestive tract, in ways that will explore stress response to improve decision making and reduction of long-term trauma response with a combined sensing and stimulation ingestible device. The program aims to understand behavioral and physiological biomarkers of stress and inform device design parameters. Ultimately, the technology will enable improved memory retention, reduced effects of post-traumatic stress disorder (PTSD), maintenance of immune defense, and regulation of metabolism in austere domains.		8.105	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>		R-1 Program Element (Number/Name) PE 0601117E / <i>BASIC OPERATIONAL MEDICAL SCIENCE</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.				
Title: Simulating Microbial Systems (SMS) Description: The Simulating Microbial Systems (SMS) program is developing the general capability to simulate microbial cell function and behavior to prevent strategic surprise. SMS is investigating combinations of computational advances with data acquisition techniques to advance the state of the art of cellular modeling. Bringing together communities in cell modeling, computational techniques, and wet lab measurements, this program will build from partial physics-based models of a minimal synthetic cell to more complete and faithful simulations of a DoW-relevant organism. SMS will experimentally evaluate simulations via pressure tests that relate to both general cell simulation and tailored stakeholder use cases. SMS is postured to be a valuable tool for acceleration of medical countermeasure developments and biomanufacturing. Additionally, the program will provide a baseline for further simulation development and importantly will provide the DoW and its stakeholders the ability to run microbial simulations to prevent surprise. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.		18.424	-	-
Title: Combatting Anti-Microbial Resistant Pathogens Description: The Combatting Anti-Microbial Resistant Pathogens program is investigating fundamental methods for using preexisting host machinery as a technology to create medical countermeasures that degrade or deactivate pathogen targets. The DoW has long recognized the warfighter's outsized risk of exposure to biological threat agents and to infectious disease, including the increasing prevalence of antimicrobial-resistant (AMR) organisms that are ranked as a Tier 1 threat to the U.S. military. Similarly, the danger posed by bacterial biothreats persists with few countermeasures available. Key advances expected from this research include identifying methods to discover and develop new classes of chimeric therapeutics against AMR bacteria, bacterial biothreats, and other DoW-relevant diseases and threats. These approaches represent a significant departure from conventional therapeutics, which typically rely on a limited number of small molecules with a narrow set of targets and mechanism of action. Advances in this area may be used to help mitigate known, new, and emerging diseases that impact military readiness and pose a global health threat. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.		10.320	-	-
Title: Physiological Overmatch Description: Warfighters operate under extreme physiological conditions, sometimes with limited resources and manpower, and must acclimate quickly to changing operational needs. The Physiological Overmatch program is investigating innovative approaches to allow the warfighter to adapt rapidly to operational challenges during deployment by developing novel detection		9.131	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>	R-1 Program Element (Number/Name) PE 0601117E / <i>BASIC OPERATIONAL MEDICAL SCIENCE</i>			
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
and treatment systems. The program will initiate work in aiding the deployed soldier's ability to defend against biological pathogens, resist fatigue, combat sleep deprivation, and maintain a high capacity for operational performance. This program will seek to develop technology devices for in vivo release of therapies as needed by the warfighter and to understand the biological mechanisms of fatigue, which will enable improvements to warfighter health and operational performance. This approach represents a significant enhancement to warfighter performance by providing protection from impacts to operational readiness and provides information related to fatigue states and the ability to operate with prolonged sleep deprivation. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.				
Title: Objective Prediction of Team Effectiveness via Models of Performance Outcomes (OP TEMPO) Description: The Objective Prediction of Team Effectiveness via Models of Performance Outcomes (OP TEMPO) program seeks to advance technologies to drive efficiency and efficacy of military operator team preparation and expertise building. This program seeks to understand fundamental biological processes to support real-time physiological assessment, performance diagnostics, and objective prediction of warfighter and team proficiency, with the ultimate goal of improved DoW mission readiness and execution. Advances in this program will result in a significant enhancement to warfighter team performance by providing methods to determine teaming potential and actionable paths to optimal teaming. Beginning in FY 2026, this program will be funded in PE 0601122E, Project EMR-04.		8.517	-	-
Accomplishments/Planned Programs Subtotals		86.255	-	-
D. Other Program Funding Summary (\$ in Millions) N/A				
Remarks				
E. Acquisition Strategy N/A				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity	R-1 Program Element (Number/Name)											
0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>	PE 0601122E / <i>EMERGING OPPORTUNITIES</i>											
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	345.053	387.633	-	387.633	423.322	427.802	436.873	445.964	-	-
EMR-01: <i>MATH AND COMPUTER SCIENCES</i>	0.000	0.000	179.647	195.459	-	195.459	197.436	201.239	206.468	212.625	-	-
EMR-02: <i>DISRUPTIVE MICROTECHNOLOGIES SCIENCES</i>	0.000	0.000	42.697	13.453	-	13.453	3.597	0.000	0.000	0.000	-	-
EMR-03: <i>MATERIALS SCIENCES</i>	0.000	0.000	24.677	53.283	-	53.283	57.233	57.856	60.968	61.902	-	-
EMR-04: <i>BASIC OPERATIONAL MEDICAL SCIENCE</i>	0.000	0.000	77.908	96.554	-	96.554	130.959	133.084	138.540	140.540	-	-
EMR-07: <i>EMERGING OPPORTUNITIES SUPPORT</i>	0.000	0.000	20.124	28.884	-	28.884	34.097	35.623	30.897	30.897	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$345,053 thousand of discretionary and \$12,500 thousand of mandatory (reconciliation) for a total of \$357,553.

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Basic Research associated with the Emerging Opportunities Program that provides the technical foundation for long-term National Security enhancement through the discovery of new phenomena and the exploration of the potential of such phenomena for Department of War applications. This PE supports the scientific study and experimentation that is the basis for more advanced knowledge and understanding in quantum, electronic, mathematical, information, computer, materials, and medical sciences. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Math and Computer Sciences project supports scientific study and experimentation on new mathematical and computational algorithms, models, and mechanisms in support of long-term national security objectives. Modern information technologies, both classical and quantum, and analytic technologies, enable important new military capabilities and drive the productivity gains essential to U.S. economic competitiveness. Conversely, new classes of threats require the creation of fundamentally new mathematical and computational approaches to enhance the software-intensive systems upon which our society depends. The basic research conducted under the Math and Computer Sciences project will produce breakthroughs that enable new capabilities for national and homeland security.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES
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The Disruptive Microtechnologies Sciences project supports investigations into materials, devices, and architectures that will enable dramatic improvements versus the current state-of-the-art. This includes developing technology to enable fundamentally new ways of creating useful circuits to enable the next generation of microtechnologies. Areas of focus include photonic circuits that harness the power of light at the chip-scale, quantum circuits that take advantage of the unique properties that quantum offers versus classical computing and sensing, and organic circuits that integrate organic molecules into microtechnologies.

The Materials Sciences project provides the fundamental research that underpins the design, development, assembly, and optimization of advanced materials, devices, and systems for DoW applications in areas such as design and simulation of novel materials and structures, energetics, complex hybrid systems, and advanced metrology.

The Basic Operational Medical Science project will explore and develop basic research in medical-related information and technology leading to fundamental discoveries, tools, and applications critical to overcoming DoW challenges. This project will address the Department's identified warfighter medical care related to prevention and treatment of infectious disease, real-time healthcare interventions of acute and chronic illness and injury, and interventions for improved warfighter resilience and performance against operational stressors.

The Emerging Opportunities Support project contains management costs in support of DARPA functions and activities across the entire Emerging Opportunities PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0601101E, Defense Research Sciences, and PE 0601117E, Basic Operational Medical Science.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	360.456	0.000	-	0.000
Current President's Budget	0.000	345.053	387.633	-	387.633
Total Adjustments	0.000	-15.403	387.633	-	387.633
• Congressional General Reductions	0.000	-11.379			
• Congressional Directed Reductions	0.000	-4.024			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	387.633	-	387.633

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity	R-1 Program Element (Number/Name)	
0400: Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research	PE 0601122E / EMERGING OPPORTUNITIES	
Change Summary Explanation FY 2025: N/A FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation. FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES				Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EMR-01: MATH AND COMPUTER SCIENCES	0.000	0.000	179.647	195.459	-	195.459	197.436	201.239	206.468	212.625	-	-
A. Mission Description and Budget Item Justification												
The Math and Computer Sciences project supports scientific study and experimentation on new mathematical and computational algorithms, models, and mechanisms in support of long-term national security objectives. Modern information technologies, both classical and quantum, and analytic technologies, enable important new military capabilities and drive the productivity gains essential to U.S. economic competitiveness. Conversely, new classes of threats require the creation of fundamentally new mathematical and computational approaches to enhance the software-intensive systems upon which our society depends. The basic research conducted under the Math and Computer Sciences project will produce breakthroughs that enable new capabilities for national and homeland security.												
Prior to FY 2026, this project was funded in PE 0601101E, Project CCS-02.												
B. Accomplishments/Planned Programs (\$ in Millions)										FY 2025	FY 2026	FY 2027
Title: Foundational Artificial Intelligence (AI) Science										-	16.435	16.054
Description: The Foundational Artificial Intelligence (AI) Science thrust is developing a fundamental scientific basis for understanding and quantifying performance expectations and limits of AI technologies. Current AI technologies are challenged in handling uncertainty and incompleteness of training protocols and data. This has prevented the successful integration of AI technology into many transformative Department of War (DoW) applications. To address these limitations, the Foundational AI Science thrust focuses on the development of new learning architectures that enhance AI systems' ability to handle uncertainty, reduce vulnerabilities, and improve robustness for Department of War AI systems. One focus area of this thrust is the ability to detect and accommodate novelty - i.e., violations of implicit or explicit assumptions - in AI applications. Another focus area is the development of a model framework for quantifying performance expectations and limits of AI systems as trusted human partners and collaborators. A third focus area is the development of new tools and methodologies that enable AI approaches for accelerated scientific discovery. The technology advances achieved under the Foundational AI Science thrust will ultimately remove technical barriers to exploiting AI technologies for scientific discovery, human-AI collaboration, accommodating novelty, and other DoW relevant applications.												
Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02.												
FY 2026 Plans:												
- Develop methods to incorporate complex logical knowledge and reasoning into AI systems, enabling assurance in decision support.												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Begin creation of scalable, highly reusable, methods for high assurance in AI systems of systems. FY 2027 Plans: - Instantiate and extend methods for high assurance in AI systems of systems to military relevant domains. - Develop a software composition library in support of developing methods for high assurance in AI systems. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects minor program repricing.			
Title: Young Faculty Award (YFA) Description: The goal of the Young Faculty Award (YFA) program is to encourage junior faculty at universities and their equivalent at non-profit science and technology research institutions to participate in sponsored research programs that will augment capabilities for future defense systems. This program focuses on cutting-edge technologies for greatly enhancing microsystems technologies, biological technologies, and defense sciences. The long-term goal for this program is to develop the next generation of scientists, engineers, and mathematicians in key disciplines who will focus a significant portion of their careers on Department of War (DoW) and national security issues. The aim is for YFA recipients to receive deep interactions with DARPA program managers, programs, performers, and the user community. Current activities include research in fifteen topic areas including Machine Learning and Many Body Physics, as well as Wideband Transmitter-Antenna Interfaces and Multi-Scale Models of Infectious Disease Dynamics. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Award FY 2026 cooperative agreements for new two-year research efforts across YFA topic areas, establishing a new set of scientific approaches to solve current DoW challenges. - Continue FY 2025 research on new concepts for microsystem, biological, strategic, and tactical technologies; information innovation; and defense sciences by exercising second year funding and by providing continued mentorship by program managers. - Award Director's Fellowships for top FY 2024 participants to refine technology further and align to DoW needs. FY 2027 Plans: - Award FY 2027 cooperative agreements for new two-year research efforts across YFA topic areas, establishing a new set of scientific approaches to solve current DoW challenges. - Continue FY 2026 research on new concepts for microsystem, biological, strategic, and tactical technologies; information innovation; and defense sciences by exercising second year funding and by providing continued mentorship by program managers.		-	16.165
			16.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Award Director's Fellowships for top FY 2025 participants to refine technology further and align to DoW needs.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects minor program repricing.			
Title: Exponentiating Mathematics (expMath) Description: The Exponentiating Mathematics (expMath) program is developing and applying artificial intelligence (AI) techniques to enable breakthrough advances in national security-related mathematical fields. At present, important problems in mathematics take decades or centuries to resolve, but recent advances in generative AI (genAI) and large language models (LLMs) offer the promise of accelerating advances in mathematics. expMath aims to use AI to automate the discovery of new mathematical knowledge. Creating an AI with these mathematical capabilities will require advances along two fronts: decomposition/abstraction and formalization/informalization. expMath focuses on mathematical domains of particular importance for national security. If successful, expMath will provide the basis for potential leap-ahead mathematical capabilities for the U.S. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Develop automated techniques for translating natural language lemmas into formal language, executing proof steps when possible, and translating the derived proof back to natural language. - Formulate mathematical problems in domains relevant to national security in terms amenable to automated analysis by an AI system specialized to discover mathematical knowledge. - Explore architectural implementation options for an AI system specialized to perform the decomposition/abstraction and formalization/informalization functions required for the discovery of mathematical knowledge. FY 2027 Plans: - Integrate automated mathematical techniques in an AI system capable of performing complex mathematical derivations and proofs. - Perform initial demonstrations of mathematical knowledge discovery on mathematical problem domains/sub-domains of particular importance for national security. - Demonstrate math automation capabilities at a major mathematics and/or AI conference and to U.S. Government and Department of War stakeholders. FY 2026 to FY 2027 Increase/Decrease Statement:		-	13.899
			14.131

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Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects continued development of math automation techniques and additional efforts to implement and demonstrate math automation system.					
Title: Translating All C To Rust (TRACTOR) Description: The Translating All C To Rust (TRACTOR) program aims to automate the process of transitioning legacy C to Rust, with the same quality and style that a skilled Rust developer would employ. At present, buffer overflow vulnerabilities and other related memory safety software flaws allow an attacker to inject messages that hijack control of a computer. TRACTOR combines programming language and machine learning techniques, including large language models, to eliminate memory safety vulnerabilities. TRACTOR researchers are building C to Rust translation systems for single-threaded application translation, and, subsequently, multi-threaded application translation. TRACTOR is intended to enable the wholesale translation of C programs to Rust, thereby strengthening security in mission-critical and other high-priority software-reliant systems. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Research the foundations of programming language translation in multi-threaded settings using formal analysis tools. - Evaluate tools in terms of percentages of C functions translated to safe and correct Rust programs, and in terms of the performance of the translation tools themselves. FY 2027 Plans: - Build tools to assist verification of program behavior equivalence in multi-threaded settings. - Scale up tools to translate C programs to safe and correct Rust while ensuring performance of the translation tools. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from developing C-to-Rust automated translation technologies to assessing the performance of the translation tools.			-	10.807	9.803
Title: Intrinsic Cognitive Security (ICS) Description: The Intrinsic Cognitive Security (ICS) program is extending computational formal methods with cognitive guarantees and models to protect mixed reality (MR) users from cognitive attack. MR systems integrate virtual and real worlds in real time, and, given their increasing prevalence in military missions, they must be resilient against a wide variety of adversary attacks that may seek to exploit the intimate connection between users and MR equipment. Formal methods are rigorous, mathematics-based approaches that provide guarantees about computer-based systems, such as the absence of exploitable weaknesses. Cognitive models represent aspects of human perception, action, memory, and reasoning. The ICS program extends formal methods by explicitly creating and analyzing cognitive models as part of MR system development to protect the user from cognitive attacks.			-	10.253	7.539

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Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
ICS will create cognitive guarantees that address MR vulnerabilities and are expressed in languages suitable for proofs from models; build cognitive models for reasoning about users of MR systems with sufficient fidelity relative to human behaviors; and evaluate model, proof, and guarantee validity using automated reasoning tools and prototype implementations of proved guarantees. The cognitive protections being developed under ICS will prevent exploitation of MR systems by adversaries. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Construct models that support establishing specific guarantees using engineering-relevant elements of the cognitive stack and validate that models are sufficiently high-fidelity. - Validate that the guarantees can be implemented in real, lab-bench quality MR systems and evaluate the effectiveness of cognitive protections. FY 2027 Plans: - Harden MR protection technologies and transition to Department of War (DoW) system development activities and acquisition programs. - Facilitate adoption of MR protection technology by commercial communities to enable broad benefits across US Government, DoW, and civilian sectors. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from developing MR protection technologies to hardening and transition of the technologies.				
Title: Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) Description: The Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) project aims to capture and solve selected artificial intelligence (AI) research challenges to accelerate the development and fielding of frontier AI for national security. At present, a relatively small number of frontier AI companies drive most innovation in AI with a strong focus on commercialization and return-on-investment. AI FORGE seeks to make national security considerations more prominent in the generative AI research and development (R&D) community. These considerations primarily involve the life- and mission-critical nature of national security generative AI use cases, which span information-centric intelligence analysis to grounded planning of logistics and tactical operations to embodiment in autonomous systems, and the highly adversarial nature of battlefield operational environments. For such use cases and environments, challenges of assurance, predictability, reliability, resilience, and transparency are of heightened importance. AI FORGE is conducting high-risk, high-reward efforts that address cross-cutting challenges that arise when using frontier AI for national security applications, systems, and missions. AI FORGE combines the		-	4.784	6.567

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
talents of universities with resources from frontier AI companies to create an R&D ecosystem focused on national security. The AI FORGE program is also funded in PE 0602025E, Project MSL-05. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Finalize an acquisition strategy and management plan for a joint government-university-industry project that features R&D efforts that solve critical challenges associated with the use of frontier AI for national security. - Execute formal agreements with government and industry partners to establish processes for initiation, approval, and review of R&D efforts focused on the use of frontier AI for national security. FY 2027 Plans: - Execute collaborative R&D efforts that address critical technical challenges associated with the use of frontier AI for national security. - Coordinate R&D efforts with government activities seeking to use frontier AI for national security applications, systems, and missions. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from planning activities to basic research addressing fundamental issues of assurance, predictability, reliability, resilience, and transparency.				
Title: Scientific Feasibility (SciFy) Description: The Scientific Feasibility (SciFy) program is developing computational methods to measure the feasibility of claims to enable accurate assessments of scientific content. Automated scientific content generation, via rapidly improving large pre-trained models, has the potential to disrupt the U.S. technology base in times of crisis and to distort the global race for technological dominance in key areas. Similarly, false capability claims can have significant negative implications for national security and international relations. To address these threats, SciFy focuses on methods for assessing the scientific feasibility of claims by using automated reasoning to decompose claims into constituent, verifiable parts. Assessing each component involves referencing existing technological advancements, foundational scientific principles, data, software, models, simulation results, and industry standards or benchmarks. SciFy will create methods that go beyond automated fact-checking by addressing complex component interactions and operational constraints, and evaluating logical consistency, system integration, and compatibility considerations. Success with the SciFy program will enable the U.S. to reliably determine whether claimed scientific and technological capabilities are practical and realistic when considered as a whole, even when theoretically possible in parts. The SciFy program is also funded in PE 0602025E, Project MSL-05.		-	8.843	4.459

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02 and PE 0602303E, Project IT-04.				
FY 2026 Plans: - Demonstrate techniques for automated feasibility assessment across multiple scientific domains, such as materials science, artificial intelligence, quantum computing, and other domains and sub-domains. - Collaborate with Department of War and Intelligence Community transition partners to evaluate utility of the feasibility assessments and to refine techniques.				
FY 2027 Plans: - Collaborate with transition partners to demonstrate feasibility assessment tools in selected domains of interest.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from development of feasibility assessment techniques to demonstration and transition.				
Title: Logic Understanding for Complex Information Dissemination (LUCID)		-	9.221	23.673
Description: The Logic Understanding for Complex Information Dissemination (LUCID) program will create artificial intelligence (AI) architectures that demonstrate that proofs encode software system knowledge to enable users to understand and adapt systems in the field without needing to be experts in software architecture or formal logic. At present, system operators rely on static, often difficult to navigate documentation, checklists, and training materials to understand their systems. Opaque/inaccessible system behaviors and limited access to system architecture information can inhibit adaptation or problem solving in the field. LUCID's insight is that formal proofs encode software system knowledge in a compact symbolic form, often beyond what the specifications or requirements detail. This information can be used to problem-solve within the known operating envelope and to identify corner and edge cases that lead to potential new operating points outside the currently accepted envelope. LUCID will develop AI reasoning architectures that address semantic gaps, temporal reasoning, and compositional reasoning at scale to make proofs a trusted source of system information and adaptation/problem-solving advice. If successful, LUCID technology will enable warfighters to more effectively use software systems during high-stakes scenarios.				
FY 2026 Plans: - Begin the proof-of-concept to demonstrate that proofs encode software system knowledge to enhance system understanding and adaptation. - Initiate development of test scenarios and proof artifacts.				
FY 2027 Plans: - Demonstrate that proof explicitly encodes key semantic information that can be mapped to the operational context to enhance the operator's insight. - Develop AI architectures that enable users to enhance system understanding.				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Develop test scenarios and proof artifacts and perform test and evaluation within a realistic gaming environment. - Develop formal proof artifacts for use in case-specific software tool components. - Develop interactive, visual, or immersive approaches such as haptics and virtual/augmented reality that facilitate users exploratory interaction with software systems. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of proof-based techniques for software understanding, adaptation, and interaction and additional efforts to demonstrate and evaluate the techniques in realistic environments and systems.				
Title: Mapping Machine Learning to Physics (ML2P) Description: The Mapping Machine Learning to Physics (ML2P) program aims to develop energy-aware machine learning (ML) technology to enable the military to adapt ML on the battlefield under conditions of limited power resources. The capability to adapt ML systems in the field will be important for the military to keep ML systems up-to-date and current with evolving threats. ML2P will map ML efficiency to physics by using precise granular measurements in joules and metrics that are directly comparable across hardware architectures. ML2P will construct energy-aware ML optimized for power and performance for a given task and candidate hardware for the life cycle of the model. ML2P will develop objective functions that are optimal, feasible, and provide the desired trade-off for power and performance for a diversity of objective functions and applications. ML2P will discover the power-performance interactions between local optimizations and facilitate energy-aware ML construction and optimization. Success in the ML2P program will enable the military to create ML systems that can be sustained under conditions of limited power resources such as are likely to be encountered in battlefield environments. FY 2026 Plans: - Map machine learning (ML) energy efficiency to physics using precise granular measurements in joules and metrics that are directly comparable across hardware architectures. FY 2027 Plans: - Construct energy-aware ML and evaluate power usage and performance for a given task. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from development to evaluation of energy-efficient ML models.		-	4.411	2.789
Title: Advanced Tools for Modeling and Simulation Description: The Advanced Tools for Modeling and Simulation thrust is developing foundational mathematical, computational, and multi-physics theories, approaches, and tools to better represent, quantify, and model complex Department of War (DoW) systems from multimodal data analysis through part/system design and fabrication. One focus area of this thrust is developing a unified mathematical framework to enable better visualization and analysis of massive, complex data sets. Rigorous mathematical		-	6.771	1.751

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
theories are also being developed to address uncertainty in the modeling and design of complex multi-scale physical and engineering systems, incorporating capabilities to handle noisy data and model uncertainty beyond the scope of previously-existing capabilities. Other work in this thrust focuses on developing the mathematical and computational tools required to generate and better manage the enormous complexity of design, ultimately allowing designers to more easily discover non-intuitive (yet realizable) designs that fully leverage new materials and advanced manufacturing approaches now available. Outcomes from this thrust seek to improve the speed and accuracy of modeling and simulation, as well as enable management of complexity across DoW devices, parts, and systems. Another focus area of this thrust is the development of quantum-assisted approaches for predicting behavior and non-intuitive failure pathways for complex, dynamic physical systems. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Demonstrate artificial intelligence-based predictive modeling of test problems in turbulence guided by tabletop quantum simulators. - Validate predictive models of test problems with computational fluid dynamical techniques. - Examine the limits and modes of failure of useful mathematical transformations. - Assess if it is feasible to provide low fidelity models with high fidelity properties. - Demonstrate discovery of new algorithms for DoW relevant applications. - Develop mathematical methods to systematically find transformations that simplify DoW relevant modeling problems. FY 2027 Plans: - Explore techniques to mitigate singularities in mathematical transformations. - Develop tools to detect and mitigate singularities of discovered transformations. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.					
Title: Artificial Intelligence Studies and Concepts Description: This thrust will investigate methods to allow artificial intelligence (AI) logical inferencing to become massively parallel and distributed. Such AI logical inferencing includes extended logic programs used for deduction (e.g., query-answering), induction (i.e., machine learning), and abduction (e.g., argumentation and causal explanation finding). Enabling the computational scaling up of a knowledge supply chain in AI: to harness accumulated human knowledge (notably expertise), then learn beyond that, with high levels of abstraction, generality, and reusability, will require new algorithms, and tools to create distributed knowledge bases that efficiently yet coherently communicate and cooperate, dynamically.			-	34.344	38.273

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Prior to FY 2026, this program was funded in PE 0601011E, Project CCS-02.					
FY 2026 Plans: - Begin creation of methods for AI to support peer-to-peer groups with doing argumentation reasoning, to increase asynchrony, speed, size, and quality. - Begin exploration of methods for logical AI deep reasoning to become massively parallel and distributed. - Formulate mathematical and computational theory and methods to understand, assess, and predict the arrival and consequences of artificial general intelligence. - Develop physics-based methods that establish high assurance properties for AI-based perception in autonomous systems. - Enable tasking an AI or robot using the same sort of dialogue that a commander would use to task a human. - Create neurosymbolic foundation models for sparse-data domains where physical laws constrain variability.					
FY 2027 Plans: - Enable the automated generation of quantum algorithms with computational quantum advantage over classical computing paradigms. - Develop a domain-agnostic approach to enable accelerating training through logical, Socratic style, specialist AI tutors. - Develop an approach for constructing common shared knowledge bases about science with automated logical justifications/ analyses for all claimed conclusions and their critical challenges. - Explore secure ad-hoc teaming with previously unseen agents. - Formulate controls to prevent insider threat behaviors by rogue AI agents.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, basic research in quantum algorithms, unseen agents, and insider threat behaviors by rogue AI agents.					
Title: Math and Algorithms Studies and Concepts			-	22.323	54.420
Description: This thrust will develop new mathematical tools for analyzing large data sets for the discovery of new/hidden relationships, for the prediction of phenomena and (rare) events, for the disentanglement of multimodal data, as well as for the discrimination between valuable and invaluable information. This thrust includes the development of novel algorithms to search data, identify and quantify their relationships for task-specific applications and enable the generalizable abstractions directly from experimental data. Emerging opportunities in this thrust will explore quantum algorithms for modeling complex systems for the simulation of highly correlated materials and systems that are intractable with current classical approaches, i.e. modeling magnetism, superconductivity and nature.					
Prior to FY 2026, this program was funded in PE 0601011E, Project CCS-02.					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Begin exploration of new mathematics to automatically disentangle multimodal data, perceive and learn their relationships, and discard superfluous information. - Develop and evaluate scientific artificial intelligence (AI) tools ability to discover principles for DoW-relevant applications. FY 2027 Plans: - Investigate approaches for enabling cross-platform entanglement across disparate architectures for distributed quantum sensing and computing applications. - Design training for AI experts to profess a given science domain and advance its research and knowledge frontiers. - Enable scientific AI to create useful, communicable, generalizable abstractions directly from experimental data. - Discover, through optimization, pairwise -or larger neighborhood- agent interaction rules that lead to desired system behaviors. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, basic research in AI tools, abstractions that will enable scientific AI and applications that will support distributed quantum sensing and computing.				
Title: In The Moment (ITM) Description: In the Moment (ITM) supports the development of algorithms that are trusted to make decisions independently in difficult domains. Military operations, such as mass casualty triage, and disaster relief require complex and rapid decision-making in dynamic situations where there is often no human consensus and no ground truth. The ITM program generates a quantitative alignment framework for trusted human decision-makers and algorithms. ITM investigates these decision-making problems in the context of medical triage in military environments. If the ITM program is successful, the alignment framework will support the development of algorithms that can be tuned to align with specific, trusted humans. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Validate computational approaches for quantifying alignment and measuring the impact of alignment on trust of algorithmic decision-makers for complex use cases. - Evaluate an algorithmic decision-maker's ability to align with a reference group of human decision-makers. - Collaborate with military stakeholders to evaluate the utility of the human-algorithm alignment techniques in the context of medical triage in military environments. FY 2026 to FY 2027 Increase/Decrease Statement:		-	14.071	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects program completion.					
Title: Enhanced SBOM for Optimized Software Sustainment (E-BOSS) Description: The Enhanced SBOM for Optimized Software Sustainment (E-BOSS) program is creating enhanced software bill of materials (eSBOM) technologies with new types of rich metadata and is developing cyber reasoning algorithms and tools that leverage eSBOMs to defend against potential flaws during the software development process and to triage and remediate flaws found in operation. The global impacts of flawed software deployed at scale, such as the Log4Shell vulnerability found in Log4j cloud and web app deployments, where mitigations took from one week to months, motivated new SBOM requirements in Executive Order 14028. E-BOSS is developing software technologies integrated with modern software build chains to enable rapid triage and remediation of vulnerabilities at enterprise scale. The enhanced metadata incorporated in the eSBOMs will enable trace back of discovered flaw evidence to derive the vulnerability triggers, starting from a crash and tracing back through complex inter-component interactions, transfers, and transformations. Successful E-BOSS technologies will enable cyber-reasoning for improved remediation and sustainment of large-scale software systems. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Extend eSBOM with additional types of metadata and design use cases that are relevant to both open-source communities and Department of War software factories. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			-	5.131	-
Title: Advanced Research Concepts (ARC) Description: The Advanced Research Concepts (ARC) program will rapidly explore revolutionary, nascent high-risk/high-reward ideas to accelerate the discovery of paradigm-shifting technologies. The program will focus on answering fundamental research questions to investigate the boundaries of what is possible and push the state of the art of science and technology. By exploring a high volume of ideas through targeted and limited-scope investments, this program will identify promising ideas early and assess the impact of further investment on problems of importance to the Department of War. The program will cover a broad spectrum of science and engineering disciplines that have applications for national security research. Individual efforts will explore numerous emerging technologies in various thrusts including, but not limited to, artificial intelligence and machine learning, mathematics, biology, chemistry, quantum technologies, social science, material science, electronics, sensing and measurement, and modeling. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02.			-	2.189	-

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Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-01 / MATH AND COMPUTER SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
FY 2026 Plans: - Complete efforts to expand data science techniques for non-ergodic systems.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	179.647
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES				Project (Number/Name) EMR-02 / DISRUPTIVE MICROTECHNOLOGIES SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EMR-02: DISRUPTIVE MICROTECHNOLOGIES SCIENCES	0.000	0.000	42.697	13.453	-	13.453	3.597	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Disruptive Microtechnologies Sciences project supports investigations into materials, devices, and architectures that will enable dramatic improvements versus the current state-of-the-art. This includes developing technology to enable fundamentally new ways of creating useful circuits to enable the next generation of microtechnologies. Areas of focus include photonic circuits that harness the power of light at the chip-scale, quantum circuits that take advantage of the unique properties that quantum offers versus classical computing and sensing, and organic circuits that integrate organic molecules into microtechnologies.												
Prior to FY 2026, efforts in this Project were funded in PE 0601101E, Project ES-02.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Compartmentalization and Privilege Management (CPM)									-	10.804	13.453	
Description: The Compartmentalization and Privilege Management (CPM) program is developing new system frameworks, architectures, and tooling to provide fine-grained, least-privileged compartmentalization that enables prevention and containment of cyber attacks. Today's information systems are structured around a monolithic core, the kernel, that operates within a single protection domain at a single high privilege-level. A monolithic kernel contains many separate components without protection boundaries between them. A single compromise anywhere in the system allows attackers effectively unlimited access through an extended sequence of exploits, privilege escalation, and lateral motion. CPM is developing technologies and tools to compartmentalize large, legacy software systems automatically and designing processor architectures and system software to enforce a compartment and privilege-level regime. CPM tools and architectures will prevent initial penetrations from propagating into successful cyber attacks.												
Prior to FY 2026, this program was funded in PE 0601101E, Project ES-02.												
FY 2026 Plans:												
- Produce code compartmentalization and processor and policy enforcement implementations.												
- Develop initial attack campaign test cases for Department of War (DoW)-relevant applications.												
- Evaluate effectiveness and overhead of compartmentalization and privilege management systems using operating system attack campaign test cases.												
FY 2027 Plans:												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-02 / DISRUPTIVE MICROTECHNOLOGIES SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Produce final processor designs and performance models for enforcing code compartmentalization policies. - Develop enhanced attack campaign test cases for DoW-relevant applications. - Apply compartmentalization and privilege management tools and processor enforcement mechanism to an application nominated by a DoW partner. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of compartmentalization techniques and additional work to test and demonstrate compartmentalization on military systems.				
Title: Joint University Microelectronics Program 2.0 (JUMP 2.0) Description: The Joint University Microelectronics Program 2.0 (JUMP 2.0) program is developing and demonstrating innovative next-generation microelectronics technologies through a public-private consortium with universities, the defense industrial base, and the semiconductor industry. The JUMP 2.0 program addresses the grand technical challenges of our increasingly connected world that must be overcome including: the need for innovation in analog hardware, increasing demand for more memory and data storage, the imbalance between data generation and communication capacity, the emerging security vulnerabilities in highly interconnected Artificial Intelligence systems, and the unsustainable growth in energy demands for computing. Therefore, the JUMP 2.0 program sponsors academic research teams focused on related key technology areas that will not only impact future defense and national security capabilities but also strengthen U.S. leadership in information and communication technology. The JUMP 2.0 program will push fundamental technology research themes in cognition, communications, sensing to action, computing and processing, memory and storage, integration and packaging, and high-performance energy efficient devices to enable key disruptive advances in microelectronic technology. Prior to FY 2026, this program was funded in PE 0601101E, Project ES-02. FY 2026 Plans: - Demonstrate promising materials, devices, and integration and packaging technologies. - Integrate components to demonstrate sub functions of next-generation artificial intelligence, efficient communication, intelligent storage, novel sensing-to-action, and distributed computing architectures. - Perform research on next generation materials, devices, and integration and packaging technologies. - Integrate subfunctions to demonstrate next-generation artificial intelligence, efficient communication, intelligent storage, novel sensing-to-action, and distributed computing architectures. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	26.000	-
Title: Disruptive Microtechnologies Sciences Studies and Concepts		-	5.893	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-02 / DISRUPTIVE MICROTECHNOLOGIES SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Description: The Disruptive Microtechnologies Sciences Studies and Concepts project examines the fundamental science needed to advance disruptive microtechnologies. This program also examines ways to take advantage of commercial scaling while building in superior capabilities for defense systems through design, integration, and hardware security. These studies may lead to the development of new programs in these areas. FY 2026 Plans: - Examine current and burgeoning technologies for creation of disruptive microtechnologies. - Investigate potential areas for leveraging commercial microsystems scaling for relevant defense applications. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the strategic prioritization of concepts and studies investments.			
Accomplishments/Planned Programs Subtotals		-	42.697
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES				Project (Number/Name) EMR-03 / MATERIALS SCIENCES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EMR-03: MATERIALS SCIENCES	0.000	0.000	24.677	53.283	-	53.283	57.233	57.856	60.968	61.902	-	-

A. Mission Description and Budget Item Justification

The Materials Sciences project provides the fundamental research that underpins the design, development, assembly, and optimization of advanced materials, devices, and systems for DoW applications in areas such as design and simulation of novel materials and structures, energetics, complex hybrid systems, and advanced metrology.

Prior to FY 2026, efforts in this Project were funded in PE 0601101E, Project MS-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Fundamental Limits Description: Understanding the Fundamental Limits (i.e., achievable boundaries) of scientific principles, processes and technologies is critical to better anticipate technological surprise for our adversaries and ourselves. This thrust explores boundaries across fields such as physics, chemistry, mathematics, biology, and engineering to address critical questions for national security, addressing foundational theory and approaches that include, for example, the fundamental limitations of optical technologies, potential implications for basic biology on national security, and the ability for modeling and simulation to provide a better understanding of complex systems. Prior to FY 2026, this program was funded in PE 0601101E, Project MS-01. FY 2026 Plans: - Realize and characterize correlated multiphoton sources for sensing, communication, and imaging. - Demonstrate quantum-enhanced optical metrology with correlated multiphoton sources for low-light imaging, quantum sensing, and resilient optical communications. - Demonstrate high fidelity quantum operations with error-resilient qubits created by measurement and feedback techniques. - Determine pathways to improve radioisotope yield in generating useful daughter products faster than natural decay processes. - Demonstrated levitation of heavy mass with high confinement bandwidth in a compact form factor along 3 axes. - Initiate solid-state reaction rate amplification factor analysis modeling and testing. FY 2027 Plans: - Demonstrate improved radioisotope yield in generating useful daughter products faster than natural decay processes and show a path to producing practical quantities of radioisotopes.	-	6.023	13.427

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-03 / MATERIALS SCIENCES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Complete solid-state reaction rate amplification factor analysis modeling and testing. - Demonstrate ~1000X improvement in solid-state reaction rates.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from initial design and development to finalization and demonstrations.				
Title: Materials Sciences Studies and Concepts		-	14.045	39.856
Description: The grounds for strategic surprise are often realized through the discovery of unifying principles, novel fundamental limits, and unexpected connections between nominally disparate fields. This thrust explores emergent capabilities and universal themes at the interface of quantum science, mathematics, nanoscience, and materials science to develop novel approaches to critical national security needs. Focus areas include harnessing the universal principles of turbulence from new forms of simulation for high complexity physical systems; systemic discovery of materials with desired properties; the analysis of new scientific and technological ideas of importance to national security.				
Prior to FY 2026, this program was funded as Emerging Opportunities in Materials Science in PE 0601101E, Project MS-01.				
FY 2026 Plans: - Advance methodologies for efficient data integration from isotope and trace element analysis of materials and developed multimodal isotope and trace element analysis measurement capabilities with increased robustness. - Explore acoustic properties of rapidly deployable non-solid materials for potential use in blast wave attenuation. - Begin evaluations of materials, characterization methods, and device architectures to electrically controllable altermagnetism.				
FY 2027 Plans: - Explore potential of complex optimized protocols to produce currently unattainable material properties and desired performance limits. - Refine isotope and trace element analysis capabilities and demonstrate robust and universal analysis capabilities with (sub)-ppm sensitivities. - Develop aerosol materials with high acoustic impedance to be implemented as a blast mitigation system. - Initiate the development of novel devices for terahertz memory and logic applications. - Detect and localize sources of trace chemical signals within complex and changing environments with edge computing. - Establish the foundational science for quantitative prediction of underwater extreme velocity, pressure and frequency events. - Establish chemical kinetics and thermodynamic laws governing liquid metal (alkali / alkaline earth metals) plus seawater combustion. - Explore the potential of forms of coherent excitation to generate pairs of entangled nuclear radiation to better control nuclear reaction processes.				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-03 / MATERIALS SCIENCES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Exploit quantum vacuum fluctuations to generate resource entanglement and decoherence to be utilized in the development of future communication technologies. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, basic research in aerosols for blast mitigation systems, trace chemical detection using edge computing and quantitative prediction models for sub-domain extreme velocity and pressure.			
Title: Molecular Systems and Materials Assembly Description: The Molecular Systems and Materials Assembly thrust is exploring new approaches for the synthesis, assembly, characterization and application of molecules and materials for a variety of DoW applications from the atomic to the product scale. Ultimately, materials and methods developed in this thrust will support a wide range of DoW applications that will leverage novel materials to extend the range, duration, and capabilities of DoW systems and the warfighter. Through control of the arrangement, interactions, and assembly of atoms and molecules, new materials and manufacturing processes are being developed to address long-standing challenges in supply chains, logistics, and sustainment while simultaneously enhancing the warfighter's capabilities on the battlefield. Efforts in this thrust range from fundamental science to better understand the chemistry and physics related to each application, to developing means to utilize such capabilities in future test systems and prototype devices. Prior to FY 2026, this program was funded in PE 0601101E, Project MS-01. FY 2026 Plans: - Develop and assess the capability of an integrated system to produce sufficient microbial biomass powder for 14 warfighters in one day from air, water, electricity, and microbes. - Determine that the microbial biomass powder meets DoW Nutritional Standards for Operational Rations (NSOR) requirement and meets DoW safety and stability standards. - Assess novel methods for modeling electrochemical interfaces, such as in batteries, to improve system lifetime performance. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	4.609
Accomplishments/Planned Programs Subtotals		-	24.677
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-03 / MATERIALS SCIENCES
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES				Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EMR-04: BASIC OPERATIONAL MEDICAL SCIENCE	0.000	0.000	77.908	96.554	-	96.554	130.959	133.084	138.540	140.540	-	-

A. Mission Description and Budget Item Justification

The Basic Operational Medical Science project will explore and develop basic research in medical-related information and technology leading to fundamental discoveries, tools, and applications critical to overcoming DoW challenges. This project will address the Department's identified warfighter medical care needs related to prevention and treatment of infectious disease, real-time healthcare interventions of acute and chronic illness and injury, and interventions for improved warfighter resilience and performance against operational stressors.

Prior to FY 2026, efforts in this PE were funded in PE 0601117E, Project MED-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Anesthetics for Battlefield Care (ABC)	-	10.519	10.301
Description: The Anesthetics for Battlefield Care (ABC) program aims to produce safe, battlefield-ready anesthetics to reduce the trauma associated with injury and improve combat casualty outcomes. Current therapeutics that enable life-saving interventions and wound stabilization must be used in hospitals or highly monitored settings due to their lack of safety. Prolonged peer or near-peer conflict could severely impact medical evacuation times, resulting in extended time before patients reach a hospital. The ABC program seeks to uncover mechanisms of anesthesia at multiple biological levels ranging from the molecular to the organismal. Novel treatments developed under the program will exhibit the desirable properties of anesthetics, including calming effects, loss of sensation and loss of consciousness but will have vastly improved safety profiles, making them usable in the field by warfighters with minimal medical training. Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01. FY 2026 Plans: - Identify novel molecular targets of anesthesia. - Conduct computational modeling to test efficacy of small molecules binding to established protein targets. - Define neural circuitry mechanisms that underlie anesthetic actions in an animal model. - Experimentally validate anesthesia associated pathways and molecular targets identified in screening approaches. - Validate and demonstrate predictive capabilities of physiological computational model of anesthetic drugs in single/poly-trauma physiologies. FY 2027 Plans:			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Begin to define neural circuits involved in safe anesthesia effects. - Demonstrate novel actionable molecular targets that contribute to the effects of anesthesia. - Complete computational screens and select hit drug candidates against novel molecular targets. - Initiate novel drug design for molecular targets of anesthesia.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects minor program repricing.				
Title: Synthetic Hemo-technologies to Locate & Disinfect (SHIELD) Description: Bloodstream infections (BSI) are a significant source of morbidity in service members who sustain combat-related injuries. Trauma temporarily degrades the efficacy of the host immune system thereby increasing the risk of life-threatening opportunistic infections from fungi and bacteria that enter into the blood. If unchecked, bloodborne fungi and bacteria lead to debilitating conditions such as invasive fungal infections, sepsis, and shock. The Synthetic Hemo-technologies to Locate and Disinfect (SHIELD) program is developing a systems-level approach to prevent BSI in warfighters who suffer trauma from burn or blast. Preventative medical countermeasures circulating in the blood will be developed to bind infectious particles in the blood early and label pathogens for clearance and deliver drugs to destroy pathogens and/or restore healthy physiology. Ultimately, this program will develop novel technologies that will protect service members from morbidity and mortality associated with BSI. Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01.		-	8.740	8.899
FY 2026 Plans: - Produce countermeasures at scale to support animal model testing. - Complete validation of countermeasures' ability to prevent pathogen growth and increase host survival in single fungal and single bacterial pathogen exposures. - Refine development of countermeasures that prevent pathogen growth and increase host survival in single fungal and bacterial pathogen exposures in blood following burn trauma. - Refine development of countermeasures that are non-toxic and non-immunogenic to the host following burn trauma.				
FY 2027 Plans: - Demonstrate that developed countermeasures prevent pathogen growth and increase host survival in single fungal and bacterial pathogen exposures in blood following burn trauma. - Demonstrate that countermeasures are non-toxic and do not produce a response from the immune system in the host following burn trauma. - Validate the countermeasures' ability to prevent pathogen growth and increase host survival in single fungal and single bacterial pathogen exposures following burn trauma.				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Refine development of countermeasures that prevent pathogen growth and increase host survival in mixed-pathogen (fungal and bacterial) exposures in blood following blast trauma.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects minor program repricing.				
Title: CoasterChase		-	12.963	12.075
Description: DoW warfighters are often required to perform complex tasks in a high-stress situation. Warfighters under stress can make poor decisions with immediate and/or lasting consequences that may also contribute to long-term trauma. The CoasterChase program seeks to evaluate host response to stimulation of the digestive tract, to explore stress response to improve decision making and reduction of long-term trauma response with a combined sensing and stimulation ingestible device. The program aims to understand behavioral and physiological biomarkers of stress and inform device design parameters. Ultimately, the technology will enable improved memory retention, reduced effects of post-traumatic stress disorder (PTSD), maintenance of immune defense, and regulation of metabolism in austere domains.				
Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01.				
FY 2026 Plans:				
- Initiate the development and encapsulation of stimulation hardware, including electrical, mechanical, or thermal modalities.				
- Initiate experiments to develop sensing hardware to identify minimum sensitivity and detection limits for chosen biomarkers.				
- Initiate acute animal subject research to determine appropriate stimulation and sensing ranges within the small intestine.				
FY 2027 Plans:				
- Integrate and miniaturize neuromodulation and sensing technologies within a single ingestible form factor for closed-loop control.				
- Further develop, test, and finalize biomarker sensing for identified biomarker(s).				
- Finalize integration of stimulation modalities (electrical and/or mechanical) to modulate the systemic stress response.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a transition from a broad developmental phase with a large parameter space to a more focused period of technology refinement.				
Title: Network of Optimal Dynamic Energy Signatures (NODES)		-	11.499	14.351
Description: Proteins are essential building blocks of life, playing a crucial role in countless biological processes. The universality of proteins across all lifeforms makes them a powerful tool in medicine, where their unique properties can be harnessed to develop life-saving therapies. The Network of Optimal Dynamic Energy Signatures (NODES) program will develop a biophysics-				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
guided deep learning tool that can scan protein sequences and, from that scan, infer biological functions based on signatures of protein movements. In addition to providing deeper understanding of potential therapies, the NODES program will provide the ability to characterize the function of potential biological threats, expanding threat assessment accuracy and speed to keep pace with rapidly advancing capabilities. The NODES efforts will bolster the Department of War's ability to capitalize on the emerging field of novel protein design by providing a tool to expedite threat characterization when the nation or warfighters are introduced to an unknown agent and shorten the time to developing medical countermeasures. FY 2026 Plans: - Design rapid computational models on a small subset of protein families to incorporate the dynamics and function of proteins. - Evaluate models for the correlation of dynamics and function of known protein sequences to assess model extensibility. - Evaluate computational model performance in a capability demonstration to predict the unknown molecular function of novel sequences. FY 2027 Plans: - Generalize computational models by incorporating available data from public protein databases. - Incorporate machine learning methods with physics-based models to provide speed and accuracy of function prediction. - Evaluate the optimized model performance in capability demonstrations to predict molecular function from folding, binding and cooperativity. - Validate the developed computational models on select datasets to assess overall model performance in drug-resistance, antibody and enzyme design. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects testing and evaluation efforts to integrate program models and perform additional capability demonstrations.				
Title: Basic Operational Medical Science Studies and Concepts Description: The DoW will accelerate discovery and development by leveraging recent fundamental advances in biology to identify new capabilities and address medical science challenges encountered by warfighters. The Basic Operational Medical Science Studies and Concepts thrust will explore basic research for fundamental discoveries, tools, and applications critical to overcoming medical-related DoW challenges. This thrust will investigate approaches to addressing the Department's identified warfighter medical needs as they pertain to warfighter health, resilience, and performance against operational stressors. Technologies in this thrust will provide predictive and diagnostic tools, as well as develop prevention and recovery techniques, for threats that affect the warfighter. FY 2026 Plans:		-	6.491	50.928

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Initiate the development of platform technologies that address military medical science challenges. - Initiate exploration of approaches to effectively leverage artificial intelligence in military medical science contexts. FY 2027 Plans: - Initiate exploration of approaches to improve assessments and diagnostics for predictive and preventative purposes. - Identify assessment paradigm for biological and artificial intelligence problem solving separately and in tandem. - Investigate high-throughput characterization technologies to examine molecular interactions and generate large datasets for biological foundational models. - Initiate the development of algorithms to accelerate medical decision making to enhance medical personnel triage performance. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, basic research in assessment paradigms in tandem biological and AI problem solving, improved creation of biological foundation models and advanced algorithms to accelerate medical triage decision making.					
Title: Simulating Microbial Systems (SMS) Description: The Simulating Microbial Systems (SMS) program is developing the general capability to simulate microbial cell function and behavior to prevent strategic surprise. SMS is investigating combinations of computational advances with data acquisition techniques to advance the state of the art of cellular modeling. Bringing together communities in cell modeling, computational techniques, and wet lab measurements, this program will build from partial physics-based models of a minimal synthetic cell to more complete and faithful simulations of a DoW-relevant organism. SMS will experimentally evaluate simulations via pressure tests that relate to both general cell simulation and tailored stakeholder use cases. SMS is postured to be a valuable tool for acceleration of medical countermeasure developments and biomanufacturing. Additionally, the program will provide a baseline for further simulation development and importantly will provide the DoW and its stakeholders the ability to run microbial simulations to prevent surprise. Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01. FY 2026 Plans: - Begin pressure tests to experimentally validate simulations for tailored DoW use cases. - Investigate and evaluate new approaches and combinations thereof to develop simulations of fundamental biological processes and responses to stimuli or stress. - Begin validating simulations of fundamental biological processes and responses to stimuli or stress. - Develop approaches to evaluate computational parameter space to optimize simulations of fundamental biological processes. FY 2026 to FY 2027 Increase/Decrease Statement:			-	18.148	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 1	R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES	Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects program completion.			FY 2027
Title: Objective Prediction of Team Effectiveness via Models of Performance Outcomes (OP TEMPO) Description: The Objective Prediction of Team Effectiveness via Models of Performance Outcomes (OP TEMPO) program seeks to advance technologies to drive efficiency and efficacy of military operator team preparation and expertise building. This program seeks to understand fundamental biological processes to support real-time physiological assessment, performance diagnostics, and objective prediction of warfighter and team proficiency, with the ultimate goal of improved DoW mission readiness and execution. Advances in this program will result in a significant enhancement to warfighter team performance by providing methods to determine teaming potential and actionable paths to optimal teaming. Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01. FY 2026 Plans: - Complete data collection of biobehavioral signatures of warfighter and team performance. - Validate bio-behavioral signatures using predictive models of team performance. - Conduct Critical Task Analysis for two additional teaming-related use cases and develop custom metrics for each use case. - Create testbeds for two additional teaming-related use cases to explore generalizability of bio-behavioral signatures and predictive models. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	4.163
Title: Combatting Anti-Microbial Resistant Pathogens Description: The Combatting Anti-Microbial Resistant Pathogens program is investigating fundamental methods for using preexisting host machinery as a technology to create medical countermeasures that degrade or deactivate pathogen targets. The DoW has long recognized the warfighter's outsized risk of exposure to biological threat agents and to infectious disease, including the increasing prevalence of antimicrobial-resistant (AMR) organisms that are ranked as a Tier 1 threat to the U.S. military. Similarly, the danger posed by bacterial biothreats persists with few countermeasures available. Key advances expected from this research include identifying methods to discover and develop new classes of chimeric therapeutics against AMR bacteria, bacterial biothreats, and other DoW-relevant diseases and threats. These approaches represent a significant departure from conventional therapeutics, which typically rely on a limited number of small molecules with a narrow set of targets and mechanism of action. Advances in this area may be used to help mitigate known, new, and emerging diseases that impact military readiness and pose a global health threat.		-	3.897

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 1		R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES		Project (Number/Name) EMR-04 / BASIC OPERATIONAL MEDICAL SCIENCE	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01.					
FY 2026 Plans: - Demonstrate medical countermeasures against pathogenic bacteria and/or their toxins with greater efficacy than state-of-the-art antibiotic therapies.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.					
Title: Physiological Overmatch			-	1.488	-
Description: Warfighters operate under extreme physiological conditions, sometimes with limited resources and manpower, and must acclimate quickly to changing operational needs. The Physiological Overmatch program is investigating innovative approaches to allow the warfighter to adapt rapidly to operational challenges during deployment by developing novel detection and treatment systems. The program will initiate work in aiding the deployed soldier's ability to resist fatigue, combat sleep deprivation, and maintain a high capacity for operational performance. This program will seek to understand the biological mechanisms of fatigue, which will enable improvements to warfighter health and operational performance. This approach represents a significant enhancement to warfighter performance by providing protection from impacts to operational readiness and provides information related to fatigue states and the ability to operate with prolonged sleep deprivation.					
Prior to FY 2026, this program was funded in PE 0601117E, Project MED-01.					
FY 2026 Plans: - Identify molecules in the blood that correspond to sleep loss and its effects on cognitive function. - Identify potential intervention targets for mitigating deleterious effects of sleep loss.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.					
Accomplishments/Planned Programs Subtotals			-	77.908	96.554
C. Other Program Funding Summary (\$ in Millions)					
N/A					
Remarks					
D. Acquisition Strategy					
N/A					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 1					R-1 Program Element (Number/Name) PE 0601122E / EMERGING OPPORTUNITIES				Project (Number/Name) EMR-07 / EMERGING OPPORTUNITIES SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EMR-07: EMERGING OPPORTUNITIES SUPPORT	0.000	0.000	20.124	28.884	-	28.884	34.097	35.623	30.897	30.897	-	-

A. Mission Description and Budget Item Justification

The Emerging Opportunities Support project contains management costs in support of DARPA functions and activities across the entire Emerging Opportunities PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0601101E, Projects CCS-02, ES-02 and MS-01, and PE 0601117E, Project MED-01.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602023E / <i>ACCESS AND AWARENESS</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	98.269	110.096	-	110.096	192.689	224.907	232.945	237.009	-	-
ACS-01: <i>ALL DOMAIN TACTICAL TECHNOLOGIES</i>	0.000	0.000	92.642	101.892	-	101.892	176.309	208.067	218.178	222.242	-	-
ACS-07: <i>ACCESS AND AWARENESS SUPPORT</i>	0.000	0.000	5.627	8.204	-	8.204	16.380	16.840	14.767	14.767	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$98,269 thousand of discretionary and \$7,000 thousand of mandatory (reconciliation) for a total of \$105,269.

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Access and Awareness Program that is directed toward the development and application of technologies that ensure physical or virtual presence where and when necessary and that collect, integrate, or disseminate data and/or knowledge. There is strong interest in the development and application of advanced, innovative autonomy, computing, and communications technologies. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The All Domain Tactical Technologies project examines and evaluates emerging technologies and system concepts for developing technologies for enhancing U.S. military effectiveness and survivability in operations across all domains. Enabling and novel technologies include concepts for expanding the envelope of operational naval capabilities to include the entire sea column such as improved situational awareness over large maritime environments, ship self-defense techniques, novel underwater propulsion modalities, high speed underwater vessels, improved techniques for underwater object detection and discrimination, long endurance unmanned surface vehicles, methods and techniques for servicing assets throughout the sea column, and high bandwidth communications. This project will also examine methods and architectures for distributing maritime operations to enable a more agile, survivable, and cost-effective fleet. Technologies will be studied for enhancing U.S. military effectiveness and survivability in ground operations ranging from traditional threats to military operations against irregular forces that can employ disruptive or catastrophic capabilities, or disrupt stabilization operations, including competing in underground spaces. Programs in this project will break the relative symmetry of land combat to strengthen and maintain U.S. forces' decisive advantages in the current and future ground battlefield. Aeronautics and space technology efforts studied will address high payoff opportunities that dramatically reduce costs associated with advanced aeronautical and space systems and/or provide revolutionary new system capabilities for satisfying current and projected military mission requirements. This includes advanced technology studies of revolutionary propulsion, vehicle, and launch concepts, sophisticated fabrication methods, and examination of novel materials and enabling technologies for aeronautics and space system applications. Studies that also fundamentally change the calculus of battle including consideration of a mix of assets, platforms that are potentially disposable or with limited lifespans, and autonomous integration of space and air platforms in the tactical battlespace are included.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602023E / <i>ACCESS AND AWARENESS</i>
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The Access and Awareness Support project contains management costs in support of DARPA functions and activities across the entire Access and Awareness PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0602702E, Tactical Technology.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	100.791	0.000	-	0.000
Current President's Budget	0.000	98.269	110.096	-	110.096
Total Adjustments	0.000	-2.522	110.096	-	110.096
• Congressional General Reductions	0.000	-1.376			
• Congressional Directed Reductions	0.000	-1.146			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	110.096	-	110.096

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS				Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ACS-01: ALL DOMAIN TACTICAL TECHNOLOGIES	0.000	0.000	92.642	101.892	-	101.892	176.309	208.067	218.178	222.242	-	-

A. Mission Description and Budget Item Justification

The All Domain Tactical Technologies project examines and evaluates emerging technologies and system concepts for developing technologies for enhancing U.S. military effectiveness and survivability in operations across all domains. Enabling and novel technologies include concepts for expanding the envelope of operational naval capabilities to include the entire sea column such as improved situational awareness over large maritime environments, ship self-defense techniques, novel underwater propulsion modalities, high speed underwater vessels, improved techniques for underwater object detection and discrimination, long endurance unmanned surface vehicles, methods and techniques for servicing assets throughout the sea column, and high bandwidth communications. This project will also examine methods and architectures for distributing maritime operations to enable a more agile, survivable, and cost-effective fleet. Technologies will be studied for enhancing U.S. military effectiveness and survivability in ground operations ranging from traditional threats to military operations against irregular forces that can employ disruptive or catastrophic capabilities, or disrupt stabilization operations, including competing in underground spaces. Programs in this project will break the relative symmetry of land combat to strengthen and maintain U.S. forces' decisive advantages in the current and future ground battlefield. Aeronautics and space technology efforts studied will address high payoff opportunities that dramatically reduce costs associated with advanced aeronautical and space systems and/or provide revolutionary new system capabilities for satisfying current and projected military mission requirements. This includes advanced technology studies of revolutionary propulsion, vehicle, and launch concepts, sophisticated fabrication methods, and examination of novel materials and enabling technologies for aeronautics and space system applications. Studies that also fundamentally change the calculus of battle including consideration of a mix of assets, platforms that are potentially disposable or with limited lifespans, and autonomous integration of space and air platforms in the tactical battlespace are included.

Prior to FY 2026, efforts in this Project were funded in PE 0602702E, Projects TT-03, TT-04, and TT-07.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Cyber Physical systems Executing in Real time (CyPhER) Forge	-	13.820	30.410
Description: Cyber Physical systems Executing in Real time (CyPhER) Forge will revolutionize and accelerate defense Test & Evaluation (T&E) by breaking the direct link between physical system complexity and test duration. This program seeks to reduce required test points by an order of magnitude for developmental systems through two core innovations: 1) a real-time digital twin incorporating multi-physics-informed surrogate modeling, uncertainty quantification, and continuous data assimilation, and 2) an Artificial Intelligence (AI) test agent that applies the digital twin to identify novel, knowledge-maximizing maneuvers, and executes maneuvers with machine precision autonomy while providing statistical safety guarantees. This integrated CyPhER will include demonstration of an accelerated flight sciences test campaign using an experimental aircraft as the primary validation platform.			
FY 2026 Plans:			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Establish the Government Integration, Validation, and Verification (IV&V) team and begin executing core tasks including developing government-owned Application Programming Interfaces and Graphical User Interfaces, conducting standards governance, and establishing the initial challenge-cycle evaluation framework. - Design and implement Government High Performance Computing (HPC) backbone and data storage infrastructure. FY 2027 Plans: - Conduct iterative risk reduction through challenge cycles supporting prototype digital twin design validation. - Evaluate prototype AI test agents using a faster than real time evaluation environment and real-world test data from an active experimental flight test program. - Initiate design development of complete CyPhER prototype (digital twin + AI test agent) for an experimental air vehicle. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects iterative increase of digital twin challenge cycle complexity, initiation of complete CyPhER prototype design work, and progression into Phase 2 of the program including preparation for flight test.				
Title: Lunar Assay via Small Satellite Orbiter (LASSO) Description: The Lunar Assay via Small Satellite Orbiter (LASSO) program will optimize autonomous navigation and propulsion in support of advancing Space Situational Awareness capabilities in cislunar space. The program will build a low-cost satellite, launched as a secondary payload, to demonstrate these capabilities in Very Low Lunar Orbit (VLLO), which is the most challenging environment in cislunar space due to continually destabilizing orbits, lack of navigation aids (i.e., GPS), variable gravity and operation in close proximity to terrain. While operating in VLLO, LASSO will collect high-resolution data regarding accessible water ice on the moon by capturing data about resource distribution and quantities to support future in-situ resource utilization (ISRU) investigation. Such discoveries support future lunar exploration. With improved navigation, propulsion, and autonomous control, the LASSO satellite will achieve persistent VLLO while building a statistically significant understanding of lunar resources. The LASSO program is also funded in PE 0603468E, Project ACX-01. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-07. FY 2026 Plans: - Conduct feasibility analysis for optimized spacecraft navigation, propulsion, and sensing for cislunar operations, including resource exploration. - Conduct System Requirements Review (SRR) and Conceptual Design Review (CoDR) for the LASSO spacecraft. - Build lunar mining lifecycle economic model. - Conduct prototype testing of critical sub-systems for lunar resource exploration spacecraft.		-	19.636	19.900

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Build very low lunar orbital simulation with full lunar variable gravity model. FY 2027 Plans: - Build software- in-the-loop simulations (SWIL) to verify mission planning and subsystem performance. - Conduct closed-loop SWIL testing with applied vehicle and payload concept constraints. - Verify that sufficient control authority is maintained by autonomous navigation and enhanced efficiency propulsion systems. - Verify spacecraft payload viability for sensing resolution and coverage through orbital mission simulation. - Conduct Preliminary Design Review (PDR) and subsystem validation testing. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from CoDR and prototype testing to PDR and validation testing.			
Title: Albatross Description: The Albatross program will explore new technologies to enable perpetually flying aircraft. Albatross approaches this by creating a bird-like soaring capability for small-unmanned aircraft systems (s-UAS) that leverages weather forecast-informed mission planning and real-time sensing of dynamic wind conditions to autonomously navigate s-UAS and extend flight endurance and range. Current estimates indicate that Albatross soaring techniques could reduce average flight power by 75% for relevant missions. Albatross is also developing autonomous guidance, navigation, and controls to harness soaring sources while maintaining safe separation from terrain. Albatross will use a series of real-world flight events to mature and demonstrate capability incrementally. Technologies developed under this program will be demonstrated on mature platforms to support rapid transition to the Services. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-07. FY 2026 Plans: - Conduct practice test events for incremental integration and development selection. - Demonstrate flight planning tool for soaring scenarios. - Conduct test events through mission-relevant scenarios. - Develop low-power flight path mission planning software to account for land and ocean soaring conditions. - Develop software/hardware algorithms for aircraft to respond to ocean soaring conditions and harness energy. - Conduct additional practice test events for final integration. FY 2027 Plans: - Demonstrate the combination of land and ocean soaring techniques in a mission relevant environment. - Improve soaring prediction capabilities and algorithms to ensure operational relevance and enable extended range and endurance in operational applications.		-	22.207
			11.124

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS		Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Integrate capabilities and technologies to enable new broad, long-term, and perpetual flight techniques for s-UAS applications.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the transition from software development to final integration and demonstration.					
Title: Next Generation Electronic Surveillance Technology (NGEST) Description: The Next Generation Electronic Surveillance Technology (NGEST) program is creating a new Radio Frequency (RF) surveillance system for sensing in complex signal environments. NGEST is defining a new surveillance paradigm to find critical signals of interest in tactically relevant timelines. Contemporary electronic surveillance systems face increasing RF complexity in the spatial (emitter proliferation), power (range), and frequency (agility, congestion) dimensions due to their static hardware configuration. Additionally, current systems need to employ exquisite hardware to deal with geometrical challenges. NGEST overcomes the complexity and geometry challenges by autonomously reconfiguring diverse frequency and spatial resources. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-07. FY 2026 Plans: - Integrate hardware components of prototype RF surveillance system. - Integrate orchestration algorithms into prototype RF surveillance system and test in a laboratory environment. - Field test initial prototype system in a complex signal environment. FY 2027 Plans: - Conduct additional field and laboratory testing in complex signal environments. - Upgrade system prototype based on lessons learned from field and laboratory testing. - Finalize system design and open architecture interfaces. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from system integration to final testing.			-	7.389	6.911
Title: All Domain Tactical Technologies Studies and Concepts Description: The All Domain Tactical Technologies Concepts and Studies will investigate system concepts for developing technologies to enhance U.S. military effectiveness and survivability in operations across all domains. Efforts will surmount key challenges associated with redefining access and timely delivery of effects by using targeted investments that explore the feasibility of novel technical solutions, force capabilities, and innovations in logistics and manufacturing. These may include feasibility studies of novel or emergent materials; sensors, effectors, and tactics for air, space, ground, and sea platforms; launch vehicles; satellites; manufacturing and implementation approaches; and hardware demonstrations of key enabling technologies.			-	8.786	33.547

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Support the issuance of standards recommendation reports for lunar power infrastructure, communications infrastructure, and Position, Navigation, and Timing (PNT) infrastructure. - Initiate research into a program that will explore the combination of adaptive maneuver warfare with sustained, surging assets for faster-than-human offensive and defensive behaviors to enable a new way of fighting. - Initiate research into the novel application of artificial technology techniques, like foundation models and Graph Neural Networks (GNN), to enable a small number of humans to command and control significant numbers of cross-domain assets and effectors. - Collect data to support passive noise radar approach investigations and applications. - Investigate algorithmic development on radars. - Research environmental noise data. FY 2027 Plans: - Explore research of novel tactical designs and build low-cost platforms that can be rapidly produced at scale and employ distributed effects on the battlefield, optimized for the environment of choice. - Investigate new propulsion technologies that offer improved thermal and thrust characteristics that will open new avenues of tactical maneuver efficiencies. - Explore new munition and launch capabilities that can be strategically placed forward into existing operating locations across domains. - Research reconfigurable, morphing robotic AI smart structures can be easily deployed from existing systems such as artillery launchers, ground vehicles, small surface craft, and other platforms. - Explore advanced system power capabilities that enable long-duration, untethered, all-weather transportation from sea to space. - Collect environmental noise data in austere, contested, or otherwise challenging environments. - Utilize collected environmental noise data to validate assumptions of physics models and phenomenology with novel applications for operational capabilities. - Develop algorithms and software to process collected data. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in new propulsion technologies, reconfigurable/morphing robotic AI smart deployable structures and advanced power systems.				
Title: Oversight Description: Oversight will develop and demonstrate a suite of autonomy technologies to provide constant custody of targets as a service for tactical operations in contested environments. Existing and emerging space systems will be evaluated. Proliferated Low Earth Orbit (p-LEO) satellite constellations and payloads will be leveraged due to their high-bandwidth, processing-on-the-edge capabilities in support of tactical, efficient, integrated missions at scale. Oversight will develop autonomous technology to		-	12.273	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
enable advanced collaboration among constellations of satellites for target custody in contested environments where the numbers of targets is far greater than the number of satellites and sensors over the operating area. The Oversight program includes a demonstration using existing on-orbit p-LEO assets combined with live, virtual and constructive terrestrial assets. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-07. FY 2026 Plans: - Complete objective capabilities. - Conduct further live demonstrations. - Transition applicable technologies to stakeholders. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Title: Expeditionary Carbon Utilization for energy Resilience and Stabilization (ExCURSion) Description: Two solutions to expeditionary energy currently exist: fuel and electric systems. Fuel has a high energy density but requires supply lines that are logistically complicated and pose strategic and tactical weaknesses. Electric systems are operationally flexible and rechargeable, but their energy density is an order of magnitude lower than fuel. The Expeditionary Carbon Utilization for Energy Resilience and Stabilization (ExCURSion) program aims to overcome key technical challenges to enable a closed-loop energy storage system with the rechargeability of electric systems and the high energy density of fuel. The program will accomplish this goal by aggressively exploring and evaluating foundational technologies of carbon dioxide reduction, capture, and storage. Technologies developed under this program will serve as key components for a future self-enclosed system, addressing the need for energy resilience in an expeditionary setting, independent of a fuel supply line, while delivering a rechargeable fuel cell with far greater energy density than batteries. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-04. FY 2026 Plans: - Complete development of carbon dioxide sorption systems that can withstand typical impurities found in post-combustion streams under a variety of burn conditions. - Integrate carbon dioxide reduction catalysts with reactor platforms to produce a carbon electrolysis system with a high fuel production rate in a compact form factor. - Complete a benchtop reversible carbon fuel cell as a first step towards a future fully rechargeable carbon-based battery system. FY 2026 to FY 2027 Increase/Decrease Statement:		-	2.811
			-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects program completion.			
Title: Robotic Autonomy in Complex Environments with Resiliency (RACER) Description: Multi-domain operations (MDO) present complex and challenging environments to ground combat platforms. Ground combat platforms must operate in a more distributed manner in these environments to gain a sustained tactical advantage and enhance warfighter survivability. The Army intends to deploy autonomous robotic combat vehicles and optionally manned fighting vehicles to accomplish this objective. In order to meet the demands of an MDO environment, significant advances in perception, planning, and control algorithms are required to autonomously maneuver faster and more resiliently in complex and novel off-road situations. Maneuver environments are characterized by three-dimensional surfaces of highly compliant soils and vegetation, hundreds of positive and negative obstacle classes, no defined road networks or driving rules, and where use of terrain for survivability is critical. In order to achieve operationally relevant speeds and resilience to novel situations on the battlefield, while simultaneously reducing the soldier's cognitive and communications burden and increasing battle space awareness, Robotic Autonomy in Complex Environments with Resiliency (RACER) will develop and demonstrate game-changing autonomous ground combat vehicle mobility using a combination of simulation and advanced platforms. RACER will deliver autonomy algorithms using the latest in Artificial Intelligence (AI) and machine-learning techniques, a code repository, an off-road simulation environment tailored for military off-road autonomy development, tactical route planning methods, and field-demonstrated off-road autonomous capabilities. The RACER program will demonstrate fully autonomous maneuver on a military Unmanned Ground Vehicle (UGV) in a variety of militarily relevant environments. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-04. FY 2026 Plans: - Integrate combat relevant payloads onto both the large-scale and small-scale platforms to experiment with US Army units. - Complete exploration of new sensing methods and perception models to determine the implications on speed for off-road autonomy. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	5.720
Accomplishments/Planned Programs Subtotals		-	92.642
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENE SS	Project (Number/Name) ACS-01 / ALL DOMAIN TACTICAL TECHNOLOGIES
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602023E / ACCESS AND AWARENESS				Project (Number/Name) ACS-07 / ACCESS AND AWARENESS SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ACS-07: ACCESS AND AWARENESS SUPPORT	0.000	0.000	5.627	8.204	-	8.204	16.380	16.840	14.767	14.767	-	-

A. Mission Description and Budget Item Justification

The Access and Awareness Support project contains management costs in support of DARPA functions and activities across the entire Access and Awareness PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0602702E, Projects TT-03, TT-04, TT-07, and TT-13.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFORMANCE
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	307.727	364.141	-	364.141	381.379	407.087	412.853	419.202	-	-
WP-01: BIOLOGICALLY BASED MATERIALS AND DEVICES	0.000	0.000	125.704	147.045	-	147.045	142.755	164.493	166.704	170.053	-	-
WP-02: BIOMEDICAL TECHNOLOGY	0.000	0.000	166.496	189.962	-	189.962	202.565	205.750	208.389	211.389	-	-
WP-07: WARFIGHTING PERFORMANCE SUPPORT	0.000	0.000	15.527	27.134	-	27.134	36.059	36.844	37.760	37.760	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$307,727 thousand of discretionary and \$87,500 thousand of mandatory (reconciliation) for a total of \$395,227.

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Warfighting Performance program that is focused on developing biologically based materials and biological technologies that make possible a wide range of new military capabilities. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Biologically Based Materials and Devices project will leverage the growing application space of the biological sciences for the development of new DoW capabilities to improve the sustainability of warfighters and operational platforms in varied domains. This project will develop solutions for critical resource processing, materials development, threat detection and characterization, and warfighter resilience to infectious disease. The materials and technologies developed through this project will protect and sustain warfighters and operations in austere conditions, enhancing warfighter resiliency and lethality.

The Biomedical Technology project focuses on applied research for medical related technologies that will maintain warfighter health and performance before, during, or after operations. Successful technologies within this project will maintain warfighter health against emerging threats through novel biothreat detection, rapid medical countermeasure identification and development, and distributed production of effective therapeutics. In-theater, warfighter health will be maintained through the development of field-relevant technologies such as reliable and accessible critical medical resources, novel detection and protection capabilities for traumatic brain injury, and rapid, effective triage of battlefield injuries. Additionally, this project will improve warfighter readiness by characterizing and assaying physical and cognitive performance to drive data-driven awareness.

The Warfighting Performance Support project contains management costs in support of DARPA functions and activities across the entire Warfighting Performance PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602024E / <i>WARFIGHTING PERFORMANCE</i>
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transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0602715E, Materials and Biological Technology, and PE 0602115E, Biomedical Technology.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	278.121	0.000	-	0.000
Current President's Budget	0.000	307.727	364.141	-	364.141
Total Adjustments	0.000	29.606	364.141	-	364.141
• Congressional General Reductions	0.000	-1.805			
• Congressional Directed Reductions	0.000	-3.589			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	35.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	364.141	-	364.141

Congressional Add Details (\$ in Millions, and Includes General Reductions)

Project: WP-01: *BIOLOGICALLY BASED MATERIALS AND DEVICES*

Congressional Add: *Heavy Rare Earth Elements (REE) Separation*

Congressional Add: *Rare Earth Elements (REE) From Valorized Waste*

Congressional Add Subtotals for Project: WP-01

Congressional Add Totals for all Projects

FY 2025	FY 2026
-	15.000
-	20.000
-	35.000
-	35.000

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation offset by Congressional adds for Heavy Rare Earth Elements Separation and Rare Earth Elements From Valorized Waste.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE				Project (Number/Name) WP-01 / BIOLOGICALLY BASED MATERIALS AND DEVICES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
WP-01: BIOLOGICALLY BASED MATERIALS AND DEVICES	0.000	0.000	125.704	147.045	-	147.045	142.755	164.493	166.704	170.053	-	-

A. Mission Description and Budget Item Justification

The Biologically Based Materials and Devices project will leverage the growing application space of the biological sciences for the development of new DoW capabilities to improve the sustainability of warfighters and operational platforms in varied domains. This project will develop solutions for critical resource processing, materials development, threat detection and characterization, and warfighter resilience to infectious disease. The materials and technologies developed through this project will protect and sustain warfighters and operations in austere conditions, enhancing warfighter resiliency and lethality.

Prior to FY 2026, efforts in this Project were funded in PE 0602715E, Project MBT-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Turning Upcycled Waste into Novel, Sustainable Materials Description: Currently, the DoW relies on critical materials and commodity molecules, such as petroleum-, rubber- and wood-derived products, which are needed to protect and provide mobility to our warfighters in an austere, expeditionary setting. Unfortunately, providing these materials to our warfighters also entails risks, such as fragile supply chains, foreign sourcing, or costly shipping to points of need. These critical materials also contribute to DoW waste streams with no further value while also creating logistical challenges. To address this, the Turning Upcycled Waste into Novel, Sustainable Materials program is investigating the feasibility of converting abundant DoW waste stream products (e.g., tires, scrap wood, and paper) into durable and sustainable materials. Approaches will be investigated to develop materials suitable for use in applications ranging from contingency construction materials to commodity molecules. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Develop methods for biology-based switching between feedstocks or products for manufacturing chemicals. - Identify techniques for stabilizing biological systems for continuous chemical production. - Evaluate approaches to temporarily and reversibly switch the chemical product of an industrial scale biomanufacturing system. - Explore the technoeconomic advantages of switchable biomanufacturing systems compared to state-of-the-art single-input, single-output batch production. - Demonstrate highly efficient production and scalability of DoW-relevant materials from wood waste feedstocks. FY 2027 Plans:	-	15.110	11.672

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-01 / BIOLOGICALLY BASED MATERIALS AND DEVICES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Expand biology-based switching to include multiple pathways for consuming different feedstocks or producing different chemicals. - Extend the stability of biological systems for continuous production. - Implement approaches for switchable production in an industrially-relevant biomanufacturing system. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects increased focus on final production efforts.				
Title: eX Virentia (eXVi) Description: The eX Virentia (eXVi) program is investigating the use of existing unmodified vegetation as a sensor to identify activities of DoW interest by utilizing natural plant responses to man-made chemical exposures. The eXVi program will bring together the research communities of plant biology, remote sensing, and computational modeling to uncover the internal biochemical response of plant species to chemical exposure, and how they relate to external observable characteristics. The program will grow plant species across various types of vegetation, measure internal and external responses, and model the responses with the goal of scaling predicted responses to similar species across phylogenies. If successful, discovery of predictable and distinct cause-and-effect relationships between exposure and plant response will significantly improve situational awareness of hazards and materials activity, thereby limiting harmful exposure to the warfighter. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Initiate plant acquisition, growth, and stabilization for selected species. - Initiate plant chemical exposure experiments to identify biochemical, molecular, and physiological patterns associated with exposure of plants to target stressors. - Perform plant exposure evaluations using multi-omics and external observable measurements. - Develop plant biological process model(s) to understand biochemical, molecular, and physiological patterns associated with exposure of plants to selected stressors. FY 2027 Plans: - Complete long term plant growth and stabilization for selected species to support scaling approach. - Complete chemical exposure experiments to identify biochemical, molecular, and physiological patterns associated with exposure of plants to target stressors. - Evaluate plant exposure using multi-omics and external observable measurements. - Determine whether external plant observables provide a distinct, predictable, and scalable indicator of chemical exposure. FY 2026 to FY 2027 Increase/Decrease Statement:		-	12.487	12.839

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE		Project (Number/Name) WP-01 / BIOLOGICALLY BASED MATERIALS AND DEVICES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects minor program repricing.					
Title: Biological Undersea Energy Description: The Biological Undersea Energy program aims to develop emerging technologies that ensure the DoW has the capability to maintain a presence in austere oceanic conditions to provide advanced knowledge of resources and conditions and achieve desired mission effects. Approaches will be developed that utilize biological processes and products to provide energy for improved endurance and performance capabilities while reducing the reliance on servicing or resupply. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Perform a Biosafety Design and Analysis for each initial brassboard system in development to assess all required biosafety control measures employed by the systems. - Finalize initial brassboard system designs that will be used for field demonstration. - Complete a technology field demonstration to demonstrate conversion of organic matter to electrical power in the marine domain. - Test and evaluate initial brassboard system. FY 2027 Plans: - Increase power generation from conversion of organic matter compared to initial brassboard systems. - Validate that final brassboard systems can perform in stakeholder domains. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift in focus from technology development and prototype building activities to final field testing and analysis.			-	13.647	5.427
Title: Strengthening Resilient Emotions and Nimble Cognition Through Engineering Neuroplasticity (STRENGTHEN) Description: The Strengthening Resilient Emotions and Nimble Cognition Through Engineering Neuroplasticity (STRENGTHEN) program aims to overcome the limitations of focusing on descriptions of individual disease effects and suicide risk factors by adopting a transdiagnostic approach that addresses the mechanisms (i.e., predictors or causes) of mental health and wellbeing. STRENGTHEN will optimize the brain networks essential for Cognitive Flexibility and Emotional Regulation, establishing dose response, time-to-onset, and duration-of-effect curves to quantify the impact of change in Cognitive Flexibility and Emotional Regulation on validated measures of suicidality, behavioral health, and wellbeing within DoW. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02.			-	8.969	0.625

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-01 / BIOLOGICALLY BASED MATERIALS AND DEVICES		
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Revise the mechanistic models of Cognitive Flexibility and Emotional Regulation to re-test predictive validity and optimize the model's ability to predict a behavioral response. - Assess brain network models in all three risk populations (low risk, at risk, high risk). - Refine and tailor hybrid interventions to maximize synergistic and sustaining mental health benefits for each target population (low risk, at risk, high risk). FY 2027 Plans: - Conduct large-scale validation studies of optimized Cognitive Flexibility and Emotional Regulation interventions across diverse warfighter populations to establish deployment readiness. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from model optimization to validation studies.					
Title: Biotechnology for Challenging Environments Description: The Biotechnology for Challenging Environments program is developing novel biological solutions to enable warfighter operations in remote and extreme conditions. As the DoW expands operations into previously inaccessible domains, new and unique logistical constraints imposed by extreme conditions and resource scarcity threaten warfighter and warfighting platform readiness. This program will develop technologies using biological approaches to protect and maintain performance of warfighters and warfighting platforms, such as electronics and infrastructure, from challenging conditions. Technology advances developed in this effort will extend mission duration and enhance operational capabilities in emerging domains. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Initiate studies of interactions with biological systems for demonstrating safety. - Optimize molecules and materials for performance in extreme cold weather. - Scale-up molecules and materials for laboratory scale testing and evaluation. FY 2027 Plans: - Perform operationally relevant field testing of selected test materials. - Conduct post-field-testing laboratory analysis to characterize performance for key stakeholder use cases. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift to field testing and post analysis of limited, high priority applications.			-	6.126	4.979
Title: Field Forward Biothreat Storage Solutions for Force Protection			-	11.585	2.500

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: Warfighters are currently deployed to emerging disease hotspots with increasing pathogen spillovers. Biosurveillance groups tasked with force health protection rely on cold chains and transport media to maintain sample viability for characterization in a laboratory setting, but these methods are unreliable, sometimes inaccessible, or limited in their utility. The Field Forward Biothreat Storage Solutions for Force Protection program will offer expanded capabilities for microbial threat characterization by developing systems capable of long-term, cold chain-free storage of microbial samples. Systems that are able to reliably store and retrieve viable microbes over long timescales will ensure that collected samples reach the lab for study in a usable state, allowing the DoW to better leverage its field-forward laboratories to perform pathogenicity assessments for countermeasure development. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Review demonstrator system design(s) and down-select technologies for continued development. - Acquire a greater diversity of microbes as well as real-world samples to test storage and retrieval methods. - Demonstrate ability to store and retrieve multiple microbes without a cold chain for a sustained duration. - Demonstrate that the system can operate in at least one austere condition. FY 2027 Plans: - Down-select technologies for continued development based on test results and demonstrator performance. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift toward infield testing and targeted tech stress testing.				
Title: Genetic Utilization for Advanced Regulation and Defense of Indigenous And Native Species (GUARDIAN) Description: Eradicating invasive species is critical to safeguarding the homeland, protecting DoW assets, and demonstrating resilience that can help deter potential adversaries. The Genetic Utilization for Advanced Regulation and Defense of Indigenous And Native Species (GUARDIAN) program aims to rapidly adapt and deploy modern genome editing tools to eliminate invasive species like the New World Screwworm and Brown Tree Snake precisely and controllably, meeting all applicable regulatory standards. The accidental introduction of the Brown Tree Snake to DoW installations has caused ecological devastation and millions of dollars in annual infrastructure damage and poses strategic risks to military operations, costing millions of dollars annually in unsuccessful eradication efforts. GUARDIAN's modular, extendable, and adaptable approach will provide the DoW and interagency partners with a toolset for rapidly defending against these pests, with potential applications for other invasive species, thereby enhancing warfighter readiness. FY 2026 Plans:		-	8.988	25.283

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Identify candidate biological targets that demonstrate highly selective metal interactions or properties that may be engineered to induce highly selective metal interactions. - Baseline the ability of biological candidates to successfully produce a metal or mineral product from simulated feedstocks. - Explore model-driven workflows for rapid design-build-test cycles of biological candidates. FY 2027 Plans: - Demonstrate model-driven, rational design of biological candidates and increase strength and selectivity of metal/mineral interactions. - Evaluate top biological metal/mineral processing candidates under real-world processing workflows and conditions. - Perform process durability assessments. - Refine the list of candidate biological targets. - Produce a high purity metal/mineral product from a real-world feedstock. - Produce a high purity metal/mineral product from a traditionally uneconomical waste feedstock. - Demonstrate potential for technological scale-up using real world feedstocks. - Initiate a techno-economic analysis to validate the commercial viability and industrial potential of the technology. - Assess the potential for adapting the developed bioprocessing platforms to process other metals and minerals of strategic importance to national security. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from proof-of-concept demonstrations and model development to engineering biologics for industrial processing of real-world feedstocks at scale.					
Title: Fleetwood Description: The Fleetwood program aims to enhance the DoW's capability to produce critical materials and molecules from biomass sources, including agricultural and forestry residues. The program will explore solutions to provide the critical materials to protect, support, and sustain warfighters in austere terrestrial conditions. This program will investigate novel catalytic materials for the conversion of domestic and global biomass sources into critical materials and commodity molecules for DoW applications and use. This program will leverage chemical and biological methods to provide enhanced materials that support the warfighter and mission execution. The technologies developed will improve domestic manufacturing resiliency and reduce reliance on fragile supply chains and foreign sources of these chemicals. FY 2027 Plans: - Investigate approaches for the development of chemical and biological processes and biologically derived materials to support and sustain the warfighter in all conditions.			-	-	20.000

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Initiate formulation and characterization of materials chemistries for the catalytic separation of domestic agricultural and forestry residues at the laboratory scale. - Initiate formulation and characterization of materials chemistries for the catalytic conversion of domestic agricultural and forestry residues to value-added chemicals at the laboratory scale. - Explore methods that leverage and combine recent advances in catalyst material development to improve the stability, activity, and selectivity of materials for biomass processing at the laboratory scale. - Explore formulation and characterization of materials chemistries for the catalytic separation and conversion of global agricultural and forestry residues to value-added chemicals at the laboratory scale. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects program initiation to enhance the DoW's capability to produce critical materials and molecules from biomass sources, including agricultural and forestry residues.				
Title: Biologically Based Materials and Devices Studies and Concepts Description: The Biologically Based Materials and Devices Studies and Concepts thrust aims to enhance the DoW's capability to conduct operations in varied conditions. The DoW operates in all domains, which creates a myriad of difficult burdens to the warfighter and sustained operations. This thrust will develop solutions to protect, support, and sustain warfighters and operational capabilities in austere conditions, whether terrestrial, aquatic, in the far atmosphere, or beyond. Technologies developed in this thrust will leverage biological or bio-inspired components to provide enhanced materials and devices that support the warfighter and mission execution. Research conducted will support biosurveillance and biosecurity with advanced detection systems, predictive analytics, and innovative countermeasure development to identify, monitor, and mitigate biological threats, ensuring global health security and resilience. FY 2027 Plans: - Initiate technology development for the integration of dynamic and responsive biological organisms or bio-inspired molecules into composite materials. - Investigate approaches for the development of biological processes and biologically derived materials to support and sustain the warfighter in austere conditions. - Explore methods that leverage and combine recent advances in biological sensors, interventions, and technologies to develop novel platforms for threat detection and analysis, in both humans and agricultural crops. - Investigate the ability of biological molecules and processes to perform calculations to enable highly adaptable and efficient computational systems and sensors. - Initiate technology development for biohybrid systems capable of enhanced threat sensing, protection, and remediation. FY 2026 to FY 2027 Increase/Decrease Statement:		-	-	30.114

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in biologically based computational systems and sensors, and novel biological sensors for threat detection and analysis.					
Title: Materiel Protection through Biologics Description: Military infrastructure and systems are expected to function years beyond their original intended lifetime but are subject to degradation by external factors. For instance, the formation of biofilms is ubiquitous, corroding and biofouling many military systems, such as aircraft, fuel tanks, ships, medical devices, and filtration systems for water and air, ultimately costing the DoW billions of dollars annually to repair and maintain. The Materiel Protection through Biologics thrust is developing approaches to sustain military infrastructure and systems by developing biological or bio-inspired technologies to imbue beneficial functions into existing systems, resulting in benefits such as, but not limited to, reducing drag and mitigating corrosion. These bio-inspired interventions will protect and sustain equipment and infrastructure, reducing operation costs and increasing service lifetime. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Validate that models can predict microbial communities and/or fiber-based approaches after two months. - Identify microbial communities and/or fiber-based approaches that produce and maintain a desired function in the field. - Verify microbial communities and/or fiber-based approaches are either resistant or resilient to multiple perturbations in the field. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			-	5.797	-
Title: Environmental Microbes as a Bioengineering Resource (EMBER) Description: The Environmental Microbes as a Bioengineering Resource (EMBER) program aims to develop novel, bio-based technologies to overcome key challenges facing the domestic supply of Rare Earth Elements (REEs) critical to the U.S. and DoW. This program is leveraging the diversity, specificity, and customizability of environmental microbiology to enable new domestic biomining methods for the separation, purification, and conversion of REEs into manufacturing-ready forms. Advances in this area will deliver capabilities to assure access to DoW-critical materials domestically and in operational settings. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-02. FY 2026 Plans: - Demonstrate separation and efficient recovery of at least eight individual REEs with high purity starting from mining partner source material using a biomining approach. FY 2026 to FY 2027 Increase/Decrease Statement:			-	1.745	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	90.704
		FY 2025	FY 2026
Congressional Add: Heavy Rare Earth Elements (REE) Separation		-	15.000
FY 2026 Plans: - Scale and validate REE separations primarily utilizing end-of-life (EOL) magnet feedstocks and optimize operational and economic performance for commercial scale production.			
Congressional Add: Rare Earth Elements (REE) From Valorized Waste		-	20.000
FY 2026 Plans: - Develop and optimize advanced heavy and medium REE separations technologies to unlock domestically available feedstocks.			
Congressional Adds Subtotals		-	35.000
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE				Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
WP-02: BIOMEDICAL TECHNOLOGY	0.000	0.000	166.496	189.962	-	189.962	202.565	205.750	208.389	211.389	-	-
A. Mission Description and Budget Item Justification												
The Biomedical Technology project focuses on applied research for medical related technologies that will maintain warfighter health and performance before, during, or after operations. Successful technologies within this project will maintain warfighter health against emerging threats through novel biothreat detection, rapid medical countermeasure identification and development, and distributed production of effective therapeutics. In-theater, warfighter health will be maintained through the development of field-relevant technologies such as reliable and accessible critical medical resources, novel detection and protection capabilities for traumatic brain injury, and rapid, effective triage of battlefield injuries. Additionally, this project will improve warfighter readiness by characterizing and assaying physical and cognitive performance to drive data-driven awareness.												
Prior to FY 2026, efforts in this Project were funded in PE 0602115E, Project BT-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: BioElectronics to Sense and Treat (BEST)									-	16.725	14.180	
Description: The BioElectronics to Sense and Treat (BEST) program seeks to improve warfighter recovery from battlefield injuries by developing technologies to diagnose, prevent, and treat wound infections in real time. This program will generate high-resolution sensors that continually monitor a wound by assessing the community composition of contaminating microorganisms and/or the host immune response. Data from these sensors will be used to predict if a wound will become infected and to regulate the targeted administration of known and novel treatments to prevent infections or resolve existing infections. Ultimately, the sensor and treatment elements will be combined into a closed-loop smart bandage that can provide rapid diagnostics and precise treatments at all levels of deployed military medical care.												
Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01.												
FY 2026 Plans:												
- Finalize the characterization of microorganisms present in tissue samples from infected wounds.												
- Continue optimization and improvement of algorithms to accurately predict the healing outcome of individual wounds.												
- Start sensor development to identify and characterize microorganisms within wounds in real time.												
- Initiate the development of an electronic platform to produce, dispense, or provide antimicrobial therapeutics.												
- Initiate the development of approaches to determine the appropriate treatments for different types of infections.												
FY 2027 Plans:												

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Demonstrate accuracy in identifying single and mixed groups of microorganisms in vitro. - Demonstrate efficacy for at least one novel wound infection treatment approach in vitro. - Determine the appropriate regulatory pathway(s) for the wound infection sensing and treatment devices with the Food and Drug Administration. - Finalize the engineering designs or schematics for the wound infection sensing and treatment devices. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift in focus from early technology development and in vitro testing to in vivo validation.			
Title: Novel Delivery Technology for Medical Countermeasures Description: The DoW requires rapid development of medical countermeasures (MCMs) to ensure force health protection and improve our ability to respond to emerging and novel biological threats. Despite recent advancements in development of new MCMs, challenges with delivery limit their current therapeutic potential. While emerging targeted delivery systems such as polymer/lipid nanoparticles and viral vectors have enabled the delivery of large, complex MCM molecules, they are still plagued by lack of widespread availability and effectiveness. Investing in efficient, adaptable delivery technology is crucial for strengthening biosecurity preparedness and will enable rapid response to the evolving biological threat landscape, whether the threat is natural or manmade. The Novel Delivery Technology for Medical Countermeasures program is developing minimally invasive MCM delivery systems, in which any therapeutic can be quickly formulated and administered to treat or prevent any disease. Developing novel delivery platforms will maintain warfighter health and readiness and enable rapid response to existing and novel biothreats. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Demonstrate that the delivery chassis and formulation strategies are capable of delivering countermeasures to multiple cell types experimentally and in whole organisms. - Deliver countermeasure cargo in vivo using a simple administration method. - Assess the safety of the delivery chassis (e.g., minimal adverse effects) in vivo. - Measure the duration of protection following countermeasure administration using the delivery chassis. FY 2027 Plans: - Demonstrate delivery of countermeasure cargo in additional cell types. - Demonstrate systemic delivery of countermeasure cargo. - Develop and demonstrate scale-up manufacturing of the highest-performing delivery vehicles. - Demonstrate delivery of two or more MCM modalities. FY 2026 to FY 2027 Increase/Decrease Statement:		-	19.037
			12.121

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects a reduction in research and development activities to conduct verification and validation studies of candidate countermeasure delivery vehicles.			
Title: Alert WARfighter Enablement (AWARE) Description: The Alert WARfighter Enablement (AWARE) program is developing a combination drug and device to non-invasively increase alertness following sleep loss in humans without negative side effects and with reduced addictive potential. The AWARE program aims to develop candidate photoswitchable (light responsive) molecules that can be reversibly activated in the presence of near infrared (NIR) light and wearable device elements that emit NIR light to activate the candidate molecules. The combination of an ingested photoswitchable molecule and a NIR-emitting device will selectively activate neural pathways responsible for executive function, working memory, and decision making, wherever and whenever both drug and light are simultaneously present. If successful, AWARE technology will enable warfighters to maintain cognitive function and alertness during long-range missions and to obtain restorative sleep when needed without negative side effects. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Demonstrate activation of at least half of the candidate molecules using NIR light. - Demonstrate that candidate molecules experience conformational change and maximum absorbance in the desired wavelength of NIR light. - Characterize distribution and how long candidate molecules remain in the brain. - Integrate NIR light sources into a wearable device while minimizing weight. - Develop and integrate a rechargeable or replaceable power supply platform into the wearable device without a significant increase to overall weight. FY 2027 Plans: - Refine NIR light delivery parameters to improve activation of candidate molecules. - Design and initiate safety and efficacy studies to support an Investigational New Drug (IND) submission to the U.S. Food and Drug Administration. - Characterize the effect of NIR light-activated candidate molecule(s) on cognitive performance following sleep loss in an animal model. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from technology development to regulatory approval activities.		-	12.116
Title: Reengineering Enabling Sleep Transitions in Operationally Restrictive Environments (RESTORE)		-	6.474
			5.684

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Description: The Reengineering Enabling Sleep Transition in Operationally Restrictive Environments (RESTORE) program aims to enhance the efficiency of sleep to optimize warfighter performance following sleep restriction, as commonly occurs in combat operations. Sleep is a critical factor in cognitive function and decision-making, yet the tempo of military operations often results in acute and chronic sleep restriction. To optimize sleep recovery in such conditions, RESTORE seeks to develop a multi-modal system that enables precision control of sleep macro- (i.e., orchestration of transition between sleep phases) and micro-architectures (i.e., efficiency within sleep phases). By optimizing efficiency of recovery despite sleep restriction, RESTORE seeks to enable warfighter to perform at their best, even in the most challenging operational environments, and to enhance their overall readiness and effectiveness. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Continue design of a multi-modal system that orchestrates and optimizes sleep architectures. - Design an evaluation protocol to evaluate changes in sleep architectures and their effect on sleep quality. - Design an evaluation protocol to evaluate effects of changes in brain activity and sleep enhancement on cognitive performance. FY 2027 Plans: - Conduct trials to validate system performance and refine interventions for optimal effectiveness. - Assess gaps in system readiness and outline steps for military integration and deployment. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from protocol design to prototype evaluation.			
Title: Controlled Genome Protection Description: The Controlled Genome Protection program is developing advanced capabilities to control and tune the activity of gene editing technologies. Advances in synthetic biology have significantly expanded the number and types of genome editors and modulators available. Many of the new genome editors have been identified from rare, slow-growing microorganisms with unique metabolic capabilities. New tools across these new classes of genome editors are required to advance our understanding, control, and use of gene editing technologies across all types of life. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Demonstrate improved potency genome editor inhibitor activity. - Develop methods to detect off-target effects of new inhibitors of genome editing.		-	7.368
			5.447

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Demonstrate computational platform ability to rapidly discover new inhibitors of genome editors. - Demonstrate the ability of newly developed methods to accurately characterize inhibition of gene editors in vitro. FY 2027 Plans: - Demonstrate successful end-to-end computational design, nomination, and validation of inhibitor candidates. - Demonstrate a novel inhibitor discovery pipeline. - Develop, demonstrate, and deliver a complete technology transfer package. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift in research and development activities to conduct verification and validation studies of complete end-to-end pipeline for discovery of high-performing candidate inhibitors.			
Title: Red Blood Cell Factory (RBC-Factory) Description: Warfighters operate in extreme, austere, and dangerous domains that stress physiology, contain pathogens, and compromise health and performance. Protective equipment often hinders performance (e.g., loss of dexterity, rapid overheating, or visual obstruction) and pharmacological approaches can carry unpleasant-to-dangerous side effects. The Red Blood Cell Factory (RBC-Factory) program is exploring the limits of what biologically active components, or cargoes, can be placed inside red blood cells (RBCs) to help protect the warfighter from within. Cargoes can include natural or synthetic products and must not perturb the physiological characteristics that allow RBCs to safely operate in a body. RBC-Factory aims to deliver a method capable of high-throughput modification of RBCs and a knowledge product identifying what kinds of cargo and how many can be put inside an RBC. RBC-Factory will determine if RBCs are a suitable vector for medium-term passive protection for warfighters against threats. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Develop non-genetic RBC modification technology prototype devices for delivery of cargo to RBCs. - Complete an in vitro RBC cargo survey to assess the impact of the RBC modification process on RBC survival and function. - Compare changes in RBC functions between normal RBCs and RBCs with cargoes in a laboratory environment. - Determine if RBC circulation time is changed by the addition of different cargoes. FY 2027 Plans: - Evaluate the physiological changes and immunological effects of modified RBCs based on cargo parameters. - Validate RBC modification and determine if the approach can be scaled.		-	10.053
			5.313

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Assess the impact of the RBC-modified cargos and device platforms on RBCs.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift to focus on the device validation and data gathering of RBC cargo capacity and physiology.			
Title: GOLDeN hour extended EVACuation (GOLDEVAC) Description: Survival from combat trauma is predicated on a rapid sequence of events that escalate care as soon as possible. The GOLDeN hour extended EVACuation (GOLDEVAC) program is developing new technologies that will provide medical care closer to the point of injury and facilitate effective casualty evacuation. This will extend the time available to evacuate injured service members to higher levels of care. This program will explore technologies that enable autonomous care of patients and/or which act as force multipliers that enable caregivers to effectively take care of a larger number of patients. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Model the physiological effects of long-duration autonomous care to inform hardware design. - Build and begin testing of autonomous care technologies that combine oxygen and hemorrhage support prior to use in animal models. - Collect end user (field and aviation medic) feedback to identify key needs in austere domains to support design goals for portability. - Construct testbeds for evaluating polytrauma conditions (i.e., bleeding and oxygenation support capabilities) using animal models. - Develop a regulatory submission plan that provides a pathway for future human evaluation and testing. FY 2027 Plans: - Demonstrate the usefulness of care technologies that directly integrate oxygen and hemorrhage support during polytrauma (i.e., injury conditions requiring both types of support). - Develop a proof-of-concept hardware feasibility assessment illustrating how oxygen and hemorrhage support hardware could be combined into a portable form factor. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects completion of laboratory data collection efforts and a shift to proof-of-concept demonstrations.		-	18.496
Title: Generative Optogenetics (GO) Description: The Generative Optogenetics (GO) program is aimed at developing a protein complex that can be expressed within a living cell that responds to light waves to synthesize a corresponding nucleic acid sequence. This groundbreaking technology		-	15.580
			4.915

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
will leverage novel light-controlled cells, advanced protein engineering, and computational design to create a molecular interface between information systems and biological systems. The GO program will enable precise programming of genetic sequences in living cells, which can be leveraged to develop advanced biosensors, on-demand therapeutics, and cellular systems that enhance warfighter resilience, threat detection, and operational sustainment in austere conditions. FY 2026 Plans: - Initiate development of a light-controlled molecular machine capable of producing a nucleic acid sequence. - Establish biosecurity and regulatory policy working groups as well as an independent commercialization and consulting group to ensure responsible development, regulation, and commercialization. - Identify key technical risks and outline mitigation strategies for a functional molecular machine. - Engage with the broader scientific community to communicate scientific knowledge and establish collaborative technical partnerships. FY 2027 Plans: - Generate molecular components that respond to distinct wavelengths of light. - Generate molecular components capable of producing nucleic acid sequences in response to light. - Initiate the development of a mechanism capable of detecting errors in a nucleotide sequence. - Initiate the integration of the developed components to generate a molecular machine capable of responding to light to produce a short nucleic acid sequence. - Begin integration of the error detection mechanism into the molecular machine. - Evaluate the molecular machine for accuracy and efficiency to produce short nucleotide products. - Initiate transition of the molecular machine and error detection mechanism into a living cell. - Optimize the molecular machine to increase the length of nucleic acid sequence produced. - Optimize the molecular machine to enhance the rate of production of nucleic acid sequences. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 funding increase reflects the expansion of efforts from concept initiation and tool design to molecular machine integration and validation.			
Title: Medics Autonomously Stopping Hemorrhage (MASH) Description: In large-scale combat operations, the inability to rapidly provide damage control surgery and lifesaving intervention results in preventable warfighter casualties. The Medics Autonomously Stopping Hemorrhage (MASH) program will automate a critical damage control surgery step for non-compressible torso hemorrhage (NCTH) by autonomously finding and stopping bleeding in the torso. MASH will provide the necessary technology to guide robotic surgery platforms to navigate to internal hemorrhage sites to stop the bleed and extend the time available for warfighters to access higher levels of care. This program		-	14.005

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
will explore technologies that enable autonomous surgical care of patients, which enables hemorrhage control across a range of relevant wound patterns.				
FY 2026 Plans: - Initiate experiments to anatomically identify the location of a bleed in simulated physical models and animal models. - Evaluate sensor suites capable of autonomous identification of a bleed location in compact form factors. - Begin development of a computational model to guide automated identification and navigation to bleed in a simulated environment. FY 2027 Plans: - Assess mechanisms to achieve localization of a bleed in animal models. - Develop systems capable of autonomous identification and navigation in desired form factors. - Demonstrate successful navigation of a sensor suite and end effector to key anatomical regions in animal models. - Demonstrate performance of context-aware bleed localization technologies that will enable end effector placement for NCTH control. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects efforts to develop demonstrator devices for end of phase demonstrations.				
Title: Protean Description: The acceleration of AI-based technologies has vastly expanded the opportunity space for malicious actors to rapidly discover and develop novel chemical weapons. The Protean program will develop candidate medical countermeasures (MCMs) that render entire chemical threat classes non-toxic by protecting human protein function. This approach shifts away from traditional symptom control and agent-specific interventions by addressing the root cause of intoxication at the human protein level. By leveraging emerging insights into understanding protein dynamics, Protean aims to design novel drug compounds that engage with unappreciated structures of various enzymes, receptors, and ion channels, which are the protein classes most commonly impacted by chemical agents and biological toxins. This program aims to provide warfighters with complete protection against chemical threats. FY 2026 Plans: - Establish an experimental structural biology technique with a protein target. - Initiate experimental and computational characterization of protein dynamics. FY 2027 Plans: - Predict binding sites within a protein dynamic trajectory.		-	6.470	15.729

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Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Identify differences in protein dynamics between functional and dysfunctional protein targets that are relevant to chemical threat mechanisms. - Determine the structure and behavior of uncommon protein states. - Identify and validate energetically linked sites on the protein. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the expansion of efforts from predictions of simple, common, and accessible proteins to characterizing more complex, uncommon, and difficult-to-capture proteins.				
Title: Smart Red Blood Cells (Smart-RBC) Description: The Smart-Red Blood Cells (Smart-RBC) program aims to engineer red blood cells to contain novel biological features that can safely, temporarily, and reliably alter function. Specifically, the Smart-RBC program will revolutionize blood products by creating smart red blood cells (SRBCs). SRBCs will contain additional biological circuits consisting of three layers: Sense a target; Decide next steps; Act by creating an output that can alter metabolism or function. The program will identify the art of the possible to create SRBCs to enhance human performance and maximize survival during life-threatening bleeding. FY 2026 Plans: - Initiate studies on editing cells to produce, position, and retain novel proteins in mature red blood cells (RBCs). - Run a computer simulation of the gene editing impact on RBC metabolism. FY 2027 Plans: - Continue studies on editing cells to produce, position, and retain novel proteins in mature RBCs. - Continue computer simulation of gene editing impact on RBC metabolism and virtually optimize protein position/retention on mature RBCs. - Initiate experiments to produce, position, and retain novel proteins in mature RBCs. - Identify any changes in RBC function due to the genetic modifications when compared to unmodified cells. - Initiate simulation experiments to optimize protein position/retention on a mature RBC. - Assess the type and character of circuits possible in RBCs based on gene editing experiments and protein expression. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the expansion of studies on editing cells to produce, position, and retain novel proteins within and on the surface of RBCs.		-	4.003	21.247
Title: Biomedical Technology Studies and Concepts Description: The Biomedical Technology Studies and Concepts thrust will ensure force effectiveness by advancing technologies that prepare, sustain, and restore the operational health and wellness of the warfighter. This thrust will seek to develop new		-	10.338	57.199

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Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
classes of medical care technologies that will support the warfighter during all stages of the operational cycle. Advanced platforms in this thrust will protect overall force health against the multitude of operationally relevant biothreats, injuries, and physiological stressors. Research conducted will support biosurveillance and biosecurity in the development of fundamental models and technologies for rapid detection, characterization, and countermeasure production to protect the warfighter against emerging classes of biothreats. This thrust will also result in the development of biomedical technologies that explore new capabilities of biology to enhance warfighter preparation and recovery. FY 2026 Plans: - Investigate approaches to develop novel robotics platforms capable of identifying, accessing, and treating injured warfighters. - Investigate methods to enhance warfighter awareness, performance, and survivability through the coupling of novel technology platforms and analysis of warfighter physiology. - Initiate development of biological, bio-inspired, or bio-hybrid platforms that enhance warfighter preparation and recovery in austere environments. FY 2027 Plans: - Initiate development of biological, bio-inspired, or bio-hybrid platforms that enhance the detection, characterization, and protection against emerging threats, both in warfighters and agricultural crops. - Investigate cellular engineering approaches that maintain physiology and viability and do not result in immunotoxicity. - Investigate new methods for identifying and characterizing warfighter physiology and generating datasets to train computational tools such as large language models. - Initiate development of high-throughput models, tools, and methods to rapidly assess biological, bio-inspired, or bio-hybrid molecules relevant to DoW challenges. - Investigate safe and responsible modification of biological materials and molecules to enhance warfighter preparation and recovery in austere environments and in the event of pathogenic exposures. - Initiate maturation of select robotics platforms capable of identifying, accessing, and treating injured warfighters. - Create predictive mental health monitoring systems through durable biomarker patches with machine learning for early intervention capabilities and diagnostic predictions. - Generate technology capable of producing hemi-sleep in humans. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in biological, bio-inspired, or bio-hybrid platforms to protect against emerging threats, and in medical care technologies that will support the warfighter.			
Title: Rapid Battlefield Triage		-	20.136

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Description: The Rapid Battlefield Triage program is advancing capabilities to quickly triage warfighters requiring urgent life-saving medical intervention and enable medical resources to provide an appropriate response in current and future battlefields. Today, triage at point of injury is limited by subjective assessments, tools that are manually intensive, and physiological signatures with little diagnostic and prognostic value. This program will build on recent biomarker discoveries and innovations in sensing platforms to develop field-portable technologies that support triage in the most challenging operational domains. By optimizing allocation of scarce medical resources and scaling to multiple casualties, these devices will help far-forward units maximize their fighting strength against potential adversaries that could inflict large numbers of casualties and constrain evacuation to advanced medical facilities. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01. FY 2026 Plans: - Conduct a second large-scale field experimentation, integrating additional autonomy and operating challenges in a competition format. - Finalize a database of trauma signatures with additional novel cases, incorporating sensors developed in previous years. - Evaluate field-portable triage solutions with increased operational challenges. - Evaluate approaches for stand-off capture of injury signature by fully autonomous systems in a real-world (physical) or virtual simulation. - Conduct a third baseline design, development, and integration of triage solutions in a larger scale integration exercise with additional operational challenges. - Conduct a third (final) large-scale field experimentation, integrating additional operating challenges in a competition format. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Title: Next-Generation Combat Casualty Care Description: The Next-Generation Combat Casualty Care program is developing advances in critical efforts to preserve warfighter life and well-being in the battlefields of the future. This research will directly address a leading cause of potentially preventable battlefield casualties by investigating new approaches for developing whole blood substitutes for traumatic injury that can be deployed on the battlefield in far-forward settings. Additional potential uses apply to disaster relief, mass casualty events, and stabilization missions. Advances within this program will ensure that the U.S. remains able to care for service members in peer and near-peer conflict by addressing gaps in combat casualty care. Prior to FY 2026, this program was funded in PE 0602115E, Project BT-01.		-	10.457

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFOR MANCE	Project (Number/Name) WP-02 / BIOMEDICAL TECHNOLOGY	

B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
<i>FY 2026 Plans:</i> - Complete efficacy and safety studies with optimized therapeutic formulation against hemorrhage in complex trauma using animal models. - Provide the initial proof of concept for easy reconstitution and packaging suitability. - Provide the final demonstration of scaled-up manufacturing. - Complete final stability tests to demonstrate minimal loss of function. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals	-	166.496	189.962

C. Other Program Funding Summary (\$ in Millions) N/A	Remarks	
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602024E / WARFIGHTING PERFORMANCE				Project (Number/Name) WP-07 / WARFIGHTING PERFORMANCE SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
WP-07: WARFIGHTING PERFORMANCE SUPPORT	0.000	0.000	15.527	27.134	-	27.134	36.059	36.844	37.760	37.760	-	-

A. Mission Description and Budget Item Justification

The Warfighting Performance Support project contains management costs in support of DARPA functions and activities across the entire Warfighting Performance PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0602715E, Project MBT-02, and PE 0602115E, Project BT-01.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602025E / <i>MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	1,318.411	1,624.523	-	1,624.523	1,484.421	1,121.759	1,166.547	1,185.521	-	-
MQB-01: <i>QUANTUM TECHNOLOGIES</i>	0.000	0.000	358.118	603.185	-	603.185	487.838	102.000	108.000	112.000	-	-
MSL-01: <i>MATERIALS PROCESSING TECHNOLOGY</i>	0.000	0.000	182.511	133.199	-	133.199	133.434	161.837	171.231	175.626	-	-
MSL-02: <i>DISRUPTIVE MICROTECHNOLOGIES</i>	0.000	0.000	354.608	368.276	-	368.276	186.898	121.300	124.300	125.300	-	-
MSL-03: <i>ELECTRONIC TECHNOLOGY</i>	0.000	0.000	87.791	65.867	-	65.867	101.813	121.178	136.944	138.523	-	-
MSL-04: <i>CYBER SECURITY</i>	0.000	0.000	139.470	157.134	-	157.134	225.500	236.533	260.557	269.185	-	-
MSL-05: <i>ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS</i>	0.000	0.000	120.707	138.950	-	138.950	202.398	237.185	248.368	251.368	-	-
MSL-06: <i>MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS</i>	0.000	0.000	0.000	36.862	-	36.862	27.751	21.391	3.628	0.000	-	-
MSL-07: <i>MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS SUPPORT</i>	0.000	0.000	75.206	121.050	-	121.050	118.789	120.335	113.519	113.519	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$1,318,411 thousand of discretionary and \$250,000 thousand of mandatory (reconciliation) for a total of \$1,568,411.

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Making, Maintaining, Supply Chain and Logistics Program that is directed toward needed technical innovations involved with preserving a healthy, responsive technology supply chain and its associated logistics. This PE also supports the applied research associated with novel techniques in materials processing to include the discovery of said materials and doing so in an environmentally stable fashion. It also includes the development of advanced electronics and disruptive microtechnologies to include advanced manufacturing and fabrication tools, built-in-test, and new circuit materials and topologies.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	
The Quantum Technologies project addresses the Applied Research associated with the development and evaluation of emerging technologies and system concepts that exploit or employ quantum technology and phenomenon for quantum computing and networking to include hardware, software, relevant protocols, and commercialization prospects. This PE also supports innovation and incubation of advanced quantum hardware, architectures, software, and design automation for computing, networking, and sensing. The major goal of the Materials Processing Technology project is to develop novel materials, fabrication and processing techniques, models, devices and components that will increase the performance, and/or enable new missions for military platforms and systems that can be manufactured at commercial scales. Included in this project are efforts across a wide range of technology areas including manufacturing, electronics, sensors, optics, and complex and autonomous systems. The Disruptive Microtechnologies project continues advancement in microelectronics, including electronic and optoelectronic devices, semiconductor device design and fabrication, and new materials and material structures. This includes developing next generation photonics, quantum, and organic circuits; developing new additive, subtractive, and combination fabrication tools that minimize waste and optimize performance; and new concepts in design, integration, and hardware security that leverages commercial scaling while preserving the superiority of defense systems. In doing so, this area will create disruptive opportunities and prevent the surprise associated with the unintended consequences of technology development. The Electronic Technology project pursues materials, devices, and architectures that will enable dramatic improvements over the current state-of-the-art. Areas of particular emphasis of this work include the development of techniques for advanced fabrication and test and the development of technologies that support electronics reuse and recycling. These technologies will reduce the barriers to designing and fabricating advanced technologies and exploit improved manufacturing techniques to provide these advanced technologies faster and cheaper to the warfighter. The Cyber Security project is developing the computing, networking, and cyber security technologies required to protect Department of War, U.S. Government, and U.S. civilian information, information infrastructure, cyber-physical and embedded systems, critical infrastructure, and other computation-intensive mission-critical systems. Information technologies enable important existing and new military capabilities and drive the productivity gains essential to U.S. industry. Meanwhile, cyber threats grow in sophistication and number, and put data, computer programs, key information systems, and U.S. economic competitiveness at risk. The technologies developed in this project will enhance the resilience of information systems to current and emerging cyber threats; enable broad situational awareness of the cyber domain; and provide the basis for accurate, calibrated, and safe cyber response. The Artificial Intelligence and Human-Machine Symbiosis project develops technologies to enable machines to function not only as tools that facilitate human action but also as trustworthy partners to human operators. Of particular interest are systems that can understand human language, extract information, and reliably categorize content contained in diverse media; answer questions, reach conclusions, and propose explanations; and learn, reason, and apply knowledge gained through experience to respond intelligently to new and unforeseen events. Enabling computing systems with such human-like intelligence matters because the tempo of military operations in emerging domains can exceed that at which unaided humans can orient, understand, and act. The technologies developed in this project will enable warfighters to make better decisions in complex, time-critical, battlefield environments with greater efficiency and confidence; and help autonomous systems and intelligent computational agents to safely and reliably perform critical missions in contested and adversarial physical and virtual environments.		

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602025E / <i>MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS</i>
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The Making, Maintaining, Supply Chain and Logistics Support project contains management costs in support of DARPA functions and activities across the entire Making, Maintaining, Supply Chain and Logistics PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0602715E, Materials and Biological Technology, PE 0602716E, Electronics Technology, and PE 0602303E, Information & Communications Technology.

<u>B. Program Change Summary (\$ in Millions)</u>	<u>FY 2025</u>	<u>FY 2026</u>	<u>FY 2027 Base</u>	<u>FY 2027 OOC</u>	<u>FY 2027 Total</u>
Previous President's Budget	0.000	1,347.049	0.000	-	0.000
Current President's Budget	0.000	1,318.411	1,624.523	-	1,624.523
Total Adjustments	0.000	-28.638	1,624.523	-	1,624.523
• Congressional General Reductions	0.000	-13.263			
• Congressional Directed Reductions	0.000	-15.375			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	1,624.523	-	1,624.523

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity					R-1 Program Element (Number/Name)				Project (Number/Name)			
0400 / 2					PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS				MQB-01 / QUANTUM TECHNOLOGIES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MQB-01: QUANTUM TECHNOLOGIES	0.000	0.000	358.118	603.185	-	603.185	487.838	102.000	108.000	112.000	-	-

A. Mission Description and Budget Item Justification

The Quantum Technologies project addresses the Applied Research associated with the development and evaluation of emerging technologies and system concepts that exploit or employ quantum technology and phenomenon for quantum computing and networking to include hardware, software, relevant protocols, and commercialization prospects. This PE also supports innovation and incubation of advanced quantum hardware, architectures, software, and design automation for computing, networking, and sensing.

Prior to FY 2026, efforts in this Project were funded in PE 0602716E, Project ELT-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Quantum Benchmarking Initiative (QBI)	-	282.000	481.000
Description: The preponderance of evidence shows that commercial quantum computers will be a key economic driver in the 21st century. It is plausible that these machines will be constructed by commercial companies in the near future. An appropriate hedging strategy is needed to prevent strategic surprise from unanticipated commercial breakthroughs, ensure privileged access to disruptive capability, and maximize U.S. economic benefit. To quantify and reduce this risk, the Quantum Benchmarking initiative (QBI) program is executing an incremental investment plan that rigorously evaluates commercial activities, aggressively develops and tests key prototypes for activities that show credible potential and prepares for full-scale deployment when feasible. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Expand evaluations of plausible utility-scale concepts and expand development of applications to include additional approaches. - Continue examining Research and Development (R&D) plans, including detailed execution plans addressing the major risks associated with building a utility-scale quantum computer, planned risk mitigation steps, and plans for prototypes that will demonstrate risk reduction. - Expand examination of R&D plans to include additional approaches. - Identify and procure long-lead hardware items needed to perform component, subsystem, and prototype research and development. - Begin to validate critical prototypes, components, and subsystems needed for building a utility-scale quantum computer. - Begin performing R&D to demonstrate burning down the major risks associated with building a utility-scale quantum computer.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MQB-01 / QUANTUM TECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Begin to implement test infrastructure needed to rigorously test a set of utility-scale quantum computer prototypes. FY 2027 Plans: - Expand evaluations of plausible utility-scale concepts as needed to prevent strategic surprise. - Finalize R&D plans, including detailed execution plans addressing the major risks associated with building a utility-scale quantum computer, planned risk mitigation steps, and plans for prototypes that will demonstrate risk reduction. - Continue to expand examination of R&D plans to include additional approaches. - Continue procuring long-lead hardware items needed to perform component, subsystem, and prototype research and development. - Validate critical prototypes, components, and subsystems needed for building a utility-scale quantum computer. - Perform R&D to demonstrate burning down the major risks associated with building a utility-scale quantum computer. - Implement test infrastructure needed to rigorously test a set of utility-scale quantum computer prototypes. - Begin working with external partners to evaluate applications and co-design system prototypes. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from component research to prototype development.			
Title: Macaroni Description: Measurement and control of the electromagnetic spectrum is a key area of research for the Department of War (DoW). Spectrum dominance requires quick and efficient control of electromagnetic radiation from low frequencies to X-rays. In classical antenna theory, the sensitivity-bandwidth product is fundamentally limited by the physical shape and size of the antenna. This performance degrades significantly as the antenna becomes electrically small, that is, the physical size becomes much smaller than the electromagnetic wavelength of operation. The Macaroni program seeks to develop electrically-small receivers and transmitters with performance that exceeds the current state-of-the-art (SoA). Recent advances in quantum sensors, materials science, electromagnetic shielding, laser technology, resonators, cryogenic systems, and vacuum components have pushed the SoA in sensing technologies. For transmitters, new insights in active antenna technology, control schemes, methods of impedance matching, and strategies for volume filling present new opportunities. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Demonstrate electrically-small receiver performance in a relevant environment. - Demonstrate electrically-small transmitter performance in a relevant environment. - Develop plan for validation of system performance with government partners. FY 2027 Plans:		-	18.781
			5.940

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MQB-01 / QUANTUM TECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Identify transition paths to wide DoW applicability for electrically-small receivers. - Identify transition paths to wide DoW applicability for electrically-small transmitters. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from concept validation to demonstration of the electrically-small receiver and transmitter.				
Title: Robust Quantum Sensors (RoQS) Description: The Robust Quantum Sensors (RoQS) program seeks to bring quantum sensors to DoW platforms. While quantum sensors have demonstrated exceptional laboratory performance in several modalities (magnetic and electric field, acceleration, rotation, and gravity, etc.), their performance degrades once the sensor is placed on moving platforms due to electric and magnetic fields, field gradients, and system vibrations. RoQS seeks to overcome these challenges through innovative physics approaches to quantum sensing. RoQS aims to develop and demonstrate quantum sensors that inherently resist performance degradation from platform interferers and demonstrate them on a DoW platform. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01. FY 2026 Plans: - Validate program of record needs and requirements for quantum sensors. - Demonstrate robust quantum sensors outside of a laboratory environment. - Validate theory and principles of operation of robust quantum sensors. - Identify specific requirements to integrate robust quantum sensors on specific platforms. FY 2027 Plans: - Demonstrate robust quantum sensors on government platforms. - Validate theory and limits of operation of robust quantum sensors. - Develop integration requirements and methodology for system demonstrations. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from design, development, and initial demonstration to the final program demonstration.		-	12.682	10.904
Title: Quantum Augmented Network (QuANET) Description: The Quantum Augmented Network (QuANET) program is developing quantum-augmented networks that add novel security and other properties inherent in quantum communications to classical, non-quantum, network infrastructures that currently trade security against interoperability. Today, all digital communication paradigms use a network stack that consists of a layered set of software protocols. The higher layers are closer to applications on computers and servers, commonly called nodes, while the bottom layers are closer to the physical implementation, i.e., network cables. State-of-the-art networks commonly rely		-	11.549	17.971

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MQB-01 / QUANTUM TECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
on security at the top layers of the stack, assuming this security also mitigates potential attacks on lower layers. Unfortunately, advanced persistent threat (APT) attacks are increasing cyber defense costs to U.S. commercial and government entities. The QuANET program seeks to augment existing software infrastructure and network protocols with quantum properties to mitigate attack vectors. QuANET is developing the hardware, protocols, and software tools required to operationalize quantum communications for sensitive missions and critical infrastructure and demonstrate these capabilities in an operational, quantum-augmented network. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Integrate fabricated quantum-network interface cards (qNICs) in a quantum-augmented network and demonstrate quantum communication, sensing, and timing algorithms. - Demonstrate advanced quantum-augmented network functionality by embedding quantum sensing and quantum time transfer into classical data streams. - Test and evaluate security capabilities of a prototype, quantum-augmented network to detect and mitigate network attacks. FY 2027 Plans: - Reduce Size, Weight and Power (SWaP) of qNICs towards deployable hardware package. - Test, evaluate, and document developed quantum-augmented network protocols for standardization of protocols. - Develop and evaluate proof of concept over air quantum-augmented links by modifying qNICs. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from design and demonstration of initial components to the design and demonstration of deployable hardware and networks.				
Title: H6 Description: The H6 program is developing the first tactical-grade clock. Tactical-grade clocks are ultra-small, low power, fieldable and can maintain the timing needed for DoW-relevant applications in challenging environments. Precise timing in a tactical package will decouple operations from GPS dependence, addressing a significant operational need for the warfighter. Precise tactical-grade clocks from H6 will enable increased signal assurance and pervasive communications security in high-jamming regions. Additionally, H6 will enable real-time, physical awareness of warfighters and will play a critical role in search and rescue through the ability to maintain precise time over a long mission duration without having to re-establish external communications. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02.		-	6.560	3.144

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MQB-01 / QUANTUM TECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Demonstrate aging-insensitive operation in realistic environments. - Initiate construction of fully-integrated clock. FY 2027 Plans: - Demonstrate a fully-integrated tactical-grade clock. - Demonstrate use of tactical-clock with DoW communications equipment. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from initial design to initiating construction of tactical-grade clock components.				
Title: Heterogeneous Architectures for Quantum (HARQ) Description: The Heterogeneous Architectures for Quantum (HARQ) program aims to demonstrate the scaling advantages of heterogeneous architectures for quantum computers. Large-scale quantum computers have long been lauded as a potentially revolutionary capability. While the race to build a quantum computer is drawing substantial attention and investment, all on-going work is focused on homogeneous architectures that essentially place a bet on a single qubit modality. Unfortunately, this handicaps any such future machine's ultimate scale and accessible computation space due to the inability for any single qubit platform to optimally perform all necessary functions for quantum computation. Heterogeneous architectures may offer a path to scale systems beyond present-day industry roadmaps and enable transformative commercial and national security opportunities, such as the discovery of superior anti-corrosion coatings for defense platforms, energetics for propulsion, and pharmaceutical drug delivery. HARQ seeks to create a modeling and simulation framework that determines the computational efficiency enhancement afforded by the ability to allocate the execution of quantum applications to diverse qubit technologies. Additionally, HARQ will drive the development of hardware components that rapidly and efficiently transfer quantum states between diverse qubit technologies, which will be necessary to interface between disparate qubit modules in a future heterogeneous architecture. FY 2026 Plans: - Develop initial designs for components of heterogeneous quantum interconnects. - Develop initial modeling & simulation frameworks for partitioning quantum circuits in heterogeneous architecture. - Establish an independent verification and validation capable of performing technoeconomic analysis on heterogeneous quantum computer architecture. FY 2027 Plans: - Demonstrate initial components of heterogeneous quantum interconnects. - Demonstrate heterogeneous quantum circuit compilers on small reference circuits.		-	14.500	15.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
- Develop preliminary heterogenous quantum computer architecture specifications.			
<i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 increase reflects the shift from initial design and modeling activities to finalizing design and architecture of a heterogeneous quantum system.			
<i>Title:</i> Quantum Technologies Studies and Concepts <i>Description:</i> Quantum Technologies Studies and Concepts examines emerging technologies and system concepts that employ quantum technology or exploit quantum phenomena for computation, networking, or sensing. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, is also evaluated. The results from these studies and initiatives are used to formulate future programs. <i>FY 2026 Plans:</i> - Evaluate design concepts for making quantum computing circuits more efficient. - Investigate technologies to make quantum sensors readily manufacturable. <i>FY 2027 Plans:</i> - Investigate new concepts in combined quantum sensing with quantum networking. - Explore hardware interconnectivity between heterogeneous qubit platforms. - Evaluate heterogeneous quantum architectures for relevant use cases. - Initiate design activities for manufacturable quantum sensors. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research to ensure quantum information dominance in the areas of quantum sensing and networking, and quantum sensor manufacturability.	-	12.046	69.226
Accomplishments/Planned Programs Subtotals	-	358.118	603.185

C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A
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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-01 / MATERIALS PROCESSING TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-01: MATERIALS PROCESSING TECHNOLOGY	0.000	0.000	182.511	133.199	-	133.199	133.434	161.837	171.231	175.626	-	-

A. Mission Description and Budget Item Justification

The major goal of the Materials Processing Technology project is to develop novel materials, fabrication and processing techniques, models, devices and components that will increase the performance, and/or enable new missions for military platforms and systems that can be manufactured at commercial scales. Included in this project are efforts across a wide range of technology areas including manufacturing, electronics, sensors, optics, and complex and autonomous systems.

Prior to FY 2026, this Project was funded in PE 0602715E, Project MBT-01.

B. Accomplishments/Planned Programs (\$ in Millions)

Title: Functional Materials and Devices Description: The Functional Materials and Devices thrust is developing advanced materials, components and systems to improve device performance for DoW structural, sensing, imaging and communication applications. One focus of this thrust involves development of advanced transductional materials that convert one form of energy to another for DoW-relevant applications. While promising transduction materials are known for a variety of applications, integration into devices is not as advanced. Another focus area is the development of physics-based models that predict material behavior when illuminated by high peak power electromagnetic interference. A third focus area involves development of new multi-functional materials and device designs that will radically decrease the size, weight and power requirements of high-energy photon sources. Such devices should enable fieldable detection units for sensing, non-destructive evaluation of parts, and detection of DoW-relevant targets. A fourth focus area is developing new liquid-based, large-aperture imaging systems such as telescopes. Such telescopes would break the unfavorable, exponential scaling between aperture size and cost for normal telescopes and enable low-cost imaging platforms for ground- and space-based applications. Another focus area under this thrust involves novel nano-architected materials to enhance device-relevant properties for applications to quantum-enhanced sensors. Finally, novel design optimization approaches will be explored where material composition and microstructure are included as explicit, continuous variables alongside shape optimization. This co-optimization of shape and material together will enable new combinations of structural performance and sustainability for a variety of DoW applications. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Perform full test and design system demonstrations on material testing platforms and design optimization routines.	FY 2025	FY 2026	FY 2027
	-	55.555	41.192

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-01 / MATERIALS PROCESSING TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Screen new materials driven by design optimization routines and incorporate newly measured properties into design processes in real time. - Demonstrate and prepare manufacturing plans for continuous material shape optimized components. - Design and fabricate functional quantum metamaterial-based nanoelectronic devices for applications in quantum sensing, computing and communications. - Validate and benchmark enhanced quantum effects in metamaterials used in nanoelectronic devices for applications in quantum sensing, computing and communications. - Experimentally validate feasibility of scaling up liquid mirror design and all required component technologies in a specific meter effective aperture size scale range. - Develop detailed design and complete hardware-informed critical design review for a liquid mirror telescope system in a laboratory environment. - Develop atomically-referenced, tunable, narrow-linewidth sources of terahertz radiation to enable calibration-free and matched transmitter-receiver systems. - Develop wafer-scale atomic vapor cells with the requisite homogeneity, manufacturability, and optical quality to enable higher performance and more robust electrometry. - Test high energy displacement threshold materials capable of converting radiation into energy over an extended lifetime. - Implement embedded sensors on a combustor system, a key component of jet engines. - Engineer phononic structures to enable quantum sensing of interaction of subatomic particles (i.e. neutrinos) with atoms in a material. - Explore novel chemical structures that enable the controlled tunability of energetic materials. FY 2027 Plans: - Perform a coupled optimization demonstration between high-throughput material testing and design optimization frameworks balancing performance, cost, and sustainability. - Establish a proof-of-concept manufacturing demonstration for a continuous material shape optimized component. - Initiate incorporation of enhanced quantum metamaterials in multifunctional nanoelectronic devices for applications to quantum computing, sensing, and signal processing. - Initiate scalable wafer-level fabrication and application-specific development of quantum metamaterial-based nanoelectronic devices. - Conduct out-of-plane liquid mirror hardware demonstration in a particular aperture size scale range and begin advancing designs from laboratory setting to on-sky demonstrations. - Develop detailed design and complete hardware-informed critical design review for a larger scale liquid mirror telescope system that will be capable of astronomical and Space Situational Awareness (SSA) observations without spin-stabilization.				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS		Project (Number/Name) MSL-01 / MATERIALS PROCESSING TECHNOLOGY	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Scale up studies of cell designs using materials with demonstrated high energy displacement thresholds and self-healing, annealing techniques. - Perform proof-of-concept demonstrations of quantum sensing of neutrinos with greatly increased sensitivity compared to currently used neutrino detection methods. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from design and development to system demonstration.					
Title: Materials for Extreme Environments Description: The Materials for Extreme Environments thrust is exploring new materials, innovative architectures, and development processes that will significantly enhance the performance and persistence of DoW platforms operating in extremely harsh environments. Materials with superior strength, functionality, and resiliency are critical for enabling DoW platforms, weapons and other components to operate and persist under conditions including, but not limited to, extremely high or low temperatures, turbulence, ionizing radiation, and/or corrosive environments. Recent developments in materials such as high entropy alloys, superconducting materials, and infiltrated carbon fiber composites hold promise for achieving material solutions for improved survivability in a wide range of harsh environmental conditions. Similarly, advancements in material design, processing and manufacturing are enabling novel material architectures that can further enhance performance and resilience in structures such as leading edges, windows and apertures, propulsion systems, and space structures. Exemplar areas of research within the Materials for Extreme Environments thrust include the following: 1) high temperature materials for hypersonic platforms, 2) high temperature window and aperture materials, 3) radiation and/or electromagnetic pulse (EMP) hardened electronics for space platforms, 4) coatings for platform survivability in corrosive environments, 5) active and passive cooling methods for apertures and forward-facing vehicle features, and 6) superconducting and magnetic materials for novel propulsion systems. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Deliver experimental payloads for in-space demonstration of polymer composite manufacturing and radio frequency antenna structure assembly. - Launch and conduct material manufacturing and assembly technology demonstrations in low earth orbit. - Complete thruster testing for sustained operations in very low earth orbit (VLEO). - Test optimized design of air-collecting inlet and use experimental data for model validation. - Test low-drag materials for use in low Earth orbit to understand drag and degradation characteristics. - Test low-drag coatings applied to actual spacecraft components for preservation of functional performance to de-risk near-term inclusion on flight spacecraft. - Complete design for the prototype and scalability study for an operationally relevant magnetohydrodynamic pump.			-	34.349	20.076

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: The Burn n' Go program will increase the production rate and availability of solid rocket motors (SRMs) via development of novel materials and methods for tailorable, post-manufacturing thrust control. Complementary to efforts to improve supply chain resilience being developed under the Open Price Exploration for National security (OPEN) program funded in this PE, Project MSL-05, Burn n' Go will enable a paradigm shift in solid rocket motor production by creating a multi-mission standardized motor design for use across multiple existing and future weapons systems. Utilizing a standard design for SRMs in set diameter classes, Burn n' Go will enable suppliers to limit line switching times, improve tooling, accelerate quality control, reduce machining costs, and enable a more stable procurement stream, which can be flexed across Services and weapons. Burn n' Go will validate the developed technology through a series of component-level feasibility tests and integrated system-level static fire tests. Technology developed under this program will be transitioned to the Services and commercial partners. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Develop physics models to validate component-level feasibility analysis. - Resolve trade studies for conceptual design. - Validate concept viability for testing and integration phase. - Perform feasibility tests of components for proof-of-concept validation. - Understand system requirements of conceptual design. FY 2027 Plans: - Conduct assessment of designs to understand ability to meet mass production requirements. - Develop preliminary design of integrated system. - Test sub-scale solid rocket motor assembly. - Compare preliminary designs to potential transition partner requirements. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from feasibility testing to sub-scale motor assembly testing.					
Title: Carbon Crunch Description: Carbon Crunch revolutionizes manufacturing for one of the longest lead time and highest risk items in the hypersonic system bill of materials: the aeroshell. Manufacturing Carbon-Carbon (C-C) composites is a slow, arduous process supported by a small, risk-intolerant industrial base, and thus not scalable to end user demands. Carbon Crunch will overcome this limitation by developing and applying cutting edge C-C manufacturing methods that are inherently faster and more scalable. By developing and demonstrating innovative manufacturing processes and techniques targeting the most constraining			-	1.470	16.169

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
bottlenecks, Carbon Crunch will enable scalable, fast, and agile production of hypersonic aeroshells. A successful Carbon Crunch program will enable hypersonic production at rates and costs truly commensurate with tactical assets. FY 2026 Plans: - Design 3D hypersonic-representative parts that mimic design features and manufacturing challenges seen in real, full-scale aeroshell. FY 2027 Plans: - Produce and test C-C 2D coupons and C-C 3D hypersonic-representative parts that mimic design features and manufacturing challenges seen in real, full-scale aeroshell. - Develop processes for high throughputs of C-C hypersonic aeroshells. - Design aeroshell production lines - Perform independent testing and assessment (i.e., material performance, manufacturing readiness assessment, techno-economic analysis) by government team. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial modeling and simulations to hardware development and sample part fabrication and testing.			
Title: Chemical Processing for Force Protection Description: Research in the Chemical Processing for Force Protection thrust is focused on the development of new chemical approaches and technologies across a broad spectrum of DoW needs. One area involves development of innovative approaches for scalable small molecule synthesis coupled with predictive tools for route design, possibly offering a new strategy to discover how to make new molecules such as pharmaceuticals and explosives. A second area includes qualification of new molecules made using agile manufacturing platforms. A third focus leverages advances in automation to develop safe, reproducible experimental approaches for systematic development of energetic materials. Another focus is to develop the tools and processing methods to safely turn locally available materials in a given location into reliable, point/time-of-need energetics, for battlefield requirements. In addition, investments in this thrust will advance chemical characterization, information management and analysis, and automation. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Demonstrate system versatility with exposure to a broader range of usable, raw material inputs to generate an increased number of energetic material outputs.		-	28.607
			11.957

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Administer energetic material synthesis demonstrations at an increasing scale. - Execute additional end-to-end validation testing on energetic materials developed from system level demonstrations. - Conduct a critical design review based on outcomes from demonstrations and validation tests on modular platforms. - Demonstrate sub-scale production capability on modular platforms. - Synthesize energetic materials from program-developed database of precursors at a pilot scale. - Develop direct nitrogen oxidation catalysts capable of achieving a high reaction rate at low energy. FY 2027 Plans: - Integrate nitrogen oxidation catalysts into a small-scale reactor capable of delivering nitric acid at a particular rate. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from heavy system development to integration and testing.					
Title: Grainy Day Description: The Grainy Day program will develop innovative approaches and new methods to address challenges in passive accurate placement of unattended sensors from Unmanned Aerial Systems (UAS) and military aircraft. The range of initial deployment altitudes requires compensation for jet streams, trade winds, and low-level weather in real time, and the sensing devices will be ruggedized to withstand extreme environments in flight as well as ground impact while maintaining functionality. Because of the large number of devices delivered, scalability and manufacturability will be established via rapid fabrication and commercial chip foundries. Program success will bring the Department of War (DoW) a new affordable, scalable, and resilient cross-domain capability to accurately place sensing devices with flexible mission sets deployed from a distance. Technology developed under this program will be transitioned to the Services and other government partners. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Establish subject matter government team to finalize test site requirements. - Create initial concept designs and begin fabrication cycle. - Fly prototypes in first of four testing cycles, collect data, analyze and modify designs to optimize performance. - Review initial capabilities with transition partners. FY 2027 Plans: - Complete first of four testing cycles, collect data, analyze and modify designs to optimize performance. - Begin second testing cycle with increased complexity at release conditions. - Complete second testing cycle and begin third testing cycle with increased complexity at release conditions.			-	15.850	10.500

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Review improved capabilities with transition partners.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift to fewer performers participating in the later testing cycles.				
Title: Reconfigurable Systems Description: The Reconfigurable Systems thrust is developing new approaches to enable more rapid and robust adaptation of defense systems and systems-of-systems to changing mission requirements and unpredictable environments. This includes development of capabilities across sensing, perception, planning and control for autonomous, high-speed operation in cluttered environments without Global Positioning System (GPS) information. This also includes development of capabilities to manipulate and control sensory perception and/or situational awareness. Additional work in this thrust focuses on how sensing systems and military systems-of-systems are designed for real-time resilient response to dynamic, unexpected signals and contingencies. Research is developing a more unified view of system behavior that allows better understanding and exploitation of complex interactions among components, including development of formal mathematical approaches to complex adaptive system composition and design. These capabilities will impact autonomous systems and systems-of-systems, including those that involve humans, in a variety of DoW-relevant contexts. Prior to FY 2026, this program was funded in PE 0602715E, Project MBT-01. FY 2026 Plans: - Develop a fully operational high-performance, environmentally-robust portable optical clock device with picosecond timing precision. - Develop a fully operational high-performance, environmentally-robust transportable optical clock with month-long nanosecond holdover. - Demonstrate proof-of-concept vacuum ultraviolet laser system for preliminary spectroscopy and validation of a nuclear clock. FY 2027 Plans: - Conduct field exercises and DoW-relevant application demonstrations of high-performance, environmentally-robust optical clocks FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from system development to field demonstrations.		-	10.644	0.900
Title: Making, Maintaining, Supply Chains and Logistics Studies and Concepts Description: The Making, Maintaining, Supply Chains and Logistics Studies and Concepts program will develop technologies that untether forces from supply chains by enabling expeditionary production of molecules, materials, and parts from local or		-	7.247	13.736

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
non-optimized feedstock, while also enabling better measurement and communications among mission systems. Concurrently, we will create novel measurement and assessment tools to understand and control complex physical environments, complex dynamic systems, and to improve new platform and system design. Furthermore, we will generate low-cost, actionable solutions for enhanced detection and characterization of adversarial systems and events by using improved processing techniques, models, and signals of opportunity for better sensing and assessment. FY 2026 Plans: - Advancing nuclear reactor physics models in order to enhance reactor control capabilities and safer operations with predictive methods. - Assess potential for autonomous control of Hall thruster spacecraft electric propulsion systems by validating sensor optimization using manifold estimation to robustly identify plasma dynamics. - Investigate quantum sensor networks with enhanced sensitivity arising from an interplay between quantum many-body interactions and noise-engineering techniques. - Initiate development of gravity defying levitated Inertial Measurement Units. - Explore multifunctional materials, new fabrication methods, and design techniques for radio-frequency systems capable of identifying signals of interest, rapidly tuning to their carrier frequency, and amplifying their message by dynamically adjusting the receiver's signal-to-noise-ratio. - Select critical hardware assemblies for production analysis in order to drastically shrink the time from developing a new prototype to manufacturing it at mass scale. FY 2027 Plans: - Finalize experiment plan for autonomous control using nuclear physics predictions and advanced, combined neutronics models. - Develop air-breathing secondary fuel cells with far greater gravimetric energy density than current comparable energy solutions, while remaining safe, rechargeable, and easily manufactured. - Assess feasibility of re-purposing hot, high-density plasma used in fusion experiments for electric propulsion applications. - Develop laboratory autonomous control of Hall thruster spacecraft electric propulsion systems, tying state identification to qualification of data-driven digital twins. - Advance predictive modeling tools and experimental techniques to forecast the degradation mechanisms and lifespan of new infrastructure materials. - Initiate development of tabletop quantum simulators for data-informed predictive modeling and aeronautical design. - Perform proof-of-concept demonstrations of quantum sensor networks with beyond state-of-the-art spatiotemporal resolution for extreme environments. - Perform initial proof-of-concept demonstrations of the core mechanisms for levitated Inertial Measurement Unit.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Perform initial proof-of-concept demonstrations of radio-frequency (RF) systems capable of identifying signals of interest, rapidly tuning to their carrier frequency, and amplifying their message by dynamically adjusting the receiver s signal-to-noise-ratio (SNR). - Integrate computational and hardware capabilities into complete capability for accelerating timelines of making and manufacturing novel designs of military systems. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in development and manufacturability of advanced fuel cells and novel spacecraft propulsion systems, and exploration of metamaterials and fabrication methods for advanced RF systems.			
Accomplishments/Planned Programs Subtotals		-	182.511
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-02: DISRUPTIVE MICROTECHNOLOGIES	0.000	0.000	354.608	368.276	-	368.276	186.898	121.300	124.300	125.300	-	-

A. Mission Description and Budget Item Justification

The Disruptive Microtechnologies project continues advancement in microelectronics, including electronic and optoelectronic devices, semiconductor device design and fabrication, and new materials and material structures. This includes developing next generation photonics, quantum, and organic circuits; developing new additive, subtractive, and combination fabrication tools that minimize waste and optimize performance; and new concepts in design, integration, and hardware security that leverages commercial scaling while preserving the superiority of defense systems. In doing so, this area will create disruptive opportunities and prevent the surprise associated with the unintended consequences of technology development.

Prior to FY 2026, efforts in this Project were funded in PE 0602716E, Project ELT-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Faithful Integration Reverse-engineering and Emulation (FIRE) Description: The Faithful Integration Reverse-engineering and Emulation (FIRE) program is developing tools to find and patch vulnerabilities within cyber-physical systems. A cyber-physical system operates in the physical world using hardware sensors to perceive the analog environment, digital software for processing, and actuators to interact with the environment. Cyber-physical vulnerabilities arise from the composition of hardware, software, and physical components where each component may not be vulnerable in-and-of itself. FIRE is developing novel modeling and simulation techniques to help expedite finding and patching vulnerabilities in cyber-physical systems. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Scale approaches and tools to real-world demonstrations. - Demonstrate tools on a variety of cyber-physical systems. - Evaluate performance of the tools for realistic application scenarios. FY 2027 Plans: - Develop features to the tools and systems that use autonomous and/or probabilistic components. - Demonstrate tools on a medium-complexity cyber-physical systems with autonomous components. FY 2026 to FY 2027 Increase/Decrease Statement:	-	19.538	11.804

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS		Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects a shift from initial development to proof-of-concept of tools.					
Title: Next Generation Microelectronics Manufacturing (NGMM) Description: Next Generation Microelectronics Manufacturing (NGMM) creates new software design tools to enable the development of novel three-dimensional heterogeneous integration (3DHI) microsystems and associated test vehicles with the NGMM program. This program addresses deficiencies in the capabilities and flexibility of current electronic design automation (EDA) workflows and simulation tools by establishing and refining the NGMM EDA capability, 3D assembly design kit (3D-ADK), and designs for devices and test vehicles. The design tools developed will be validated through design challenges. These design challenges provide the opportunity to explore approaches that will improve and accelerate the adoption of 3DHI standardized chip-to-chip interfaces and package optimization. Leading-edge chip designs will be fabricated, and subsequently integrated into 3DHI designs in multi-project demonstration runs. Additional research related to this effort is funded within PE 0603469E, Project AET-02. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Refine material property analysis and characterization to improve accuracy of modeling and simulation of 3DHI stacks. - Demonstrate open-access, EDA infrastructure that enables EDA tool interoperability, compatibility with 3D-ADKs, and modular workflows. - Release updated 3D-ADK, which includes standards, training documentation, revision control, and predictive modeling. FY 2027 Plans: - Develop thermal modeling and characterization capabilities to improve fidelity of thermal analysis of 3DHI stacks. - Establish reference EDA flow for improved design accuracy and reduced design cycle time. - Release alpha version of 3D-ADK, which includes standards, training documentation, revision control, and predictive modeling.			-	25.000	25.000
Title: Ultra-Wide BandGap Semiconductors (UWBGS) Description: The Ultra-Wide BandGap Semiconductors (UWBGS) program is developing and optimizing ultra-wide bandgap (UWBG) materials and fabrication processes required to enable the next revolution in semiconductor electronics. UWBGS will establish the foundation for the creation of producible and reliable, high performance UWBG devices for a variety of DoW (and commercial) applications. These include but are not limited to: high power radio frequency (RF) switches; high power density RF amplifiers; high RF power protection device; high voltage switches for power electronics; high temperature electronics and deep ultraviolet light-emitting diodes and lasers. The program will address the key technical challenges that are limiting the performance of UWBG device. These challenges include realizing high quality UWBG materials, ability to tailor electrical characteristics of UWBG materials; ability to create homo- and heterostructures with abrupt junctions and low defect density; and the realization of			-	17.845	8.611

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
ultra-low resistance electrical contacts. UWBGs will fabricate device test structures to quantify the improvements in these areas. To be successful, the program will leverage recent advances in UWBG materials. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Further optimize UWBG material synthesis approaches to maintain low defect density and further improve doping efficiency and uniformity while simultaneously increasing wafer area. - Quantify improvement in material quality by designing, fabricating, and characterizing test structures. - Further optimize fabrication process to create robust, low resistance electrical contacts to UWBG materials and fabricate and characterize test structures to quantify robustness. - Continue to evaluate characterization results versus current state-of-the-art to quantify the improvement possible with UWBG devices. FY 2027 Plans: - Demonstrate large diameter UWBG material with low defect density and improved doping efficiency and uniformity. - Quantify improvement in material quality by designing, fabricating, and characterizing test structures. - Demonstrate optimized microfabrication processing with low resistance electrical contacts to UWBG materials with high robustness and reliability. - Continue to evaluate characterization results and compare to current state-of-the-art to quantify the performance improvement possible with UWBG devices. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from processes of interest to materials and device development and optimization.				
Title: Optomechanical Thermal Imaging (OpTIm) Description: Advanced infrared (IR) detectors and thermal imaging systems underpin a vast DoW application space including biochemical detection; infrared Search-and-Track; and terrestrial and space-based Intelligence, Surveillance, and Reconnaissance. Current IR detectors face limitations in either sensitivity, signal bandwidth, or the need for expensive cryogenic cooling. The Optomechanical Thermal Imaging (OpTIm) program is developing a new modality of low size, weight, and power, room temperature IR detectors capable of quantum-level sensitivity, thereby enabling transformative enhancements to DoW capabilities including, but not limited to, night vision, surveillance, multispectral detection, and remote detection of trace industrial pollutants and greenhouse gases. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02.		-	21.836	17.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Explore applications and initiate application-specific development of optomechanical IR detectors. - Demonstrate and optimize all-optical interrogation and readout performance of novel optomechanical IR detector devices. - Initiate fabrication process development and optimization for optomechanical IR detectors. - Develop device-level integration protocols and packaging recipes for optomechanical IR detectors. FY 2027 Plans: - Demonstrate complete device-level integration and characterization of scalable optomechanical infrared detectors. - Demonstrate preliminary testing and benchmarking of application-specific optomechanical infrared detectors for multiple applications. - Demonstrate field testing and environmental resilience of fully packaged optomechanical infrared detectors. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from device development to application-specific testing, validation, and optimization of infrared detectors.					
Title: Digital RF Battlespace Emulator (DRBE) Description: The Digital RF Battlespace Emulator (DRBE) program is developing a large-scale, interactive, emulated radio frequency (RF) environment, providing the DoW with the capability to cost-effectively evaluate adaptive, intelligent, and spatially distributed next-generation RF systems. DRBE is leveraging advances in massively multi-core computing hardware and high-bandwidth digital cross-connects to emulate realistic RF environments accounting for RF platform movement, signal propagation effects and delays, signal interference, and interactions between RF systems. DRBE is pursuing three technical thrust areas: architecture, massively multi-core computing, and scenario modeling. The resulting test environment will allow plug-and-play connections for hundreds of RF systems in a battlespace test. Multi-system exercises will then be quickly executed through many different combat scenarios and variations. DRBE is serving to develop concept of operations (CONOPS), inform operational and contingency planning, and fine-tune the performance of both individual and large groups of RF systems. Additional development started in 2024 greatly expands the input/output bandwidth of DRBE to support for much larger RF scenarios. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Define and develop hybrid wafer-scale bond process. - Assemble and test mechanical prototype to support large-scale integrated photonics. FY 2027 Plans:			-	19.274	29.186

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Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Deliver preliminary design review (PDR) for high-performance compute system. - Complete optical wafer design, tape out, and manufacturing. - Integrate optical wafer with digital high-performance compute wafer. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from the initial design of the expanded input/output subsystem High-Performance Computer to the final design and fabrication.			
Title: Technologies for Heat Removal in Electronics At the Device Scale (THREADS) Description: The Technologies for Heat Removal in Electronics At the Device Scale (THREADS) program is developing technologies to overcome transistor thermal limits to realize robust, high power density transistors that operate near their fundamental electronic limit of radio-frequency (RF) output power. DoW's RF transmitters increasingly use transistors that quintuple RF power output compared to legacy technology. Achieving high RF power output while maintaining a transistor operating temperature below the nominal maximum reliable operation temperature faces two challenges. The first challenge is reducing thermal resistance within the device. The second challenge is more efficiently moving heat away from the transistor hot spots. THREADS will address these challenges by leveraging recent scientific and technological advances in order to demonstrate high efficiency X-band transistors and power amplifier (PA) test vehicles with an output power density of 16X higher than current production amplifiers. THREADS technology will enable increased range for radar, communications, and electronic warfare systems. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Design and fabricate thermal resistance test structures with a 5X reduction in thermal resistance compared to the state-of-the-art (SOA). - Design and fabricate robust transistors and PAs with a 10X increase in output power density, compared to the SOA. - Demonstrate robust operation of RF transistors and PAs with a 10X increase in output power density. FY 2027 Plans: - Reduce transistor thermal resistance by 8x compared to SOA, and design thermal resistance test structures to demonstrate this reduction. - Design and fabricate RF PAs with a 16X increase in output power density, compared to SOA, close to electrical limit. - Demonstrate robust operation of RF transistors and PAs with a 16X increase in output power density, compared to the SOA. FY 2026 to FY 2027 Increase/Decrease Statement:		-	13.800
			14.100

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects a shift from initial development to design and fabrication.					
Title: Optimum Processing Technology Inside Memory Arrays (OPTIMA) Description: The Optimum Processing Technology Inside Memory Arrays (OPTIMA) program aims to realize a high throughput, compact, energy-efficient, and adaptable compute-in-memory (CIM) accelerator that is compatible with very large-scale integration (VLSI) fabrication. Traditional accelerators based on von Neumann architecture have limitations in terms of computational power efficiency and speed. By demonstrating a CIM accelerator with an array of Multiply Accumulate Macros (MAMs) consisting of a large number of Multiply Compute Elements (MCE) to perform the matrix multiply operating within memory elements, the power and speed challenges can be overcome. The program goal is to showcase high-performance CIM accelerator with innovative signal processing circuitry and architectures, with a focus on optimizing both area and power efficiency. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Evaluate performance of compact MAM with high number of MCEs. - Demonstrate a high throughput compute-in-memory accelerator with high energy and area efficiencies. FY 2027 Plans: - Demonstrate small, scalable and power-efficient MAMs for high-performance CIM. - Evaluate power-efficient models run on CIM hardware. - Demonstrate new capabilities compact, power efficient CIM offers in warfighter relevant use-cases. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from algorithm and hardware design to subsystem development and design.			-	15.586	21.058
Title: Minitherms3D Description: Minitherms3D is developing thermal management solutions for the three-dimensional heterogeneous integration (3DHI) of microelectronics to accelerate the growth of compact, high-performance microsystems. 3DHI microsystems are enabling technologies for phased array systems and dense computing for artificial intelligence and machine learning applications. Minitherms3D seeks to reduce the size, weight and power (SWaP) of high-performance 3DHI microsystems by developing novel methods to transfer heat from within the 3D stack to its boundaries, transmit it to a remote location, and reject it to ambient air. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans:			-	11.553	6.339

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Begin development of five-tier test vehicle for intended usage environment. - Demonstrate operational five-tier test vehicle to meet target thermal management capabilities in a laboratory. FY 2027 Plans: - Evaluate five-tier stack high power test vehicle models for intended usage environment. - Demonstrate operational five-tier test vehicle to meet target thermal management capabilities in intended usage environment. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from development and demonstration to test and evaluation.				
Title: Heterogenous Adaptively Produced Photonic Interfaces (HAPPI) Description: The Heterogeneous Adaptively Produced Photonic Interfaces (HAPPI) program aims to revolutionize information transmission within microsystems by achieving a 1000x increase in connectivity density through advanced photonic solutions. The program focuses on developing three-dimensional optical routing capabilities both within and between chips, enabling unprecedented levels of information movement and processing throughout integrated systems. This includes the potential to interface with both electronics and other optical elements, enhancing the flexibility and reach of the system. HAPPI's advancements will enable large-scale photonic circuits with thousands of components unlocking new microsystem architectures for applications such as signal processing, free-space communications, remote sensing, digital computing, and atomic sensing. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Develop fabrication flows and assembly processes capable of 3D integration of dense arrays of photonic links. - Demonstrate an out-of-plane guided optical path originating in the waveguide fabric of one discrete chip and terminating in the waveguide fabric of another discrete chip stacked on the initial chip. - Deliver a preliminary design review (PDR) for a large-scale dense photonic array capturing simulated array performance. FY 2027 Plans: - Deliver a critical design review (CDR) for a large-scale dense photonic array with updated simulated array performance. - Demonstrate a non-blocking routing network connecting nodes on a three-dimensional photonic grid. - Fabricate the large-scale photonic array design. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the move from intensive design and demonstration to design review and fabrication.		-	11.037	9.846
Title: Advanced Sources for Single-event Effect Radiation Testing (ASSERT)		-	7.847	11.866

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: 3D heterogeneously integrated (3DHI) microelectronics will be a key driver of the next wave in electronics performance. However, the nation's current single-event effect (SEE) radiation testing infrastructure lacks the ability to analyze and qualify emerging 3D devices for operation in high radiation environments. To fill this gap, the Advanced Sources for Single-event Effect Radiation Testing (ASSERT) program is developing new source technologies to create charge tracks with deep penetration depths for SEE qualification of 3DHI topologies and packaging, providing the means to selectively probe device topologies to inform engineering design, and generate data to validate developing models and codes and to provide training sets for optimization. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Refine design and fabricate radiation sources. - Initiate experimental and computational simulation campaigns to correlate single-event effects created by the new radiation sources with conventional heavy-ion data. FY 2027 Plans: - Finalize design and fabricate radiation sources. - Conduct experimental and computational simulation campaigns to correlate single-event effects created by the new radiation sources with conventional heavy-ion data. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from development to design finalization and initiating fabrication of the sources.					
Title: Machine Learning and Optimization-guided Compilers for Heterogeneous Architectures (MOCHA) Description: The Machine Learning and Optimization-guided Compilers for Heterogeneous Architectures (MOCHA) program is developing a new generation of compiler technology that uses machine learning to automatically generate and optimize compilers that realize the full potential of heterogenous architectures. The ending of Dennard scaling, which has limited the increase of clock speeds of digital circuits, has led to computing architectures that include a variety of co-processors and accelerators to enable high performance in specific domains. Traditional compilers, software that turns source code into machine instructions, do a poor job of optimizing code for heterogenous ensembles of processing units and accelerators, and a portion of the potential benefit of accelerators is lost. Extending compilers to handle heterogeneity is at present a manual task that is both time consuming and error prone, thus limiting the ability to upgrade mission-critical systems in a timely manner. MOCHA is applying machine learning techniques to the problems of compiler adaptation and extension by semi-automatically generating the main components of compilers for heterogeneous architectures. The use of machine learning will reduce human effort and development time while improving the quality of the machine code emitted by the compiler in terms of measures such as performance, memory size, and			-	11.208	13.084

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
power consumption. If successful, MOCHA will speed production of the compilers needed to capitalize on emerging specialized processor hardware in heterogeneous computing architectures for high performance military systems. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Design and implement selected programming-language-specific compiler front ends for transforming source code into a target intermediate representation. - Develop new techniques to model data movement costs and to factor these costs into the compiler optimizations. - Implement an evaluation framework for compilation based on measures such as performance, memory size, and power consumption. FY 2027 Plans: - Integrate selected approaches for machine-learning-driven compilation in an end-to-end capability for optimization of compiled software for heterogeneous architectures. - Develop techniques to automatically recognize source code patterns that can be allocated to specific hardware accelerators. - Extend evaluation framework to accommodate systems under test that comprise multiple types of hardware accelerators. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of compilation techniques for heterogeneous architectures and expanded work to test and evaluate the techniques on complex systems.					
Title: NanoWatt Platforms for Sensing, Analysis, and Computation (NaPSAC) Description: Efficient, high-speed scientific computing architectures are a ubiquitous requirement for applications including modeling of complex physical systems, advanced device designs, and multiscale computations of dynamical phenomena such as climate models or turbulence. Current state-of-the-art computing systems require prohibitive amounts of energy and time to perform such calculations. The NanoWatt Platforms for Sensing, Analysis, and Computation (NaPSAC) program aims to develop a novel computational architecture for massively parallel, ultralow power "in-memory" computation. NaPSAC-based computing architectures can potentially yield transformative impact by enabling beyond-state-of-the-art computational speed and accuracy. Applications of immediate relevance to the DoW include simulations of turbulent flows, multiscale electromagnetic simulations of plasma dynamics, advanced semiconductor device design, and the modeling of high-performance materials. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans:			-	11.945	6.072

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Initiate development of two-dimensional novel nanoresonator-based computing engine arrays for high speed, energy efficient scientific computations. - Initiate optimization and benchmarking of the performance of nanoresonator-based computing cores against state-of-the-art solvers for complex physical problems of interest to the DoW. FY 2027 Plans: - Demonstrate optimized algorithms and computational performance of nanoresonator-based computing cores for complex physical problems of interest to the DoW. - Initiate size weight and power reduction and systems-level packaging of nanoresonator-based computing cores for DoW relevant applications. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from system development to application-specific demonstration and refinement.				
Title: Intelligent Generation of Tools for Security (INGOTS) Description: The Intelligent Generation of Tools for Security (INGOTS) program is developing techniques to accelerate cyber defense activities by identifying, triaging, and assessing exploitability of chainable vulnerabilities. Today, sophisticated cyber attacks link multiple vulnerabilities together into exploit chains that enable them to compromise various systems. Accurately understanding risk is critical for both developers and defenders within cyberspace, but the metrics currently in use do not account for the multiple factors that differentiate an innocuous software flaw from a chainable vulnerability. INGOTS is developing semi-automated tools and techniques to characterize and measure the interdependent exploitability of vulnerabilities and a new vulnerability severity metrology that characterizes and measures interdependent exploitability. With the INGOTS vulnerability measurement capability, developers and defenders will improve software and hardware resiliency by rapidly identifying and prioritizing the most dangerous flaws. The INGOTS program is also funded in PE 0602025E, Project MSL-04. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02 and PE 0602303E, Project IT-03. FY 2026 Plans: - Develop techniques for automated replacement and repair of exploit chains in complex systems. - Automate the analysis of the scale of applicability of an exploit chain across a corpus of software and firmware versions. - Extend the exploit chain model to incorporate and represent the system context. FY 2027 Plans: - Demonstrate techniques for fully automated exploit chain synthesis, including link replacement and repair, in complex systems.		-	11.019	8.308

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Demonstrate the capability to reproduce exploit chains in software of interest to transition partners and integrate tools and capabilities into operational contexts. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects ramping down of development of exploit chain techniques and focus shifting to demonstration and transition.				
Title: Material Synthesis Technologies for Universal and Diverse Integration Opportunities (M-STUDIO) Description: The M-STUDIO program will realize a universal defect-free heterogeneous integration methodology, informed by emerging nano-scale material growth with surface free-energy driven defect termination and non-thermal-equilibrium synthesis, to achieve defect-free multi-layer heterogeneous materials with nanometer dimensions and atomically sharp interfaces. Specifically, M-STUDIO program will achieve: (1) Material synthesis with one heterogeneous interface: a semiconductor layer on lattice mismatch substrate with a total thickness 10 nm (> 100x reduction from the state-of-the-art (SOA)) and with low defect density, and (2) demonstrate the synthesis scalability with multiple heterogeneous interfaces: multiple 10 nm heterogenous semiconductor layers on lattice mismatch substrate with atomically sharp transition. In addition, M-STUDIO integration technology is compatible with leading edge and future advanced node semiconductor manufacturing process. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Initiate multiple lattice mismatched nano-meter heterogeneous semiconductor layers with high electrical performance on silicon substrate. - Evaluate quality of initial multiple lattice mismatched nano-meter heterogeneous semiconductor layers. FY 2027 Plans: - Demonstrate sharp atomic layer growth. - Demonstrate multiple atomically flat, compositionally sharp thin heterogenous layers. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the change from initial scientific evaluation to actual technology demonstration.		-	8.000	5.014
Title: Intensity-Squeezed Photonic Integration for Revolutionary Detectors (INSPIRED) Description: The Intensity-Squeezed Photonic Integration for Revolutionary Detectors (INSPIRED) program is developing compact, ultra-low-noise optical detectors. Low-noise detection is vital to all optical science and technology, but the quantum nature of light imposes a fundamental quantum limit on a conventional optical detector's noise performance. Recent experiments have demonstrated that exotic quantum states can be harnessed to overcome the quantum limit, albeit from bench-scale		-	7.287	2.341

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
apparatuses that ultimately restrict the application of enhanced detectors to esoteric applications such as particular forms of astronomy. The INSPIRED program is leveraging recent advances in chip-scale quantum optics and materials to realize optical detector modules operating well below the quantum noise limit in form factors that enable deployment in applications such as biosensing, navigation, and communications. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Experimentally demonstrate generator and interferometer circuit subsystems. - Complete overall system design for compact optical detector modules the provide enhanced sensitivity. FY 2027 Plans: - Experimentally demonstrate optical detector modules for enhanced sensitivity. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from design to fabrication and demonstration.					
Title: COmpact Front-end Filters at the EIEment-level (COFFEE) Description: The COmpact Front-end Filters at the EIEment-level (COFFEE) program is developing and demonstrating compact, high frequency radio frequency (RF) filter technology without compromising performance, specifically low insertion loss and high-power handling. The new filtering technology will enable interference rejection capability, efficient spectral management, and coexistence with commercial 5G applications. It is projected that COFFEE filter technology will enhance the resilience of military microwave and mm-wave radar and communication systems for DoW spectral dominance into the future. For commercial applications, COFFEE will result in more efficient use of mm-wave frequency allocations for 5G networks. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Demonstrate breakthrough filter technology through transition focused exemplars. FY 2027 Plans: - Transition filter technology to DoW radar systems and high-volume communications market. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from technology integration and demonstration to readying the technology for transition.			-	4.396	2.000
Title: High Operational Temperature Sensors (HOTS)			-	12.038	2.000

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: Many commercial, industrial, and defense systems experience thermal environments that are beyond the performance and survivability of today's physical sensors. The High Operational Temperature Sensors (HOTS) program is developing a high-performance sensor platform for operation at extremely high temperatures for high-priority mission-relevant timescales. Among other uses, HOTS technology will find application in combustion engine research, superhot rock geothermal energy development, and petrochemical processing. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Complete design and simulation of integrated sensor module. - Complete fabrication of integrated sensor module. FY 2027 Plans: - Begin functional test and performance characterization at high temperature. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from design and fabrication of the discrete high temperature components to the completion of fabrication and initial testing.					
Title: PROtein SEquencing (PROSE) Description: The PROtein SEquencing (PROSE) program seeks to bring protein sequencing capabilities to relevant missions. A protein's sequence largely determines its structure and ultimately, its function. PROSE will demonstrate molecular readers that can accurately read a broad array range of amino acids and post translational modifications (i.e., letters) in sequence for unknown protein samples. The program will seek hardware demonstrations of novel read element designs integrated with microsystem architectures. These approaches must overcome the challenges associated with reading long protein sequences with high accuracy at high throughput. Additionally, given the scale of the current alphabet, and scaling estimates of future alphabet size protein letter space, approaches must also demonstrate the ability to read a relevant alphabet size. This alphabet size is chosen to provide confidence that PROSE technologies can scale, with future commercial development, to read the entire protein alphabet. FY 2026 Plans: - Demonstrate the ability to read protein sequences of sufficient length and complexity. - Initiate microsystem design plan and develop software to predict performance of integrated system. FY 2027 Plans:			-	7.930	17.400

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Demonstrate increased read ability for proteins of greater length and complexity. - Complete microsystem design and prepare for tape out. - Initiate development of integrated system.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial design to comprehensive design, demonstration, and development.					
Title: Crystal Palace			-	13.535	27.070
Description: The Crystal Palace program aims to develop new inorganic material growth tools and techniques capable of producing complex and customizable inorganic materials with atomic precision and control that are unrealizable today. This program is designed to serve as a foundation to bridge the gap between artificial intelligence-driven material design and the ability to grow any combination of inorganic materials from the periodic table. Crystal Palace aims to accelerate microsystem innovation and create new tools and techniques to enable the rapid development of new complex inorganic materials at single crystal quality and at a relevant scale, ensuring that the demands of future DoW systems are met.					
FY 2026 Plans: - Start development of new material growth tools or modification of existing growth tools to be used to achieve new complex single crystal materials. - Identify and evaluate key technical risks with the functionality of these novel growth tools and techniques, and plan appropriate mitigation strategies, such as system upgrades or procurement of spare parts.					
FY 2027 Plans: - Demonstrate fully functional growth tools/techniques with precise control for complex material growth. - Achieve small-scale demonstrations of four new complex inorganic materials. - Scale up growth to at least 2 inches at single crystal quality for one complex material. - Demonstrate that growth process is uniform at large-scale and repeatable across multiple runs.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial development to small-scale demonstrations.					
Title: Promethean Clay			-	7.540	18.095
Description: The Promethean Clay program will produce a new type of energy storage device boasting significantly greater capacity, broad integrability, and intrinsic safety that will overtake current technologies and allow the United States to reclaim the battery supply chain. Promethean Clay deliverables will exceed the performance of state-of-the-art (SoA) lithium-ion batteries by thinking of the device not as a collection of independently-improved components, but as a complete system. Due to the removal					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS		Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
of parasitic mass of packaging, it is expected that Promethean Clay devices will be integrable into a large number of critical DoW platforms, significantly improving performance.					
FY 2026 Plans: - Perform initial safety testing on base materials for energy storage devices. - Demonstrate surrogate device operability. - Perform initial manufacturing feasibility analysis. FY 2027 Plans: - Delivery of first energy storage device prototypes. - Perform function and safety tests of initial prototypes. - Report on manufacturing improvements and economic cost drivers. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial feasibility analysis to prototype delivery and test.					
Title: Smash Description: The Smash program will develop and demonstrate highly efficient, near-zero-waste systems to process diverse raw materials and extract a wide array of elements with high purity. The new processing technology will reduce American reliance on foreign sources for critical minerals by creating new, efficient methods for processing domestic resources. Smash technology will enhance the resilience of the American supply chain and secure an asymmetric advantage in yield, purity, and energy utilization. FY 2026 Plans: - Develop initial standard feedstock formulations and test/validation methodology. - Conduct initial procurement of long-lead items and establish experimental capabilities. FY 2027 Plans: - Develop initial models capturing the physics of the separation processes to inform projections of prototype efficiency and throughput. - Perform initial experimental demonstration of technique achieving medium purity separation of simple feedstock. - Deliver initial commercialization plan that outlines value proposition, market analysis, intellectual property strategy, and risk mitigation to facilitate capital and transition partnerships. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial formulation and procurement to modeling and experimentation.			-	7.342	17.620
Title: Photonic Integrated Circuit Architectures for Scalable System Objectives (PICASSO)			-	6.158	18.475

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS		Project (Number/Name) MSL-02 / DISRUPTIVE MICROTECHNOLOGIES	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: The aim of the Photonic Integrated Circuit Architectures for Scalable System Objectives (PICASSO) program is to revolutionize photonic integrated circuit architectures. PICASSO seeks to expand the field's focus from individual components to include a more holistic, circuit- and system-level orientation. This will catalyze future photonic capabilities by enabling functional expansion with predictable, system-level performance. Scaling photonic circuits is challenged by the fundamental physics of signaling with light. As optical signals traverse long processing chains, they inevitably degrade and accumulate noise that simple amplification cannot fix. Signal fidelity is also highly sensitive to spurious wave interactions such as scattering and back-reflections, which become unmanageable in large systems. While today's systems mitigate these issues by converting light to electricity, this workaround negates the latency, efficiency, and bandwidth advantages inherent to photonics. PICASSO confronts these challenges by embracing a new paradigm: creating perfect photonic circuits with imperfect components. Drawing inspiration from modern electronics, in which clever circuit design overcomes the limitations of individual transistors, the program will foster innovative circuit-level strategies to achieve unprecedented system performance and stability. FY 2026 Plans: - Complete initial Interface Control Document. - Deliver initial circuit design. FY 2027 Plans: - Complete Preliminary Design Reviews for photonic circuits. - Complete Critical Design Reviews for photonic circuits. - Complete tapeout of initial photonic circuit designs. - Perform analysis of system-level performance including designed photonic circuit. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from initial design activities to completing design and preparing it for fabrication.					
Title: Disruptive Microtechnologies Studies and Concepts Description: The Disruptive Microtechnologies Studies and Concepts thrust will develop technologies to ensure a sustainable domestic supply chain of advanced microsystems. This includes the design, assembly, fabrication, and testing technologies for advanced microsystems that exploits and extends beyond commercial activities. It takes advantage of innovations in photonics, optics, materials, energy, biology, and advanced three dimensional heterogeneous integration (3DHI) for the highest performance technology. In doing so, the goal is to revolutionize domestic industry and enable safe and reliable access to disruptive technology. FY 2026 Plans: - Develop methodology for very large-scale photonic integration.			-	32.363	45.592

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Initiate a trade study on new hardware architectures that can leverage commercial scaling while preserving their security and lead to superior systems. - Evaluate technologies for the control of advanced materials down to atomic levels. - Perform analysis of application benefits due to advanced microsystems design, architectures, and materials. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in design techniques for next generation capabilities, and the evaluation of technologies for the control of advanced materials.				
Title: EElectronics for G-band ARrays (ELGAR) Description: The EElectronics for G-band ARrays (ELGAR) program is developing the integration technologies needed to create compact, high-performance G-band (220 GHz) array front-end electronics to enable phased array antenna systems for DoW communications and sensing. ELGAR will address the key technical challenges that prevent III-V electronics from realizing high-performance G-band arrays, namely achieving efficient, compact G-band III-V monolithic microwave/millimeter wave integrated circuit power amplifiers (MMIC PAs) with high output power density and achieving low loss off-chip interconnects between adjacent G-band array components. In particular, ELGAR will develop III-V compatible, silicon-like fabrication and integration approaches to enable compact, high-power density, high efficiency G-band MMICs and arrays. The technologies developed will support applications including high data rate communications in size, weight, and power-constrained platforms. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Maximize the output power of compact G-band III-V MMIC PAs that use the silicon-like multilayer interconnects while optimizing efficiency. - Design, fabricate and test circularly-polarized, high-power transmit and ultra-low noise receive array test articles. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	2.600	-
Title: Space Power Conversion Electronics (SPCE) Description: The Space Power Conversion Electronics (SPCE) program seeks to develop highly-efficient, radiation-tolerant point of load (POL) converters for low-earth-orbit satellites. In today's space power systems, power transistors for POL converters have derated operating voltage to maintain radiation tolerance, resulting in decreased POL efficiency and limiting the satellite's available power, capabilities, and battery lifetime. To address this deficiency, SPCE is developing high-performance, radiation-tolerant high voltage switches and compact passives by exploiting advanced wide-bandgap semiconductor advanced material		-	13.179	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
synthesis, novel device architectures, and advanced manufacturing processes. High efficiency and compact radiation tolerant POL will be realized with advanced gate drive/control circuitries and novel 3D heterogeneous integration technology. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Experimentally demonstrate gate drive and control circuitries for high efficiency and compact radiation tolerant POL converters. - Demonstrate device integration technologies which enables high-efficiency, high-energy-density POL converters. - Demonstrate fully integrated, high efficiency and compact radiation tolerant POL converters. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: Additive Manufacturing of MicrosystEms (AMME) Description: The Additive Manufacturing of MicrosystEms (AMME) program is revolutionizing microsystem manufacturing by leveraging selective material synthesis and 3D patterning to enable a new class of microsystems. Additive Manufacturing (AM) has enabled complex single-material geometries that were previously impossible to produce via traditional manufacturing methods. However, microsystem manufacturing has not exploited AM due to fundamental limits of material quality, resolution, and print throughput. The AMME program is using selective material synthesis to create high-quality material precursors that permit simultaneous printing of conductors and insulators with high-resolution and high-volume throughput. Additionally, AMME is focusing on commercialization of this technology such that the Department of War and intelligence community can quickly adopt the productized system to fabricate novel microsystems. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Deliver finalized commercialization plan for advanced additive manufacturing technology. - Finalize additive manufacturing synthesis technique. - Demonstrate a functional microsystem that is fabricated with AMME technology. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	11.647	-
Title: Lasers for Universal Microscale Optical Systems (LUMOS) Description: The Lasers for Universal Microscale Optical Systems (LUMOS) program is integrating high-performance light sources into silicon integrated photonics enabling compact, rugged, high-performance systems for positioning, navigation,		-	1.903	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
communications, 3D imaging, and quantum technologies. Silicon photonics today enable microscale integration of complex optical systems, but the platforms lack of optical gain precludes the creation of lasers and amplifiers through foundry processes. LUMOS will deliver the missing capability to provide compact optical sources at wavelengths from the visible to the infrared and will create a universal manufacturing platform that builds upon the current photonics ecosystem. To drive innovation and maintain DoW access to leading-edge deployable photonic solutions, LUMOS will establish a technology pathway connecting government, academic, commercial, and defense users of integrated photonics, and will provide multi-project wafer runs through an open-access foundry. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Construct system demonstrators utilizing visible-wavelength integrated platforms. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: Scalable On-Array Processing (SOAP) Description: The Scalable On-Array Processing (SOAP) program is designed to achieve scalable algorithms and processing architectures to overcome the inherent digital bottlenecks that severely limit today's wideband operation on arbitrarily large elemental digital phased arrays. SOAP aims to reduce the computational complexity of array processing as a function of element count, from exponential to linear scaling. SOAP also seeks to move the processing from physically separated back-end processors to processors integrated into the array, in order to fully process all the information generated at the element level, with no elemental information loss. To achieve these aims, SOAP will design processors that can be distributed within the array, as close to the elements as possible. These processors should be connected and networked in such a way that the data from any element can be processed by any processor. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-02. FY 2026 Plans: - Finalize adaptive array processing algorithms to meet SOAP goals. - Complete co-design of high-level hardware architectures to prove out the algorithms on-array. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	1.466	-
Accomplishments/Planned Programs Subtotals		-	354.608	368.276

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
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C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-03 / ELECTRONIC TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-03: ELECTRONIC TECHNOLOGY	0.000	0.000	87.791	65.867	-	65.867	101.813	121.178	136.944	138.523	-	-

A. Mission Description and Budget Item Justification

The Electronic Technology project pursues materials, devices, and architectures that will enable dramatic improvements over the current state-of-the-art. Areas of particular emphasis of this work include the development of techniques for advanced fabrication and test and the development of technologies that support electronics reuse and recycling. These technologies will reduce the barriers to designing and fabricating advanced technologies and exploit improved manufacturing techniques to provide these advanced technologies faster and cheaper to the warfighter.

Prior to FY 2026, efforts in this Project were funded in PE 0602716E, Project ELT-01.

B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Microsystem Induced CAtalysis (MICA) Description: The Microsystem Induced CAtalysis (MICA) program is developing advanced concepts for microsystem control of biological function. The program seeks hardware demonstrations of molecular catalysts immobilized to microsystem surfaces so that catalyst activity is controlled by physical forces generated by the microsystem. Additionally, the program seeks high-accuracy modeling and simulation of such integrated molecular microsystems. Through these demonstrations, MICA aims to answer three critical questions: (1) how can microsystems be used to actively control molecules? (2) what are the different microsystem physics that can be used to drive catalyst function? and (3) what co-design approaches can be used to integrate the different physics of microsystems and molecules? Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01. FY 2026 Plans: - Demonstrate and integrate initial set of catalysts and microsystems. - Characterize catalyst attachment physics to microsystem and control of catalytic activity by microsystem and feedback performance characteristics to simulation teams. - Feed forward simulation results to experimental teams, and demonstrate and integrate second set of catalysts and microsystems. FY 2027 Plans: - Demonstrate control of a single catalyst type with different control physics. - Demonstrate size and complexity scaling.	-	20.640	18.197

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-03 / ELECTRONIC TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Increase the accuracy of both models and fabrication approaches over a larger number of attachment sites and catalyst types.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from demonstration and characterization of a new microsystem to refining its accuracy and scaling its size.				
Title: Scalable Analog Neural networks (ScAN) Description: The Scalable Analog Neural networks (ScAN) program is increasing neural network (NN) inferencing capabilities at the edge and simultaneously reducing the size, weight, and power (SWaP) needed to support inferencing on edge platforms. Currently, sensor outputs are digitized at the edge and then transmitted to the command center for processing. ScAN aims to skip the digitization step and instead perform analog inferencing onboard the edge platform by operating directly on the analog sensor outputs. ScAN objectives are to enable 2000-fold reduction in the power needed for inferencing. ScAN technology will enable intelligence generation at the edge for missions that collect large amounts of sensor data, such as hyper-spectral imaging for unmanned aerial systems. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01. FY 2026 Plans: - Perform hardware and algorithm co-design analysis at intermediate scales. - Develop tapeout-ready design for analog hardware at intermediate scales. - Develop analog feature extraction and classification techniques at intermediate scales. - Develop training, inferencing and compression algorithms at intermediate scales. FY 2027 Plans: - Demonstrate analog hardware at intermediate scales. - Demonstrate analog feature extraction and classification techniques at intermediate scales. - Demonstrate training, inferencing and compression algorithms at intermediate scales. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects minor program repricing.		-	12.255	12.265
Title: Fast and Curious Description: The Fast and Curious program aims to define the next generation of advanced processing technology by developing and demonstrating new logic devices that compete with Complementary Metal-Oxide-Semiconductor (CMOS) performance in speed and density while achieving 100-1000x greater energy efficiency. Such devices will enable processing at the edge for DoW-relevant applications such as processing in space, in autonomous platforms and in SWaP limited processors which are limited		-	9.173	25.867

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
today due to CMOS's power and cooling requirements. These technical challenges included the design of logic devices that exploit the attributes of new materials, along with the development of scalable fabrication and growth techniques that create defect free, ultra-thin interfaces. Fast and Curious will focus on developing new scalable fabrication and growth techniques to create defect free, engineered interfaces required to make new logic circuits as fast-switching as CMOS systems. FY 2026 Plans: - Fabricate and engineer non-traditional transistor heterostructures with ultralow energy and high-speed switching characteristics. - Design and develop circuits using non-traditional transistor heterostructures for basic logic gates. - Perform theoretical analysis and modeling of non-traditional transistor materials and heterostructures to predict performance in logic circuits. FY 2027 Plans: - Fabricate and engineer non-traditional heterostructures with ultralow energy and high-speed switching characteristics. - Design and develop basic computational circuits using non-traditional transistor heterostructures. - Begin radiation hardness testing. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from initial design to design refinement and fabrication.			
Title: Electronic Technology Studies and Concepts Description: The Electronic Technology Studies and Concepts thrust is developing technologies that can enable a new generation of integrated circuits. Technologies such as advanced organic-electronic hybrid circuits, new photonic / optical / quantum components, and novel concepts in electronics and Micro-Electro-Mechanical Systems (MEMS) will be developed to address applications that are beyond the reach of traditional electronics. The feasibility and impact of the magnitude of the improvements these non-traditional circuits can achieve will also be evaluated. FY 2026 Plans: - Perform study of current state-of-the-art for the integration of organic and electronic devices to include potential use cases. - Perform study of the current state-of-the-art materials systems and wavelengths used in photonic and optical components. - Evaluate applications areas that could be enabled using non-traditional photonic and optical wavelengths. - Initiate design activity for the integration of organic and electronic devices for relevant use cases. FY 2027 Plans: - Perform initial design photonic and optical components using optimized materials systems and device wavelengths. - Evaluate applications areas that could be enabled using novel concepts in electronics that are inaccessible by traditional electronics.		-	26.718
			9.538

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Investigate the benefits of non-traditional circuits as compared to traditional electronics. - Initiate the design and development of photonic, electronics, MEMS, or quantum hybrid circuits.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the strategic prioritization of concepts and studies investments.			
Title: Highly Accelerated Learnings of Vibratory Systems (HALOVS) Description: The Highly Accelerated Learning of Vibratory Systems (HALOVS) portfolio investigates the foundational limits and transformative potential of vibratory sensors for positioning and navigation in GPS-denied environments. By leveraging advances in 3D micromachining, material science, and dynamic modeling, HALOVS explores novel architectures and principles through efforts examining ultrafast microsystems, higher-order composite resonators for extra resilience, and chemistries and monolayers for anti-aging kinematics. These efforts aim to address critical challenges in sensor performance, including operating beyond linear regimes to achieve high-velocity motion, leveraging nonlinearities for enhanced resilience to shock and vibration, and extending sensor longevity by addressing drift and aging at the molecular level. The insights and technologies developed within HALOVS promise to redefine the design space and practical limits of vibratory sensors, delivering capabilities essential to national security and global positioning system (GPS)-denied missions. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01. FY 2026 Plans: - Achieve high velocity motion with a tethered microsystem on chip exceeding the fracture limit of polycrystalline silicon. - Perform preliminary experiments to demonstrate the shock and vibration immunity of mechanical frequency combs. - Demonstrate stability of surfaces to extend the long-term bias stability and aging of sensors.		-	16.022
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Title: Emon Description: While the technical sophistication of radar systems has advanced tremendously since their inception, specifically with the use of digital circuitry that enables fine-tuned control over both waveform generation and received signal processing, the basic principles upon which radar operates have not varied greatly. Some properties of the classical electromagnetic (EM) field have not yet been exploited fully, whether due to past limitations in sensor capabilities or lack of insight into the possible benefits of leveraging the more subtle aspects of the EM field. One of these underutilized properties is the fact that the EM field has a tensorial nature. The Emon program is investigating the potential utility of constructing radar systems that benefit more completely		-	2.983

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B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
from the tensorial nature of the EM field, which may offer a variety of ways that could make radar sensing more useful to the warfighter. Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01. <i>FY 2026 Plans:</i> - Explore the interaction of new waveforms and sensing targets to determine properties at stand-off distances. - Use experimental data to refine the theory and assess utility of these novel methods for future defense purposes. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals	-	87.791	65.867

C. Other Program Funding Summary (\$ in Millions) N/A	Remarks	
D. Acquisition Strategy N/A		

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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-04 / CYBER SECURITY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-04: CYBER SECURITY	0.000	0.000	139.470	157.134	-	157.134	225.500	236.533	260.557	269.185	-	-

A. Mission Description and Budget Item Justification

The Cyber Security project is developing the computing, networking, and cyber security technologies required to protect Department of War, U.S. Government, and U.S. civilian information, information infrastructure, cyber-physical and embedded systems, critical infrastructure, and other computation-intensive mission-critical systems. Information technologies enable important existing and new military capabilities and drive the productivity gains essential to U.S. industry. Meanwhile, cyber threats grow in sophistication and number, and put data, computer programs, key information systems, and U.S. economic competitiveness at risk. The technologies developed in this project will enhance the resilience of information systems to current and emerging cyber threats; enable broad situational awareness of the cyber domain; and provide the basis for accurate, calibrated, and safe cyber response.

Prior to FY 2026, efforts in this Project were funded in PE 0602303E, Project IT-03.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Constellation	-	28.135	25.000
Description: The Constellation program is comprised of multiple projects that are developing technologies, capabilities, and prototype systems to enable full spectrum military cyberspace operations to defend the U.S. and deter, disrupt, and defeat adversary cyber actors. Technologies of interest include, but are not limited to, artificial intelligence, machine learning, and data science; resilient software, networking, and computing systems; formal methods and program analysis; data and information assurance; and cyber threat intelligence. High relevance is achieved through close coordination with U.S. cyber operators and the use of development, security, and operations (DevSecOps) and other collaborative development processes. High velocity is achieved through streamlined acquisition, assessment, approval, and deployment processes. Constellation development and deployment pipelines enable the rapid and continuous delivery of cyber technologies, capabilities, and prototype systems into operational use for the Department of War (DoW). The Constellation program is also funded in PE 0603469E, AET-03 to facilitate rapid transition of cyber technologies and laboratory prototypes from applied research to operational prototypes. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-05, and PE 0602303E, Project IT-03. FY 2026 Plans: - Refine and integrate cyber operational prototypes with established DoW programs and systems of record to create a cohesive, robust, and effective cyber warfighting architecture. - Develop, test, evaluate, and operationalize cyber technologies, prototypes, and capabilities in collaboration with DoW cyber stakeholders.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Coordinate with U.S. cyber operators to formulate cyber operations challenges as problems that research programs can address through new projects/prototypes. - Coordinate with cyber acquisition offices to expand pipelined development of cyber operational prototypes using DevSecOps and other collaborative development processes and through streamlined acquisition, assessment, approval, and deployment processes. FY 2027 Plans: - Refine and integrate cyber operational prototypes with established DoW programs and systems to create a cohesive, robust, and effective cyber warfighting architecture. - Develop, test, evaluate, and operationalize cyber technologies, prototypes, and capabilities in collaboration with DoW cyber stakeholders. - Coordinate with U.S. cyber operators to formulate cyber operations challenges as problems that research programs can address through new projects/prototypes. - Coordinate with cyber acquisition offices to expand pipelined development of cyber operational prototypes using DevSecOps and other collaborative development processes and through streamlined acquisition, assessment, approval, and deployment processes. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the focus shifting to prototyping activities.			
Title: Intelligent Generation of Tools for Security (INGOTS) Description: The Intelligent Generation of Tools for Security (INGOTS) program is developing techniques to accelerate cyber defense activities by identifying, triaging, and assessing exploitability of chainable vulnerabilities. Today, sophisticated cyber attacks link multiple vulnerabilities together into exploit chains that enable them to compromise various systems. Accurately understanding risk is critical for both developers and defenders within cyberspace, but the metrics currently in use do not account for the multiple factors that differentiate an innocuous software flaw from a chainable vulnerability. INGOTS is developing semi-automated tools and techniques to characterize and measure the interdependent exploitability of vulnerabilities and a new vulnerability severity metrology that characterizes and measures interdependent exploitability. With the INGOTS vulnerability measurement capability, developers and defenders will improve software and hardware resiliency by rapidly identifying and prioritizing the most dangerous flaws. The INGOTS program is also funded in PE 0602025E, Project MSL-02. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03 and PE 0602716E, Project ELT-02. FY 2026 Plans: - Develop and demonstrate techniques for automated replacement and repair of exploit chains in complex systems.		-	11.146
			12.582

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-04 / CYBER SECURITY	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Automate the analysis of the scale of applicability of the exploit chain across a corpus of software and firmware versions. - Extend the exploit chain model to incorporate and represent the system context. FY 2027 Plans: - Develop and demonstrate techniques for fully automated exploit chain synthesis, including link replacement and repair, in complex systems. - Advance the analysis and modelling of exploit chains to forecast exploit chain lifecycle in complex software systems that have state-of-the-art defenses. - Demonstrate the capability to reproduce exploit chains in software of interest to transition partners and integrate tools and capabilities into operational contexts. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of exploit chain technologies and additional efforts related to demonstration and transition.			
Title: Business Process Logic (BPL) Description: The Business Process Logic (BPL) program is developing techniques to characterize and resolve vulnerabilities in business logic systems to protect and assure defense-critical workflows for government and business. Automated workflows written in business logic (BL) control much of the world's enterprises, from administration and operation of seaports to the assembly of weapons systems. Losses due to BL faults and vulnerabilities can range from annoyances to business-threatening outcomes. Therefore, it is important to identify and correct potentially problematic logic issues, such as one-way actions or lost resources as early as possible. The BPL program is developing tools to extract workflow representations from BL and use those representations to identify, characterize, and mitigate faults and vulnerabilities in BL scripts and templates automatically. The technologies developed by BPL will enable increased assurance for manufacturing and assembly and greater efficiency for logistics and supply chain management. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Automatically generate business process models through ingest of diverse artifacts of real-world BL systems. - Demonstrate assured continuity of operation of BL systems via a functioning, end-to-end, artifact ingest, analysis, and fault mitigation capability. - Assess the performance of techniques for rapid identification and resolution of BL faults in defense manufacturing and logistics systems. FY 2027 Plans:		-	11.466
			11.800

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Extend ingest and process model generation to handle noisy or incomplete system information of real-world BL systems. - Demonstrate automated fault identification capability on an enterprise system of interconnected BL systems. - Evaluate the performance of techniques for identification and resolution of BL faults to existing and new real-world defense industrial base BL systems. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects minor program repricing.			
Title: Provably Weird Network Deployment and Detection (PWND²) Description: The Provably Weird Network Deployment and Detection (PWND²) program is developing formal models of emergent communication pathways to fundamentally improve the deployment and detection of resilient hidden networks in the real world. Authoritarian regimes are increasingly able to monitor and target internet communications, leaving many populations unable to communicate freely and safely. In response, the internet freedom and national security communities manually design hidden communication systems that use existing/available infrastructure and then validate these systems via empirical testing and analysis. These tests have difficulty capturing the range of real-world network environments and require infeasible amounts of testing to assess comprehensively at scale. PWND² is developing principled approaches for creating hidden communications systems with formal guarantees and for defending against use of these techniques by an adversary. If successful, PWND² technologies would enable the rapid and automatic generation and deployment of hidden communications systems with high assurance and strong guarantees. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Develop approaches for quantifying the scalability, security, and privacy of weird networks in the context of a capable adversary alerted to the possibility of hidden or obfuscated communications. - Analyze weird networks as components of larger systems that can support hidden communications at global scales. - Demonstrate weird network use cases, such as internet freedom and others where the fact that entities are communicating needs to be hidden or obfuscated. FY 2027 Plans: - Refine approaches for formally modeling adversaries that actively hunt for hidden or obfuscated communications. - Demonstrate the scalability of approaches using highly realistic network environments. FY 2026 to FY 2027 Increase/Decrease Statement:		-	14.307
			9.390

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects a shift from demonstration to refinement and transition.					
Title: Reclaiming Bus-based Systems During Compromise (Red-C)			-	5.633	6.182
Description: The Reclaiming Bus-based Systems During Compromise (Red-C) program focuses on developing and demonstrating novel approaches for resilient, self-healing, bus-based systems. The embedded computing systems common to modern vehicles and other cyber-physical systems typically use a bus-based architecture for communication among the various sensor, processor, actuator, and controller components organic to the platform. Due to the physical isolation of embedded computing systems, most standard buses currently in use were developed under an implicit assumption that all bus-connected components are secure and can be trusted, but such assumptions were unwise. Red-C aims to address this by creating resilient, self-healing, bus-based systems through advances to instrumentation and response. For instrumentation, Red-C aims to enhance the individual components on the bus to enable them to function as sensors that collectively monitor the bus and peer bus components to detect a cyber attack. Then, having identified that an attack is underway, Red-C-enhanced bus peers will respond by collaborating on software patch creation to remediate bus vulnerabilities and enable recovery of bus functionality. Red-C will implement these bus resilience techniques as firmware updates that can be deployed to existing bus-based systems and components. Successful Red-C technologies will mitigate bus vulnerabilities in military and civilian platforms and cyber-physical systems. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Formulate approaches for instrumenting bus-based systems to enable collective monitoring and detection of cyber-attacks by bus peers. - Explore techniques for responding to attack and compromise of bus-based systems through collaborative software patch creation and real-time repair by bus peers. FY 2027 Plans: - Implement techniques for peer-based monitoring, detection, and remediation of bus cyber-attacks. - Conduct demonstrations of bus resilience technologies implemented as firmware updates suitable for deployment to common bus-based systems and components. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from development of bus resilience technologies to demonstration.					
Title: Pipelined Reasoning of Verifiers Enabling Robust Systems (PROVERS)			-	25.803	6.110

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Description: The Pipelined Reasoning of Verifiers Enabling Robust Systems (PROVERS) program is creating scalable mathematically based technologies, tools, and practices to achieve continuous reasoning about complex systems that can support software development pipelines. These mathematically based techniques, or formal methods, enable rigorous modeling, reasoning, and proving of diverse properties of software code or design models, such as the absence of a specific type of defect or security vulnerability. PROVERS integrates formal methods into a modern incremental and iterative development process by running tools at each code commit and delivering results to developers when they can most effectively remediate discovered issues. PROVERS focuses on creating and sustaining a body of evidence that can co-evolve with the system under change to support continuous assessment and ensure that the system remains free of identified categories of defects and security vulnerabilities through its lifetime. Key PROVERS objectives include enabling proof maintenance and repair capabilities at a cost that is proportionate to code change; integration of formal methods with code, properties, and proofs in a single workflow that reduces human involvement; providing improved explanations to facilitate proof repair; and automating formal methods-based software analysis to support software developers that are not formal methods experts. PROVERS technologies will facilitate the agile development and continuous improvement of mission-critical software systems that meet the high security and quality standards required by the Department of War (DoW). Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Create novel system-architecture modelling and analysis tools that enable integration of formal-methods early in the development lifecycle. - Evaluate the barriers-to-adoption of formal-methods-based tooling into DoW software development workflows. - Collaborate with DoW stakeholders on controlled formal methods-based experiments on selected high-assurance and mission-critical military software systems. - Implement formal-methods-based tools that are capable of the specification and verification of DoW-relevant cybersecurity properties. - Quantify cost of incorporating formal-methods-based proof maintenance and repair capabilities in software development workflow. FY 2027 Plans: - Demonstrate the ability of regular DoW development staff to incorporate formal-methods-based tooling into development processes at minimal cost. - Demonstrate cybersecurity benefits of formal-methods development integration. FY 2026 to FY 2027 Increase/Decrease Statement:			

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects a shift from development to demonstration.				
Title: Cyber Agents for Security Testing and Learning Environments (CASTLE) Description: The Cyber Agents for Security Testing and Learning Environments (CASTLE) program is developing an artificial intelligence (AI) toolkit to instantiate realistic network environments and train AI cyber agents to enable resilient network operations against advanced persistent threats (APTs). CASTLE formulates network hardening as a reinforcement learning (RL) problem and teaches RL agents to operate through the post-breach behavior of widely available penetration testing tools. Over progressive rounds of attack and defense, agents explore defensive actions to proactively stop ongoing attacks while maintaining operationally relevant workflows. Environments execute agents inside instrumented subnets that are deployed to live networks and simulate defensive actions that counter APT tools. Agent execution produces calibrated datasets for progressively improving simulations. The defensive AI cyber agents developed under CASTLE will provide the Department of War with continual security assessments of critical networks and real-time response to cyber attacks. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Extend agent training in larger network environments and automate defensive responses to attacks while maintaining operations. - Automate instantiation of exemplar network environments and execution of agent strategies. - Demonstrate automated system security of realistic network environments. FY 2027 Plans: - Refine environment simulations and agent training to transition partner operational priorities. - Extend environments to support training goals of defensive cyber operators. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from demonstration to refinement and transition.		-	13.512	6.087
Title: Cyber Security and Resilience Studies and Concepts Description: Efforts conducted under this thrust study and refine technologies for cyber security and resilience of U.S. software systems. Security and resilience of the software in weapon, combat, and information systems is essential for warfighters who rely on the correct functioning of this software for success on the battlefield. Thrust emphasis is on dramatically enhancing the security and resilience of both new and legacy software systems that may be subject to cyber attack by adversaries. Focus areas include enabling the confident creation of software systems that are secure by design against exploits; the rapid remediation of software defects once discovered; and the cost-effective improvement of the legacy software on which critical missions often depend.		-	11.203	40.496

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Technical approaches of interest include but are not limited to automated vulnerability discovery and remediation, binary analysis, reverse engineering, formal methods, artificial intelligence-based approaches, and other advanced software generation, analysis, and understanding technologies. Research is performed to explore new software security and resilience concepts and techniques, and evaluation and refinement of the concepts and techniques is performed in collaboration with software developers and cyber red teams. This thrust creates the new concepts and techniques required for the U.S. to develop and sustain software systems that are secure and resilient in the face of cyber attack by adversaries.				
FY 2026 Plans: - Explore the use of formal reasoning to verify configurations of time-sensitive networking components to ensure performance and security. - Develop techniques for automated contextual software vulnerability analysis. - Formulate approaches for verifying configurations of networks to ensure performance and security.				
FY 2027 Plans: - Develop interactive tools that facilitate understanding of formal mathematical proofs for high assurance software systems. - Create formal compilation and verification techniques that improve the resistance of computing systems to soft errors. - Develop techniques and tools that integrate verification and validation with model-based systems engineering to enable timely systems accreditation. - Develop digital twins for mission resiliency. - Enable secure collaboration among autonomous agents. - Develop protections for markets facing nation-state attacks. - Enable rapid creation of photonic hybrid specialty devices. - Enable software optimization to exploit advances in optical/analog hardware.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in developing cyber security interactive tools and techniques, and in protection against attacks.				
Title: Cyber Operations Studies and Concepts		-	9.026	39.487
Description: Efforts conducted under this thrust study and refine technologies to achieve decisive and enduring operational advantages in the cyber domain. The thrust has two focus areas: Offensive Cyberspace Operations (OCO) that aim to project power and deliver effects through actions in the cyber domain, and Defensive Cyberspace Operations (DCO) that aim to defend U.S. cyberspace. OCO interests span strategic, which includes attacks on adversary military infrastructure, to tactical, which includes attacks on adversary weapons, combat, and information systems. In addition to overt denial/disruption/destruction of adversary systems, OCO offers the potential for action that can undermine adversary confidence in their systems and thereby				

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
deter adversary aggression. DCO focuses on defeating adversary malicious cyber activity by preventing adversary access to U.S. networks and systems and by promptly discovering and remediating any compromise of U.S. systems. For both OCO and DCO, research explores new cyber operations concepts, and the evaluation and refinement of these concepts is performed in collaboration with cyber operators. This thrust creates the new concepts required for the U.S. to achieve and maintain superior cyber warfighting capabilities. FY 2026 Plans: - Explore tools to predict disruptive innovation to help anticipate strategic surprise. - Explore channels that emerge from standard voice communications. FY 2027 Plans: - Develop techniques for automated identification of latent behaviors for heterogenous mega-systems to enable novel defenses. - Develop techniques for digital continuity of operations. - Use on-board sensor data to enable timely detection of cyber attacks. - Discover and prove logic flaws in security protocol implementations. - Learn to generate disruptive counter-strategies. - Develop techniques to prevent information reconstruction. - Enable network applications to function under severely impaired internet connectivity. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in developing techniques for digital continuity of operations, prevention of information reconstruction, and addressing logic flaws in security protocol implementations.			
Title: Hardening Development Toolchains Against Emergent Execution Engines (HARDEN) Description: The Hardening Development Toolchains Against Emergent Execution Engines (HARDEN) program is developing techniques and tools to anticipate, isolate, and mitigate emergent system behaviors and thereby improve security of complex integrated software. Today's software development toolchains and testing methodologies provide very limited means for reasoning about adversarial reuse of code as written and designed. This limitation results in unwitting creation of stable, reliable patterns of emergent behaviors within systems that adversaries can reuse in attacks. The HARDEN approach to preventing adversarial code reuse is to create techniques, tools, metadata, and instrumentation for reasoning about emergent execution at all stages of the software development life cycle (SDLC) and to flag code segments and design patterns where there is high potential for adversarial reuse and emergent execution. The utility of HARDEN technologies will be assessed by applying them to integrated software systems and critical system elements, such as bootloaders. Successful technologies developed by HARDEN will		-	6.790
			-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
facilitate efficient mitigation of complex code-reuse and emergent-execution vulnerabilities at early SDLC stages and provide the stronger roots-of-trust required by zero trust architectures and high assurance integrated military software systems. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: - Demonstrate and evaluate the effectiveness of mitigations against unintended system behaviors to reduce risk of adversarial reuse and emergent execution. - Refine tools for emergent execution mitigation to enhance potential integration across the SDLC. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Title: Signature Management using Operational Knowledge and Environments (SMOKE) Description: The Signature Management using Operational Knowledge and Environments (SMOKE) program is developing signature management technologies that generate evasive cyber infrastructure which minimizes signatures as a source of attribution. SMOKE technologies incorporate counter-attribution techniques into the design process, quantitatively measure attribution risk in real-time, and maintain evasiveness after infrastructure changes. SMOKE data-driven tools automate the planning and execution of threat-emulated cyber infrastructure needed for network security assessments by red teams. SMOKE data-driven tools automate the discovery of cyber threat infrastructure signatures. SMOKE prototypes enable red teams to plan, build, and deploy cyber infrastructure that is informed by machine-readable signatures of sophisticated cyber threats. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-03. FY 2026 Plans: Harden cyber planning and risk management tools and transition capabilities to operational partners. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	2.449
Accomplishments/Planned Programs Subtotals		-	139.470
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			

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D. Acquisition Strategy N/A		

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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-05: ARTIFICIAL INTELLIGENCE AND HUMAN- MACHINE SYMBIOSIS	0.000	0.000	120.707	138.950	-	138.950	202.398	237.185	248.368	251.368	-	-
A. Mission Description and Budget Item Justification												
The Artificial Intelligence and Human-Machine Symbiosis project develops technologies to enable machines to function not only as tools that facilitate human action but also as trustworthy partners to human operators. Of particular interest are systems that can understand human language, extract information, and reliably categorize content contained in diverse media; answer questions, reach conclusions, and propose explanations; and learn, reason, and apply knowledge gained through experience to respond intelligently to new and unforeseen events. Enabling computing systems with such human-like intelligence matters because the tempo of military operations in emerging domains can exceed that at which unaided humans can orient, understand, and act. The technologies developed in this project will enable warfighters to make better decisions in complex, time-critical, battlefield environments with greater efficiency and confidence; and help autonomous systems and intelligent computational agents to safely and reliably perform critical missions in contested and adversarial physical and virtual environments.												
Prior to FY 2026, efforts in this Project were funded in PE 0602303E, Project IT-04.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Securing Artificial Intelligence for Battlefield Effective Robustness (SABER)									-	13.667	16.400	
Description: The Securing Artificial Intelligence for Battlefield Effective Robustness (SABER) program aims to advance understanding of the military's use of artificial intelligence (AI) enabled systems in battlefield environments to enhance the robustness of AI-enabled systems against real-world adversarial attacks. AI-enabled systems can be subject to adversarial AI attacks, but most attacks studied to date are not physically realizable or assume the adversary already has some level of system access. SABER anticipates adversaries will develop combined AI-cyber-physical attacks and formalize these attacks in tactics, techniques, and procedures (TTPs). To address such capable AI adversaries, SABER will evaluate AI-enabled military systems in realistic operational settings to quantify the risks of deploying AI-enabled systems on the battlefield. Simultaneously, SABER will develop technologies and TTPs for continuous monitoring, integration, and deployment of counter-AI techniques to enhance the robustness of AI-enabled systems. SABER technologies will enable the formation of Department of War (DoW) AI red teams that continuously integrate and employ emerging counter-AI techniques and tools, establishing a sustainable model for operational AI red teaming. If successful, SABER will empower military AI red teams to confidently assess and secure U.S. AI-enabled battlefield systems against attack by sophisticated adversaries.												
Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04.												
FY 2026 Plans:												

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Formulate a comprehensive approach to adversarial AI to serve as the basis for the development of risk analysis and robustness enhancement of AI-enabled systems. - Create techniques and tools for use by developers and red teams to perform risk analysis of AI-enabled systems during development and operations. - Collaborate with the Joint Force on a series of high-fidelity operational exercises to iteratively assess and red team AI-enabled military systems in battlefield settings, to demonstrate utility of AI robustness techniques across multiple use cases, and to formulate AI robustness TTPs. FY 2027 Plans: - Refine AI risk analysis techniques and tools based on the results of initial operational exercises and red team experiences. - Quantify the effectiveness of alternative robustness enhancements and TTPs across diverse use cases. - Conduct high-fidelity operational and red team exercises targeted to high-priority DoW AI-robustness use cases, and capture results in AI-robustness TTPs. - Explore the combined AI-cyber-physical attack and counter-measure spaces to anticipate and prepare for sophisticated adversarial AI challenges. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of AI risk analysis techniques and tools and expanded efforts to evaluate the technologies on diverse DoW use cases.				
Title: Artificial Intelligence Quantified (AIQ) Description: The Artificial Intelligence Quantified (AIQ) program is developing technologies for assessing the capabilities of generative artificial intelligence (genAI) to enable mathematical guarantees on performance. Current state-of-the-art methods for genAI quantification and assessment are ad hoc, deal with only the simplest of capabilities, and are not properly grounded in a rigorous theory. The AIQ approach combines mathematical methods with advances in measurement and modeling to enable rigorous quantification of genAI capabilities. The program addresses three capability levels: 1) the specific problem level, which addresses quantification for a given problem; 2) the classes of problem level, which address quantification of a given problem class; and 3) the natural class level which addresses the reverse question of the class of problems quantifiable for a given genAI architecture. Success with AIQ will provide guarantees on the performance of genAI technologies. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Develop mathematical methods for assessment at the natural class level for a given genAI architecture.		-	9.412	13.352

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Formalize best practices for specific problem level assessments to facilitate adoption by the genAI user community. FY 2027 Plans: - Conduct the first at-scale evaluation of genAI assessment methods for compositions of classes of problems. - Integrate mathematical guarantees into assessment methods at scale. - Formalize best practices for classes of problems and compositions of classes to facilitate adoption by the genAI user community. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects continued development of AI assessment techniques and expanded work to evaluate techniques.					
Title: Anticipatory and Adaptive Anti-money Laundering (A3ML) Description: The Anticipatory and Adaptive Anti-money Laundering (A3ML) program aims to develop graph search-based algorithms that enable anticipation of and adaptation to money laundering tactics, techniques, and procedures (TTPs) in order to prevent money laundering in the global financial system. Today, anti-money laundering (AML) practitioners accumulate evidence of money laundering largely via manual searches of a myriad of commercial and financial intelligence (FININT) databases and rule-based AML systems that lack flexibility to adapt to changing TTPs. Because of the static nature of current AML processes and the limited capabilities of AML tools, novel money laundering schemes can go on for years. To address this challenge, A3ML will develop and apply graph search and subgraph isomorphism techniques that reference high-level models of money laundering to extract specific money laundering TTPs from heterogeneous transaction data and to anticipate and adapt to new money laundering TTPs in a timely fashion. Successful A3ML technologies and tools will enhance AML practices across U.S. Government (USG) and the private sector. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Formulate adaptive approaches for anticipating and extracting novel money laundering TTPs from heterogeneous FININT and transaction data. - Explore the potential for the graph-based TTP models to support collaborative AML by organizations across USG and the private sector without sharing classified information or sensitive private data. - Begin incorporating feedback from U.S. Treasury and financial institutions that would pave the way for algorithm-based AML reporting compliance. FY 2027 Plans: - Improve TTP detection performance through adaptive generative approaches and demonstrate capability to anticipate and extract novel money laundering TTPs from heterogeneous FININT and transaction data.			-	13.173	10.224

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Collaborate with AML stakeholders from across USG and the private sector to evaluate techniques and tools in operationally relevant timescales on operationally relevant databases and graphs. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the ramping down of development of AML techniques and expanded efforts to evaluate techniques on operationally relevant data.			
Title: Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) Description: The Fostering Research and Growth in Emerging Artificial Intelligence (AI FORGE) project aims to capture and solve selected artificial intelligence (AI) research challenges to accelerate the development and fielding of frontier AI for national security. At present, a relatively small number of frontier AI companies drive most innovation in AI with a strong focus on commercialization and return-on-investment. AI FORGE seeks to make national security considerations more prominent in the generative AI research and development (R&D) community. These considerations primarily involve the life- and mission-critical nature of national security generative AI use cases, which span information-centric intelligence analysis to grounded planning of logistics and tactical operations to embodiment in autonomous systems, and the highly adversarial nature of battlefield operational environments. For such use cases and environments, matters of assurance, predictability, reliability, resilience, and transparency are of heightened importance. AI FORGE is conducting high-risk, high-reward efforts that address cross-cutting challenges that arise when using frontier AI for national security applications, systems, and missions. AI FORGE combines the talents of universities with resources from frontier AI companies to create an R&D ecosystem focused on national security. This applied research funding supports applied research efforts that are part of the larger basic research AI FORGE activity under PE 0601122E, Project EMR-01. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02. FY 2026 Plans: - Complete an acquisition strategy and management plan for a joint government-university-industry project that features R&D efforts that solve critical challenges associated with the use of frontier AI for national security. - Formalize agreements with government and industry partners to establish processes for initiation, approval, and review of R&D efforts focused on the use of frontier AI for national security. FY 2027 Plans: - Evaluate R&D efforts addressing critical technical challenges associated with the use of frontier AI for national security and initiate new efforts as appropriate.		-	4.783
			6.567

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Launch additional collaborative R&D efforts that address critical technical challenges associated with the use of frontier AI for national security.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects ramping up of R&D efforts that address critical challenges associated with the use of frontier AI for national security.				
Title: Kallisti Description: The Kallisti program seeks to develop new capabilities for modeling adversarial behaviors. Actions taken by independent state and non-state actors are determined by their beliefs about the state of their environment, the risks and rewards associated with these beliefs, and the strategy they use to react to the risks or rewards they believe exist in their environment. This program is developing an algorithmic model of behaviors by independent actors through inferring and understanding the belief states that might deter or result in distributions of actions. Kallisti's modeling will allow projection of an actor's future behavior based on their perceived strategy, enabling external actions to be taken. The program seeks not only to understand an actor's current strategy but also to find a decomposed version of the strategy into relevant basis vectors to track strategy changes under non-stationary assumptions. The primary goal is to utilize algorithmic modeling of beliefs and behaviors to deter aggression by military adversaries. Technology developed under this program will transition to the Services. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Develop relevant context of beliefs and system dynamics to constrain belief space. - Actively manage multiple reference frames using less data and computational complexity. - Use novel methods in moment matching to improve cross-entropy of observations. FY 2027 Plans: - Extend core capabilities to include the estimation of risk and reward structures associated with belief states. - Transition Kallisti technologies. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from developing models that estimate actor comprehension to developing models that estimate actor strategies.		-	10.619	6.000
Title: National Security Economic Theory (NASCENT) Description: The National Security Economic Theory (NASCENT) program will perpetuate a policy and market feedback loop between the United States Government (USG) and private-sector executives to support more effective geoeconomic actions with		-	17.442	5.195

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
precisely defined positive national security outcomes. Current approaches to geoeconomics are based largely on qualitative ideas, drawn primarily from international relations and legal backgrounds, and are often non-falsifiable with limited scope. Furthermore, the lack of a body that facilitates neutral and open discussion and iterative development of geoeconomic theory has hindered innovation and renders the development of theory and tools of economic statecraft cost ineffective. NASCENT is addressing these challenges by creating a membership-based forum where the USG, academia, and the private sector can collaboratively and iteratively design and rigorously evaluate geoeconomic mechanisms. NASCENT will combine techniques from economic mechanism design and experimental economics to work backwards from precisely described national security outcomes, elicited from across the USG, to understand what economic incentives and mechanisms enable those outcomes. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Employ precisely described national security outcomes to iteratively design and rigorously evaluate geoeconomic mechanisms. - Hold symposia with USG and industry to precisely define national security objectives and their related economic incentives to inform geoeconomic research objectives. - Generate theoretically valid economic mechanisms that concurrently produce positive national security outcomes and U.S. economic benefits. - Test the empirical validity of proposed economic mechanisms to ensure they meet desired national security outcomes while beneficially incentivizing U.S. economic activity. FY 2027 Plans: - Attract private capital to support efficient operation of the forum and operationally relevant development of geoeconomic theory and playbooks. - Develop geoeconomic playbooks derived from geoeconomic research. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the transition of the forum from DARPA-provided funding to private capital.				
Title: Transfer from Imprecise and Abstract Models to Autonomous Technologies (TIAMAT) Description: The Transfer from Imprecise and Abstract Models to Autonomous Technologies (TIAMAT) program is developing techniques to robustly transfer learned autonomy from fast abstract simulations to autonomous platforms in real-world environments. Autonomous systems that heavily depend on modeling and simulation (M&S) training environments are relatively constrained in their abilities because those training environments insufficiently account for the data domain shift common when translating M&S outcomes to the real world, a phenomenon sometimes referred to as the sim2real gap. The TIAMAT approach integrates symbolic structures with neural structures to transfer learned autonomy more realistically and robustly. TIAMAT		-	14.965	2.656

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
will enable the use of fast abstract simulations by anchoring the learning and transfer of autonomy on semantically consistent components shared across simulations and real environments. For TIAMAT, semantically consistent components of particular importance include militarily relevant phenomena that remain consistent in the source and target environments, such as mission objectives, special instructions, subject matter expert guidance, rules of engagement, and the laws of physics. Autonomy transfer using TIAMAT methods and techniques will reduce the complexity of the autonomy learning and transfer problems to the comparatively simpler points of reference in the anchored representation. Successful TIAMAT transfer of M&S-based learning will enable more rapid and robust training and deployment of autonomous systems at higher levels of autonomy. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Develop methods for measuring the sim2sim and sim2real gaps, as well as robustness to these gaps. - Demonstrate a capability to transfer autonomy from abstract simulations to live platforms for scenarios of interest to potential transition partners. - Conduct a sim2real competition in collaboration with private sector stakeholders to evaluate the robustness of simulated autonomy in a real-world platform subject to large sim2real gaps. FY 2027 Plans: - Transition techniques to Department of War program offices that can benefit from enhanced capabilities to simulate autonomy. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from development to transition of techniques.				
Title: Scientific Feasibility (SciFy) Description: The Scientific Feasibility (SciFy) program is developing computational methods to measure the feasibility of claims to enable accurate assessments of scientific content. Automated scientific content generation, via rapidly improving large pre-trained models, has the potential to disrupt the U.S. technology base in times of crisis and to distort the global race for technological dominance in key areas. Similarly, false capability claims can have significant negative implications for national security and international relations. To address these threats, SciFy focuses on methods for assessing the scientific feasibility of claims by using automated reasoning to decompose claims into constituent, verifiable parts. Assessing each component involves referencing existing technological advancements, foundational scientific principles, data, software, models, simulation results, and industry standards or benchmarks. SciFy methods will go beyond automated fact-checking by addressing complex component interactions and operational constraints as well as evaluating logical consistency, system integration, and compatibility considerations. Success with the SciFy program will enable the U.S. to reliably determine whether claimed scientific and technological capabilities are practical and realistic when considered as a whole, even when theoretically possible in parts. This		-	2.607	1.168

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
applied research funding supports applied research efforts that are part of the larger basic research SciFy activity under PE 0601122E, Project EMR-01. Prior to FY 2026, this program was funded in PE 0601101E, Project CCS-02 and PE 0602303E, Project IT-04. FY 2026 Plans: - Collaborate with Department of War and Intelligence Community transition partners to evaluate the utility of the feasibility assessments and to refine techniques. FY 2027 Plans: - Collaborate with transition partners to leverage feasibility assessment tools in operational settings. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from tool evaluation to transition.				
Title: Artificial Intelligence Engineering and Assurance Studies and Concepts Description: Efforts conducted under this thrust study and refine artificial intelligence (AI), human-machine symbiosis (HMS), and machine learning (ML) concepts and technologies to facilitate the engineering and assurance of AI-based systems for mission-critical national security applications. Mission-critical AI use cases span information-centric intelligence analysis to grounded planning of logistics and tactical operations to embodiment in autonomous systems. For each of these use cases, the highly adversarial, dynamic, and uncertain nature of the battlefield environment both complicates and greatly heightens the need for assurance, predictability, reliability, and resilience. This thrust studies and refines abstractions, patterns, architectures, assurance techniques, and iterative processes to facilitate the creation and sustainment of complex mission-critical systems that rely on AI-based components. The capability to engineer and assure AI/HMS/ML systems that meet the safety, trustworthiness, transparency, and security requirements for mission-critical applications with efficiency and timeliness is essential for the Department of War. FY 2027 Plans: - Develop formal approaches for software adaptation and system behavior prediction in the field. - Create physics-based models of perception for high reliability AI/ML-based sensing in autonomous systems. - Integrate distributed control, game-theoretic incentives, and strategic reasoning to enable robust, controllable, and adaptive distributed intelligence. - Develop techniques to detect, contain, and neutralize harmful behavior by AI agents. - Develop AI systems to translate between human language descriptions and provably correct software. - Develop multi-level security for foundation models.		-	-	39.650

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Enable generative, adaptive engineering of complex systems that redesign themselves to maintain effectiveness.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in developing multi-level security for foundation models, techniques to neutralize harmful behavior by AI agents, and formal approaches for software adaptation and system behavior prediction.				
Title: Artificial Intelligence Operations and Effects Studies and Concepts Description: Efforts conducted under this thrust study and refine artificial intelligence (AI), human-machine symbiosis, and machine learning concepts and technologies to enable a broad range of military operations and effects. Kinetic operations in traditional physical conflict domains are being reimagined due to the rapid advances in modern AI which enable drones and other autonomous warfighting capabilities to be more efficient and effective in their assigned missions. Modern AI also opens the door to new possibilities for non-kinetic operations and effects in advance of and in conjunction with kinetic operations. Prime examples include AI agents for dynamically optimized planning/replanning that are responsive to real-time developments in the joint battlespace, and generative AI for coordinated operations at multiple scales. This thrust aims to ensure that the U.S. is at the forefront of emerging AI operations across all warfighting domains. FY 2027 Plans: - Enable enhanced collaborative reasoning by integrated human-machine warfighting formations. - Automatically generate high-fidelity digital representations and behavioral models of adversary groups. - Develop AI for automated reverse engineering during active cyber warfare. - Develop AI that continually improves its model as it operates in the field. - Develop AI that accelerates targeted mineral exploration and secures supply chains. - Develop reinforcement learning agents that are sensitive to risk due to uncertainty. - Develop AI systems that autonomously reflect, self-assess, and adapt. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, applied research in developing AI for automated reverse engineering, advanced AI systems, and accelerated targeted mineral exploration.		-	-	37.738
Title: Open Price Exploration for National security (OPEN) Description: The Open Price Exploration for National security (OPEN) program aims to increase supply chain resilience and enable more efficient critical mineral markets by leveraging advances in artificial intelligence (AI) prediction and forecasting to increase structural price, supply, and demand transparency. OPEN constructs structural price predictions from fundamental and observable critical mineral input costs and increases the accuracy and precision of supply and demand forecasts by leveraging		-	18.735	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
this structural price in conjunction with advances in AI and economic modeling. Today, critical mineral markets and supply chains are subject to international supply shocks that can lead to large and rapid critical mineral price spikes with immediate economic ramifications, and commodities purchase transactions (e.g., offtake agreements) are negotiated leveraging a mix of opaque and flawed pricing data. OPEN leverages a decomposition of a critical mineral price into four components (input costs, supply/ demand shocks, distortions due to noncompetitive behavior, and stochastic fluctuation) to construct transparent estimations of an approximate marginal cost for critical minerals indexed by time and geographic location, and estimates supply and demand forecasts for critical minerals that take into account geopolitical factors, energy fluctuations, and technological innovations in recycling and supply chain management. Technology developed under this program will transition to applicable government entities and commercial partners. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Complete final version of structural price prediction models and extend to additional minerals. - Complete final supply and demand forecast models and extend to additional minerals. - Conduct evaluation of completed structural price prediction and supply and demand forecast models. - Conduct research surrounding application of transparent structural price and supply/demand forecast methodology to additional areas. - Transition final models to government and commercial partners. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: Learning Introspective Control (LINC) Description: The Learning Introspective Control (LINC) program is developing machine introspection and learning technologies to characterize a modified or damaged military platform from its behavior and update the control law to maintain stability and control. The current approach to handling platform modification or damage places the burden of recovery and control on the operator, whether the operator is human or an autonomous controller. In contrast, a platform equipped with LINC technology continually compares the real-time behavior of the platform as measured by on-board sensors with a learned model, determines if the current observed behavior of the platform differs from that model in ways that might compromise stability and control, and implements an updated control law when required. The LINC capability would aid operators in maintaining effective control of military platforms that suffer damage or have been modified in the field to address emergent requirements identified during operations. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04.		-	7.300	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Demonstrate cooperative operation of multiple controller instances in a single, complex physical system. - Collaborate with transition partners to demonstrate safe operation and functionality on real-world, operational platforms despite damage or malfunction. - Formalize mathematical foundations of the recovery and control techniques for adoption by industry and academia for robust control system design.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.					
Title: Assured Neuro Symbolic Learning and Reasoning (ANSR) Description: The Assured Neuro Symbolic Learning and Reasoning (ANSR) program is developing new hybrid artificial intelligence (AI) algorithms that deeply integrate symbolic reasoning with data driven learning to create trustworthy AI-based systems. For purposes of this program, an AI-based system is considered trustworthy if it is: (a) robust to domain informed and adversarial perturbations; (b) supported by an assurance framework that creates and analyzes heterogenous evidence towards safety and risk assessments; and (c) predictable with respect to some specification and model of fitness. ANSR develops hybrid AI algorithms for which it is possible to develop evidence-based techniques that support confident assurance judgments. The key idea is to interleave symbolic and neural representations in hybrid AI algorithms that can acquire symbolic knowledge through learning and perform symbolic reasoning at scale to deliver robust inference, generalize to new situations, and provide evidence for assurance and trust. ANSR technologies will be demonstrated and evaluated on Department of War (DoW) use cases, such as autonomy, where trustworthiness is essential. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04.			-	4.164	-
FY 2026 Plans: - Finalize AI neurosymbolic datasets and benchmarks and conduct demonstrations on selected use cases. - Transition methods to DoW.					
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.					
Title: Artificial Intelligence Cyber Challenge (AlxCC) Description: The Artificial Intelligence Cyber Challenge (AlxCC) program is developing and demonstrating techniques for automated discovery and remediation of software vulnerabilities at speed and at scale. Current automated vulnerability discovery and remediation tools are based on techniques such as fuzzing, logical reasoning, and genetic algorithms, but are limited in			-	3.840	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS	Project (Number/Name) MSL-05 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
terms of effectiveness and user support. AlxCC leverages recent dramatic advances in artificial intelligence (AI) and machine learning, such as large pre-trained models and neurosymbolic AI, as the basis for new automated cyber security technologies and tools. AlxCC employs a contest model where teams use their automation and tooling to complete vulnerability discovery and remediation challenges. Performer teams are selected for the AlxCC competition based on their capability to leverage advances in AI to create usable, automated tools for vulnerability discovery and remediation and focus on tools suitable for broad deployment and applicable to critical infrastructure sectors. AlxCC competitors will train and develop their systems to find and fix vulnerabilities in widely used open-source software, focusing on software used in critical infrastructure. Each competitor system is evaluated on real-world critical infrastructure software suites and is scored based on their results in terms of absolute performance and performance relative to other competitor systems. Winning teams will receive cash awards. Success with AlxCC will create novel AI-enabled cyber vulnerability remediation technology and tools for securing code at the scale and speed needed to defend U.S. critical infrastructure from cyber attacks. Prior to FY 2026, this program was funded in PE 0602303E, Project IT-04. FY 2026 Plans: - Organize intellectual property generated for use by the U.S. Government and Department of War. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	120.707
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-06 / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-06: MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS	0.000	0.000	0.000	36.862	-	36.862	27.751	21.391	3.628	0.000	-	-

A. Mission Description and Budget Item Justification

This project funds classified DARPA programs that are reported in accordance with the SAP Annual Report to Congress.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602025E / MAKING, MAINTAINING, S UPPLY CHAIN AND LOGISTICS				Project (Number/Name) MSL-07 / MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MSL-07: MAKING, MAINTAINING, SUPPLY CHAIN AND LOGISTICS SUPPORT	0.000	0.000	75.206	121.050	-	121.050	118.789	120.335	113.519	113.519	-	-

A. Mission Description and Budget Item Justification

The Making, Maintaining, Supply Chain and Logistics Support project contains management costs in support of DARPA functions and activities across the entire Making, Maintaining, Supply Chain and Logistics PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0602715E, Project MBT-01, PE 0602716E, Projects ELT-01 and ELT-02, and PE 0602303E, Projects IT-03 and IT-04.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602026E / <i>EFFECTS</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	13.730	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
EFF-01: <i>EFFECTS</i>	0.000	0.000	12.598	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
EFF-07: <i>EFFECTS SUPPORT</i>	0.000	0.000	1.132	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the examination and evaluation of emerging technologies and system concepts that employ either physical or non-physical means to degrade or deny targeted adversary capabilities. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Effects project will examine and evaluate emerging technologies and system concepts that employ physical or non-physical means to degrade or deny targeted adversary capabilities. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, is also evaluated. The results from these studies and initiatives are used, in part, to formulate future prototype development programs or refocus ongoing work. Topics for this thrust may include but are not limited to: methods of defeating enemy anti-aircraft attacks; munition technologies to increase precision, range, endurance, and hard or soft-kill weapons for a variety of mission sets; digital design methodologies that are compatible with surged production.

The Effects Support project contains management costs in support of DARPA functions and activities across the entire Effects PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0602716E, Electronics Technology.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602026E / <i>EFFECTS</i>
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B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	20.275	0.000	-	0.000
Current President's Budget	0.000	13.730	0.000	-	0.000
Total Adjustments	0.000	-6.545	0.000	-	0.000
• Congressional General Reductions	0.000	-6.385			
• Congressional Directed Reductions	0.000	-0.160			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602026E / EFFECTS				Project (Number/Name) EFF-01 / EFFECTS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EFF-01: EFFECTS	0.000	0.000	12.598	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Effects project will examine and evaluate emerging technologies and system concepts that employ physical or non-physical means to degrade or deny targeted adversary capabilities. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, is also evaluated. The results from these studies and initiatives are used, in part, to formulate future prototype development programs or refocus ongoing work. Topics for this thrust may include but are not limited to: methods of defeating enemy anti-aircraft attacks; munition technologies to increase precision, range, endurance, and hard or soft-kill weapons for a variety of mission sets; digital design methodologies that are compatible with surged production.												
Prior to FY 2026, efforts in this Project were funded in PE 0602716E, Project ELT-01.												
B. Accomplishments/Planned Programs (\$ in Millions)										FY 2025	FY 2026	FY 2027
Title: Humboldt										-	4.872	-
Description: The Humboldt program seeks to develop directed energy (DE) devices to produce disruptive effects in electronic systems. The devices have potential for dual-use as sources to characterize the susceptibility of commercial electronics to electromagnetic interference (EMI).												
Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01.												
FY 2026 Plans:												
- Develop a range of application-specific antennas.												
- Experimentally characterize Humboldt devices integrated with antennas in relevant, simulated environments.												
FY 2026 to FY 2027 Increase/Decrease Statement:												
The FY 2027 decrease reflects program completion.												
Title: Sync										-	4.824	-
Description: The Sync program is creating systems to defend against uncrewed underwater vehicles (UUVs) entering U.S. harbors and ports. Today's counter-UUV approaches rely on complex detection and localization methods to find UUVs prior to engagement. The Sync program looks at orthogonal approaches to solving this problem to defeat UUVs prior to detection. Sync is investigating several different approaches to defeat UUVs, starting with modeling and simulation eventually moving to hardware in the loop and pool testing.												
Prior to FY 2026, this program was funded in PE 0602716E, Project ELT-01.												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602026E / <i>EFFECTS</i>	Project (Number/Name) EFF-01 / <i>EFFECTS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
<i>FY 2026 Plans:</i> - Demonstrate techniques on a UUV in laboratory environment in pool demonstration. - Demonstrate techniques on additional UUVs in simulation. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.			
<i>Title:</i> Effects Studies and Concepts <i>Description:</i> The Effects Studies and Concepts thrust examines emerging technologies and system concepts that employ either physical or non-physical means to degrade or deny targeted adversary capabilities. Topics under investigation include high pulsed power laser systems and design methodologies for effects-based technologies. The results from these studies and initiatives are used to formulate future programs. <i>FY 2026 Plans:</i> - Evaluate design of high pulsed power laser systems to be used for non-kinetic effects. - Investigate digital design methodologies for effects-based technologies. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects completion of Effects Studies and Concepts.		-	2.902
Accomplishments/Planned Programs Subtotals		-	12.598
C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602026E / EFFECTS				Project (Number/Name) EFF-07 / EFFECTS SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
EFF-07: EFFECTS SUPPORT	0.000	0.000	1.132	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Effects Support project contains management costs in support of DARPA functions and activities across the entire Effects PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0602716E, Project ELT-01.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity	R-1 Program Element (Number/Name)											
0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>											
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	128.307	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
BT-01: <i>BIOMEDICAL TECHNOLOGY</i>	0.000	128.307	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Biomedical Technology Program Element (PE) focuses on applied research for medical related technologies that will maintain warfighter health and performance before, during, or after operations. Successful technologies within this Program Element will maintain warfighter health against emerging threats through novel biothreat detection, rapid medical countermeasure identification and development, and distributed production of effective therapeutics. In-theater, warfighter health will be maintained through the development of field-relevant technologies such as reliable and accessible critical medical resources, novel detection and protection capabilities for traumatic brain injury, and rapid, effective triage of battlefield injuries. Technologies are also being developed to provide new capabilities for warfighter recovery from sustained injury including but not limited to spinal cord injury. Additionally, this PE will improve warfighter readiness by characterizing and assaying physical and cognitive performance to drive data-driven awareness. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA-funded technologies take root in the U.S. and provide new capabilities for national defense.

Beginning in FY 2026, efforts in this PE will be funded in PE 0602024E, Warfighting Performance.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	132.601	0.000	0.000	-	0.000
Current President's Budget	128.307	0.000	0.000	-	0.000
Total Adjustments	-4.294	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	-4.294	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>		R-1 Program Element (Number/Name) PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Novel Delivery Technology for Medical Countermeasures Description: The DoW requires rapid development of medical countermeasures (MCMs) to ensure force health protection and improve our ability to respond to emerging and novel biological threats. Despite recent advancements in development of new MCMs, challenges with delivery limit their current therapeutic potential. While emerging targeted delivery systems such as polymer/lipid nanoparticles and viral vectors have enabled the delivery of large, complex MCM molecules, they are still plagued by lack of widespread availability and effectiveness. Investing in efficient, adaptable delivery technology is crucial for strengthening biosecurity preparedness and will enable rapid response to the evolving biological threat landscape, whether the threat is natural or manmade. The Novel Delivery Technology for Medical Countermeasures program is developing minimally invasive MCM delivery systems, in which any therapeutic can be quickly formulated and administered to treat or prevent any disease. Developing novel delivery platforms will maintain warfighter health and readiness and enable rapid response to existing and novel biothreats. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		18.287	-	-
Title: GOLDeN hour extended EVACuation (GOLDEVAC) Description: Survival from combat trauma is predicated on a rapid sequence of events that escalate care as soon as possible. The GOLDeN hour extended EVACuation (GOLDEVAC) program is developing new technologies that will provide medical care closer to the point of injury and facilitate effective casualty evacuation. This will extend the time available to evacuate injured service members to higher levels of care. This program will explore technologies that enable autonomous care of patients and/or increase the ability of caregivers to effectively take care of a larger number of patients. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		14.701	-	-
Title: Red Blood Cell Factory (RBC-Factory) Description: Warfighters operate in extreme, austere, and dangerous domains that stress physiology, contain pathogens, and compromise health and performance. Protective equipment often hinders performance (e.g., loss of dexterity, rapid overheating, visual obstruction, etc.) and pharmacological approaches can carry unpleasant-to-dangerous side effects. The Red Blood Cell Factory (RBC-Factory) program is exploring the limits of what biologically active components, or cargoes, can be placed inside red blood cells (RBCs) to help protect the warfighter from within. Cargoes can include natural or synthetic products and must not perturb the physiological characteristics that allow RBCs to safely operate in a body. RBC-Factory aims to deliver a method capable of high-throughput modification of RBCs and a knowledge product identifying what kinds of cargo and how many can be put inside an RBC. RBC-Factory will determine if RBCs are a suitable vector for medium-term passive protection for warfighters against threats.		8.622	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>		R-1 Program Element (Number/Name) PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.				
Title: Controlled Genome Protection Description: The Controlled Genome Protection program is developing advanced capabilities to control and tune the activity of gene editing technologies. Advances in synthetic biology have significantly expanded the number and types of genome editors and modulators available. Many of the new genome editors have been identified from rare, slow-growing microorganisms with unique metabolic capabilities. New tools across these new classes of genome editors are required to advance our understanding, control, and use of gene editing technologies across all types of life. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		5.680	-	-
Title: Alert Warfighter Enablement (AWARE) Description: The Alert Warfighter Enablement (AWARE) program is developing a combination drug and device to non-invasively increase alertness following sleep loss in humans, without negative side effects and with reduced addictive potential. The AWARE program aims to develop candidate photoswitchable (light responsive) molecules that can be reversibly activated in the presence of near infrared (NIR) light and wearable device elements that emit NIR light to activate the candidate molecules. The combination of an ingested photoswitchable molecule and a NIR-emitting device will selectively activate neural pathways responsible for executive function, working memory, and decision making, wherever and whenever both drug and light are simultaneously present. If successful, AWARE technology will enable warfighters to maintain cognitive function and alertness during long-range missions and to obtain restorative sleep when needed without negative side effects. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		5.091	-	-
Title: BioElectronics to Sense and Treat (BEST) Description: The BioElectronics to Sense and Treat (BEST) program seeks to improve warfighter recovery from battlefield injuries by developing technologies to diagnose, prevent, and treat wound infections in real time. This program will generate high-resolution sensors that continually monitor a wound by assessing the community composition of contaminating microorganisms and/or the host immune response. Data from these sensors will be used to predict if a wound will become infected and to regulate the targeted administration of known and novel treatments to prevent infections or resolve existing infections. Ultimately, the sensor and treatment elements will be combined into a closed-loop smart bandage that can provide rapid diagnostics and precise treatments at all levels of deployed military medical care. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		3.534	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>		R-1 Program Element (Number/Name) PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Reengineering Enabling Sleep Transition in Operationally Restrictive Environments (RESTORE) Description: The Reengineering Enabling Sleep Transition in Operationally Restrictive Environments (RESTORE) program aims to enhance the efficiency of sleep to optimize warfighter performance following sleep restriction, as commonly occurs in combat operations. Sleep is a critical factor in cognitive function and decision-making, yet the tempo of military operations often results in acute and chronic sleep restriction. To optimize sleep recovery in such conditions, RESTORE seeks to develop a multi-modal system that enables precision control of sleep macro- (i.e., orchestration of transition between sleep phases) and micro-architectures (i.e., efficiency within sleep phases). By optimizing efficiency of recovery despite sleep restriction, RESTORE seeks to enable warfighter to perform at their best, even in the most challenging operational environments, and to enhance their overall readiness and effectiveness. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		0.700	-	-
Title: Rapid Battlefield Triage Description: The Rapid Battlefield Triage program is advancing capabilities to quickly triage warfighters requiring urgent life-saving medical intervention and enable medical resources to provide an appropriate response in current and future battlefields. Today, triage at point of injury is limited by subjective assessments, tools that are manually intensive, and physiological signatures with little diagnostic and prognostic value. This program will build on recent biomarker discoveries and innovations in sensing platforms to develop field-portable technologies that support triage in the most challenging operational domains. By optimizing allocation of scarce medical resources and scaling to multiple casualties, these devices will help far-forward units maximize their fighting strength against potential adversaries that could inflict large numbers of casualties and constrain evacuation to advanced medical facilities. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.		23.250	-	-
Title: Next-Generation Combat Casualty Care Description: The Next-Generation Combat Casualty Care program is developing advances in critical efforts to preserve warfighter life and well-being in the battlefields of the future. This research will directly address a leading cause of potentially preventable battlefield casualties by investigating new approaches for developing whole blood substitutes for traumatic injury that can be deployed on the battlefield in far-forward settings. Additional potential uses apply to disaster relief, mass casualty events, and stabilization missions. Advances within this program will ensure that the U.S. remains able to care for service members in peer and near-peer conflict by addressing gaps in combat casualty care.		13.167	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>		R-1 Program Element (Number/Name) PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-02.				
Title: Improved Personnel Placement (IPP) Description: The Improved Personnel Placement (IPP) program aimed to improve force lethality and overmatch by developing assays to determine physical/cognitive states in order to maximize performance and resilience while minimizing attrition. IPP identified and measured biomarkers for unique physical, cognitive, and behavioral traits associated with a broad spectrum of military specialties. This knowledge will help individualize training and provide novel measures of physical/cognitive states for specialized roles, while providing training cadres greater precision for predicting candidate readiness without bias. Measuring an individual's biological system will ensure that they achieve their maximum potential while facilitating readiness and resilience for the DoW.		8.731	-	-
Title: Neurological Assessment and Protection from Brain Injury Description: The Neurological Assessment and Protection from Brain Injury program transformed our current detection and protection strategies against traumatic brain injury (TBI), such as injury from blast exposure. This program characterized early molecular signals that lead to brain injury and seeks to de-risk drug discovery. Available tools in far-forward operating domains for these injuries were lacking, especially those that effectively discriminate between mild- and medium-level trauma. These novel technologies changed the paradigm for treatment of TBI.		17.170	-	-
Title: Bridging the Gap after Spinal Cord Injury Description: The Bridging the Gap after Spinal Cord Injury program developed and integrated technologies to heal and restore function associated with spinal cord injuries. This program significantly advanced treatment technologies by developing implantable, adaptive devices to address different stages of spinal cord injury. For early phases of injury, this program developed technologies for real-time biomarker tracking and delivery of therapies to stabilize or rebuild nerve connections at the injury site. For the chronic phase of injury, the Bridging the Gap after Spinal Cord Injury program developed and integrated devices deployed across the body to effectively "bridge the gap" of the spinal cord injury to restore multiple functions. The Bridging the Gap after Spinal Cord Injury program dramatically pushed the field forward in improving the quality of life for wounded warfighters and veterans suffering from spinal cord injuries.		5.428	-	-
Title: Deployable Medical Countermeasures for Warfighter Readiness Description: Maintaining robust protection and treatment against infectious disease threats during stabilization operations (e.g., Humanitarian and Disaster Relief) requires rapid drug discovery and reducing manufacturing and supply chain burdens. A major limitation of our current response to emerging biological and chemical threats is the lack of immediate availability of ideal medical countermeasures (MCMs) for rapid response, which includes high quality nucleic acid templates for MCM manufacturing. These		3.946	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602115E / <i>BIOMEDICAL TECHNOLOGY</i>			
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
nucleic acids are also critical for research and development (R&D) applications ranging from synthetic biology to the testing and development of MCMs. Current U.S.-based DNA production capabilities are limited to less than a handful of manufacturers; it takes weeks to months to produce adequate quality and quantity of DNA at these manufacturing sites and ship them to downstream partners. The Deployable Medical Countermeasures for Warfighter Readiness program aimed to develop an on-demand deployable platform to manufacture nucleic acid drugs safely at scale in short timeframes. The platform is comprised of a fully contained system capable of selectively manufacturing relevant doses of current Good Manufacturing Process (cGMP) grade nucleic acid therapeutics at or near the point of care. This effort also developed high quality gene-length DNA for R&D. This on-demand platform will enable countermeasures capable of combating novel threats, allowing a small force to prevent regional outbreaks from becoming global emergencies.				
Accomplishments/Planned Programs Subtotals		128.307	-	-
D. Other Program Funding Summary (\$ in Millions) N/A				
Remarks				
E. Acquisition Strategy N/A				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>					R-1 Program Element (Number/Name) PE 0602303E / <i>INFORMATION & COMMUNICATIONS TECHNOLOGY</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	352.961	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
IT-02: <i>HIGH PRODUCTIVITY, HIGH-PERFORMANCE RESPONSIVE ARCHITECTURES</i>	0.000	28.998	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
IT-03: <i>CYBER SECURITY</i>	0.000	159.749	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
IT-04: <i>ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS</i>	0.000	164.214	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Information and Communications Technology Program that is directed toward the application of advanced, innovative computing systems and communications technologies. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The High Productivity, High-Performance Responsive Architectures project focuses on developing the computer hardware and associated software technologies required for future computationally- and data-intensive national security applications. Powerful new approaches are needed to manage the rapid growth in available sensor data, to leverage advances in machine learning, artificial intelligence, and quantum computing, and to maintain the security of DoW information systems. The project therefore aims not only to create new computing platforms to include quantum technology, but also to efficiently extract information out of large and chaotic data sets with embedded and low-size, weight, and power systems. Advances in these areas will allow for DoW electronic systems to collaboratively manage scarce resources, such as the electromagnetic spectrum, and to adapt to new requirements and situations. Further, the resulting technologies, by being accessible to a wide range of application developers, will support new, sustainable computing systems for a broad spectrum of scientific and engineering applications.

The Cyber Security project is developing the computing, networking, and cyber security technologies required to protect Department of War, U.S. Government, and U.S. civilian information, information infrastructure, cyber-physical and embedded systems, critical infrastructure, and other computation-intensive mission-critical systems. Information technologies enable important existing and new military capabilities and drive the productivity gains essential to U.S. industry. Meanwhile, cyber threats grow in sophistication and number, and put data, computer programs, key information systems, and U.S. economic competitiveness at risk. The technologies developed in this project will enhance the resilience of information systems to current and emerging cyber threats; enable broad situational awareness of the cyber domain; and provide the basis for accurate, calibrated, and safe cyber response.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602303E / <i>INFORMATION & COMMUNICATIONS TECHNOLOGY</i>
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The Artificial Intelligence and Human-Machine Symbiosis project develops technologies to enable machines to function not only as tools that facilitate human action but also as trustworthy partners to human operators. Of particular interest are systems that can understand human language, extract information, and reliably categorize content contained in diverse media; answer questions, reach conclusions, and propose explanations; and learn, reason, and apply knowledge gained through experience to respond intelligently to new and unforeseen events. Enabling computing systems with such human-like intelligence matters because the tempo of military operations in emerging domains can exceed that at which unaided humans can orient, understand, and act. The technologies developed in this project will enable warfighters to make better decisions in complex, time-critical, battlefield environments with greater efficiency and confidence; and help autonomous systems and intelligent computational agents to safely and reliably perform critical missions in contested and adversarial physical and virtual environments.

Beginning in FY 2026, efforts in this PE will be funded in PE 0602025E, Making, Maintaining, Supply Chain and Logistics.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	364.775	0.000	0.000	-	0.000
Current President's Budget	352.961	0.000	0.000	-	0.000
Total Adjustments	-11.814	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	-11.814	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY				Project (Number/Name) IT-02 / HIGH PRODUCTIVITY, HIGH-PERFORMANCE RESPONSIVE ARCHITECTURES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
IT-02: HIGH PRODUCTIVITY, HIGH-PERFORMANCE RESPONSIVE ARCHITECTURES	0.000	28.998	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The High Productivity, High-Performance Responsive Architectures project focused on developing the computer hardware and associated software technologies required for future computationally- and data-intensive national security applications. Powerful new approaches were needed to manage the rapid growth in available sensor data, to leverage advances in machine learning, artificial intelligence, and quantum computing, and to maintain the security of DoW information systems. The project therefore aimed not only to create new computing platforms to include quantum technology, but also to efficiently extract information out of large and chaotic data sets with embedded and low-size, weight, and power systems. Advances in these areas may allow for DoW electronic systems to collaboratively manage scarce resources, such as the electromagnetic spectrum, and to adapt to new requirements and situations. Further, the resulting technologies, by being accessible to a wide range of application developers, may support new, sustainable computing systems for a broad spectrum of scientific and engineering applications.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Underexplored Systems for Utility-Scale Quantum Computing (US2QC)									28.998	-	-	
Description: It has been credibly hypothesized - but not proven - that a fault-tolerant quantum computer of sufficient size would revolutionize multiple commercial industries and scientific disciplines. Quantum computers are shown to have transformative potential for critical problems facing the United States, it is in the Government's interest to foster and accelerate commercial progress towards a truly useful, "utility-scale" quantum computer. Initiated under Alternative Computing to both reduce strategic risk and realize transformative opportunity, the US2QC thrust began work to (1) evaluate disruptive designs for utility-scale, fault-tolerant quantum computers, specifically, systems that can be constructed in less than 10 years; (2) demonstrate each of the enabling sub-systems and components for these designs; and (3) construct a prototype fault-tolerant quantum computer that demonstrates that utility-scale design is viable. This program will continue as the Quantum Benchmarking Initiative (QBI) in PE 0602716E, Project ELT-02 in FY 2025 and to PE 0602025E, Project MQB-01 in FY 2026.												
Accomplishments/Planned Programs Subtotals									28.998	-	-	
C. Other Program Funding Summary (\$ in Millions)												
N/A												
Remarks												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY	Project (Number/Name) IT-02 / HIGH PRODUCTIVITY, HIGH-PERFORMANCE RESPONSIVE ARCHITECTURES
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY				Project (Number/Name) IT-03 / CYBER SECURITY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
IT-03: CYBER SECURITY	0.000	159.749	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Cyber Security project is developing the computing, networking, and cyber security technologies required to protect Department of War, U.S. Government, and U.S. civilian information, information infrastructure, cyber-physical and embedded systems, critical infrastructure, and other computation-intensive mission-critical systems. Information technologies enable important existing and new military capabilities and drive the productivity gains essential to U.S. industry. Meanwhile, cyber threats grow in sophistication and number, and put data, computer programs, key information systems, and U.S. economic competitiveness at risk. The technologies developed in this project will enhance the resilience of information systems to current and emerging cyber threats; enable broad situational awareness of the cyber domain; and provide the basis for accurate, calibrated, and safe cyber response.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0602025E, Project MSL-04.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Constellation									44.000	-	-	
Description: The Constellation program is comprised of multiple projects that are developing technologies, capabilities, and prototype systems to enable full spectrum military cyberspace operations to defend the U.S. and deter, disrupt, and defeat adversary cyber actors. Technologies of interest include, but are not limited to, artificial intelligence, machine learning, and data science; resilient software, networking, and computing systems; formal methods and program analysis; data and information assurance; and cyber threat intelligence. High relevance is achieved through close coordination with U.S. cyber operators and the use of development, security, and operations (DevSecOps) and other collaborative development processes. High velocity is achieved through streamlined acquisition, assessment, approval, and deployment processes. Constellation development and deployment pipelines enable the rapid and continuous delivery of cyber technologies, capabilities, and prototype systems into operational use for the Department of War. The Constellation program is also funded in PE 0603760E, Project CCC-05 to facilitate rapid transition of cyber technologies and laboratory prototypes from applied research to operational prototypes.												
Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-03 and PE 0602025E, Project MSL-04.												
Title: Cyber Agents for Security Testing and Learning Environments (CASTLE)									18.500	-	-	
Description: The Cyber Agents for Security Testing and Learning Environments (CASTLE) program is developing an artificial intelligence (AI) toolkit to instantiate realistic network environments and train AI cyber agents to enable resilient network operations against advanced persistent threats (APTs). CASTLE formulates network hardening as a reinforcement learning (RL) problem and teaches RL agents to operate through the post-breach behavior of widely available penetration testing tools. Over												

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
progressive rounds of attack and defense, agents explore defensive actions to proactively stop ongoing attacks while maintaining operationally relevant workflows. Environments execute agents inside instrumented subnets that are deployed to live networks and simulate defensive actions that counter APT tools. Agent execution produces calibrated datasets for progressively improving simulations. The defensive AI cyber agents developed under CASTLE will provide the Department of War (DoW) with continual security assessments of critical networks and real-time response to cyber attacks.				
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.				
Title: Pipelined Reasoning of Verifiers Enabling Robust Systems (PROVERS)		18.200	-	-
Description: The Pipelined Reasoning of Verifiers Enabling Robust Systems (PROVERS) program is creating scalable mathematically based technologies, tools, and practices to achieve continuous reasoning about complex systems that can support software development pipelines. These mathematically based techniques, or formal methods, enable rigorous modeling, reasoning, and proving of diverse properties of software code or design models, such as the absence of a specific type of defect or security vulnerability. PROVERS integrates formal methods into a modern incremental and iterative development process by running tools at each code commit and delivering results to developers when they can most effectively remediate discovered issues. PROVERS focuses on creating and sustaining a body of evidence that can co-evolve with the system under change to support continuous assessment and ensure that the system remains free of identified categories of defects and security vulnerabilities through its lifetime. Key PROVERS objectives include enabling proof maintenance and repair capabilities at a cost that is proportionate to code change; integration of formal methods with code, properties, and proofs in a single workflow that reduces human involvement; providing improved explanations to facilitate proof repair; and automating formal methods-based software analysis to support software developers that are not formal methods experts. PROVERS technologies will facilitate the agile development and continuous improvement of mission-critical software systems that meet the high security and quality standards required by the Department of War.				
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.				
Title: Business Process Logic (BPL)		15.500	-	-
Description: The Business Process Logic (BPL) program is developing techniques to characterize and resolve vulnerabilities in business logic systems to protect and assure defense-critical workflows for government and business. Automated workflows written in business logic (BL) control much of the world's enterprises, from administration and operation of seaports to the assembly of weapons systems. Losses due to BL faults and vulnerabilities can range from annoyances to business-threatening outcomes. Therefore, it is important to identify and correct potentially problematic logic issues such as one-way actions or lost resources as early as possible. The BPL program is developing tools to extract workflow representations from BL and use those				

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
representations to identify, characterize, and mitigate faults and vulnerabilities in BL scripts and templates automatically. The technologies developed by BPL will enable increased assurance for manufacturing and assembly and greater efficiency for logistics and supply chain management.				
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.				
Title: Signature Management using Operational Knowledge and Environments (SMOKE) Description: The Signature Management using Operational Knowledge and Environments (SMOKE) program is developing signature management technologies that generate evasive cyber infrastructure which minimizes signatures as a source of attribution. SMOKE technologies incorporate counter-attribution techniques into the design process, quantitatively measure attribution risk in real-time, and maintain evasiveness after infrastructure changes. SMOKE data-driven tools automate the planning and execution of threat-emulated cyber infrastructure needed for network security assessments by red teams. SMOKE data-driven tools automate the discovery of cyber threat infrastructure signatures. SMOKE prototypes enable red teams to plan, build, and deploy cyber infrastructure that is informed by machine-readable signatures of sophisticated cyber threats.		15.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.				
Title: Hardening Development Toolchains Against Emergent Execution Engines (HARDEN) Description: The Hardening Development Toolchains Against Emergent Execution Engines (HARDEN) program is developing techniques and tools to anticipate, isolate, and mitigate emergent system behaviors and thereby improve security of complex integrated software. Today's software development toolchains and testing methodologies provide very limited means for reasoning about adversarial reuse of code as written and designed. This limitation results in unwitting creation of stable, reliable patterns of emergent behaviors within systems that adversaries can reuse in attacks. The HARDEN approach to preventing adversarial code reuse is to create techniques, tools, metadata, and instrumentation for reasoning about emergent execution at all stages of the software development life cycle (SDLC) and to flag code segments and design patterns where there is high potential for adversarial reuse and emergent execution. The utility of HARDEN technologies will be assessed by applying them to integrated software systems and critical system elements, such as bootloaders. Successful technologies developed by HARDEN will facilitate efficient mitigation of complex code-reuse and emergent-execution vulnerabilities at early SDLC stages and provide the stronger roots-of-trust required by zero trust architectures and high assurance integrated military software systems.		15.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.				
Title: Intelligent Generation of Tools for Security (INGOTS)		11.500	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: The Intelligent Generation of Tools for Security (INGOTS) program is developing techniques to accelerate cyber defense activities by identifying, triaging, and assessing exploitability of chainable vulnerabilities. Today, sophisticated cyber attacks link multiple vulnerabilities together into exploit chains that enable them to compromise various systems. Accurately understanding risk is critical for both developers and defenders within cyberspace, but the metrics currently in use do not account for the multiple factors that differentiate an innocuous software flaw from a chainable vulnerability. INGOTS is developing semi-automated tools and techniques to characterize and measure the interdependent exploitability of vulnerabilities and a new vulnerability severity metrology that characterizes and measures interdependent exploitability. With the INGOTS vulnerability measurement capability, developers and defenders will improve software and hardware resiliency by rapidly identifying and prioritizing the most dangerous flaws. The INGOTS program is also funded in PE 0602716E, Project ELT-02. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02 and MSL-04.					
Title: Provably Weird Network Deployment and Detection (PWND²) Description: The Provably Weird Network Deployment and Detection (PWND²) program is developing formal models of emergent communication pathways to fundamentally improve deployment and detection of resilient hidden networks in the real world. Authoritarian regimes are increasingly able to monitor and target internet communications, leaving many populations unable to communicate freely and safely. In response, the internet freedom and national security communities manually design hidden communication systems that use existing/available infrastructure and then validate these systems via empirical testing and analysis. These tests have difficulty capturing the range of real-world network environments and require infeasible amounts of testing to assess comprehensively at scale. PWND² is developing principled approaches for creating hidden communications systems with formal guarantees and for defending against use of these techniques by an adversary. If successful, PWND² technologies would enable the rapid and automatic generation and deployment of hidden communications systems with high assurance and strong guarantees. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.			10.000	-	-
Title: Reclaiming Bus-based Systems During Compromise (Red-C) Description: The Reclaiming Bus-based Systems During Compromise (Red-C) program focuses on developing and demonstrating novel approaches for resilient, self-healing, bus-based systems. The embedded computing systems common to modern vehicles and other cyber-physical systems typically use a bus-based architecture for communication among the various sensor, processor, actuator, and controller components organic to the platform. Due to the physical isolation of embedded computing systems, most standard buses currently in use were developed under an implicit assumption that all bus-connected components are secure and can be trusted, but such assumptions were unwise. Red-C aims to address this by creating resilient,			2.049	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
self-healing, bus-based systems through advances to instrumentation and response. For instrumentation, Red-C aims to enhance the individual components on the bus to enable them to function as sensors that collectively monitor the bus and peer bus components to detect a cyber attack. Then, having identified that an attack is underway, Red-C-enhanced bus peers will respond by collaborating on software patch creation to remediate bus vulnerabilities and enable recovery of bus functionality. Red-C will implement these bus resilience techniques as firmware updates that can be deployed to existing bus-based systems and components. Successful Red-C technologies will mitigate bus vulnerabilities in military and civilian platforms and cyber-physical systems. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-04.			
Title: Verified Security and Performance Enhancement of Large Legacy Software (V-SPELLS) Description: The Verified Security and Performance Enhancement of Large Legacy Software (V-SPELLS) program created methods and tools to recover succinct models of domain data abstractions and logic from source code, add enhancements to the models, and convert them to performant new component implementations verified to be compatible and secure. The Department of War (DoW) has a critical need for replacing or reworking components of existing software with more secure and more performant code, including cases where a key performance or security benefit comes from moving parts of the software to new hardware, such as utilizing hardware accelerators, isolation enclaves, offload processors, and distributed computation. At present, the enhancement of legacy software components faces high risk that the new software will not be fully compatible with the existing, larger environment. Verified software is currently written from scratch, starting with a formal specification, rather than incrementally added to a system as provably compatible enhancements. V-SPELLS addressed these problems by combining novel concepts in verified programming with recent developments in domain specific languages and systems architecture. V-SPELLS technology enables piecewise, compatible-by-construction improvement of software components in legacy DoW systems, which provides incremental software (re)engineering the benefits of formal software verification currently available only to clean-slate development efforts.		10.000	-
Accomplishments/Planned Programs Subtotals		159.749	-
C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A			

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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY				Project (Number/Name) IT-04 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
IT-04: ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS	0.000	164.214	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Artificial Intelligence and Human-Machine Symbiosis project develops technologies to enable machines to function not only as tools that facilitate human action but also as trustworthy partners to human operators. Of particular interest are systems that can understand human language, extract information, and reliably categorize content contained in diverse media; answer questions, reach conclusions, and propose explanations; and learn, reason, and apply knowledge gained through experience to respond intelligently to new and unforeseen events. Enabling computing systems with such human-like intelligence matters because the tempo of military operations in emerging domains can exceed that at which unaided humans can orient, understand, and act. The technologies developed in this project will enable warfighters to make better decisions in complex, time-critical, battlefield environments with greater efficiency and confidence; and help autonomous systems and intelligent computational agents to safely and reliably perform critical missions in contested and adversarial physical and virtual environments.

Beginning in FY 2026, efforts in this Project will be funded in PE 0602025E, Project MSL-05.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Open Price Exploration for National security (OPEN) Description: The Open Price Exploration for National security (OPEN) program aims to increase supply chain resilience and enable more efficient critical mineral markets by leveraging advances in artificial intelligence (AI) prediction and forecasting to increase structural price, supply, and demand transparency. OPEN constructs structural price predictions from fundamental and observable critical mineral input costs and increases the accuracy and precision of supply and demand forecasts by leveraging this structural price in conjunction with advances in AI and economic modeling. Today, critical mineral markets and supply chains are subject to international supply shocks that can lead to large and rapid critical mineral price spikes with immediate economic ramifications, and commodities purchase transactions (e.g., offtake agreements) are negotiated leveraging a mix of opaque and flawed pricing data. OPEN leverages a decomposition of a critical mineral price into four components (input costs, supply/demand shocks, distortions due to noncompetitive behavior, and stochastic fluctuation) to construct transparent estimations of an approximate marginal cost for critical minerals indexed by time and geographic location, and estimates supply and demand forecasts for critical minerals that take into account geopolitical factors, energy fluctuations, and technological innovations in recycling and supply chain management. Technology developed under this program will transition to applicable government entities and commercial partners. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.	42.000	-	-

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Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY	Project (Number/Name) IT-04 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Artificial Intelligence Cyber Challenge (AlxCC) Description: The Artificial Intelligence Cyber Challenge (AlxCC) program is developing and demonstrating techniques for automated discovery and remediation of software vulnerabilities at speed and at scale. Current automated vulnerability discovery and remediation tools are based on techniques such as fuzzing, logical reasoning, and genetic algorithms, but are limited in terms of effectiveness and user support. AlxCC leverages recent dramatic advances in artificial intelligence (AI) and machine learning, such as large pre-trained models and neurosymbolic AI, as the basis for new automated cyber security technologies and tools. AlxCC employs a contest model where teams use their automation and tooling to complete vulnerability discovery and remediation challenges. Performer teams are selected for the AlxCC competition based on their capability to leverage advances in AI to create usable, automated tools for vulnerability discovery and remediation and focus on tools suitable for broad deployment and applicable to critical infrastructure sectors. AlxCC competitors will train and develop their systems to find and fix vulnerabilities in widely used open-source software, focusing on software used in critical infrastructure. Each competitor system is evaluated on real-world critical infrastructure software suites and is scored based on their results in terms of absolute performance and performance relative to other competitor systems. Winning teams will receive cash awards. Success with AlxCC will create novel AI-enabled cyber vulnerability remediation technology and tools for securing code at the scale and speed needed to defend U.S. critical infrastructure from cyber-attacks. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		38.000	-	-
Title: Assured Neuro Symbolic Learning and Reasoning (ANSR) Description: The Assured Neuro Symbolic Learning and Reasoning (ANSR) program is developing new hybrid artificial intelligence (AI) algorithms that deeply integrate symbolic reasoning with data driven learning to create trustworthy AI-based systems. For purposes of this program, an AI-based system is considered trustworthy if it is: (a) robust to domain-informed and adversarial perturbations; (b) supported by an assurance framework that creates and analyzes heterogenous evidence towards safety and risk assessments; and (c) predictable with respect to some specification and model of fitness. ANSR develops hybrid AI algorithms for which it is possible to develop evidence-based techniques that support confident assurance judgments. The key idea is to interleave symbolic and neural representations in hybrid AI algorithms that can acquire symbolic knowledge through learning and perform symbolic reasoning at scale to deliver robust inference, generalize to new situations, and provide evidence for assurance and trust. ANSR technologies will be demonstrated and evaluated on Department of War (DoW) use cases, such as autonomy, where trustworthiness is essential. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		17.000	-	-
Title: Learning Introspective Control (LINC)		16.600	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: The Learning Introspective Control (LINC) program is developing machine introspection and learning technologies to characterize a modified or damaged military platform from its behavior and update the control law to maintain stability and control. The current approach to handling platform modification or damage places the burden of recovery and control on the operator, whether the operator is human or an autonomous controller. In contrast, a platform equipped with LINC technology would continually compare the real-time behavior of the platform as measured by on-board sensors with a learned model, determine if the current observed behavior of the platform differs from that model in ways that might compromise stability and control, and implement an updated control law when required. The LINC capability would aid operators in maintaining effective control of military platforms that suffer damage or have been modified in the field to address emergent requirements identified during operations. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.				
Title: Transfer from Imprecise and Abstract Models to Autonomous Technologies (TIAMAT) Description: The Transfer from Imprecise and Abstract Models to Autonomous Technologies (TIAMAT) program is developing techniques to robustly transfer learned autonomy from fast abstract simulations to autonomous platforms in real-world environments. Autonomous systems that heavily depend on modeling and simulation (M&S) training environments are relatively constrained in their abilities because those training environments insufficiently account for the data domain shift common when translating M&S outcomes to the real world, a phenomenon sometimes referred to as the sim2real gap. The TIAMAT approach integrates symbolic structures with neural structures to transfer learned autonomy more realistically and robustly. TIAMAT will enable the use of fast abstract simulations by anchoring the learning and transfer of autonomy on semantically consistent components shared across simulations and real environments. For TIAMAT, semantically consistent components of particular importance include militarily relevant phenomena that remain consistent in the source and target environments, such as mission objectives, special instructions, subject matter expert guidance, rules of engagement, and the laws of physics. Autonomy transfer using TIAMAT methods and techniques will reduce the complexity of the autonomy learning and transfer problems to the comparatively simpler points of reference in the anchored representation. Successful TIAMAT transfer of M&S-based learning will enable more rapid and robust training and deployment of autonomous systems at higher levels of autonomy. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		16.000	-	-
Title: Anticipatory and Adaptive Anti-money Laundering (A3ML) Description: The Anticipatory and Adaptive Anti-money Laundering (A3ML) program aims to develop graph search-based algorithms that enable anticipation of and adaptation to money laundering tactics, techniques, and procedures (TTPs) in order to prevent money laundering in the global financial system. Today, anti-money laundering (AML) practitioners accumulate evidence		8.600	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
of money laundering largely via manual searches of a myriad of commercial and financial intelligence (FININT) databases and rule-based AML systems that lack flexibility to adapt to changing TTPs. Because of the static nature of current AML processes and the limited capabilities of AML tools, novel money laundering schemes can go on for years. To address this challenge, A3ML will develop and apply graph search and subgraph isomorphism techniques that reference high-level models of money laundering to extract specific money laundering TTPs from heterogeneous transaction data and to anticipate and adapt to new money laundering TTPs in a timely fashion. If successful, A3ML technologies and tools would enhance AML practices across U.S. Government (USG) and the private sector. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.				
Title: Artificial Intelligence Quantified (AIQ) Description: The Artificial Intelligence Quantified (AIQ) program is developing technologies for assessing the capabilities of generative artificial intelligence (genAI) to enable mathematical guarantees on performance. Current state-of-the-art methods for genAI quantification and assessment are ad hoc, deal with only the simplest of capabilities, and are not properly grounded in a rigorous theory. The AIQ approach combines mathematical methods with advances in measurement and modeling to enable rigorous quantification of genAI capabilities. The program addresses three capability levels: 1) the specific problem level, which addresses quantification for a given problem; 2) the classes of problem level, which address quantification of a given problem class; and 3) the natural class level which addresses the reverse question of the class of problems quantifiable for a given genAI architecture. Success with AIQ will provide guarantees on the performance of genAI technologies. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		8.000	-	-
Title: Kallisti Description: The Kallisti program seeks to develop new capabilities for modeling adversarial behaviors. Actions taken by independent state and non-state actors are determined by their beliefs about the state of their environment, the risks and rewards associated with these beliefs, and the strategy they use to react to the risks or rewards they believe exist in their environment. This program is developing an algorithmic model of behaviors by independent actors through inferring and understanding the belief states that might deter or result in distribution of actions. Kallisti's modeling will allow projection of an actor's future behavior based on their perceived strategy, enabling external actions to be taken. The program seeks not only to understand an actor's current strategy but also to find a decomposed version of the strategy into relevant basis vectors to track strategy changes under non-stationary assumptions. The primary goal is to utilize algorithmic modeling of beliefs and behaviors to deter aggression by military adversaries. Technology developed under this program will transition to the Services.		6.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.			
Title: Securing Artificial Intelligence for Battlefield Effective Robustness (SABER) Description: The Securing Artificial Intelligence for Battlefield Effective Robustness (SABER) program aims to advance understanding of the military's use of artificial intelligence (AI) enabled systems in battlefield environments to enhance the robustness of AI-enabled systems against real-world adversarial attacks. AI-enabled systems can be subject to adversarial AI attacks, but most attacks studied to date are not physically realizable or assume the adversary already has some level of system access. SABER anticipates adversaries will develop combined AI-cyber-physical attacks and formalize these attacks in tactics, techniques, and procedures (TTPs). To address such capable AI adversaries, SABER is evaluating AI-enabled military systems in realistic operational settings to quantify the risks of deploying AI-enabled systems on the battlefield. Simultaneously, SABER is developing technologies and TTPs for continuous monitoring, integration, and deployment of counter-AI techniques to enhance the robustness of AI-enabled systems. SABER technologies will enable the formation of Department of War (DoW) AI red teams that continuously integrate and employ emerging counter-AI techniques and tools, establishing a sustainable model for operational AI red teaming. If successful, SABER will empower military AI red teams to confidently assess and secure U.S. AI-enabled battlefield systems against attack by sophisticated adversaries. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		4.000	-
Title: Scientific Feasibility (SciFy) Description: The Scientific Feasibility (SciFy) program is developing computational methods to measure the feasibility of claims to enable accurate assessments of scientific content. Automated scientific content generation, via rapidly improving large pre-trained models, has the potential to disrupt the U.S. technology base in times of crisis and to distort the global race for technological dominance in key areas. Similarly, false capability claims can have significant negative implications for national security and international relations. To address these threats, SciFy is focused on methods for assessing the scientific feasibility of claims by using automated reasoning to decompose claims into constituent, verifiable parts. Assessing each component involves referencing existing technological advancements, foundational scientific principles, data, software, models, simulation results, and industry standards or benchmarks. SciFy is creating methods that go beyond automated fact-checking by addressing complex component interactions and operational constraints as well as evaluating logical consistency, system integration, and compatibility considerations. Success with the SciFy program will enable the U.S. to reliably determine whether claimed scientific and technological capabilities are practical and realistic when considered as a whole, even when theoretically possible in parts. This applied research funding supports applied research efforts that are part of the larger basic research SciFy activity under PE 0601101E, Project CCS-02.		2.900	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05 and PE 0601122E, Project EMR-01.			
Title: National Security Economic Theory (NASCENT) Description: The National Security Economic Theory (NASCENT) program will perpetuate a policy and market feedback loop between the United States Government (USG) and private-sector executives to support more effective geoeconomic actions with precisely defined positive national security outcomes. Current approaches to geoeconomics are based largely on qualitative ideas, drawn primarily from international relations and legal backgrounds, and are often non-falsifiable with limited scope. Furthermore, the lack of a body that facilitates neutral and open discussion and iterative development of geoeconomic theory has hindered innovation and renders the development of theory and tools of economic statecraft cost ineffective. NASCENT is addressing these challenges by creating a membership-based forum where the USG, academia, and the private sector can collaboratively and iteratively design and rigorously evaluate geoeconomic mechanisms. NASCENT will combine techniques from economic mechanism design and experimental economics to work backwards from precisely described national security outcomes, elicited from across the USG, to understand what economic incentives and mechanisms enable those outcomes. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-05.		2.000	-
Title: Accelerating Artificial Intelligence (AAI) Description: The Accelerating Artificial Intelligence (AAI) program sought to go beyond commercially-driven advances in artificial intelligence (AI) and to address important national security challenge applications. Trustworthy AI, which is AI that is safe, reliable, accurate, explainable, and resilient to attacks, was a major focus. Technical challenges included robustness of AI systems in novel, uncertain, and/or unanticipated situations; efficiency and timeliness of AI development, test, evaluation, approval, and certification processes; and identification of tasks or sub-tasks for which greater automation through the use of AI/machine learning (ML) was appropriate. Approaches to addressing these challenges leveraged recent advances in transfer learning, causal reasoning, reinforcement learning, generative AI, large pre-trained models, and large language models. AAI sought to significantly accelerate AI innovation in many important Department of War (DoW) domains while also reducing the time and cost needed to transition and deploy new AI technologies.		3.114	-
Accomplishments/Planned Programs Subtotals		164.214	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602303E / INFORMATION & COMMUNICATIONS TECHNOLOGY	Project (Number/Name) IT-04 / ARTIFICIAL INTELLIGENCE AND HUMAN-MACHINE SYMBIOSIS
D. Acquisition Strategy N/A		

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>											
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	112.715	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
TT-04: <i>ADVANCED LAND SYSTEMS TECHNOLOGY</i>	0.000	22.197	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
TT-07: <i>AERONAUTICS AND SPACE TECHNOLOGY</i>	0.000	60.573	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
TT-13: <i>INFORMATION ANALYTICS TECHNOLOGY</i>	0.000	29.945	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Tactical Technology Program that supports the advancement of concepts and technologies to enhance the next generation of tactical systems. This PE funds a number of projects in the areas of Advanced Land Systems, Aeronautics and Space Technology and Information Analytics Technology. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA-funded technologies take root in the U.S. and provide new capabilities for national defense.

The Advanced Land Systems Technology project is developing technologies for enhancing U.S. military effectiveness and survivability in operations ranging from traditional threats to military operations against irregular forces that can employ disruptive or catastrophic capabilities, or disrupt stabilization operations, including competing in underground spaces. Programs in this project will break the relative symmetry of land combat to strengthen and maintain U.S. forces' decisive advantages in the current and future ground battlefield. The emphasis is on developing affordable technologies that reduce reliance on consolidated forward-operating bases and required lines of communication, and provide small units and individual warfighters with hyper-mobility and hyper-lethality. This project will develop methods and technologies to expand the maneuver trade space to include the vertical dimension, including subterranean environments, as well as underground spaces. It will leverage advances in artificial intelligence to enable integrated manned-unmanned operations and decrease warfighter exposure through the use of autonomous agents.

Aeronautics and Space Technology efforts will address high payoff opportunities that dramatically reduce costs associated with advanced aeronautical and space systems and/or provide revolutionary new system capabilities for satisfying current and projected military mission requirements. This includes advanced technology studies of revolutionary propulsion, vehicle, and launch concepts, sophisticated fabrication methods, and examination of novel materials and enabling technologies for aeronautics and space system applications. Studies that also fundamentally change the calculus of battle including consideration of a mix of assets, platforms that are potentially disposable or with limited lifespans, and autonomous integration of space and air platforms in the tactical battlespace are included.

The Information Analytics Technology project developed technology for analyzing data and information arising from: 1) intelligence networks; 2) open sources, social and broadcast media, and other external sources; 3) sensors and signal/image processors; and 4) collection platforms and weapon systems. Technical challenges included processing huge volumes of diverse, incomplete, and uncertain data in tactically relevant timeframes and countering the information operations of sophisticated adversaries who seek to deceive, degrade, deny, and disrupt the U.S. information enterprise. Benefits sought include a deeper understanding of the evolving operational

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>
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environment tailored to the needs of commanders at every echelon; an enhanced capability to plan, monitor, and control diverse military operations ranging from stabilization and information operations to combat engagements; and increased efficiency of core military functions such as national and homeland security, warfighter protection, and defense support of law enforcement and civil authorities.

Beginning in FY 2026, efforts in this PE will be funded in PE 0602023E, Access and Awareness.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	117.935	0.000	0.000	-	0.000
Current President's Budget	112.715	0.000	0.000	-	0.000
Total Adjustments	-5.220	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	-1.400	0.000			
• SBIR/STTR Transfer	-3.820	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602702E / TACTICAL TECHNOLOGY				Project (Number/Name) TT-04 / ADVANCED LAND SYSTEMS TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
TT-04: ADVANCED LAND SYSTEMS TECHNOLOGY	0.000	22.197	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Advanced Land Systems Technology project is developing technologies for enhancing U.S. military effectiveness and survivability in operations ranging from traditional threats to military operations against irregular forces that can employ disruptive or catastrophic capabilities, or disrupt stabilization operations, including competing in underground spaces. Programs in this project will break the relative symmetry of land combat to strengthen and maintain U.S. forces' decisive advantages in the current and future ground battlefield. The emphasis is on developing affordable technologies that reduce reliance on consolidated forward-operating bases and required lines of communication, and provide small units and individual warfighters with hyper-mobility and hyper-lethality. This project will develop methods and technologies to expand the maneuver trade space to include the vertical dimension, including subterranean environments, as well as underground spaces. It will leverage advances in artificial intelligence to enable integrated manned-unmanned operations and decrease warfighter exposure through the use of autonomous agents.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0602023E, Project ACS-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Expeditionary Carbon Utilization for energy Resilience and Stabilization (ExCURSion)									10.163	-	-	
Description: Two solutions to expeditionary energy currently exist: fuel and electric systems. Fuel has a high energy density but requires supply lines that are logistically complicated and pose strategic and tactical weaknesses. Electric systems are operationally flexible and rechargeable, but their energy density is an order of magnitude lower than fuel. The Expeditionary Carbon Utilization for Energy Resilience and Stabilization (ExCURSion) program, originally conceived under the Molecular Systems and Materials Assembly program in PE 0601101E, Project MS-01, aims to overcome key technical challenges to enable a closed-loop energy storage system with the rechargeability of electric systems and the high energy density of fuel. The program will accomplish this goal by aggressively exploring and evaluating foundational technologies of carbon dioxide reduction, capture, and storage. Technologies developed under this program will serve as key components for a future self-enclosed system, addressing the need for energy resilience in an expeditionary setting, independent of a fuel supply line, while delivering a rechargeable fuel cell with far greater energy density than batteries.												
Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.												
Title: Robotic Autonomy in Complex Environments with Resiliency (RACER)									12.034	-	-	
Description: Multi-domain operations (MDO) present complex and challenging environments to ground combat platforms. Ground combat platforms must operate in a more distributed manner in these environments to gain a sustained tactical advantage and												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>	Project (Number/Name) TT-04 / <i>ADVANCED LAND SYSTEMS TECHNOLOGY</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
enhance warfighter survivability. The Army intends to deploy autonomous robotic combat vehicles and optionally manned fighting vehicles to accomplish this objective. In order to meet the demands of an MDO environment, significant advances in perception, planning, and control algorithms are required to autonomously maneuver faster and more resiliently in complex and novel off-road situations. Maneuver environments are characterized by three-dimensional surfaces of highly compliant soils and vegetation, hundreds of positive and negative obstacle classes, no defined road networks or driving rules, and where use of terrain for survivability is critical. In order to achieve operationally relevant speeds and resilience to novel situations on the battlefield, while simultaneously reducing the soldier's cognitive and communications burden and increasing battle space awareness, Robotic Autonomy in Complex Environments with Resiliency (RACER) will develop and demonstrate game-changing autonomous ground combat vehicle mobility using a combination of simulation and advanced platforms. RACER will deliver autonomy algorithms using the latest in Artificial Intelligence (AI) and machine-learning techniques, a code repository, an off-road simulation environment tailored for military off-road autonomy development, tactical route planning methods, and field-demonstrated off-road autonomous capabilities. The RACER program will demonstrate fully autonomous maneuver on a military Unmanned Ground Vehicle (UGV) in a variety of militarily relevant environments. Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.			
Accomplishments/Planned Programs Subtotals		22.197	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602702E / TACTICAL TECHNOLOGY				Project (Number/Name) TT-07 / AERONAUTICS AND SPACE TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
TT-07: AERONAUTICS AND SPACE TECHNOLOGY	0.000	60.573	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
Aeronautics and Space Technology efforts is addressing high payoff opportunities that dramatically reduce costs associated with advanced aeronautical and space systems and/or provide revolutionary new system capabilities for satisfying current and projected military mission requirements. This includes advanced technology studies of revolutionary propulsion, vehicle, and launch concepts, sophisticated fabrication methods, and examination of novel materials and enabling technologies for aeronautics and space system applications. Studies that also fundamentally change the calculus of battle including consideration of a mix of assets, platforms that are potentially disposable or with limited lifespans, and autonomous integration of space and air platforms in the tactical battlespace are included.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0602023E, Project ACS-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Oversight									21.192	-	-	
Description: Oversight is developing and demonstrating a suite of autonomy technologies to provide constant custody of targets as a service for tactical operations in contested environments. Existing and emerging space systems are being evaluated. Proliferated Low Earth Orbit (p-LEO) satellite constellations and payloads will be leveraged due to their high-bandwidth, processing-on-the-edge capabilities in support of tactical, efficient, integrated missions at scale. Oversight is developing autonomous technology to enable advanced collaboration among constellations of satellites for target custody in contested environments where the numbers of targets is far greater than the number of satellites and sensors over the operating area. The Oversight program includes a demonstration using existing on-orbit p-LEO assets combined with live, virtual and constructive terrestrial assets.												
Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.												
Title: Next Generation Electronic Surveillance Technology (NGEST)									9.947	-	-	
Description: The Next Generation Electronic Surveillance Technology (NGEST) program is creating a new Radio Frequency (RF) surveillance system for sensing in complex signal environments. NGEST is defining a new surveillance paradigm to find critical signals of interest in tactically relevant timelines. Contemporary electronic surveillance systems face increasing RF complexity in the spatial (emitter proliferation), power (range), and frequency (agility, congestion) dimensions due to their static hardware configuration. Additionally, current systems need to employ exquisite hardware to deal with geometrical challenges. NGEST overcomes the complexity and geometry challenges by autonomously reconfiguring diverse frequency and spatial resources.												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>	Project (Number/Name) TT-07 / <i>AERONAUTICS AND SPACE TECHNOLOGY</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.			
Title: Albatross Description: The Albatross program is exploring new technologies to enable perpetually flying aircraft. Albatross approaches this by creating a bird-like soaring capability for small-unmanned aircraft systems (s-UAS) that leverages weather forecast-informed mission planning and real-time sensing of dynamic wind conditions to autonomously navigate s-UAS and extend flight endurance and range. Current estimates indicate that Albatross soaring techniques could reduce average flight power by 75% for relevant missions. Albatross is also developing autonomous guidance, navigation, and controls to harness soaring sources while maintaining safe separation from terrain. Albatross will use a series of real-world flight events to mature and demonstrate capability incrementally. Technologies developed under this program will be demonstrated on mature platforms to support rapid transition to the Services. Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.		21.865	-
Title: Lunar Assay via Small Satellite Orbiter (LASSO) Description: The Lunar Assay via Small Satellite Orbiter (LASSO) program will optimize autonomous navigation and propulsion in support of advancing Space Situational Awareness capabilities in cislunar space. The program will build a low-cost satellite, launched as a secondary payload, to demonstrate these capabilities in Very Low Lunar Orbit (VLLO), which is the most challenging environment in cislunar space due to continually destabilizing orbits, lack of navigation aids (i.e., GPS), variable gravity and operation in close proximity to terrain. While operating in VLLO, LASSO will collect high-resolution data regarding accessible water ice on the moon by capturing data about resource distribution and quantities to support future in-situ resource utilization (ISRU) investigation. Such discoveries support future lunar exploration. With improved navigation, propulsion, and autonomous control, the LASSO satellite will achieve persistent VLLO while building a statistically significant understanding of lunar resources. Beginning in FY 2026, this program will be funded in PE 0602023E, Project ACS-01.		2.156	-
Title: Persistent Optical Wireless Energy Relay (POWER) Description: The Persistent Optical Wireless Energy Relay (POWER) program designed, built, and validated foundational components for effective optical energy relays. These relays will enable a ground-based laser to efficiently transmit energy over 100s of kilometers leveraging a high-altitude transmission layer which minimizes atmospheric absorption and scattering. The high-altitude energy relay nodes will redirect, correct, and selectively harvest energy from the optical energy source and then beam		3.892	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>	Project (Number/Name) TT-07 / <i>AERONAUTICS AND SPACE TECHNOLOGY</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
that energy back down to the surface for conversion to electricity. These relays are the core building blocks to construct a flexible, resilient, reconfigurable, persistent, and distributed energy network. POWER also produced conceptual designs for new platform capabilities realized by offboarding power storage and generation. These platforms will have range, endurance, and payload power performance that is no longer tied to platform size enabling a new class of small but high-performance platforms.			
Title: Advanced Aeronautics and Space Technologies Description: The Advanced Aeronautics and Space Technologies studies examined and evaluated aeronautical and space technologies and concepts through applied research. These included feasibility studies of novel or emergent materials, sensors and tactics for air and space platforms, launch vehicles, satellites, manufacturing and implementation approaches, and hardware demonstrations of key enabling technologies. The areas of interest ranged from propulsion and power, guidance and control, concepts to enable novel air platforms, to innovative technologies and platform concepts to enable new missions and resilient operations for space systems, from low earth orbit to cislunar space. Aeronautics interest areas included hybrid electric/ combustion propulsion concepts, small-scale air mobility solutions, and networking of both piloted and unpiloted air vehicles. Space interest areas included advanced or novel power and propulsion systems, novel sensors, advanced lightweight structures, advanced miniature radio frequency (RF) technology, precision navigation and timing technologies, ground and space-based space domain awareness, avionics, structures, and novel approaches to support terrestrial operations. These studies led to the development of new programs, components or subsystems to enhance future aerospace platforms, or improvement of existing systems.		1.521	-
Accomplishments/Planned Programs Subtotals		60.573	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602702E / TACTICAL TECHNOLOGY				Project (Number/Name) TT-13 / INFORMATION ANALYTICS TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
TT-13: INFORMATION ANALYTICS TECHNOLOGY	0.000	29.945	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Information Analytics Technology project developed technology for analyzing data and information arising from: 1) intelligence networks; 2) open sources, social and broadcast media, and other external sources; 3) sensors and signal/image processors; and 4) collection platforms and weapon systems. Technical challenges included processing huge volumes of diverse, incomplete, and uncertain data in tactically relevant timeframes and countering the information operations of sophisticated adversaries who seek to deceive, degrade, deny, and disrupt the U.S. information enterprise. Benefits sought included a deeper understanding of the evolving operational environment tailored to the needs of commanders at every echelon; an enhanced capability to plan, monitor, and control diverse military operations ranging from stabilization and information operations to combat engagements; and increased efficiency of core military functions such as national and homeland security, warfighter protection, and defense support of law enforcement and civil authorities.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Resilient Supply-and-Demand Networks (RSDN)									15.545	-	-	
Description: The Resilient Supply-and-Demand Networks (RSDN) program is developing supply-chain risk management analytics to detect systemic challenges and improve resilience in supply and demand networks (SDN). At present, the separation of supply-chain information into confidential silos obscures a system-wide view, inhibiting comprehensive risk-focused analysis of supply and demand networks. RSDN is developing techniques for modeling and stress-testing real-world SDNs, including network analytics and visualizations, to reveal emerging fragilities and enable deep situational awareness of systemic risks and potential disruptions. An RSDN stress-testing framework enables repeatable scenario analysis of strategic issues in supply and demand networks, automated analysis and discovery of risk patterns, and evaluation of alternative risk mitigation strategies.												
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.												
Title: Defense Applications of Innovative Remote Sensing (DAIRS)									14.400	-	-	
Description: The Defense Applications of Innovative Remote Sensing (DAIRS) program was intended to enable new approaches for the persistent long-range tracking of maritime and air targets. Specifically, DAIRS focused on the use of surface wave over-the-horizon radar (SWOTHR) with operation in low latitudes. The program explored passive remote sensing using endemic noise sources as a highly disruptive method for environmental and target sensing. A variety of processing and sensing techniques provided the baseline technologies for clutter rejection, and this program combined those techniques with technologies to uncover												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602702E / <i>TACTICAL TECHNOLOGY</i>	Project (Number/Name) TT-13 / <i>INFORMATION ANALYTICS TECHNOLOGY</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
spatial-temporal correlation, to try to suppress spread-Doppler clutter. The program investigated these methods for various conditions that affect SWOTHR, including the day/night cycle, sea state, forward scatter and backscatter, and bistatic range. Additionally, the program took a first-principles approach for classifying high frequency wavelengths for total disruption of the field.			
Accomplishments/Planned Programs Subtotals		29.945	-
C. Other Program Funding Summary (\$ in Millions) N/A			
Remarks			
D. Acquisition Strategy N/A			

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>					R-1 Program Element (Number/Name) PE 0602715E / <i>MATERIALS AND BIOLOGICAL TECHNOLOGY</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	304.669	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MBT-01: <i>MATERIALS PROCESSING TECHNOLOGY</i>	0.000	186.816	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MBT-02: <i>BIOLOGICALLY BASED MATERIALS AND DEVICES</i>	0.000	117.853	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Materials and Biological Technology Program that is focused on developing materials and biological technologies that make possible a wide range of new military capabilities. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Materials Processing Technology project will develop novel materials, fabrication and processing techniques, models, devices and components that will lower the cost, increase the performance, and/or enable new missions for military platforms and systems. Included in this project are efforts across a wide range of technology areas including manufacturing, electronics, sensors, optics, and complex and autonomous systems.

The Biologically Based Materials and Devices project will leverage the growing application space of the biological sciences for the development of new DoW capabilities to improve the sustainability of warfighters, and operational platforms in varied environments. This project will develop solutions for critical resource processing, materials development, threat detection and characterization, environmental remediation, and warfighter resilience to infectious disease and environmental stressors. The materials developed through this project will protect and sustain warfighters and operations in austere environments.

Beginning in FY 2026, efforts in this PE will be funded in PE 0602024E, Warfighting Performance, and PE 0602025E, Making, Maintaining, Supply Chain and Logistics.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602715E / <i>MATERIALS AND BIOLOGICAL TECHNOLOGY</i>
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B. Program Change Summary (\$ in Millions)	<u>FY 2025</u>	<u>FY 2026</u>	<u>FY 2027 Base</u>	<u>FY 2027 OOC</u>	<u>FY 2027 Total</u>
Previous President's Budget	316.221	0.000	0.000	-	0.000
Current President's Budget	304.669	0.000	0.000	-	0.000
Total Adjustments	-11.552	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	-1.311	0.000			
• SBIR/STTR Transfer	-10.241	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY				Project (Number/Name) MBT-01 / MATERIALS PROCESSING TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MBT-01: MATERIALS PROCESSING TECHNOLOGY	0.000	186.816	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Materials Processing Technology project will develop novel materials, fabrication and processing techniques, models, devices and components that will lower the cost, increase the performance, and/or enable new missions for military platforms and systems. Included in this project are efforts across a wide range of technology areas including manufacturing, electronics, sensors, optics, and complex and autonomous systems.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0602025E, Project MSL-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Materials for Extreme Environments									79.523	-	-	
Description: The Materials for Extreme Environments thrust is exploring new materials, innovative architectures, and development processes that will significantly enhance the performance and persistence of DoW platforms operating in extremely harsh environments. Materials with superior strength, functionality, and resiliency are critical for enabling DoW platforms, weapons and other components to operate and persist under conditions including, but not limited to, extremely high or low temperatures, turbulence, ionizing radiation, and/or corrosive environments. Recent developments in materials such as high entropy alloys, superconducting materials, and infiltrated carbon fiber composites hold promise for achieving material solutions for improved survivability in a wide range of harsh environmental conditions. Similarly, advancements in material design, processing and manufacturing are enabling novel material architectures that can further enhance performance and resilience in structures such as leading edges, windows and apertures, propulsion systems, and space structures. Exemplar areas of research within the Materials for Extreme Environments thrust include the following: 1) high temperature materials for hypersonic platforms, 2) high temperature window and aperture materials, 3) radiation and/or electromagnetic pulse (EMP) hardened electronics for space platforms, 4) coatings for platform survivability in corrosive environments, 5) active and passive cooling methods for apertures and forward-facing vehicle features, and 6) superconducting and magnetic materials for novel propulsion systems.												
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.												
Title: Functional Materials and Devices									70.117	-	-	
Description: The Functional Materials and Devices thrust is developing advanced materials, components and systems to improve device performance for DoW structural, sensing, imaging and communication applications. One focus of this thrust involves development of advanced transductional materials that convert one form of energy to another for DoW-relevant applications.												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-01 / MATERIALS PROCESSING TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
While promising transduction materials are known for a variety of applications, integration into devices is not as advanced. Another focus area is the development of physics-based models that predict material behavior when illuminated by high peak power electromagnetic interference. A third focus area involves development of new multi-functional materials and device designs that will radically decrease the size, weight and power requirements of high-energy photon sources. Such devices should enable fieldable detection units for sensing, non-destructive evaluation of parts, and detection of DoW-relevant targets. A fourth focus area is developing new liquid-based, large-aperture imaging systems such as telescopes. Such telescopes would break the unfavorable, exponential scaling between aperture size and cost for normal telescopes and enable low-cost imaging platforms for ground- and space-based applications. Another focus area under this thrust involves novel nano-architected materials to enhance device-relevant properties for applications to quantum-enhanced sensors. Finally, novel design optimization approaches will be explored where material composition and microstructure are included as explicit, continuous variables alongside shape optimization. This co-optimization of shape and material together will enable new combinations of structural performance and sustainability for a variety of DoW applications. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.				
Title: Chemical Processing for Force Protection Description: Research in the Chemical Processing for Force Protection thrust is focused on the development of new chemical approaches and technologies across a broad spectrum of DoW needs. One area involves development of innovative approaches for scalable small molecule synthesis coupled with predictive tools for route design, possibly offering a new strategy to discover how to make new molecules such as pharmaceuticals and explosives. A second area includes qualification of new molecules made using agile manufacturing platforms. A third focus leverages advances in automation to develop safe, reproducible experimental approaches for systematic development of energetic materials. Another focus is to develop the tools and processing methods to safely turn locally available materials in a given location into reliable, point/time-of-need energetics, for battlefield requirements. In addition, investments in this thrust will advance chemical characterization, information management and analysis, and automation. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.		21.241	-	-
Title: Reconfigurable Systems Description: The Reconfigurable Systems thrust is developing new approaches to enable more rapid and robust adaptation of defense systems and systems-of-systems to changing mission requirements and unpredictable environments. This includes development of capabilities across sensing, perception, planning and control for autonomous, high-speed operation in cluttered environments without Global Positioning System (GPS) information. This also includes development of capabilities to manipulate		11.900	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY		Project (Number/Name) MBT-01 / MATERIALS PROCESSING TECHNOLOGY	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
and control sensory perception and/or situational awareness. Additional work in this thrust focuses on how sensing systems and military systems-of-systems are designed for real-time resilient response to dynamic, unexpected signals and contingencies. Research is developing a more unified view of system behavior that allows better understanding and exploitation of complex interactions among components, including development of formal mathematical approaches to complex adaptive system composition and design. These capabilities will impact autonomous systems and systems-of-systems, including those that involve humans, in a variety of DoW-relevant contexts.					
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.					
Title: Burn n' Go			3.385	-	-
Description: The Burn n' Go program will increase the production rate and availability of solid rocket motors (SRMs) via development of novel materials and methods for tailorable, post-manufacturing thrust control. Complementary to efforts to improve supply chain resilience being developed under the Open Price Exploration for National security (OPEN) program funded in PE 0602303E, Project IT-04, Burn n' Go will enable a paradigm shift in solid rocket motor production by creating a multi-mission standardized motor design for use across multiple existing and future weapons systems. Utilizing a standard design for SRMs in set diameter classes, Burn n' Go will enable suppliers to limit line switching times, improve tooling, accelerate quality control, reduce machining costs, and enable a more stable procurement stream, which can be flexed across Services and weapons. Burn n' Go will validate the developed technology through a series of component-level feasibility tests and integrated system-level static fire tests. Technology developed under this program will be transitioned to the Services and commercial partners.					
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.					
Title: Rainy Day			0.650	-	-
Description: The Rainy Day program will develop innovative approaches and new methods to address challenges in passive accurate placement of unattended sensors from Unmanned Aerial Systems (UAS) and military aircraft. The range of initial deployment altitudes requires compensation for jet streams, trade winds, and low-level weather in real time, and the sensing devices will be ruggedized to withstand extreme environments in flight as well as ground impact while maintaining functionality. Because of the large number of devices delivered, scalability and manufacturability will be established via rapid fabrication and commercial chip foundries. Program success will bring the Department of War (DoW) a new affordable, scalable, and resilient cross-domain capability to accurately place sensing devices with flexible mission sets deployed from a distance. Technology developed under this program will be transitioned to the Services and other government partners.					
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-01.					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-01 / MATERIALS PROCESSING TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Accomplishments/Planned Programs Subtotals		186.816	-	-
C. Other Program Funding Summary (\$ in Millions) N/A				
Remarks				
D. Acquisition Strategy N/A				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY				Project (Number/Name) MBT-02 / BIOLOGICALLY BASED MATERIALS AND DEVICES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MBT-02: BIOLOGICALLY BASED MATERIALS AND DEVICES	0.000	117.853	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
A. Mission Description and Budget Item Justification												
The Biologically Based Materials and Devices project will leverage the growing application space of the biological sciences for the development of new DoW capabilities to improve the sustainability of warfighters and operational platforms in varied domains. This project will develop solutions for critical resource processing, materials development, threat detection and characterization, and warfighter resilience to infectious disease. The materials developed through this project will protect and sustain warfighters and operations in austere conditions, enhancing warfighter resiliency and lethality.												
Beginning in FY 2026, efforts in this Project will be funded in PE 0602024E, Project WP-01.												
B. Accomplishments/Planned Programs (\$ in Millions)									FY 2025	FY 2026	FY 2027	
Title: Turning Upcycled Waste into Novel, Sustainable Materials									16.914	-	-	
Description: Currently, the DoW relies on critical materials and commodity molecules, such as petroleum-, rubber- and wood-derived products, which are needed to protect and provide mobility to our warfighters in an austere, expeditionary setting. Unfortunately, providing these materials to our warfighters also entails risks, such as fragile supply chains, foreign sourcing, or costly shipping to points of need. These critical materials also contribute to DoW waste streams with no further value while also creating logistical challenges. To address this, the Turning Upcycled Waste into Novel, Sustainable Materials program is investigating the feasibility of converting abundant DoW waste stream products (e.g., tires, scrap wood, and paper) into durable and sustainable materials. Approaches will be investigated to develop materials suitable for use in applications ranging from contingency construction materials to commodity molecules.												
Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.												
Title: Biological Undersea Energy									14.440	-	-	
Description: The Biological Undersea Energy program aims to develop emerging technologies that ensure the DoW has the capability to maintain a presence in austere oceanic conditions to provide advanced knowledge of resources and conditions and achieve desired mission effects. Approaches will be developed that utilize biological processes and products to provide energy for improved endurance and performance capabilities while reducing the reliance on servicing or resupply.												
Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.												

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-02 / BIOLOGICALLY BASED MATERIALS AND DEVICES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Biotechnology for Challenging Environments Description: The Biotechnology for Challenging Environments program is developing novel biological solutions to enable warfighter operations in remote and extreme conditions. As the DoW expands operations into previously inaccessible domains, new and unique logistical constraints imposed by extreme conditions and resource scarcity threaten warfighter and warfighting platform readiness. This program will develop technologies using biological approaches to protect and maintain performance of warfighters and warfighting platforms, such as electronics and infrastructure, from challenging conditions. Technology advances developed in this effort will extend mission duration and enhance operational capabilities in emerging domains. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.		13.270	-	
Title: Strengthening Resilient Emotions and Nimble Cognition Through Engineering Neuroplasticity (STRENGTHEN) Description: The Strengthening Resilient Emotions and Nimble Cognition Through Engineering Neuroplasticity (STRENGTHEN) program aims to overcome the limitations of focusing on descriptions of individual disease effects and suicide risk factors by adopting a transdiagnostic approach that addresses the mechanisms (i.e., predictors or causes) of mental health and wellbeing. STRENGTHEN will optimize the brain networks essential for Cognitive Flexibility and Emotional Regulation, establishing dose response, time-to-onset, and duration-of-effect curves to quantify the impact of change in Cognitive Flexibility and Emotional Regulation on validated measures of suicidality, behavioral health, and wellbeing within DoW. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.		10.016	-	-
Title: eX Virentia (eXVi) Description: The eX Virentia (eXVi) program is investigating the use of existing unmodified vegetation as a sensor to identify activities of DoW interest by utilizing natural plant responses to man-made chemical exposures. The eXVi program will bring together the research communities of plant biology, remote sensing, and computational modeling to uncover the internal biochemical response of plant species to chemical exposure, and how they relate to external observable characteristics. The program will grow plant species across various types of vegetation, measure internal and external responses, and model the responses with the goal of scaling predicted responses to similar species across phylogenies. If successful, discovery of predictable and distinct cause-and-effect relationships between exposure and plant response will significantly improve situational awareness of hazards and materials activity, thereby limiting harmful exposure to the warfighter. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.		5.090	-	-
Title: Field Forward Biothreat Storage Solutions for Force Protection		11.035	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-02 / BIOLOGICALLY BASED MATERIALS AND DEVICES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: Warfighters are currently deployed to emerging disease hotspots with increasing pathogen spillovers. Biosurveillance groups tasked with force health protection rely on cold chains and transport media to maintain sample viability for characterization in a laboratory setting, but these methods are unreliable, sometimes inaccessible, or limited in their utility. The Field Forward Biothreat Storage Solutions for Force Protection program will offer expanded capabilities for microbial threat characterization by developing systems capable of long-term, cold chain-free storage of microbial samples. Systems that are able to reliably store and retrieve viable microbes over long timescales will ensure that collected samples reach the lab for study in a usable state, allowing the DoW to better leverage its field-forward laboratories to perform pathogenicity assessments for countermeasure development. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.				
Title: Environmental Microbes as a Bioengineering Resource (EMBER) Description: The Environmental Microbes as a Bioengineering Resource (EMBER) program aims to develop novel, bio-based technologies to overcome key challenges facing the domestic supply of Rare Earth Elements (REEs) critical to the U.S. and DoW. This program is leveraging the diversity, specificity, and customizability of environmental microbiology to enable new domestic biomining methods for the separation, purification, and conversion of REEs into manufacturing-ready forms. Advances in this area will deliver capabilities to assure access to DoW-critical materials domestically and in operational settings. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.		9.815	-	-
Title: Materiel Protection through Biologics Description: Military infrastructure and systems are expected to function years beyond their original intended lifetime but are subject to degradation by external factors. For instance, the formation of biofilms is ubiquitous, corroding and biofouling many military systems, such as aircraft, fuel tanks, ships, medical devices, and filtration systems for water and air. In another example, critical defense assets such as missile silos and naval piers rely on aging concrete infrastructure, ultimately costing the DoW billions of dollars annually to repair and maintain. The Materiel Protection through Biologics thrust is developing approaches to sustain military infrastructure and systems by developing biological or bio-inspired technologies to imbue beneficial functions into existing systems, resulting in benefits such as, but not limited to, reducing drag, mitigating corrosion, or repairing concrete. These bio-inspired interventions will protect and sustain equipment and infrastructure, reducing operation costs and increasing service lifetime. Beginning in FY 2026, this program will be funded in PE 0602024E, Project WP-01.		8.661	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-02 / BIOLOGICALLY BASED MATERIALS AND DEVICES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Bio-Inspired Coastal Defense Description: The Bio-Inspired Coastal Defense program developed self-sustaining, hybrid man-made and biological reef structures to fortify and defend DoW bases in low-lying coastal regions. Military assets in these coastal regions face storm-driven wave action that causes erosion, degrades infrastructure, and impedes operations. Innovative coastal defense requires major technological advances in (1) design, construction, and placement of manufactured reef primers; (2) accelerated recruitment and/or growth of reef species; and (3) sustained, zero-cost natural maintenance and improvement (e.g., increased durability after anchor and materials testing) of the defensive reef. The primary benefit of such structures is to attenuate wave height during storm events for both established and under construction coastal facilities.		13.721	-	-
Title: Signal Processing and Communication with Biotechnology Description: The DoW requires the ability to monitor complex operating domains with sufficient resolution and confidence to inform missions and protect personnel and platforms against various physical and chemical threats. The Signal Processing and Communication with Biotechnology program developed a new customizable sensing methodology using a novel microbe-based, platform technology capable of detecting a variety of input signals, processing information, and generating multiple output signal types in diverse operational conditions. Technology developed in this program will offer insight into signal processing and transmission methodologies with logistical advantages and reliable operability in contested domains.		10.964	-	-
Title: Bioremediation of Battlefields Description: The Bioremediation of Battlefields program addressed the DoW need to stabilize and remediate sites impacted by prior military activities, including contaminated combat zones, defense installations, and test ranges. This research supported the safety of service members and local communities and minimized the impact of warfare by developing biological tools that remediate soil contamination. This program's aim was to eliminate contaminants, and thus restore habitability, by identifying and optimizing organisms, such as microbes, fungi, and plants, that could detect toxic compounds, mitigate their impact, and report on the state of remediation. Bioremediation of Battlefields demonstrated technologies in the laboratory that could reduce the long-term impacts of military activities and improve the overall health and land use potential for contaminated sites.		2.325	-	-
Title: Unburdening the Warfighter from Chemical/Biological (CB) Defense Description: The Unburdening the Warfighter from Chemical/Biological (CB) Defense program aimed to increase warfighter survivability by developing improved personal protective equipment (PPE) and medical countermeasure technologies to protect against CB threats. Current methods of CB protection require significant logistical burdens, including suits that are bulky and hot, which limit operational effectiveness. These burdens increase if additional levels of protection are required. The Unburdening the Warfighter from CB Defense program investigated and designed novel biological and material approaches that provide rapid protection against multiple CB agents for the warfighter. This research innovated PPE through the discovery of compounds		1.482	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602715E / MATERIALS AND BIOLOGICAL TECHNOLOGY	Project (Number/Name) MBT-02 / BIOLOGICALLY BASED MATERIALS AND DEVICES	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
and lightweight, durable systems designed to capture, neutralize, or repel CB agents. This novel approach will provide almost immediate and lasting protection even in austere operational settings.			
Title: Preemptive Expression of Protective Alleles and Response Elements (PREPARE) Description: The Preemptive Expression of Protective Alleles and Response Elements (PREPARE) program aimed to create a transient, near immediate prophylaxis and treatment to protect military personnel and civilians against public health and national security threats. Currently, protection against Chemical, Biological, Radiological, and Nuclear (CBRN) threats relies on physical barrier technology. This program included research to develop novel transient and reversible gene modulator therapies to bolster intrinsic host defenses. Work within this program provided novel solutions that extend beyond the DoW's capabilities to respond to re-emerging, newly emerging, or engineered threats.		0.120	-
Accomplishments/Planned Programs Subtotals		117.853	-
C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A			

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>	R-1 Program Element (Number/Name) PE 0602716E / <i>ELECTRONICS TECHNOLOGY</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	621.104	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
ELT-01: <i>ELECTRONIC TECHNOLOGY</i>	0.000	100.375	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
ELT-02: <i>BEYOND SCALING TECHNOLOGY</i>	0.000	520.729	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Applied Research associated with the Electronics Technology Program that is directed towards developing electronics that make a wide range of military applications possible. The PE focuses on turning basic advancements into the underpinning technologies required to address critical national security issues and to enable an information-driven warfighter. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

Advances in microelectronic device technologies continue to significantly benefit improved weapons effectiveness, intelligence capabilities, and information superiority. The Electronic Technology project supports continued advancement in microelectronics, including electronic and optoelectronic devices, Microelectromechanical Systems (MEMS), semiconductor device design and fabrication, and new materials and material structures. Areas of particular emphasis of this work include reducing the barriers to designing and fabricating custom electronics and exploiting improved manufacturing techniques to provide low-cost, high-performance sensors. Programs in this project will also greatly improve the size, weight, power, and performance characteristics of electronic systems; support positioning, navigation, and timing in GPS-denied environments; and develop sensors more sensitive and robust than today's standards. This project has six major focus areas: Electronics, Photonics, Microelectromechanical Systems, Architectures, Algorithms, and other Electronic Technology research.

The Beyond Scaling Technology project pursues electronics performance advancements that exploit new concepts in circuit specialization and three-dimensional heterogeneous integration (3DHI) by the optimization of materials, devices, architectures, and designs to achieve specific circuit function at high performance. Because electronics advancements must simultaneously make progress in performance and secure the foundation on which our microelectronics infrastructure relies, this envisioned specialization will require incorporation of security safeguards and advancing manufacturing tools and process automation. Accordingly, programs within the Beyond Scaling Technology project will reduce barriers to making specialized circuits in today's silicon hardware and 3DHI by improving producibility. This will significantly increase the ease with which DoW can design, deliver, and eventually upgrade critical, customized microelectronics, particularly for operation in extreme environments. Programs also explore alternatives to traditional circuit architectures, for instance by exploiting 3DHI to optimize electronic devices and by incorporating novel materials and new techniques for securing DoW and commercial data and hardware.

Beginning in FY 2026, efforts in this PE will be funded in PE 0602025E, Making, Maintaining, Supply Chain and Logistics.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency					Date: April 2026
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 2: Applied Research</i>			R-1 Program Element (Number/Name) PE 0602716E / <i>ELECTRONICS TECHNOLOGY</i>		
B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	641.893	0.000	0.000	-	0.000
Current President's Budget	621.104	0.000	0.000	-	0.000
Total Adjustments	-20.789	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	-20.789	0.000			
<u>Congressional Add Details (\$ in Millions, and Includes General Reductions)</u>					FY 2025
Project: ELT-02: <i>BEYOND SCALING TECHNOLOGY</i>					FY 2026
Congressional Add: <i>Quantum Benchmarking Initiative (QBI) - Congressional</i>					135.000
Congressional Add Subtotals for Project: ELT-02					-
Congressional Add Totals for all Projects					135.000
<u>Change Summary Explanation</u>					
FY 2025: Decrease reflects SBIR/STTR transfer.					
FY 2026: N/A					
FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602716E / <i>ELECTRONICS TECHNOLOGY</i>				Project (Number/Name) ELT-01 / <i>ELECTRONIC TECHNOLOGY</i>			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ELT-01: <i>ELECTRONIC TECHNOLOGY</i>	0.000	100.375	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

Advances in microelectronic device technologies continue to significantly benefit improved weapons effectiveness, intelligence capabilities, and information superiority. The Electronic Technology project supports continued advancement in microelectronics, including electronic and optoelectronic devices, Microelectromechanical Systems (MEMS), semiconductor device design and fabrication, and new materials and material structures. Areas of particular emphasis of this work include reducing the barriers to designing and fabricating custom electronics and exploiting improved manufacturing techniques to provide low-cost, high-performance sensors. Programs in this project will also greatly improve the size, weight, power, and performance characteristics of electronic systems; support positioning, navigation, and timing in GPS-denied environments; and develop sensors more sensitive and robust than today's standards. This project has six major focus areas: Electronics, Photonics, Microelectromechanical Systems, Architectures, Algorithms, and other Electronic Technology research.

Beginning in FY 2026, efforts in this Project will be funded in PEs 0602025E, Project MSL-03 and 0602026E, Project EFF-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Humboldt Description: The Humboldt program seeks to develop directed energy (DE) devices to produce disruptive effects in electronic systems. The devices have potential for dual-use as sources to characterize the susceptibility of commercial electronics to electromagnetic interference (EMI). Beginning in FY 2026, this program will be funded in PE 0602026E, Project EFF-01.	10.000	-	-
Title: Scalable Analog Neural networks (ScAN) Description: The Scalable Analog Neural networks (ScAN) program is increasing neural network (NN) inferencing capabilities at the edge and simultaneously reducing the size, weight, and power (SWaP) needed to support inferencing on edge platforms. Currently, sensor outputs are digitized at the edge and then transmitted to the command center for processing. ScAN aims to skip the digitization step and instead perform analog inferencing onboard the edge platform by operating directly on the analog sensor outputs. ScAN objectives are to enable 2000-fold reduction in the power needed for inferencing. ScAN technology will enable intelligence generation at the edge for missions that collect large amounts of sensor data, such as hyper-spectral imaging for unmanned aerial systems. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-03.	9.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602716E / <i>ELECTRONICS TECHNOLOGY</i>	Project (Number/Name) ELT-01 / <i>ELECTRONIC TECHNOLOGY</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Title: Highly Accelerated Learnings of Vibratory Systems (HALOVS) Description: The Highly Accelerated Learning of Vibratory Systems (HALOVS) portfolio investigates the foundational limits and transformative potential of vibratory sensors for positioning and navigation in GPS-denied environments. By leveraging advances in 3D micromachining, material science, and dynamic modeling, HALOVS explores novel architectures and principles through efforts examining ultrafast microsystems, higher-order composite resonators for extra resilience, and chemistries and monolayers for anti-aging kinematics. These efforts aim to address critical challenges in sensor performance, including operating beyond linear regimes to achieve high-velocity motion, leveraging nonlinearities for enhanced resilience to shock and vibration, and extending sensor longevity by addressing drift and aging at the molecular level. The insights and technologies developed within HALOVS promise to redefine the design space and practical limits of vibratory sensors, delivering capabilities essential to national security and global positioning system (GPS)-denied missions. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-03.		14.000	-
Title: Sync Description: The Sync program is creating systems to defend against uncrewed underwater vehicles (UUVs) entering U.S. harbors and ports. Today's counter-UUV approaches rely on complex detection and localization methods to find UUVs prior to engagement. The Sync program looks at orthogonal approaches to solving this problem to defeat UUVs prior to detection. Sync is investigating several different approaches to defeat UUVs, starting with modeling and simulation eventually moving to hardware in the loop and pool testing. Beginning in FY 2026, this program will be funded in PE 0602026E, Project EFF-01.		10.000	-
Title: Microsystem Induced CAtalysis (MICA) Description: The Microsystem Induced Catalysis (MICA) program is developing advanced concepts for microsystem control of biological function. The program seeks hardware demonstrations of molecular catalysts immobilized to microsystem surfaces so that catalyst activity is controlled by physical forces generated by the microsystem. Additionally, the program seeks high-accuracy modeling and simulation of such integrated molecular microsystems. Through these demonstrations, MICA aims to answer three critical questions: (1) how can microsystems be used to actively control molecules? (2) what are the different microsystem physics that can be used to drive catalyst function? and (3) what co-design approaches can be used to integrate the different physics of microsystems and molecules? Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-03.		15.000	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Title: Emon		6.000	-
Description: While the technical sophistication of radar systems has advanced tremendously since their inception, specifically with the use of digital circuitry that enables fine-tuned control over both waveform generation and received signal processing, the basic principles upon which radar operates have not varied greatly. Some properties of the classical electromagnetic (EM) field have not yet been exploited fully, whether due to past limitations in sensor capabilities or lack of insight into the possible benefits of leveraging the more subtle aspects of the EM field. One of these underutilized properties is the fact that the EM field has a tensorial nature. The Emon program is investigating the potential utility of constructing radar systems that benefit more completely from the tensorial nature of the EM field, which may offer a variety of ways that could make radar sensing more useful to the warfighter. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-03.			-
Title: Robust Quantum Sensors (RoQS)		15.000	-
Description: The Robust Quantum Sensors (RoQS) seeks to bring quantum sensors to DoW platforms. While quantum sensors have demonstrated exceptional laboratory performance in several modalities (magnetic and electric field, acceleration, rotation, and gravity, etc.), their performance degrades once the sensor is placed on moving platforms due to electric and magnetic fields, field gradients, and system vibrations. RoQS seeks to overcome these challenges through innovative physics approaches to quantum sensing. RoQS aims to develop and demonstrate quantum sensors that inherently resist performance degradation from platform interferers and demonstrate them on a DoW platform. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MQB-01.			-
Title: Waveform Agile Radio-frequency Directed Energy (WARDEN)		8.375	-
Description: The Waveform Agile Radio-frequency Directed Energy (WARDEN) program aimed to extend the range and lethality of high-power microwave (HPM) systems by introducing flexible waveform techniques that use combinations of frequency, amplitude, and pulse-width modulations. Applications for HPM systems include vehicle and vessel disruption, electronic strike, and guided missile defense. Legacy HPM systems used oscillators to produce electromagnetic radiation. These systems were inherently narrowband and lack the frequency agility to support waveforms to maximize electromagnetic coupling and to optimally exploit electronic system vulnerabilities. Lacking the capability to use optimized waveforms, HPM oscillators had been pushed close to the physical limits of peak power generation. To develop a more efficient, lower power, waveform agile approach, the WARDEN program developed and demonstrated the first broadband HPM amplifier; created new theory and simulation tools; and developed novel agile waveform techniques to make U.S. capabilities more effective.			-

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Title: Fast Event-based Neuromorphic Camera and Electronics (FENCE)			6.000	-	-
Description: The Fast Event-based Neuromorphic Camera and Electronics (FENCE) program developed and demonstrated a low latency, low power event-based infrared (IR) camera to enable intelligent sensors for tactical DoW applications. Event-based imagers are an emerging class of sensors with major demonstrated advantages relative to traditional cameras. State-of-the-art visible event-based cameras have been shown to produce over two orders of magnitude less data in optimal conditions relative to traditional framing cameras because they transmit data only from pixels that have changed. This leads directly to two orders of magnitude lower data latency and a commensurate reduction in power consumption. Despite their inherent advantages, existing event-based cameras were not compatible with DoW applications because DoW applications regularly face conditions that are not optimal, where issues such as clutter and noise cause a large percentage of the event-based pixels to change simultaneously. When this happens, such cameras did not perform significantly better than traditional cameras. FENCE developed an infrared event-based imager, consistent with military requirements, that intelligently removes noise and clutter to maintain low power and latency operation, even when faced with all of the pixels firing simultaneously. This new class of sensors enabled by FENCE are capable of responding to fast moving targets and discriminating dim targets in noisy conditions.					
Title: Generating RF with Photonics for low Noise (GRYPHON)			5.000	-	-
Description: The Generating RF with Photonics for low Noise (GRYPHON) program developed compact sources of microwaves and millimeter waves with extremely low phase noise. Legacy compact signal sources, such as crystal oscillators, are too noisy to support advanced military radar and communications functions. Conversely, best-in-class oscillators which use optical techniques to synthesize extremely pure microwaves faced size and cost issues that inhibited deployment on the airborne systems, munitions, and other size-constrained platforms where the DoW requires high-performance capabilities. The GRYPHON program drew on recent advances in miniature optical components to replicate best-in-class optical frequency synthesis techniques in microchip form factors.					
Title: Quantum Imaging of Vector Electromagnetic Radiation (QuIVER)			2.000	-	-
Description: The Quantum Imaging of Vector Electromagnetic Radiation (QuIVER) program developed full-tensor magnetic field sensors and demonstrated them in DoW-relevant applications and concept of operations. In addition to being diagnostically relevant, such sensitive magnetometers could enable future human-machine/brain-machine interfaces. The DoW and industry also use magnetometers to detect anomalies such as mineral/oil deposits, old wellheads, or of improvised explosive devices. In addition, magnetometers offer the possibility of magnetic navigation, which may operate in GPS-denied environments.					
Accomplishments/Planned Programs Subtotals			100.375	-	-

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C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Appropriation/Budget Activity 0400 / 2					R-1 Program Element (Number/Name) PE 0602716E / ELECTRONICS TECHNOLOGY				Project (Number/Name) ELT-02 / BEYOND SCALING TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ELT-02: BEYOND SCALING TECHNOLOGY	0.000	520.729	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The Beyond Scaling Technology project pursues electronics performance advancements that exploit new concepts in circuit specialization and three-dimensional heterogeneous integration (3DHI) by the optimization of materials, devices, architectures, and designs to achieve specific circuit function at high performance. Because electronics advancements must simultaneously make progress in performance and secure the foundation on which our microelectronics infrastructure relies, this envisioned specialization will require incorporation of security safeguards and advancing manufacturing tools and process automation. Accordingly, programs within the Beyond Scaling Technology project will reduce barriers to making specialized circuits in today's silicon hardware and 3DHI by improving producibility. This will significantly increase the ease with which DoW can design, deliver, and eventually upgrade critical, customized microelectronics, particularly for operation in extreme environments. Programs also explore alternatives to traditional circuit architectures, for instance by exploiting 3DHI to optimize electronic devices and by incorporating novel materials and new techniques for securing DoW and commercial data and hardware.

Beginning in FY 2026, efforts in this Project will be funded in PE 0602025E, Projects MQB-01 and MSL-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Faithful Integration Reverse-engineering and Emulation (FIRE) Description: The Faithful Integration Reverse-engineering and Emulation (FIRE) program is developing tools to find and patch vulnerabilities within cyber-physical systems. A cyber-physical system operates in the physical world using hardware sensors to perceive the analog environment, digital software for processing, and actuators to interact with the environment. Cyber-physical vulnerabilities arise from the composition of hardware, software, and physical components where each component may not be vulnerable in-and-of itself. FIRE is developing novel modeling and simulation techniques to help expedite finding and patching vulnerabilities in cyber-physical systems. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.	27.000	-	-
Title: Macaroni Description: Measurement and control of the electromagnetic spectrum is a key area of research for the Department of War (DoW). Spectrum dominance requires quick and efficient control of electromagnetic radiation from low frequencies to X-rays. In classical antenna theory, the sensitivity-bandwidth product is fundamentally limited by the physical shape and size of the antenna. This performance degrades significantly as the antenna becomes electrically small, that is, the physical size becomes much smaller than the electromagnetic wavelength of operation. The Macaroni program seeks to develop electrically-small receivers	22.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
and transmitters with performance that exceeds the current state-of-the-art (SoA). Recent advances in quantum sensors, materials science, electromagnetic shielding, laser technology, resonators, cryogenic systems, and vacuum components have pushed the SoA in sensing technologies. For transmitters, new insights in active antenna technology, control schemes, methods of impedance matching, and strategies for volume filling present new opportunities.				
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MQB-01.				
Title: Additive Manufacturing of MicrosystEms (AMME) Description: The Additive Manufacturing of MicrosystEms (AMME) program is revolutionizing microsystem manufacturing by leveraging selective material synthesis and 3D patterning to enable a new class of microsystems. Additive Manufacturing (AM) has enabled complex single-material geometries that were previously impossible to produce via traditional manufacturing methods. However, microsystem manufacturing has not exploited AM due to fundamental limits of material quality, resolution, and print throughput. The AMME program is using selective material synthesis to create high-quality material precursors that permit simultaneous printing of conductors and insulators with high-resolution and high-volume throughput. Additionally, AMME is focusing on commercialization of this technology such that the Department of War and intelligence community can quickly adopt the productized system to fabricate novel microsystems.		15.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Space Power Conversion Electronics (SPCE) Description: The Space Power Conversion Electronics (SPCE) program seeks to develop highly-efficient, radiation-tolerant point of load (POL) converters for low-earth-orbit satellites. In today's space power systems, power transistors for POL converters have derated operating voltage to maintain radiation tolerance, resulting in decreased POL efficiency and limiting the satellite's available power, capabilities, and battery lifetime. To address this deficiency, SPCE is developing high-performance, radiation-tolerant high voltage switches and compact passives by exploiting advanced wide-bandgap semiconductor advanced material synthesis, novel device architectures, and advanced manufacturing processes. High efficiency and compact radiation tolerant POL will be realized with advanced gate drive/control circuitries and novel 3D heterogeneous integration technology.		19.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Intensity-Squeezed Photonic Integration for Revolutionary Detectors (INSPIRED) Description: The Intensity-Squeezed Photonic Integration for Revolutionary Detectors (INSPIRED) program is developing compact, ultra-low-noise optical detectors. Low-noise detection is vital to all optical science and technology, but the quantum		18.472	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
nature of light imposes a fundamental quantum limit on a conventional optical detector's noise performance. Recent experiments have demonstrated that exotic quantum states can be harnessed to overcome the quantum limit, albeit from bench-scale apparatuses that ultimately restrict the application of enhanced detectors to esoteric applications such as particular forms of astronomy. The INSPIRED program is leveraging recent advances in chip-scale quantum optics and materials to realize optical detector modules operating well below the quantum noise limit in form factors that enable deployment in applications such as biosensing, navigation, and communications. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.					
Title: Technologies for Heat Removal in Electronics At the Device Scale (THREADS) Description: The Technologies for Heat Removal in Electronics At the Device Scale (THREADS) program is developing technologies to overcome transistor thermal limits to realize robust, high power density transistors that operate near their fundamental electronic limit of radio-frequency (RF) output power. DoW's RF transmitters increasingly use transistors that quintuple RF power output compared to legacy technology. Achieving high RF power output while maintaining a transistor operating temperature below the nominal maximum reliable operation temperature faces two challenges. The first challenge is reducing thermal resistance within the device. The second challenge is more efficiently moving heat away from the transistor hot spots. THREADS will address these challenges by leveraging recent scientific and technological advances in order to demonstrate high efficiency X-band transistors and power amplifier (PA) test vehicles with an output power density of 16X higher than current production amplifiers. THREADS technology will enable increased range for radar, communications, and electronic warfare systems. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.			18.000	-	-
Title: Minitherms3D Description: Minitherms3D is developing thermal management solutions for the three-dimensional heterogeneous integration (3DHI) of microelectronics to accelerate the growth of compact, high-performance microsystems. 3DHI microsystems are enabling technologies for phased array systems and dense computing for artificial intelligence and machine learning applications. Minitherms3D seeks to reduce the size, weight and power (SWaP) of high-performance 3DHI microsystems by developing novel methods to transfer heat from within the 3D stack to its boundaries, transmit it to a remote location, and reject it to ambient air. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.			18.000	-	-
Title: High Operational Temperature Sensors (HOTS)			18.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: Many commercial, industrial, and defense systems experience thermal environments that are beyond the performance and survivability of today's physical sensors. The High Operational Temperature Sensors (HOTS) program is developing a high-performance sensor platform for operation at extremely high temperatures for high-priority mission-relevant timescales. Among other uses, HOTS technology will find application in combustion engine research, superhot rock geothermal energy development, and petrochemical processing. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Optimum Processing Technology Inside Memory Arrays (OPTIMA) Description: The Optimum Processing Technology Inside Memory Arrays (OPTIMA) program aims to realize a high throughput, compact, energy-efficient, and adaptable compute-in-memory (CIM) accelerator that is compatible with very large-scale integration (VLSI) fabrication. Traditional accelerators based on von Neumann architecture have limitations in terms of computational power efficiency and speed. By demonstrating a CIM accelerator with an array of Multiply Accumulate Macros (MAMs) consisting of a large number of Multiply Compute Elements (MCE) to perform the matrix multiply operating within memory elements, the power and speed challenges can be overcome. The program goal is to showcase high-performance CIM accelerator with innovative signal processing circuitry and architectures, with a focus on optimizing both area and power efficiency. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.		16.500	-	-
Title: Optomechanical Thermal Imaging (OpTIm) Description: Advanced infrared (IR) detectors and thermal imaging systems underpin a vast DoW application space including biochemical detection; infrared Search-and-Track; and terrestrial and space-based Intelligence, Surveillance, and Reconnaissance. Current IR detectors face limitations in either sensitivity, signal bandwidth, or the need for expensive cryogenic cooling. The Optomechanical Thermal Imaging (OpTIm) program is developing a new modality of low size, weight, and power, room temperature IR detectors capable of quantum-level sensitivity, thereby enabling transformative enhancements to DoW capabilities including, but not limited to, night vision, surveillance, multispectral detection, and remote detection of trace industrial pollutants and greenhouse gases. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.		9.000	-	-
Title: Scalable On-Array Processing (SOAP)		15.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: The Scalable On-Array Processing (SOAP) program is designed to achieve scalable algorithms and processing architectures to overcome the inherent digital bottlenecks that severely limit today's wideband operation on arbitrarily large elemental digital phased arrays. SOAP aims to reduce the computational complexity of array processing as a function of element count, from exponential to linear scaling. SOAP also seeks to move the processing from physically separated back-end processors to processors integrated into the array, in order to fully process all the information generated at the element level, with no elemental information loss. To achieve these aims, SOAP will design processors that can be distributed within the array, as close to the elements as possible. These processors should be connected and networked in such a way that the data from any element can be processed by any processor. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.					
Title: Intelligent Generation of Tools for Security (INGOTS) Description: The Intelligent Generation of Tools for Security (INGOTS) program is developing techniques to accelerate cyber defense activities by identifying, triaging, and assessing exploitability of chainable vulnerabilities. Today, sophisticated cyber attacks link multiple vulnerabilities together into exploit chains that enable them to compromise various systems. Accurately understanding risk is critical for both developers and defenders within cyberspace, but the metrics currently in use do not account for the multiple factors that differentiate an innocuous software flaw from a chainable vulnerability. INGOTS is developing semi-automated tools and techniques to characterize and measure the interdependent exploitability of vulnerabilities and a new vulnerability severity metrology that characterizes and measures interdependent exploitability. With the INGOTS vulnerability measurement capability, developers and defenders will improve software and hardware resiliency by rapidly identifying and prioritizing the most dangerous flaws. The INGOTS program is also funded in PE 0602303E, Project IT-03. Beginning in FY 2026, this program will be funded in PE 0602025E, Projects MSL-02 and MSL-04.			13.000	-	-
Title: COmpact Front-end Filters at the EIEment-level (COFFEE) Description: The COmpact Front-end Filters at the EIEment-level (COFFEE) program is developing and demonstrating compact, high frequency radio frequency (RF) filter technology without compromising performance, specifically low insertion loss and high-power handling. The new filtering technology will enable interference rejection capability, efficient spectral management, and coexistence with commercial 5G applications. It is projected that COFFEE filter technology will enhance the resilience of military microwave and mm-wave radar and communication systems for DoW spectral dominance into the future. For commercial applications, COFFEE will result in more efficient use of mm-wave frequency allocations for 5G networks.			14.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.					
Title: Digital RF Battlespace Emulator (DRBE) Description: The Digital RF Battlespace Emulator (DRBE) program is developing a large-scale, interactive, emulated radio frequency (RF) environment, providing the DoW with the capability to cost-effectively evaluate adaptive, intelligent, and spatially distributed next-generation RF systems. DRBE is leveraging advances in massively multi-core computing hardware and high-bandwidth digital cross-connects to emulate realistic RF environments accounting for RF platform movement, signal propagation effects and delays, signal interference, and interactions between RF systems. DRBE is pursuing three technical thrust areas: architecture, massively multi-core computing, and scenario modeling. The resulting test environment will allow plug-and-play connections for hundreds of RF systems in a battlespace test. Multi-system exercises will then be quickly executed through many different combat scenarios and variations. DRBE is serving to develop concept of operations (CONOPS), inform operational and contingency planning, and fine-tune the performance of both individual and large groups of RF systems. Additional development started in 2024 greatly expands the input/output bandwidth of DRBE to support for much larger RF scenarios. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.			14.000	-	-
Title: Advanced Sources for Single-event Effect Radiation Testing (ASSERT) Description: 3D heterogeneously integrated (3DHI) microelectronics will be a key driver of the next wave in electronics performance. However, the nation's current single-event effect (SEE) radiation testing infrastructure lacks the ability to analyze and qualify emerging 3D devices for operation in high radiation environments. To fill this gap, the Advanced Sources for Single-event Effect Radiation Testing (ASSERT) program is developing new source technologies to create charge tracks with deep penetration depths for SEE qualification of 3DHI topologies and packaging, providing the means to selectively probe device topologies to inform engineering design, and generate data to validate developing models and codes and to provide training sets for optimization. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.			14.000	-	-
Title: ELelectronics for G-band ARrays (ELGAR) Description: The ELelectronics for G-band ARrays (ELGAR) program is developing the integration technologies needed to create compact, high-performance G-band (220 GHz) array front-end electronics to enable phased array antenna systems for DoW communications and sensing. ELGAR will address the key technical challenges that prevent III-V electronics from realizing high-performance G-band arrays, namely achieving efficient, compact G-band III-V monolithic microwave/millimeter wave integrated circuit power amplifiers (MMIC PAs) with high output power density and achieving low loss off-chip interconnects between			13.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
adjacent G-band array components. In particular, ELGAR will develop III-V compatible, silicon-like fabrication and integration approaches to enable compact, high-power density, high efficiency G-band MMICs and arrays. The technologies developed will support applications including high data rate communications in size, weight, and power-constrained platforms. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: H6 Description: The H6 program is developing the first tactical-grade clock. Tactical-grade clocks are ultra-small, low power, fieldable and can maintain the timing needed for DoW-relevant applications in challenging environments. Precise timing in a tactical package will decouple operations from GPS dependence, addressing a significant operational need for the warfighter. Precise tactical-grade clocks from H6 will enable increased signal assurance and pervasive communications security in high-jamming regions. Additionally, H6 will enable real-time awareness of warfighters and will play a critical role in search and rescue through the ability to maintain precise time over a long mission duration without having to re-establish external communications. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MQB-01.		12.000	-	-
Title: NanoWatt Platforms for Sensing, Analysis, and Computation (NaPSAC) Description: Efficient, high-speed scientific computing architectures are a ubiquitous requirement for applications including modeling of complex physical systems, advanced device designs, and multiscale computations of dynamical phenomena such as climate models or turbulence. Current state-of-the-art computing systems require prohibitive amounts of energy and time to perform such calculations. The NanoWatt Platforms for Sensing, Analysis, and Computation (NaPSAC) program aims to develop a novel computational architecture for massively parallel, ultralow power "in-memory" computation. NaPSAC-based computing architectures can potentially yield transformative impact by enabling beyond-state-of-the-art computational speed and accuracy. Applications of immediate relevance to the DoW include simulations of turbulent flows, multiscale electromagnetic simulations of plasma dynamics, advanced semiconductor device design, and the modeling of high-performance materials. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.		10.000	-	-
Title: Ultra-Wide BandGap Semiconductors (UWBGS) Description: The Ultra-Wide BandGap Semiconductors (UWBGS) program is developing and optimizing ultra-wide bandgap (UWBG) materials and fabrication processes required to enable the next revolution in semiconductor electronics. UWBGS will establish the foundation for the creation of producible and reliable, high performance UWBG devices for a variety of DoW (and commercial) applications. These include but are not limited to: high power radio frequency (RF) switches; high power density RF		27.000	-	-

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B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
amplifiers; high RF power protection device; high voltage switches for power electronics; high temperature electronics and deep ultraviolet light-emitting diodes and lasers. The program will address the key technical challenges that are limiting the performance of UWBG device. These challenges include realizing high quality UWBG materials, ability to tailor electrical characteristics of UWBG materials; ability to create homo- and heterostructures with abrupt junctions and low defect density; and the realization of ultra-low resistance electrical contacts. UWBGs will fabricate device test structures to quantify the improvements in these areas. To be successful, the program will leverage recent advances in UWBG materials. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Heterogenous Adaptively Produced Photonic Interfaces (HAPPI) Description: The Heterogeneous Adaptively Produced Photonic Interfaces (HAPPI) program aims to revolutionize information transmission within microsystems by achieving a 1000x increase in connectivity density through advanced photonic solutions. The program focuses on developing three-dimensional optical routing capabilities both within and between chips, enabling unprecedented levels of information movement and processing throughout integrated systems. This includes the potential to interface with both electronics and other optical elements, enhancing the flexibility and reach of the system. HAPPI's advancements will enable large-scale photonic circuits with thousands of components unlocking new microsystem architectures for applications such as signal processing, free-space communications, remote sensing, digital computing, and atomic sensing. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.		15.000	-	-
Title: Machine Learning and Optimization-guided Compilers for Heterogeneous Architectures (MOCHA) Description: The Machine Learning and Optimization-guided Compilers for Heterogeneous Architectures (MOCHA) program is developing a new generation of compiler technology that uses machine learning to automatically generate and optimize compilers that realize the full potential of heterogenous architectures. The ending of Dennard scaling, which has limited the increase of clock speeds of digital circuits, has led to computing architectures that include a variety of co-processors and accelerators to enable high performance in specific domains. Traditional compilers, software that turns source code into machine instructions, do a poor job of optimizing code for heterogenous ensembles of processing units and accelerators, and a portion of the potential benefit of accelerators is lost. Extending compilers to handle heterogeneity is at present a manual task that is both time consuming and error prone, thus limiting the ability to upgrade mission-critical systems in a timely manner. MOCHA is applying machine learning techniques to the problems of compiler adaptation and extension by semi-automatically generating the main components of compilers for heterogeneous architectures. The use of machine learning will reduce human effort and development time while improving the quality of the machine code emitted by the compiler in terms of measures such as performance, memory size, and		10.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602716E / ELECTRONICS TECHNOLOGY	Project (Number/Name) ELT-02 / BEYOND SCALING TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
power consumption. If successful, MOCHA will speed production of the compilers needed to capitalize on emerging specialized processor hardware in heterogeneous computing architectures for high performance military systems.				
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Material Synthesis Technologies for Universal and Diverse Integration Opportunities (M-STUDIO) Description: The M-STUDIO program will realize a universal defect-free heterogeneous integration methodology, informed by emerging nano-scale material growth with surface free-energy driven defect termination and non-thermal-equilibrium synthesis, to achieve defect-free multi-layer heterogeneous materials with nanometer dimensions and atomically sharp interfaces. Specifically, M-STUDIO program will achieve: (1) Material synthesis with one heterogeneous interface: a semiconductor layer on lattice mismatch substrate with a total thickness 10 nm (> 100x reduction from the state-of-the-art (SOA)) and with low defect density, and (2) demonstrate the synthesis scalability with multiple heterogeneous interfaces: multiple 10 nm heterogenous semiconductor layers on lattice mismatch substrate with atomically sharp transition. In addition, M-STUDIO integration technology is compatible with leading edge and future advanced node semiconductor manufacturing process.		13.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.				
Title: Quantum Augmented Network (QuANET) Description: The Quantum Augmented Network (QuANET) program is developing quantum-augmented networks that add novel security and other properties inherent in quantum communications to classical, non-quantum, network infrastructures that currently trade security against interoperability. Today, all digital communication paradigms use a network stack that consists of a layered set of software protocols. The higher layers are closer to applications on computers and servers, commonly called nodes, while the bottom layers are closer to the physical implementation, i.e., network cables. State-of-the-art networks commonly rely on security at the top layers of the stack, assuming this security also mitigates potential attacks on lower layers. Unfortunately, advanced persistent threat (APT) attacks are increasing cyber defense costs to U.S. commercial and government entities. The QuANET program seeks to augment existing software infrastructure and network protocols with quantum properties to mitigate attack vectors. QuANET is developing the hardware, protocols, and software tools required to operationalize quantum communications for sensitive missions and critical infrastructure and demonstrate these capabilities in an operational, quantum-augmented network.		14.000	-	-
Beginning in FY 2026, this program will be funded in PE 0602025E, Project MQB-01.				
Title: Lasers for Universal Microscale Optical Systems (LUMOS)		9.000	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 2		R-1 Program Element (Number/Name) PE 0602716E / ELECTRONICS TECHNOLOGY	Project (Number/Name) ELT-02 / BEYOND SCALING TECHNOLOGY		
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
Description: The Lasers for Universal Microscale Optical Systems (LUMOS) program is integrating high-performance light sources into silicon integrated photonics enabling compact, rugged, high-performance systems for positioning, navigation, communications, 3D imaging, and quantum technologies. Silicon photonics today enable microscale integration of complex optical systems, but the platforms lack of optical gain precludes the creation of lasers and amplifiers through foundry processes. LUMOS will deliver the missing capability to provide compact optical sources at wavelengths from the visible to the infrared and will create a universal manufacturing platform that builds upon the current photonics ecosystem. To drive innovation and maintain DoW access to leading-edge deployable photonic solutions, LUMOS will establish a technology pathway connecting government, academic, commercial, and defense users of integrated photonics, and will provide multi-project wafer runs through an open-access foundry. Beginning in FY 2026, this program will be funded in PE 0602025E, Project MSL-02.					
Title: Quantum Inspired Classical Computing (QuICC) Description: The Quantum Inspired Classical Computing (QuICC) program implemented quantum-inspired algorithms using classical dynamic systems in novel computing architectures for the efficient solving of complex optimization problems. This program created frameworks for analyzing the computational advantage provided by quantum-inspired algorithms and performed the hardware and algorithm co-design needed to reduce the required energy to optimally solve mission-scale problems.			6.757	-	-
Title: Data Privacy in Virtual Environments (DPRIVE) Description: The Data Privacy in Virtual Environments (DPRIVE) program made secure processing on untrusted hardware feasible through the development of new hardware accelerators that allow the data to remain encrypted at all times, even during processing. The hardware developed under DPRIVE will accelerate several fully homomorphic encryption (FHE) schemes more than three orders of magnitude over commodity processors. The program's technology provides strong privacy protections at the tactical edge with no more than one order of magnitude penalty in computation time, and to enable very strong privacy at the enterprise level with no more than three orders of magnitude penalty compared to the corresponding unencrypted processing on commodity processors. The program enabled the development and deployment of these hardware accelerators to edge computing devices where power and time are a premium, as well as to enterprise computing facilities where the amount and sensitivity of the data requires increased protection.			5.000	-	-
Accomplishments/Planned Programs Subtotals			385.729	-	-
			FY 2025	FY 2026	
Congressional Add: Quantum Benchmarking Initiative (QBI) - Congressional			135.000	-	

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 2	R-1 Program Element (Number/Name) PE 0602716E / <i>ELECTRONICS TECHNOLOGY</i>	Project (Number/Name) ELT-02 / <i>BEYOND SCALING TECHNOLOGY</i>

	FY 2025	FY 2026
<i>FY 2025 Accomplishments:</i> - Evaluated utility-scale quantum computer concepts for plausibility, which included assessing high-risk elements of the system architecture, confirming technical feasibility, and reviewing system requirements. - Developed and evaluated utility-scale quantum computer applications and co-designed appropriate applications with planned utility-scale quantum computer concepts. - Began examining research and development (R&D) plans, including detailed execution plans addressing the major risks associated with building a utility-scale quantum computer, planned risk mitigation steps, and plans for prototypes that will demonstrate risk reduction. - Identified and began procuring long-lead hardware items needed to perform component, subsystem, and prototype research and development.		
Congressional Adds Subtotals	135.000	-

C. Other Program Funding Summary (\$ in Millions)
 N/A

Remarks

D. Acquisition Strategy
 N/A

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity	R-1 Program Element (Number/Name)											
0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	PE 0603286E / <i>ADVANCED AEROSPACE SYSTEMS</i>											
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	241.495	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
AIR-01: <i>ADVANCED AEROSPACE SYSTEMS</i>	0.000	241.495	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Advanced Aerospace Systems Program that is focused on exploiting high pay-off opportunities to provide revolutionary new system capabilities, as opposed to incremental or evolutionary advancements, in order to achieve undeterrable air presence at dramatically reduced costs. Rapid prototyping and experimentation of integrated system concepts, as well as enabling vehicle subsystems will be conducted. Programs will explore new architectural concepts that employ a mix of weapon technologies that achieve lethality through a combination of overwhelming performance and overwhelming numbers rather than through the use of singular and costly high value assets. Studies conducted under this program element include examination and evaluation of emerging aerospace threats, technologies, concepts, use of autonomy to minimize risk, and applications for missiles, munitions, and vehicle systems.

Beginning in FY 2026, efforts in this PE will be funded in PE 0603468E, Advanced Complex Systems.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	252.018	0.000	0.000	-	0.000
Current President's Budget	241.495	0.000	0.000	-	0.000
Total Adjustments	-10.523	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	-2.361	0.000			
• SBIR/STTR Transfer	-8.162	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>		R-1 Program Element (Number/Name) PE 0603286E / <i>ADVANCED AEROSPACE SYSTEMS</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: LongShot Description: The LongShot program is developing and flight demonstrating an air-launched system capable of engaging multiple adversary targets from standoff ranges using existing air-to-air missiles. LongShot will be deployed either externally from existing fighters or internally from existing bombers. This system will capitalize on a slower speed, fuel-efficient air vehicle for ingress, while retaining highly energetic air-to-air missiles for end-game target engagements, providing several key benefits that increase weapon effectiveness. This program will address the stability and control challenges of launching air-to-air missiles from a relatively small uninhabited system in an operational environment. Potential transition partners include the Navy and Air Force. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		64.213	-	-
Title: SPeed and Runway INdependent Technologies (SPRINT) X-Plane Demonstration Project Description: The SPeed and Runway INdependent Technologies (SPRINT) X-Plane Demonstration Project is developing and demonstrating the fundamental technologies needed for combined high speed and vertical take-off and landing (VTOL) capabilities in a single aircraft. This program culminates in the fabrication and flight test of a demonstrator that validates the critical technologies in a representative environment and reduces technical, schedule, and cost risk for a follow-on operational system. High speed VTOL aircraft are highly desired in a variety of military missions such as contested logistics under attack, infiltration/exfiltration, electronic warfare, personnel recovery, troop transport, and armed escort. The SPRINT Demonstrator is envisioned to transition to a Services component for further risk reduction flight testing. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		47.000	-	-
Title: Artificial Intelligence (AI) Reinforcements (AIR) Description: AI Reinforcements (AIR) is developing and demonstrating dominant tactical autonomy for multi-ship, beyond visual range (BVR), real-world air combat missions. This program is focused on developing highly accurate models that are orders of magnitude faster than the present state-of-the-art and then using those models to unlock novel and robust AI-driven autonomy approaches. An operations-centric development approach will be enabled through the use of human-on-the-loop F-16 aircraft. On piloted platforms, AIR's algorithms will automate tactical control tasks transforming junior pilots from low-level tacticians into high-level mission commanders. For unpiloted platforms, AIR will enable vehicles to perform missions with minimal human oversight. The outcome of this program will be an AI air combat capability that works in dynamic, operationally representative environments. The transition partner is the U.S. Air Force. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		51.884	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>		R-1 Program Element (Number/Name) PE 0603286E / <i>ADVANCED AEROSPACE SYSTEMS</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Control of Revolutionary Aircraft with Novel Effectors (CRANE) Description: The Control of Revolutionary Aircraft with Novel Effectors (CRANE) program is developing and demonstrating revolutionary improvements in aircraft controls technology. The program will design, build, and flight test an aircraft able to fly and maneuver at altitude relying on state-of-the-art Active Flow Control (AFC) technology. AFC is a broad term that encompasses a range of technology approaches; it includes a number of control mechanisms which alter the aerodynamic flow field through ejection or suction of fluid via an orifice on a lifting body. An emphasis of the program is on assessing AFC component technologies, risk reduction and experimentation, integrated testing, fabrication and demonstration of a relevant scale novel and innovative aircraft. Technologies, design tools, and models developed and demonstrated under this program will be made available to all Services as well as the civilian aerospace sector for application to future air systems development. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		18.313	-	-
Title: Liberty Lifter Description: The Liberty Lifter program was intended to design and demonstrate a runway-independent, large-payload, survivable, dual-flight regime aircraft capable of extended on-water operations and flight both in and out of ground effect. Critical to an effective aircraft of this type is a robust sea plane capability to operate in high sea states as well as an innovative manufacturing approach that dramatically reduces vehicle acquisition costs. The vehicle was anticipated to be survivable against peer threats due to the combination of extremely low altitude operations and speeds significantly higher than ships. The ability to deploy amphibious cargo while on the water will minimize exposure time and enable a wide variety of mission capabilities in the maritime domain including rapid contested logistics support, and search and rescue. The Liberty Lifter program was envisioned to deliver a technology demonstrator with potential to transition to military service partners for continued testing and development activities. The demonstrator was expected to be approximately 80% size and 50% maximum gross takeoff weight of a future Liberty Lifter objective system.		14.185	-	-
Title: AdvaNced airCRAFT Infrastructure-Less Launch And RecoverY (ANCILLARY) Description: The AdvaNced airCRAFT Infrastructure-Less Launch And RecoverY (ANCILLARY) program developed and flight demonstrated an X-plane with the critical technologies required for a leap-ahead in long endurance, vertical takeoff and landing (VTOL) unmanned air system (UAS) performance. The UAS can launch and recover from small ship flight decks and austere land locations in adverse weather without additional infrastructure equipment, thus enabling expeditionary deployments. Both the U.S. Marine Corps and the U.S. Central Command will conduct follow on testing, payload integration, and operations with the aircraft at the completion of the program.		28.458	-	-
Title: Rapid Experimental Missionized Autonomy (REMA)		13.828	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>		R-1 Program Element (Number/Name) PE 0603286E / <i>ADVANCED AEROSPACE SYSTEMS</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: Commercial-quality drones demonstrate surprising usefulness on the modern battlefield. Rapid Experimental Missionized Autonomy (REMA) enhanced commercially available and stock military drones with a subsystem to enable autonomous operation. The program focused on delivering autonomy without being tied to a specific drone design. REMA developed these capabilities through rapid spirals of development. New mission functionality was delivered through development spirals accelerating from three-month duration at program inception to one-month by program completion. Legacy drones are either remotely piloted via radio frequency (RF) tethers or pre-programmed with mission profiles relying on GPS waypoints; neither approach is ideal for RF-contested environments, hence the need for greater autonomy. Research and Development (R&D) programs have demonstrated autonomy capabilities for drones, but these have frequently been bespoke solutions, with software spirals of nine months or longer, too slow of a response in a dynamic battlefield. The REMA program addressed specific challenge problems, during which performers developed, collaborated, and delivered an autonomy subsystem for drones at a rapid pace.				
Title: Advanced Aerospace System Concepts Description: Studies conducted under this program examined and evaluated emerging aerospace technologies and system concepts for applicability to military use. This included the degree and scope of potential impact and improvements to military operations, mission utility, and warfighter capability. Studies were also conducted to analyze emerging aerospace threats along with possible methods and technologies to counter them. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, was also evaluated. The results from these studies were used, in part, to formulate future prototype development programs or refocus ongoing work. The topics that were researched and analyzed included: methods of defeating enemy anti-aircraft attacks; munition technologies to increase precision, range, endurance, and lethality of weapons for a variety of mission sets; novel launch systems; air vehicle control, power, propulsion, materials, and architectures; and payload and cargo handling systems.		3.614	-	-
Accomplishments/Planned Programs Subtotals		241.495	-	-
D. Other Program Funding Summary (\$ in Millions) N/A Remarks				
E. Acquisition Strategy N/A				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>					R-1 Program Element (Number/Name) PE 0603287E / <i>SPACE PROGRAMS AND TECHNOLOGY</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	175.105	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
SPC-01: <i>SPACE PROGRAMS AND TECHNOLOGY</i>	0.000	175.105	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Space Programs and Technology Program that addresses high payoff opportunities to dramatically reduce costs associated with advanced space systems and provides revolutionary new system capabilities for satisfying current and projected military missions.

A space force structure that is robust and resilient to attack represents a stabilizing deterrent against adversary attacks on space assets. This program element will examine concepts and architectures that move the U.S. away from a dependence on monolithic, ultra-capable, vulnerable, and unsustainably costly assets; replacing them with disaggregated assets that are agile, affordable, and easily replaced. Ready access to space requires the delivery of capabilities, replenishment of supplies into orbit, and rapid manufacturing of affordable space capabilities. In addition, developing space access and spacecraft servicing technologies will lead to reduced ownership costs of space systems and new opportunities for introducing technologies for the exploitation of space.

Systems development is also required to increase the interactivity and functionality of space systems, space-derived information, and services with terrestrial users. Studies under this program element include technologies and systems that will enable satellites and microsatellites to operate more effectively by increasing maneuverability, survivability, and situational awareness, and precision control of multi-payload systems. Studies will actively seek to take advantage of new commercial developments which may enable both rapid constitution/reconstitution of assets, and agility/functionality not previously available for military systems.

Beginning in FY 2026, efforts in this PE will be funded in PE 0603468E, Advanced Complex Systems.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency				Date: April 2026		
Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)		R-1 Program Element (Number/Name) PE 0603287E / SPACE PROGRAMS AND TECHNOLOGY				
B. Program Change Summary (\$ in Millions)		FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget		181.498	0.000	0.000	-	0.000
Current President's Budget		175.105	0.000	0.000	-	0.000
Total Adjustments		-6.393	0.000	0.000	-	0.000
• Congressional General Reductions		0.000	0.000			
• Congressional Directed Reductions		0.000	0.000			
• Congressional Rescissions		0.000	0.000			
• Congressional Adds		0.000	0.000			
• Congressional Directed Transfers		0.000	0.000			
• Reprogrammings		-0.515	0.000			
• SBIR/STTR Transfer		-5.878	0.000			
Change Summary Explanation						
FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.						
FY 2026: N/A						
FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.						
C. Accomplishments/Planned Programs (\$ in Millions)				FY 2025	FY 2026	FY 2027
Title: Otter				86.015	-	-
Description: The Otter program is developing and demonstrating air breathing propulsion technologies that enable operations in very low earth orbital domains that are currently inaccessible. Propulsion capabilities demonstrated will provide increased mission duration and ability to maneuver without regret. Key efforts include the development of new propulsion systems, improved ground test capabilities, and analysis tools to support system development. Otter will progress through development of analysis and test tools, design of candidate propulsion systems, ground testing, build of a demonstrator satellite, and culminate in a long duration (> 1 year) spaceflight demonstration. The anticipated transition partner is the U.S. Space Force.						
Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.						
Title: Demonstration Rocket for Agile Cislunar Operations (DRACO)				61.579	-	-
Description: Maintaining U.S. interests in cislunar space requires significant advances in propulsion technology. Current space propulsion includes electric (high efficiency but low thrust) and chemical (high thrust but low efficiency) systems. The Demonstration Rocket for Agile Cislunar Operations (DRACO) program would have developed and demonstrated a High-Assay Low-Enriched Uranium (HALEU) nuclear thermal rocket (NTR) system on orbit and would have targeted an FY 2027 launch. The NTR technology demonstrated by DRACO would have achieved thrust similar to chemical rockets, but with 2-5 times the						

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>		R-1 Program Element (Number/Name) PE 0603287E / <i>SPACE PROGRAMS AND TECHNOLOGY</i>		
C. Accomplishments/Planned Programs (\$ in Millions) efficiency. The enhanced performance afforded by NTR would have allowed the U.S. to lead operations in the cislunar volume, in particular for missions that require flexible mission planning and agile maneuverability. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		FY 2025	FY 2026	FY 2027
Title: Robotic Servicing of Geosynchronous Satellites (RSGS) Description: A large number of national security and commercial space systems operate at geosynchronous earth orbit (GEO), providing persistence and enabling ground station antennas to point in a fixed direction. Technologies for servicing of GEO spacecraft would involve a mix of highly automated and remotely operated (from Earth) robotic systems. The Robotic Servicing of Geosynchronous Satellites (RSGS) program is establishing the capability to provide robotic services in GEO suitable for a variety of potential servicing tasks, in full collaboration and cooperation with existing satellite owners and national security space operators, and with sufficient propellant for several years of follow-on capability. Key RSGS challenges include robotic tool/end effector requirements, efficient orbital maneuvering of a servicing vehicle, robotic arm systems, automation of certain spacecraft operations, and development of the infrastructure for coordinated control between the servicer and client spacecraft operations teams. The transition agreement is with a commercial partner who will provide the satellite to carry the robotic payload and who will operate the robotic servicer. To support the development of a broadly accepted satellite servicing capability, DARPA is using the Consortium for Execution of Rendezvous and Servicing (CONFERS) operations approach to bring together experts from the private sector and Government to research, develop and publish nonbinding, consensus-based standards for safe operational approaches to on-orbit servicing. Beginning in FY 2026, this program will be funded in PE 0603468E, Project ACX-01.		15.504	-	-
Title: Advanced Space Technology Concepts Description: Studies conducted under this program examined and evaluated emerging technologies and concepts with the potential to provide substantial improvement in efficiency, effectiveness, and resilience of operations in space. Topics that were researched and analyzed included the degree and scope of potential impact and improvements to military operations, mission utility, and warfighter capability. Studies also analyzed emerging threats along with possible methods and technologies for countermeasures. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, was also evaluated. The results from these studies were used, in part, to formulate future programs or refocus ongoing work. Topics of consideration included applying artificial intelligence to low earth orbit (LEO) constellation operations to enable collaboration between space, air, maritime, and ground platforms in anti-access/area denial (A2/AD) theaters; robust architectures for precision navigation and timing; enabling operations in Cislunar space; novel approaches to space domain awareness; integration of commercial capabilities into military operations; and on-orbit software environments.		12.007	-	-

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide I BA 3: Advanced Technology Development (ATD)		R-1 Program Element (Number/Name) PE 0603287E I SPACE PROGRAMS AND TECHNOLOGY		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Accomplishments/Planned Programs Subtotals		175.105	-	-
D. Other Program Funding Summary (\$ in Millions) N/A				
Remarks				
E. Acquisition Strategy N/A				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>					R-1 Program Element (Number/Name) PE 0603467E / <i>DARPA ADVANCED TECHNOLOGY DEVELOPMENT</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	1,528.100	1,568.124	-	1,568.124	1,142.182	1,198.039	1,126.657	1,068.912	-	-
DAT-01: <i>ADVANCED SYSTEMS</i>	0.000	0.000	357.490	407.285	-	407.285	341.217	586.088	682.441	696.035	-	-
DAT-02: <i>ISR AND SENSORS</i>	0.000	0.000	104.820	126.983	-	126.983	251.696	321.994	339.081	299.081	-	-
DAT-06: <i>DARPA ADVANCED TECHNOLOGY DEVELOPMENT</i>	0.000	0.000	974.036	919.617	-	919.617	505.217	245.105	59.163	27.824	-	-
DAT-07: <i>DARPA ADVANCED TECHNOLOGY DEVELOPMENT SUPPORT</i>	0.000	0.000	91.754	114.239	-	114.239	44.052	44.852	45.972	45.972	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$1,528,100 thousand of discretionary and \$90,000 thousand of mandatory (reconciliation) for a total of \$1,618,100.

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with Systems, Intelligence, Surveillance and Reconnaissance (ISR) and Sensors that addresses high payoff opportunities to develop and rapidly mature advanced technologies required for today's warfare concepts. It is imperative that future U.S. forces operate seamlessly in all domains with improved battlefield awareness, strike capability and battle damage assessment. This PE also supports innovation and robust transition and experimentation planning in the technology cycle to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Advanced Systems project will create enabling technologies for seamless all domain operations, from strategic planning to tactical and urban operations. All Domain Warfare Systems leverage current and emerging network, robotic, and information technology and provide next generation U.S. forces with greatly increased capability, lethality, and rapid responsiveness. Critical issues facing this project are: (1) U.S. opponents using systems that are flexible, robust, and difficult to neutralize; and (2) U.S. doctrine that limits the use of firepower to lessen the impact of operations on noncombatants. Meeting these challenges places a heavy burden on joint war planning. Understanding opponent networks is essential so that creative options can be developed to counter their strategies. Synchronization of air, sea, space, and ground operations to apply force only where needed and with specific effects is required. This project supports all levels of the force structure including: (1) the strategic/operational level by generating targeting options against opponents' centers of gravity that have complex networked relationships; (2) the tactical/operational level by managing highly automated forces with tight coupling between air, seas, space, and ground platforms; and (3) the focused tactical level by developing platforms and tools, which acquire targets of opportunity and cue network-based analysis of likely enemy operations.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603467E / <i>DARPA ADVANCED TECHNOLOGY DEVELOPMENT</i>
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The ISR and Sensors project develops technologies to improve surveillance and targeting systems for improved battlefield awareness, strike capability and battle damage assessment, and demonstrates these technologies and system concepts against emerging threat scenarios to support rapid technology adoption by the warfighter. Leveraging technology advancements in size, weight, power and cost and enabling with the practical application of mature machine learning and artificial intelligence, these efforts are also packaged in novel ways for rapid deployment and adoption for operational use. Capabilities are intended to overcome an adversary's intent to deny and deceive sensing and ISR systems in order to provide all-weather continuous coverage over areas of interest. This project will exploit recent advances in multispectral target phenomenology, signal processing, low-power high-performance computing, and low-cost microelectronics to develop advanced surveillance and targeting systems.

The DARPA Advanced Technology Development Support project contains management costs in support of DARPA functions and activities across the entire DARPA Advanced Technology Development PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0603760E, Command, Control and Communications Systems, PE 0603766E, Network-Centric Warfare Technology and PE 0603767E, Sensor Technology.

B. Program Change Summary (\$ in Millions)	<u>FY 2025</u>	<u>FY 2026</u>	<u>FY 2027 Base</u>	<u>FY 2027 OOC</u>	<u>FY 2027 Total</u>
Previous President's Budget	0.000	1,643.465	0.000	-	0.000
Current President's Budget	0.000	1,528.100	1,568.124	-	1,568.124
Total Adjustments	0.000	-115.365	1,568.124	-	1,568.124
• Congressional General Reductions	0.000	-97.545			
• Congressional Directed Reductions	0.000	-17.820			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	1,568.124	-	1,568.124

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT				Project (Number/Name) DAT-01 / ADVANCED SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
DAT-01: ADVANCED SYSTEMS	0.000	0.000	357.490	407.285	-	407.285	341.217	586.088	682.441	696.035	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Systems project will create enabling technologies for seamless all domain operations, from strategic planning to tactical and urban operations. All Domain Warfare Systems leverage current and emerging network, robotic, and information technology and provide next generation U.S. forces with greatly increased capability, lethality, and rapid responsiveness. Critical issues facing this project are: (1) U.S. opponents using systems that are flexible, robust, and difficult to neutralize; and (2) U.S. doctrine that limits the use of firepower to lessen the impact of operations on noncombatants. Meeting these challenges places a heavy burden on joint war planning. Understanding opponent networks is essential so that creative options can be developed to counter their strategies. Synchronization of air, sea, space, and ground operations to apply force only where needed and with specific effects is required. This project supports all levels of the force structure including: (1) the strategic/operational level by generating targeting options against opponents' centers of gravity that have complex networked relationships; (2) the tactical/operational level by managing highly automated forces with tight coupling between air, seas, space, and ground platforms; and (3) the focused tactical level by developing platforms and tools, which acquire targets of opportunity and cue network-based analysis of likely enemy operations.

Prior to FY 2026, efforts in this Project were funded in PE 0603766E, Projects NET-01 and NET-02, and PE 0603739E, Project MT-15.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Advanced Propulsor, Experimental (APEX)	-	61.827	73.391
Description: Current submarine propulsor and propeller designs impose constraints on ship layouts and maneuvering capabilities. The Advanced Propulsor, Experimental (APEX) program is developing and demonstrating a new generation of submarine propulsor designs enabling revolutionary improvements in submarine design, maneuverability, speed, and quieting that will transform future submarine designs. The anticipated transition is to the Navy. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Complete motor risk reduction assembly testing and incorporate lessons learned. - Complete testing for the smaller scale demonstrator article and incorporate lessons learned. - Execute Preliminary Design Review (PDR) for larger scale demonstrator vehicle. - Complete detailed design of the demonstrator vehicle. - Purchase long-lead items for demonstrator vehicle. - Initiate manufacture of the demonstrator vehicle.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Conduct subsystem modeling, simulation, and analysis activities. - Conduct High Performance Computing (HPC) performance modeling to validate performance. - Complete the development, building, and testing of the quarter scale demonstrator article subsystems. FY 2027 Plans: - Execute Critical Design Review (CDR) for larger scale demonstrator vehicle. - Purchase materials for manufacturing of the larger scale demonstrator vehicle. - Continue to test software control system for demonstrator. - Continue to update the software control system for demonstrator. - Continue manufacture of the demonstrator vehicle. - Continue to build the larger scale demonstrator article subsystems. - Continue to test the larger scale demonstrator article subsystems. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects increase in manufacturing larger subsystems for the larger scale demonstrator.				
Title: Embedded Entrepreneurship Initiative (EEI) Description: EEI pairs top technical talent from DARPA programs with experienced business leaders to accelerate the transition of breakthrough technologies to market. The program provides hands-on go-to-market support, equipping teams with the tools, expertise, and strategies needed to succeed in both government and commercial sectors. By fostering strong commercialization roadmaps, EEI helps DARPA awardees attract U.S. investment and scale their innovations. FY 2026 Plans: - Prevent problematic foreign investment in strategic DARPA-funded technologies. - Accelerate transition of disruptive DARPA technologies to the warfighter and other United States Government (USG) entities. - Address market inefficiencies that prevent DARPA-funded technologies from transitioning to the warfighter and/or commercial markets. FY 2027 Plans: - Continue prevention activities addressing problematic foreign investment in strategic DARPA-funded technologies. - Continue to accelerate transition of disruptive DARPA technologies to the warfighter and other USG entities. - Continue to address market inefficiencies that prevent DARPA-funded technologies from transitioning to the warfighter and/or commercial markets.		-	25.000	25.000
Title: Archer		-	23.976	20.763

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: The Archer program will enable the production of a new cross-domain resilient effector capability to provide a strategic advantage over adversaries. Current capabilities rely on high-value capital assets to employ. The Archer program will provide an alternative solution which will enable capital assets to perform their other critical taskings. Building on technologies developed in various domains, Archer will produce hardware, conduct Independent Verification and Validation (IV&V), and perform iterative testing to create this capability. Technology developed under this program will be transitioned to the Services. FY 2026 Plans: - Evaluate current state-of-the-art technologies to identify the most promising for future development. - Conduct a study to categorize the viability of components to be incorporated with the capability. - Conduct system design based on initial evaluations and studies. - Perform subscale component-level testing. - Conduct performance testing in representative environments. FY 2027 Plans: - Evaluate long duration subscale component performance. - Execute performer-based incremental test plan to demonstrate approach viability. - Conduct graded component evaluations to demonstrate progress towards critical technical challenges. - Integrate viable components into full system. - Demonstrate system capability in controlled environment with traceability to reaching program metrics. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from subscale component-level testing to preparation for final system demonstration.				
Title: Pulling Guard Description: The Pulling Guard program will revolutionize Freedom of Navigation escort operations. The program will create semi-autonomous, modular, point-defense overwatch/escort systems, and an ecosystem that commercially delivers these escort services in partnership with the Services. Pulling Guard will dramatically reduce risk and insurance costs for logistics platforms in peacetime (piracy and terror) and wartime (blockades), while increasing large combatant availability for other critical missions. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Conduct iterative risk reduction through build and test activities supporting design validation. - Mature business models to enable dual commercial/military investment for long-term sustainment. - Address key policy/legal issues through active engagements.		-	38.375	19.943

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Conduct laboratory and on-water engineering tests of critical hardware components. FY 2027 Plans: - Conduct Overwatch demonstration for the sensing and sensemaking components. - Conduct design review for Platform Transition to production. - Initiate construction of platforms with sensing and sensemaking system and conduct interoperability test. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a planned down select from five performer teams to two integrated performer teams with expectation of protection-as-a-service commercial investment.				
Title: Midway Description: The Midway program will develop and demonstrate a communication-based payload distributed onboard mobile platforms in order to enable long range transmission and reception capabilities. The program will leverage commercially available and modified commercial-off-the-shelf (COTS) components to create a resilient, rapidly deployable, and scalable architecture for data transport. This program culminates in a full-scale tactical demonstration that validates the critical technologies in a representative environment and reduces technical, schedule, and cost risk for a follow-on operational system. FY 2026 Plans: - Initiate communications architecture and vehicle sizing studies. FY 2027 Plans: - Complete communications architecture and vehicle sizing studies. - Initiate host platform fabrication and evaluation. - Conduct payload integration and ground qualification testing. - Demonstrate payload functionality in a representative environment. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from platform and payload design to fabrication and integration activities.		-	7.048	39.603
Title: Deep Thoughts Description: Deep Thoughts will expand access in the deep ocean domain by developing a new class of novel lightweight, low-cost, and rapidly designed, produced and deployed deep-ocean Autonomous Undersea Vehicles (AUVs). The program aims to harness breakthroughs in advanced materials, manufacturing, and next-generation structural and mechanical design technologies to dramatically reduce the volume and mass of deep-ocean systems, while eliminating dependence on traditional long-lead components. The foundational goal of Deep Thoughts is to achieve a neutrally buoyant, deep-ocean AUV without relying on syntactic foam. Innovation in pressure-vessel design, including alternative materials, configurations, architectures, and		-	4.128	15.800

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
manufacturing techniques are foundational elements for Deep Thoughts. Key performance metrics, aside from size (e.g., mass and volume), include significantly reducing the cost and development time for new systems while enabling rapid adaptability for incorporating diverse payloads, capabilities, and new use cases. FY 2026 Plans: - Initiate foundational pressure vessel prototype design and fabrication. FY 2027 Plans: - Conduct testing of prototype pressure vessels. - Complete designs for integration of AUV components and subsystems within pressure vessels. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects a shift from design and fabrication to integration and testing.			
Title: Time-definite Capability Delivery Description: As an outgrowth from the success of Awareness in Joint Warfighting Technology, the Time-definite Capability Delivery effort will plan, research, and execute activities designed to field capabilities based on proven technologies on relevant timelines and at sufficient quantities. The effort will achieve this objective by planning and coordinating virtual, live, constructive modelling and simulation in conjunction with in-theater demonstrations and experimentation to help orient programs on acute warfighting problems. Additionally, the effort will establish industry and policy-oriented working groups, host technical exchanges and exercises, conduct research, and develop analytic tools that will explore systemic options to speed time to field and ramp production to relevant volumes to support national security objectives. The effort will probe the boundaries of the current institutional processes through rapid experimentation, novel transition frameworks, and technology analysis to provide proof points that rationally challenge existing process boundaries and identify ways to complement, adapt, or replace institutional processes in order to deliver operationally relevant capabilities on strategically relevant timescales. Time-definite Capability Delivery will create frameworks to evaluate technology investment, policy changes, and other government levers to generate competitive advantage. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-01. FY 2026 Plans: - Continue development of a systems of systems level model of the U.S. defense industrial base, focused on key sectors using systems dynamics modeling. - Conduct technical exchange forums with industry, DoW, and interagency partners to validate modeling assumptions and co-design industrial incentives. FY 2027 Plans:		-	8.000
			8.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Expand use of the systems dynamics modeling environment to simulate the effects of regulatory changes, policy flexibilities, and capital interventions on production ramp timelines. - Continue technical exchanges and simulations to validate and refine the systems level model of the industrial base with cross-sector stakeholders.				
Title: Resilient Supply-and-Demand Networks (RSDN) Description: The Resilient Supply-and-Demand Networks (RSDN) program is developing supply-chain risk management analytics to detect systemic challenges and improve resilience in supply and demand networks (SDN). At present, the separation of supply-chain information into confidential silos obscures a system-wide view, inhibiting comprehensive risk-focused analysis of supply and demand networks. RSDN is developing techniques for modeling and stress-testing real-world SDNs, including network analytics and visualizations, to reveal emerging fragilities and enable deep situational awareness of systemic risks and potential disruptions. An RSDN stress-testing framework enables repeatable scenario analysis of strategic issues in supply and demand networks, automated analysis and discovery of risk patterns, and evaluation of alternative risk mitigation strategies. Prior to FY 2026, this program was funded in PE 0602702E, Project TT-13. FY 2026 Plans: - Optimize forward and backward stress-testing capabilities for high priority use cases. FY 2027 Plans: - Collaborate with Department of War stakeholders to transition technologies to critical supply and demand network domains.		-	4.060	4.060
Title: FLexible networking Using Intelligent Dialecting (FLUID) Description: FLexible networking Using Intelligent Dialecting (FLUID) will enable Command, Control, Computing, Communications, Cyber, Intelligence, Surveillance, Reconnaissance, and Targeting (C5ISRT) under extremely constrained internet protocol (IP) network environments, including those with link capacities in the range from 10s to single digit Kilobits per second. FLUID will research technologies that reduce network and user communications with semantically correct but condensed data, application, and/or protocol information. FLUID will apply intelligent mechanisms to reduce the link capacity required for C5ISRT systems to operate at target scale. Research will focus on extracting and communicating compressed but semantically similar bits from user data, application messages, and network protocol messages to fit within the network capacity constraints; thus, creating the potential for greatly increased efficiency at and across the tactical edge. Software developed under the program will transition to the Services. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-01.		-	13.096	2.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
FY 2026 Plans: - Continue research and design of initial semantically correct user data, application, and network reduction technologies. - Conduct data collection and analysis. - Conduct laboratory and exercise experimentation to validate support to service operations. - Integrate data plane reduction technologies into transition partner applications of interest. - Continue integration into existing Service-specific systems. FY 2027 Plans: - Demonstrate full system integration in operational demonstration with transition partners. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from integration and initial testing to conducting a final operational demonstration with transition partners.			
Title: Box Kick Description: The Box Kick program is developing a capability to engage and defeat raids of adversary subsonic cruise missiles and unmanned aerial systems at low cost while preserving more capable and expensive weapons for higher-end threats. The program will integrate this capability into existing platforms with little-to-no platform modifications required. Technology developed under this program is planned for transition to the Services. Prior to FY 2026, this program was funded in PE 0603739E, Project MT-15. FY 2026 Plans: - Complete safety of flight and safe separation activity. - Perform flight test and a mission system integration test series. - Conduct necessary development and testing activities to incorporate design modification based on constraints identified from previously executed tests. - Plan a series of Capstone Test Events with transition Stakeholders. FY 2027 Plans: - Conduct a series of Capstone Test Events with transition Stakeholders. - Complete validation of the bulk prototype process and provide Lots to transition partners. FY 2026 to FY 2027 Increase/Decrease Statement:		-	39.000
			20.000

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects the shift from system testing and initial flight tests to Capstone test events with transition Stakeholders			FY 2027
Title: Advanced Systems Studies and Concepts Description: Studies conducted under this program will explore emerging technologies, prototypes, and system concepts for applicability to military use. This includes the degree and scope of potential impact and improvements to military operations, mission utility, and warfighter capability. Research will also foster innovative system solutions and architectures at all levels of military operations and enable tight coupling of platforms across land, air, sea, and space domains. Desired effects include reducing potential adversaries' labor advantage by leveraging automation, modeling, and simulation technologies in support of system design and development. Topics include reshaping design tools and processes enabled by automation to actively manage engineering uncertainty, accelerated fixture development enabled by physics-informed modeling, and accelerated verification analytics for decisions-making, authoring, testing and certification enabled by state-of-the-art machine automation. This program also includes funding for potential classified program initiations. FY 2026 Plans: - Develop new methods of design space exploration that redefine system architectures, provide new insights into higher order interactions, and enable tailored testing. - Initiate research that enables seamless cross-domain communications into, within, and out of contested environments. - Initiate research into a program to demonstrate a semi-autonomous platform that provides sensing, tracking, targeting, and engagement capability. - Develop platforms for sub-domain access to enable novel access and operational envelopes. - Explore and refine advanced system concepts to deliver integrated, cross-domain platform autonomy at disruptive scale and unprecedented penetration capabilities. - Explore automated advanced sensing systems that leverage multi-modal computer vision and can be deployed on uncrewed vehicles and with squad sized warfighting units. - Explore concepts to provide a low-cost, easy-to-integrate, large-scale national infrastructure sensing system. - Examine concepts and technology to prevent the impacts, degradation and disruption of port operations. - Investigate rapid and ad hoc mechanisms for dynamic, composable, and adaptable uncrewed swarm systems. FY 2027 Plans: - Initiate research approaches to acquiring new defense systems, including disrupting systems engineering and reimaging test and certification. - Develop new design approaches to make manufacturable designs or to dramatically reduce human engineering effort. - Develop new approaches to fabrication that enable rapid start-up, reduced production lead times, and frequent changes to military systems.		-	80.391
			178.725

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Test and validate approaches that dramatically shorten the timeline from initial delivery to full military utility. - Explore tools and processes enabled by automation to actively manage engineering uncertainty. - Accelerate fixture development enabled by physics-informed modeling. - Explore methods to accelerate analytics, decision-making, authoring, testing and certification enabled by Large Language Model (LLM) based automation. - Investigate ways to develop a game-play engine and reinforcement learning techniques to support course of action (COA) development. - Define simulation scenarios for COA inputs. - Demonstrate game-play engine and course of action (COA) generation for surrogate operational vignettes. - Integrate higher level intelligence products in COA generation engine. - Investigate new technologies for capturing space debris to include robotics, propulsion and remote pilot operation systems. - Determine feasibility of integrating novel technologies into spacecraft architecture and mission concepts and how to enable rapid deployment of spacecraft. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to capture space debris, new tools and processes to actively manage engineering uncertainty, and novel processes and architectures to enable the rapid manufacture and deployment of spacecraft.			
Title: Goblin Description: The undersea domain has significant importance to national security and military operations, but as in every domain, manned missions have some limitations. The Goblin program will enhance U.S. autonomous capabilities in the challenging undersea domain by developing and demonstrating complex underwater systems able to search, locate, and execute mission objectives without the need for human control. Navigation approaches will focus on the use of commercial, low-cost navigation hardware combined with environmental feature-based algorithm approaches to eliminate reliance on the Global Positioning System (GPS) for long-duration missions. Key Goblin technical challenges include sensing techniques that provide high-resolution navigation without GPS, perception and effector strategies for objects with unknown parameters, long-duration autonomy approaches to support mission execution, and autonomy approaches that do not rely on human interaction. The anticipated transition is to the Navy. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Complete integration and testing of Goblin payloads. - Conduct final at-sea technology demonstration of the full Goblin system in support of transition objectives.		-	3.990
			-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Deliver the Goblin system to the Navy for further maturation and fielding.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: No Manning Required Ship (NOMARS) Description: No Manning Required Ship (NOMARS) is developing small, low-cost, disaggregated naval platforms to demonstrate the ability to perform persistent power projection and force application combat missions currently conducted from large, high value capital ships. The NOMARS program entails designing, building, and sailing a ship that can operate autonomously for long durations at sea, enabling a ship design process that eliminates considerations associated with crew. NOMARS focuses on exploring novel approaches to the design of the sea frame (the ship without mission systems) while accommodating representative payload size, weight, and power. The goal of the program is to demonstrate the feasibility of Unmanned Surface Vessels (USVs) that operate autonomously for months to years without human intervention, in large numbers, with only periodic, depot-based maintenance. This capability will enable disaggregated persistent USVs, allowing the surface fleet to credibly threaten peer adversaries and negate their investments in high-cost weapon systems designed to counter large naval targets such as aircraft carriers. A successful NOMARS program will prove feasibility of a small, unmanned ship with significantly improved reliability and functional performance over current USVs providing a pathway to fully realize the greatest potential of distributed lethality concepts: small ships, in large numbers, each of which is individually low-cost and low value, but in aggregate presents a significant deterrent and warfighting advantage. The transition partner is the Navy. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Complete at-sea demonstrations. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	4.000	-
Title: Scorpionfish Description: The Scorpionfish program will design and test novel undersea weapon technologies. Successful execution of the program will be a revolutionary step in the execution of missions in the undersea domain. The anticipated transition is to the Navy. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Conduct risk reduction activities supporting development of demonstrator systems.		-	13.100	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Perform trade studies for system integration and deployment. - Procure components and assemble demonstrator systems. - Conduct in-water engineering tests of critical hardware components. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Title: Trenton Description: The Trenton program is enabling the production of a wide range of structural objects and underwater repairs through the optimization of three-dimensional concrete printing (3DCP) formulations and fabrications at depth. Trenton is developing the first underwater 3DCP (U3DCP) formulations utilizing in-situ materials to support various marine engineering needs. Current technologies rely on land-based techniques that require forming on land then deployment by barge or ship. Trenton is overcoming this challenge by developing the formulations and fabrication technologies necessary for U3DCP. Technology developed under this program will transition to the Services and commercial partners. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Conduct capabilities testing for underwater printing of simple structures in laboratory settings. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	2.872
Title: Willow Description: The Willow program is developing innovative payloads to conduct Acoustic Warfare (AW) to counter sonars using a unique combination of acoustic hardware and waveforms provided by advanced sonar signal processing algorithms. Willow provides a robust capability to help the Navy respond to sonar threats. This is a novel method for challenging adversary sonars. Willow uses advanced hardware-in-the-loop simulations, Independent Verification and Validation (IV&V), and stressing at-sea testing to create this capability. Technology developed under this program will transition to the Navy. Prior to FY 2026, this program was funded in PE 0603766E, Project NET-02. FY 2026 Plans: - Continue IV&V activities to verify performer updates to simulations, hardware and waveforms based on lessons learned. - Conduct Final Design Review of acoustic systems and other technologies prior to at-sea demonstration. - Conduct at-sea demonstration with prototype integrated system.		-	28.627

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-01 / ADVANCED SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Conduct interim in-water assessment of updated system design to verify additional capabilities. - Conduct final at-sea demonstration with updated systems. - Coordinate with Navy partners to enable system transition.			
<i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	357.490
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-02 / ISR AND SENSORS
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
DAT-02: ISR AND SENSORS	0.000	0.000	104.820	126.983	-	126.983	251.696	321.994	339.081	299.081	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The ISR and Sensors project develops technologies to improve surveillance and targeting systems for improved battlefield awareness, strike capability and battle damage assessment, and demonstrates these technologies and system concepts against emerging threat scenarios to support rapid technology adoption by the warfighter. Leveraging technology advancements in size, weight, power and cost and enabling with the practical application of mature machine learning and artificial intelligence, these efforts are also packaged in novel ways for rapid deployment and adoption for operational use. Capabilities are intended to overcome an adversary's intent to deny and deceive sensing and ISR systems in order to provide all-weather continuous coverage over areas of interest. This project will exploit recent advances in multispectral target phenomenology, signal processing, low-power high-performance computing, and low-cost microelectronics to develop advanced surveillance and targeting systems.

Prior to FY 2026, efforts in this Project were funded in PE 0603767E, Projects SEN-01 and SEN-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Track at Big Distances with Track-Before-Detect (TBD2)	-	11.276	3.952
Description: The Track at Big Distances with Track Before Detect (TBD2) program will enable continuous space-based detection and tracking of objects in cislunar space on relevant timelines, thereby increasing the safety of cislunar commercial and civilian traffic for the benefit of all nations and enabling a sustainable space ecosystem. Existing space situational awareness capabilities are primarily focused on objects in geosynchronous orbit (or closer) and are not optimized to scan the much greater volume of space or detect objects at the greater distances required for cislunar space. Building on technologies developed in various domains, the TBD2 program will develop signal processing algorithms and payload designs necessary to enable detection from space of faint objects in space. Technology developed under this program will be transitioned to the U.S. Air Force or U.S. Space Force.			
FY 2026 Plans: - Develop low complexity algorithms for cislunar space situational awareness. - Conduct payload design trade study. - Conduct midterm assessment of algorithms.			
FY 2027 Plans: - Assess algorithms for accuracy and computational power required.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3		R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT		Project (Number/Name) DAT-02 / ISR AND SENSORS
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Assess payload design trade study for power consumption and revisit times.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects transition from technology development to final assessment.				
Title: Cancun Description: The Cancun program will create distributable nodes to measure the radio high frequency (HF) environment for improved warfighter situational awareness. Cancun will enable cost-effective wide-area deployment of low size, weight, power, and cost (SWaP-C) nodes. Cancun will also develop the command and control (C2) network and planning tools required to address the challenge of coordinating large numbers of Cancun nodes deployed over distances of well over 1000 kilometers. The Cancun nodes will measure the state of the ionosphere using a sounding function, as well as record and relay portions of the HF radio band for analysis. The mission planning tool will be developed with warfighter input to optimize functionality. Technologies developed under the Cancun program will transition to the Services. Prior to FY 2026, this program was funded in PE 0603767E, Project SEN-02. FY 2026 Plans: - Purchase hardware and scale node production for follow-on field tests. - Incorporate warfighter feedback for improvement of the mission planning tool. - Integrate full system for at-scale demonstration of hardware, software, and algorithmic techniques. - Deploy equipment and conduct field test and demonstration of integrated distributed system. FY 2027 Plans: - Deploy equipment and test in relevant environment with transition partners. - Transition Cancun technology. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects a shift from integration to final testing and transition.		-	23.908	1.700
Title: Frosty Description: The Frosty program will enable reliable long-range sensing in the Arctic radio frequency (RF) environment. The remoteness and large area of the Arctic favor the use of high frequency (HF) over-the-horizon (OTH) sensing methods, but the challenging propagation channel due to ionospheric turbulence requires new approaches to sensor configuration and signal processing to obtain the desired reliable all-day all-weather sensing. Approaches have been proposed to improve low-altitude target tracking including space-based radars and long-range skywave OTH radar (OTHR); however, because of how the Earth's upper atmosphere interacts with solar wind, the Arctic is often a hostile radio propagation environment that degrades both of these		-	14.217	18.234

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-02 / ISR AND SENSORS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
two approaches. The Frosty program seeks to provide an alternative to these traditional approaches through stand-off illumination and innovative signal processing to create a radar sensor that operates reliably in this propagation environment. Technology developed under this program will be transitioned to the Services. FY 2026 Plans: - Perform initial field data collection in relevant environment to validate physics models and phenomenology. - Develop algorithm and software development roadmap. - Using data from tests, develop algorithms and software to produce initial target tracks. FY 2027 Plans: - Complete initial algorithm development and perform independent assessment. - Develop conceptual design for a fieldable Frosty prototype. - Update algorithms and demonstrate improved performance. - Deliver prototypes of Frosty software stack. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase is due to a shift from initial development to assessments and prototyping.				
Title: ISR and Sensors Studies and Concepts Description: The ISR Sensors Studies and Concepts will design and demonstrate advanced sensing systems and countermeasure technologies that provide novel insights to inform unique future capabilities and expand capabilities into new areas of operation. Efforts will emphasize improvements to size, weight, and performance to extend endurance, range, and precision; advance autonomous operations; and reduce costs to maximize system coverage and provide operational capability at greater speed and scale. Challenges that will be overcome include extended operations without the need for supporting infrastructure, continued operations in harsh physical environments, and extended persistent operations in contested environments. Technology advancements will support resilience and enable strategic sensor countermeasures. FY 2026 Plans: - Conduct data collection to support concept development for imaging across the frequency spectrum. - Generate preliminary low-resolution and high-resolution images for further study. - Identify system models and software to integrate models with a sensor suite. - Develop robust sensor to test functionality and survivability in austere environments. - Demonstrate tactical use-case scenarios. - Investigate algorithms and modeling simulations for delivery. - Examine robust sensors to test functionality and survivability in austere environments.		-	35.604	103.097

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-02 / ISR AND SENSORS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Complete development of a curved photocathode reference design with high quantum efficiency. - Complete development of a test stand for sub-scale photocathode evaluation. - Iteratively produce and test high quantum efficiency photocathode samples. - Update and finalize a curved photocathode reference design using data from photocathode sample characterization. - Scale manufacturing of high quantum efficiency photocathodes to large areas and develop photocathode assembly techniques.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: Dynamic Optimization for Defense of Ground bases with Electromagnetic warfare (DODGEball) Description: The Dynamic Optimization for Defense of Ground bases with Electromagnetic warfare (DODGEball) program will develop algorithms for optimization of non-kinetic countermeasures for efficient and effective resource management in extended campaign warfare. DODGEball will optimize heterogeneous applications of electromagnetic warfare for the defense of surface forces and infrastructure for long-duration campaigns. Technology developed under this program will transition to the Services. Prior to FY 2026, this program was funded in PE 0603767E, Project SEN-01.		-	8.490	-
FY 2026 Plans: - Incorporate countermeasure and feedback parameters into optimization algorithms. - Finalize optimization algorithms and subsystem designs. - Complete the characterization and evaluation of countermeasures and feedback systems in hardware-in-the-loop (HWIL) testing. - Evaluate final HWIL-validated models in a government-provided laboratory simulation environment. - Coordinate technology transition to the Services.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Accomplishments/Planned Programs Subtotals		-	104.820	126.983
C. Other Program Funding Summary (\$ in Millions)				
N/A				
Remarks				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT	Project (Number/Name) DAT-02 / ISR AND SENSORS
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TECHNOLOGY DEVELOPMENT				Project (Number/Name) DAT-06 / DARPA ADVANCED TECHNOLOGY DEVELOPMENT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
DAT-06: DARPA ADVANCED TECHNOLOGY DEVELOPMENT	0.000	0.000	974.036	919.617	-	919.617	505.217	245.105	59.163	27.824	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

This project funds classified DARPA programs that are reported in accordance with the SAP Annual Report to Congress.

Prior to FY 2026, efforts in this Project were funded in PE 0603760E, Project CCC-06, PE 0603766E, Project NET-06, and PE 0603767E, Project SEN-06.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603467E / DARPA ADVANCED TEC HNOLOGY DEVELOPMENT				Project (Number/Name) DAT-07 / DARPA ADVANCED TECHNOLOGY DEVELOPMENT SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
DAT-07: DARPA ADVANCED TECHNOLOGY DEVELOPMENT SUPPORT	0.000	0.000	91.754	114.239	-	114.239	44.052	44.852	45.972	45.972	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The DARPA Advanced Technology Development Support project contains management costs in support of DARPA functions and activities across the entire DARPA Advanced Technology Development PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0603766E, Projects NET-01 and NET-02, and PE 0603767E, Projects SEN-01 and SEN-02.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>					R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	263.461	540.362	-	540.362	787.731	847.968	845.657	840.771	-	-
ACX-01: <i>ADVANCED COMPLEX SYSTEMS</i>	0.000	0.000	243.882	500.098	-	500.098	702.156	760.996	768.081	763.195	-	-
ACX-07: <i>ADVANCED COMPLEX SYSTEMS SUPPORT</i>	0.000	0.000	19.579	40.264	-	40.264	85.575	86.972	77.576	77.576	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Advanced Complex Systems Program that is focused on exploiting high pay-off opportunities to provide revolutionary new system capabilities, as opposed to incremental or evolutionary advancements, in order to achieve undeterrable air and space presence at dramatically reduced costs. Rapid prototyping and experimentation of integrated system concepts, as well as enabling vehicle subsystems will be conducted. Programs will explore new architectural concepts that employ a mix of weapon technologies that achieve lethality through a combination of overwhelming performance and overwhelming numbers rather than through the use of singular and costly high value assets. Efforts conducted under this PE include examination and evaluation of emerging aerospace threats, technologies, concepts, use of autonomy to minimize risk, and applications for missiles, munitions, and vehicle systems. This PE also supports innovation and robust transition and experimentation planning in the technology cycle to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Advanced Complex Systems project is focused on exploiting high pay-off opportunities to provide revolutionary new system capabilities, as opposed to incremental or evolutionary advancements, in order to achieve undeterrable air and space presence at dramatically reduced costs. Rapid prototyping and experimentation of integrated system concepts, as well as enabling vehicle subsystems will be conducted. Programs will explore new architectural concepts that employ a mix of weapon technologies that achieve lethality through a combination of overwhelming performance and overwhelming numbers rather than through the use of singular and costly high value assets. Efforts conducted under this program element include examination and evaluation of emerging aerospace threats, technologies, concepts, use of autonomy to minimize risk, and applications for missiles, munitions, and vehicle systems.

The Advanced Complex Systems Support project contains management costs in support of DARPA functions and activities across the entire Advanced Complex Systems PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0603286E, Advanced Aerospace Systems, and PE 0603287E, Space Programs and Technology.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>
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B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	350.695	0.000	-	0.000
Current President's Budget	0.000	263.461	540.362	-	540.362
Total Adjustments	0.000	-87.234	540.362	-	540.362
• Congressional General Reductions	0.000	-89.162			
• Congressional Directed Reductions	0.000	-3.072			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	5.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	540.362	-	540.362

Congressional Add Details (\$ in Millions, and Includes General Reductions)

Project: ACX-01: *ADVANCED COMPLEX SYSTEMS*

Congressional Add: *Autonomous intercept of live-fire cruise missiles*

Congressional Add Subtotals for Project: ACX-01

Congressional Add Totals for all Projects

FY 2025	FY 2026
-	5.000
-	5.000
-	5.000

Change Summary Explanation

FY 2025: N/A

FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation offset by a Congressional add for autonomous intercept of live-fire cruise missiles.

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS				Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ACX-01: ADVANCED COMPLEX SYSTEMS	0.000	0.000	243.882	500.098	-	500.098	702.156	760.996	768.081	763.195	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Complex Systems project is focused on exploiting high pay-off opportunities to provide revolutionary new system capabilities, as opposed to incremental or evolutionary advancements, in order to achieve undeterrable air and space presence at dramatically reduced costs. Rapid prototyping and experimentation of integrated system concepts, as well as enabling vehicle subsystems will be conducted. Programs will explore new architectural concepts that employ a mix of weapon technologies that achieve lethality through a combination of overwhelming performance and overwhelming numbers rather than through the use of singular and costly high value assets. Efforts conducted under this program element include examination and evaluation of emerging aerospace threats, technologies, concepts, use of autonomy to minimize risk, and applications for missiles, munitions, and vehicle systems.

Prior to FY 2026, efforts in this Project were funded in PE 0603286E, Project AIR-01, and PE 0603287E, Project SPC-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Valley Forge	-	4.602	37.071
Description: The Valley Forge program will develop wholly new, low-cost missiles through a novel approach that enables designing, manufacturing and producing munitions at greater speed and scale. This capability augments the nation's magazine depth and provides capacity adaptability for weapons quantity demands in future conflicts. The program pivots from traditional missile development approaches, which have focused on delivering a pre-determined missile capability, by shifting to an alternate development approach that places manufacturability and producibility first and then determines the resulting capability. Valley Forge leverages readily available commercial industry components and subsystems, advanced manufacturing processes, and automated assembly to deliver missiles at capacities and rates that support national demands.			
FY 2026 Plans: - Initiate missile designs and develop modeling and simulation validation tools.			
FY 2027 Plans: - Conduct propulsion system risk reduction testing. - Conduct avionics system risk reduction testing. - Mature missile designs based on early risk reduction activities. - Develop missile manufacturing plans and cost assessments based on missile designs.			
FY 2026 to FY 2027 Increase/Decrease Statement:			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects the shift from initial design to risk reduction testing and manufacturing plans development.				
Title: Longshot		-	52.660	151.500
Description: The LongShot program is developing and flight demonstrating an air-launched system capable of engaging multiple adversary targets from standoff ranges using existing air-to-air missiles. LongShot will be deployed either externally from existing fighters or internally from existing bombers. This system will capitalize on a slower speed, fuel-efficient air vehicle for ingress, while retaining highly energetic air-to-air missiles for end-game target engagements, providing several key benefits that increase weapon effectiveness. This program will address the stability and control challenges of launching air-to-air missiles from a relatively small uninhabited system in an operational environment. Potential transition partners include the Navy and Air Force.				
Prior to FY 2026, this program was funded in PE 0603286E, Project AIR-01.				
FY 2026 Plans:				
- Conduct weapon integration and ground testing.				
- Complete fabrication and integration of flight test vehicles.				
- Complete airworthiness certification and host aircraft integration.				
- Conduct flight testing of the vehicle on a host aircraft.				
- Conduct flight demonstrations validating vehicle release from a host aircraft.				
- Conduct flight demonstrations validating air vehicle performance, stability and control through all phases of flight.				
FY 2027 Plans:				
- Continue flight demonstrations validating vehicle release from a host aircraft.				
- Continue flight demonstrations validating vehicle performance, stability and control through all phases of flight.				
- Conduct analysis and flight testing to expand flight envelope.				
- Continue analysis and development of integrated mission systems to mature operationalized capabilities.				
- Conduct alternate payload integration analysis based on Service-desired concepts of employment.				
- Initiate alternate aircraft integration studies to include internal and external bomber weapons station integration and safe and acceptable separation computational fluids dynamics analysis.				
- Continue alternate launch options analysis and risk reduction testing to include palletized and ground launch options.				
- Conduct propulsion integration studies to explore alternate engine options and engine operability analysis.				
- Determine manufacturing approaches and design updates required for vehicle production at scale to inform Operational System design and production planning.				

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX SYSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Continue maturation of Operational System design based on lessons learned from the Demonstration System, mission systems testing, and manufacturing analysis. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from platform integration and initial demonstrations to flight test operations and follow-on development.			
Title: SPeed and Runway INdependent Technologies (SPRINT) X-Plane Demonstration Project Description: The SPeed and Runway INdependent Technologies (SPRINT) X-Plane Demonstration Project is developing and demonstrating the fundamental technologies needed for combined high speed and vertical take-off and landing (VTOL) capabilities in a single aircraft. This program culminates in the fabrication and flight test of a demonstrator that validates the critical technologies in a representative environment and reduces technical, schedule, and cost risk for a follow-on operational system. High speed VTOL aircraft are highly desired in a variety of military missions such as contested logistics under attack, infiltration/exfiltration, electronic warfare, personnel recovery, troop transport and armed escort. The SPRINT Demonstrator is envisioned to transition to a Services component for further risk reduction flight testing. Prior to FY 2026, this program was funded in PE 0603286E, Project AIR-01. FY 2026 Plans: - Complete critical design activities. - Finalize subsystem development (propulsion system, thermal management system, hydraulics, electrical system, etc.), vehicle integration, and engineering drawing release. - Continue toward final aircraft software and aircraft flight control algorithm development. - Continue flight test planning and airworthiness certification. - Integrate with the flight test facility for connectivity, data transfer, and flight safety requirements. - Execute demonstrator vehicle manufacturing and assembly, to include material acquisition and unique tooling for fabrication. - Initiate ground station and human-machine interface development. - Initiate ground system checkouts. FY 2027 Plans: - Finalize flight test planning and airworthiness certification. - Complete structural proof load test to verify the built aircraft matches structural design predictions. - Complete final software integration lab verification testing to ensure the system operates as intended including the proper vehicle responses in the presence of simulated failures. - Complete subcomponent manufacturing and assembly.		-	47.524
			62.113

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 3		R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>		Project (Number/Name) ACX-01 / <i>ADVANCED COMPLEX SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
- Complete manufacturing, assembly, and quality assurance of the demonstrator vehicle. - Complete ground station and human-machine interface development. - Complete ground system checkouts including engine runs. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the transition from critical design activities to the manufacturing and final assembly and system checkout of the demonstration vehicle.					
Title: Otter Description: The Otter program is developing and demonstrating air breathing propulsion technologies that enable operations in very low earth orbital domains that are currently inaccessible. Propulsion capabilities demonstrated will provide increased mission duration and ability to maneuver without regret. Key efforts include the development of new propulsion systems, improved ground test capabilities, and analysis tools to support system development. Otter will progress through development of analysis and test tools, design of candidate propulsion systems, ground testing, build of a demonstrator satellite, and culminate in a long duration (> 1 year) spaceflight demonstration. The anticipated transition partner is the U.S. Space Force. Prior to FY 2026, this program was funded in PE 0603287E, Project SPC-01. FY 2026 Plans: - Conduct flight hardware acceptance testing for the propulsion system. - Integrate the developed propulsion systems with the air harvesting inlets. - Conduct environmental qualification tests for the air harvesting inlets. - Continue ground software build and testing. - Continue flight software build and testing. - Kick-off the launch services partner process. FY 2027 Plans: - Continue to conduct flight hardware acceptance testing for the propulsion system. - Continue to integrate the developed propulsion systems with the air harvesting inlets. - Continue conducting environmental qualification tests for the air harvesting inlets. - Finalizing ground software build and testing. - Finalizing flight software build and testing. - Integrate the satellite onto the launch provider. - Launch of the satellite. FY 2026 to FY 2027 Increase/Decrease Statement:			-	35.788	47.712

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 3		R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX S</i> <i>YSTEMS</i>		Project (Number/Name) ACX-01 / <i>ADVANCED COMPLEX</i> <i>SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 increase reflects the shift from design and build activities to final satellite integration, launch and on orbit demonstration.					
Title: Artificial Intelligence (AI) Reinforcements (AIR)			-	51.610	32.174
Description: AI Reinforcements (AIR) is developing and demonstrating dominant tactical autonomy for multi-ship, beyond visual range (BVR), real-world air combat missions. This program is focused on developing highly accurate models that are orders of magnitude faster than the present state-of-the-art and then using those models to unlock novel and robust AI-driven autonomy approaches. An operations-centric development approach will be enabled through the use of human-on-the-loop F-16 aircraft. On piloted platforms, AIR's algorithms will automate tactical control tasks transforming junior pilots from low-level tacticians into high-level mission commanders. For unpiloted platforms, AIR will enable vehicles to perform missions with minimal human oversight. The outcome of this program will be an AI air combat capability that works in dynamic, operationally representative environments. The transition partner is the U.S. Air Force. Prior to FY 2026, this program was funded in PE 0603286E, Project AIR-01. FY 2026 Plans: - Initiate incorporation of F-16 aircraft into the AIR integration and testing pipeline and iterate development through testing. - Scale the AI-driven autonomy to four-ship operations with increased complexity. - Improve speed and predictivity of model development. - Incorporate dissimilar capabilities, non-stationaries, and deceptive environments to the multi-agent AI training. - Quantify the robustness of AI agent performance by rigorously evaluating its response to realistic operational uncertainties, such as sensor degradation and communication loss, and validate its effectiveness in critical mission scenarios like Offensive and Defensive Counter Air (OCA/DCA). FY 2027 Plans: - Expand operational testing into increasingly complex and contested environments, including electronic warfare scenarios, to rigorously evaluate system robustness and adaptability. - Advance multi-platform collaboration by integrating heterogeneous teams, such as manned-unmanned formations, to assess the scalability of the autonomous teaming capabilities. - Maintain compatibility with other related combat autonomy efforts, enabling rapid integration and testing of new capabilities within the operational framework. FY 2026 to FY 2027 Increase/Decrease Statement:					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects the shift from scaling to live flight to concentrated focus on solutions that demonstrated the strongest technical performance and operational promise.				
Title: More Opportunities with HAWC (MoHAWC) Description: MoHAWC is developing, integrating, and demonstrating technologies to increase effectiveness and producibility of an air-launched hypersonic cruise missile. These technologies include advancing hydrocarbon scramjet-powered propulsion operation, upgrading aircraft integration algorithms, and improving manufacturing approaches. Flight tests will expand the operational envelope. This program collaborates with Navy and Air Force science and technologies efforts to meet future technology insertion dates for service programs of record. This program builds off the demonstrator system design, technology advances and lessons learned under the Hypersonic Airbreathing Weapon Concept (HAWC) and supporting technology maturation programs. Prior to FY 2026, this program was funded in PE 0603286E, Project AIR-01. FY 2026 Plans: - Execute multiple flight tests. - Complete flight test data analysis. FY 2027 Plans: - Execute multiple flight tests. - Complete flight test data analysis and final program review.		-	20.000	20.000
Title: Robotic Servicing of Geosynchronous Satellites (RSGS) Description: A large number of national security and commercial space systems operate at geosynchronous earth orbit (GEO), providing persistence and enabling ground station antennas to point in a fixed direction. Technologies for servicing of GEO spacecraft would involve a mix of highly automated and remotely operated (from Earth) robotic systems. The Robotic Servicing of Geosynchronous Satellites (RSGS) program is establishing the capability to provide robotic services in GEO suitable for a variety of potential servicing tasks, in full collaboration and cooperation with existing satellite owners and national security space operators, and with sufficient propellant for several years of follow-on capability. Key RSGS challenges include robotic tool/end effector requirements, efficient orbital maneuvering of a servicing vehicle, robotic arm systems, automation of certain spacecraft operations, and development of the infrastructure for coordinated control between the servicer and client spacecraft operations teams. The transition agreement is with a commercial partner who will provide the satellite to carry the robotic payload and who will operate the robotic servicer. To support the development of a broadly accepted satellite servicing capability, DARPA is using the Consortium for Execution of Rendezvous and Servicing (CONFERS) operations approach to bring together experts from the		-	6.950	1.700

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>	Project (Number/Name) ACX-01 / <i>ADVANCED COMPLEX SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
private sector and Government to research, develop and publish nonbinding, consensus-based standards for safe operational approaches to on-orbit servicing.			
Prior to FY 2026, this program was funded in PE 0603287E, Project SPC-01.			
FY 2026 Plans:			
- Complete combined testing of integrated robotic payload and spacecraft bus. - Perform training of commercial partner and conduct mission operations rehearsals. - Conduct launch, on-orbit checkout, and calibration of integrated robotic payload.			
FY 2027 Plans:			
- Conduct on-orbit demonstrations of satellite servicing capabilities.			
FY 2026 to FY 2027 Increase/Decrease Statement:			
The FY 2027 decrease reflects the shift from hardware testing to flight operations.			
Title: Process Intensified Energetics (PIE)		-	-
Description: The Process Intensified Energetics (PIE) program will develop the tools and processing methods to safely turn locally available materials into reliable energetics in order to enable the production of fieldable warheads and rocket motors. This program will develop a flexible, robust, and agile manufacturing platform for the production of energetics from locally available material (industrial chemicals to crude oil), leveraging innovative chemical processing and manufacturing controls to dramatically drive down the timeframes and scale needed for production from days to hours and from full-scale factories to containers. PIE will work with diverse input materials and conversion methodologies to continually increase rate, diversity, and quality of energetic output, demonstrating both explosive and propellant manufacturing from a set of locally sourced materials, verified by small-scale propellant and explosive testing. Technology developed under this program will be transitioned to the Services. Applied research related to this effort is funded in PE 0602025E, Project MSL-01.			13.047
FY 2027 Plans:			
- Execute end-to-end validation testing on novel energetic materials developed from diverse materials. - Conduct a system readiness review based on outcomes from sub-scale demonstrations and validation tests on modular platforms. - Demonstrate pilot scale production capability on modular platforms.			
FY 2026 to FY 2027 Increase/Decrease Statement:			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>	Project (Number/Name) ACX-01 / <i>ADVANCED COMPLEX SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 increase reflects the shift from applied research activities under Chemical Processing for Force Protection in PE 0602025E, Project MSL-01 to advanced technology development in this PE/Project.			FY 2027
Title: Lunar Assay via Small Satellite Orbiter (LASSO) Description: The Lunar Assay via Small Satellite Orbiter (LASSO) program will optimize autonomous navigation and propulsion in support of advancing Space Situational Awareness capabilities in cislunar space. The program will build a low-cost satellite, launched as a secondary payload, to demonstrate these capabilities in Very Low Lunar Orbit (VLLO), which is the most challenging environment in cislunar space due to continually destabilizing orbits, lack of navigation aids (i.e., GPS), variable gravity and operation in close proximity to terrain. While operating in VLLO, LASSO will collect high-resolution data regarding accessible water ice on the moon by capturing data about resource distribution and quantities to support future in-situ resource utilization (ISRU) investigation. Such discoveries support future lunar exploration. With improved navigation, propulsion, and autonomous control, the LASSO satellite will achieve VLLO and remain there for long duration while building a statistically significant understanding of lunar resources. Additional research related to this effort is funded within PE 0602023E, Project ACS-01. Prior to FY 2027, this program was funded in PE 0602702E, Project TT-07 and PE 0602023E, Project ACS-01. FY 2027 Plans: - Conduct final mission design planning and verification testing of spacecraft subsystems. - Verify spacecraft and integrated sensor performance through full fidelity Software-in-the-Loop (SWIL) simulations. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from applied research activities in PE 0602023E, Project ACS-01 to advanced technology development in this PE/Project.		-	8.900
Title: Rubble to Rockets (R2) Description: The Rubble to Rockets (R2) program will create a flexible, robust, and agile platform for the production and characterization of raw material for use in rocket body fabrication in a rapidly deployable format. The program will take in indigenous materials, convert it to usable feedstock, characterize the material properties, then adaptively update a sounding rocket's structural design with predictive changes in system level performance. Leveraging material informatics and innovative processing and manufacturing techniques within an agile, rapid iteration schedule structure, R2 will dramatically drive down the timeframes and scale needed for production from months to days. R2 will overcome current limitations in processing mixed and blended scrap by developing tooling and processing approaches that can accommodate widely variable inputs, use online analysis of the upcycling process along with rapid testing to update and develop material informatics models to predict minimum material properties of blended or contaminated scrap streams, and update the design to account for the new materials while		-	7.533

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / <i>ADVANCED COMPLEX SYSTEMS</i>	Project (Number/Name) ACX-01 / <i>ADVANCED COMPLEX SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
understanding the performance impacts of the changes to enable the manufacturing of structures. Additional research related to this effort is funded within PE 0602025E, Project MSL-01.			
FY 2027 Plans: - Iterate sounding rocket design trials with more diverse, impure, and complex mixed material inputs. - Conduct a critical design review evaluating the performance of the systems based on demonstrations. - Demonstrate and validate adaptively redesigned pressure vessel formed from government-provided scrap material inputs of increased complexity.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the shift from applied research activities under Materials for Extreme Environments in PE 0602025E, Project MSL-01 to advanced technology development in this PE/Project.			
Title: Scaled Hypersonics Studies and Concepts Description: Studies conducted under this program will examine and evaluate emerging technologies that seek to disrupt the current paradigms for the development approach for supersonic cruise-missile capabilities by using rapid manufacturing to iteratively build and test new enhanced technologies. This initiative will focus on those technologies which will enable scalable, fast, and agile production of hypersonic components at significantly reduced costs.		-	-
FY 2027 Plans: - Enhanced Propulsion Efficiency: Analyze new propulsion technologies, such as Rotating Detonation Engines (RDE) among others, that offer the potential for increased fuel efficiency in flight, leading to longer range and reduced operating costs. - Improved Thrust-to-Weight Ratio: Design a higher thrust-to-weight ratio that could enable more agile and responsive aircraft. - Advanced Flight Capabilities: Analyze unique characteristics of new technologies, e.g. RDEs, that could unlock new flight capabilities and operational scenarios. - Higher Thermodynamic Efficiency: Design new propellant methods for increased pressure that allows for more work to be extracted from the combustion process. - Compact Engine Design: Analyze new design spaces for increased efficiency to allow for a more compact engine design for a given power output. - Scalable Production: Develop processes and production lines for high throughputs of hypersonic components.			56.000
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to enable scalable, fast, and agile production of hypersonic components at significantly reduced costs through enhanced			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
propulsion efficiency, improved thrust-to-weight ratio, advance flight capabilities, higher thermodynamic efficiency and compact engine design.				
Title: Advanced Complex Systems Studies and Concepts Description: Studies conducted under this program will examine and evaluate emerging technologies and system concepts for applicability to military use. This includes the degree and scope of potential impact and improvements to military operations, mission utility, and warfighter capability. Studies are also conducted to analyze emerging aerospace threats along with possible methods and technologies for countermeasures. The feasibility of achieving potential improvements, in terms of resources, schedule, and technological risk, is also evaluated. The results from these studies are used, in part, to formulate future prototype development programs or refocus ongoing work. Topics include: methods of defeating enemy anti-aircraft attacks; munition technologies to increase precision, range, endurance, and lethality of weapons for a variety of mission sets; novel launch systems; air vehicle control, power, propulsion, materials, and architectures; payload and cargo handling systems; applying artificial intelligence to low earth orbit (LEO) constellation operations to enable collaboration between space, air, maritime, and ground platforms in anti-access/area denial (A2/AD) theaters; robust architectures for precision navigation and timing; enabling operations in Cislunar space; novel approaches to space domain awareness; integration of commercial capabilities into military operations; and on-orbit software environments. FY 2026 Plans: - Explore and refine advanced platform concepts to deliver integrated, cross-domain dominance solutions at disruptive scale and unprecedented penetration capabilities. - Evaluate innovative architectures, operational concepts, and propulsion technologies spanning chemical rocket, airbreathing, and hybrid air vehicle systems to effectively counter and penetrate advanced Anti-Access/Area Denial (A2/AD) environments. FY 2027 Plans: - Initiate research approaches for technologies that enable decisive military effects at distances from the tactical to the strategic. - Explore new insertion routes and cost/fuel efficient range improvements. - Research technologies for pervasive persistence for observation and constant low-level engagement, supporting blue force and allies. - Evaluate life extension and upgrading existing systems with new technologies to confound adversary advances. - Explore novel engineering, technology, and systems approaches that have the potential to change the traditional calculus for military advantage. - Investigate low-cost mass/volume in quantity to overwhelm defense and deliver effects. - Explore decoy denial of service attack and new sensing modalities that change the game for defense operations. - Research capabilities to create doubt among adversaries and strengthen resolve of allies.		-	0.101	62.348

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Further investigate other innovative ideas to disrupt adversary perceived advantages.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to enable pervasive persistence for observation and constant low-level engagement, to develop decoy denial of service attack and new sensing modalities, to evaluate life extension and upgrading of existing systems and to explore new insertion routes and cost/ fuel efficient range improvements.				
Title: DARPA Lift Challenge Description: The DARPA Lift Challenge addresses limitations in existing unmanned aircraft system (UAS) technologies, aiming to incentivize innovative designs that achieve a 4:1 payload-to-weight ratio. This approach tackles airframe cost, payload capacity, operational cost, and overall efficiency through a combination of objective and subjective prize categories. The challenge spans 12 months, encompassing design, build, competition logistics, and post-challenge tracking. Success will be measured by milestones such as registration rates, Federal Aviation Administration (FAA) certifications, consistent team engagement, and the quality of submitted designs. Ultimately, the challenge seeks to identify and test revolutionary UAS designs, with robust participation from over 100 teams and multiple competitors demonstrating the target 4:1 payload-to-weight capability. FY 2026 Plans: - Initiate challenge logistic set up following public announcement. - Conduct challenge event including set up, competition, award announcement, and tear down. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.		-	13.360	-
Title: Control of Revolutionary Aircraft with Novel Effectors (CRANE) Description: The Control of Revolutionary Aircraft with Novel Effectors (CRANE) program is developing and demonstrating revolutionary improvements in aircraft controls technology. The program will design, build, and flight test an aircraft able to fly and maneuver at altitude relying on state-of-the-art Active Flow Control (AFC) technology. AFC is a broad term that encompasses a range of technology approaches; it includes a number of control mechanisms which alter the aerodynamic flow field through ejection or suction of fluid via an orifice on a lifting body. An emphasis of the program is on assessing AFC component technologies, risk reduction and experimentation, integrated testing, fabrication and demonstration of a relevant scale novel and innovative aircraft. Technologies, design tools and models developed and demonstrated under this program will be made available to all Services as well as the civilian aerospace sector for application to future air systems development. Prior to FY 2026, this program was funded in PE 0603286E, Project AIR-01.		-	6.287	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX SYSTEMS	Project (Number/Name) ACX-01 / ADVANCED COMPLEX SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
FY 2026 Plans: - Complete demonstrator production and subsystem integration. - Complete ground-based testing. - Execute flight test campaign. - Complete flight-testing review and data delivery. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	238.882
		FY 2025	FY 2026
Congressional Add: Autonomous intercept of live-fire cruise missiles		-	5.000
FY 2026 Plans: - Initiate research to develop autonomous intercept of live-fire cruise missiles.			
Congressional Adds Subtotals		-	5.000
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603468E / ADVANCED COMPLEX S YSTEMS				Project (Number/Name) ACX-07 / ADVANCED COMPLEX SYSTEMS SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
ACX-07: ADVANCED COMPLEX SYSTEMS SUPPORT	0.000	0.000	19.579	40.264	-	40.264	85.575	86.972	77.576	77.576	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Complex Systems Support project contains management costs in support of DARPA functions and activities across the entire Advanced Complex Systems PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0603286E, Project AIR-01, and PE 0603287E, Project SPC-01.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)					R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	0.000	317.460	331.007	-	331.007	539.812	667.235	673.690	707.458	-	-
AET-01: ADVANCED C4 SYSTEMS	0.000	0.000	30.904	32.545	-	32.545	97.192	110.019	114.884	127.139	-	-
AET-02: ADVANCED DISRUPTIVE MICROTECHNOLOGIES	0.000	0.000	212.361	212.483	-	212.483	229.016	237.140	238.502	238.502	-	-
AET-03: ADVANCED CYBER SYSTEMS	0.000	0.000	55.456	61.314	-	61.314	173.602	273.471	278.958	300.471	-	-
AET-07: ADVANCED ENABLING TECHNOLOGIES SUPPORT	0.000	0.000	18.739	24.665	-	24.665	40.002	46.605	41.346	41.346	-	-

Note

The FY 2026 spend plan amount for Research, Development, Test and Evaluation, Defense-Wide for DARPA includes \$317,460 thousand of discretionary and \$20,000 thousand of mandatory (reconciliation) for a total of \$337,460.

A. Mission Description and Budget Item Justification

The Advanced Enabling Technologies program element is focused on exploiting opportunities to leverage novel capabilities to address the challenges of the modern physical and cyber battlefields. Programs will explore new concepts across electronics, quantum, photonics, materials, communication, software, artificial intelligence (AI), machine learning (ML), and cyber while also pulling from the best available commercial technologies and combining them in ways that give the warfighter disruptive capabilities. Efforts conducted under this program element include quantum circuits, integrated photonic-electronic components, novel materials, next generation command and control, resilient software, formal methods, adversarial AI/ML, and AI/ML-based cyber. This PE also supports innovation and robust transition and experimentation planning in the technology cycle to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Advanced C4 project develops and demonstrates technologies that will provide effective communications, information integration, and decision support to U.S. forces. The success of military operations depends on timely, reliable, secure, and synchronized dissemination of command and control and relevant situational awareness information to every military echelon. Technologies are intended to support highly distributed crewed, uncrewed, and human machine teams in all domains and across the entire spectrum of conflict, specifically end-to-end kill chains. While leveraging advancements in commercial technology development, novel approaches promote adaptability, resilience and reliability to overcome environmental challenges while countering a highly capable and dynamic adversary.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>
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The Advanced Disruptive Microtechnologies project supports activities to enable and accelerate the transition of advanced microsystems advancement, including those developed under the Disruptive Microtechnologies Sciences (EMR-02) and Disruptive Microtechnologies (MSL-02) projects. Funding under this project will include quantum circuits for computation and sensing; integrated photonic-electronic components for timing and data transfer; and developing new technologies and capabilities that intelligently leverage commercial scaling while still preserving advanced capabilities for the warfighter.

The Advanced Cyber Systems project develops, implements, and demonstrates techniques, tools, and frameworks for the full range of cyber operations. Cyber is now ubiquitous to warfighting. For non-kinetic operations in advance of lethal conflict, cyber can be a powerful enabler of operations that limit adversary options and deter adversary actions. For kinetic operations during lethal conflict, cyber can be a force multiplier and provide an asymmetric advantage. The Advanced Cyber Systems project aims to create operational prototypes based on the cyber technology developed in applied research programs in the private sector and in academia. The utility of the operational prototypes that are developed in this project will be assessed, and improvements made, based on demonstrations and evaluations conducted in collaboration with warfighters, acquisition programs, and combatant commands.

The Advanced Enabling Technologies Support project contains management costs in support of DARPA functions and activities across the entire Advanced Enabling Technologies PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, efforts in this PE were funded in PE 0603760E, Command, Control and Communications Systems, and PE 0603739E, Advanced Electronics Technologies.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	335.647	0.000	-	0.000
Current President's Budget	0.000	317.460	331.007	-	331.007
Total Adjustments	0.000	-18.187	331.007	-	331.007
• Congressional General Reductions	0.000	-14.485			
• Congressional Directed Reductions	0.000	-3.702			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	331.007	-	331.007

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)		R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES
Change Summary Explanation FY 2025: N/A FY 2026: Decrease reflects a Congressional general undistributed reduction and a directed reduction for Sec. 8151 expiring appropriation. FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES				Project (Number/Name) AET-01 / ADVANCED C4 SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
AET-01: ADVANCED C4 SYSTEMS	0.000	0.000	30.904	32.545	-	32.545	97.192	110.019	114.884	127.139	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced C4 project develops and demonstrates technologies that will provide effective communications, information integration, and decision support to U.S. forces. The success of military operations depends on timely, reliable, secure, and synchronized dissemination of command and control and relevant situational awareness information to every military echelon. Technologies are intended to support highly distributed crewed, uncrewed, and human machine teams in all domains and across the entire spectrum of conflict, specifically end-to-end kill chains. While leveraging advancements in commercial technology development, novel approaches promote adaptability, resilience and reliability to overcome environmental challenges while countering a highly capable and dynamic adversary.

Prior to FY 2026, efforts in this Project were funded in PE 0603760E, Project CCC-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: DARPAVERSE	-	5.494	0.544
Description: The DARPAVERSE program will create a novel environment for the faster, iterative, development of operational concepts and capabilities by factoring, decomposing, and recomposing models of full DIMEFIL (Diplomatic, Intelligence, Military, Economic, Financial, Informational, and Legal) effects to deliver decision superiority. DARPAVERSE will enable warfighters to develop an iterative understanding of goals, capabilities, and options using adaptive modeling to explore and visualize novel capabilities and operational concepts with teams in immersive environments to reduce modeling and simulation costs and effort. Analysts will be able to explore and compare options with real-time adaptive simulation capabilities to discover optimal choices given current force structure, and the ability of new capabilities to improve outcomes. DARPAVERSE will overcome challenges in modern concept and capability development using natural language specifications with an operator. Technology developed under this program will transition to the Services.			
FY 2026 Plans: - Develop generalized and composable methods of automatic reduced order model development from existing models. - Enable artificial intelligence aided definition and orchestration of custom multi-model environments with a single operator.			
FY 2027 Plans: - Transition DARPAVERSE technology.			
FY 2026 to FY 2027 Increase/Decrease Statement:			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400 / 3		R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>		Project (Number/Name) AET-01 / <i>ADVANCED C4 SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)			FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects the shift from development to technology transition.					
Title: Advanced C4 Systems Studies and Concepts			-	4.071	32.001
Description: The Advanced C4 Systems Studies and Concepts will design and demonstrate advanced communications systems and information systems technology to provide novel concepts and advanced capabilities to access challenging new environments and overcome contested domains. Emphasis will be on concepts and approaches that increase situational awareness, command and control, communications, information infrastructure, cyber operations, information operations, artificial intelligence, and autonomous capabilities at the tactical edge. Technology advancements will support interoperability, security, and resilience. FY 2026 Plans: - Conduct multi-domain studies and assess impact to resilience. - Characterize the range and imaging performance of existing radars using fundamental targets of various sizes and shapes. FY 2027 Plans: - Investigate fully homomorphic architecture for secure artificial reasoning on platforms in the field. - Develop planning capabilities with artificial intelligence to support operational preparation. - Conduct design study on impacts to integration with current mission planning software. - Investigate forward-looking, machine-driven scenario development capable of identifying critical data sets for situational awareness. - Investigate potential to create a hybrid modeling environment where subject matters experts assess the utility and validity of forecasted scenarios. - Demonstrate feasibility of a potential secure and scalable mobile ad-hoc network for maritime vehicles. - Investigate nonlinear mission architectures to generate decision advantage and optionality overmatch by imposing complexity on the adversary. - Formally evaluate operating system for network appliances. - Generate a library of sensing target features and images that can be used for identification and classification. - Conduct state-of-the-art analysis for current information systems technology and begin tradeoff studies. - Meet with transition partners and government stakeholders to define situational awareness gaps. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to enable homomorphic architecture for secure artificial reasoning, to demonstrate a secure and scalable mobile ad-hoc network for maritime vehicles, and to support enhanced forward-looking, machine-driven scenario development.					
Title: Generating Communications Channels to Operate (GeCCO)			-	11.513	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TE CHNOLOGIES	Project (Number/Name) AET-01 / ADVANCED C4 SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: The Generating Communications Channels to Operate (GeCCO) program will enable secure communications for military operations in contested environments by creating communications paths that assure privacy and availability. This effort will develop advanced and flexible communication architectures that employ new virtual network services. GeCCO will enable communications by leveraging commercial networks. Future distributed operations across the globe will require a small logistical footprint and the flexibility to adapt to the available communication environments (commercial and military). GeCCO will address the secure use of already widespread advanced cellular networks to preserve privacy of communications by preventing pattern-of-life analysis. Technology developed under this program will transition to the Services. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-02. FY 2026 Plans: - Continue development, security, and operations (DevSecOps) to test and validate integrated network services and network architecture. - Demonstrate network services and architecture with transition partners in operationally representative scenarios. - Assess technical compatibility and ability for third parties to independently integrate framework into both complementary and transition partner systems. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects program completion.				
Title: Space domain Wide Area Tracking & Characterization (Space-WATCH) Description: The Space domain Wide Area Tracking & Characterization (Space-WATCH) program will enable real-time persistent tracking of objects in low earth orbit (LEO) and provide actionable intelligence on tactical timescales. Space-WATCH will enable detection and tracking of objects orbiting the Earth on much faster timescales than current ground-based sensors are capable of by combining proliferated, on-orbit sensors with automated data fusion. By working with commercial companies operating in LEO to host low-cost sensors on their space platforms, Space-WATCH will employ thousands of sensors on orbit to continuously gather data. Space-WATCH will utilize automated algorithms to process and fuse all the collected data for anomaly detection and false alarm reduction, making the data useful and actionable to ground-based operators. This comprehensive data set of objects in LEO and real-time information on anomalies will greatly increase the accuracy of the Department of War's space situational awareness, as well as enable appropriate responses to anomalies, such as maneuvering space assets out of the way of orbital debris. Technology developed under this program will transition to the U.S. Space Force and Space Development Agency (SDA). Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-02.		-	6.426	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-01 / <i>ADVANCED C4 SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
<i>FY 2026 Plans:</i> - Improve data processing and fusion capabilities. - Transition operational capability for real-time situational awareness. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.			
<i>Title:</i> Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency (SCEPTER) <i>Description:</i> The Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency (SCEPTER) program will develop machine-generated strategies for strategic planning. SCEPTER will discover novel and surprising courses of action (CoAs) by exploring the high complexity state-action space of military engagements at high machine speeds. High CoA exploration speed will be enabled by tailorable abstraction of trusted, expert-informed models. A few of the highest performing CoAs will be validated in higher fidelity simulators along with a thorough human review. Initially, SCEPTER will generate synthetic CoAs to identify challenges in human-generated plans. In later stages of the program, SCEPTER will apply them to develop novel plans. Ultimately, SCEPTER will continually evaluate iterations in planning and plans as changes in theater occur (such as evolutions in force laydowns and capabilities) and will identify new challenges and opportunities that will help prevent surprise from competitors. Technology developed under this program will transition to the Services. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-02. <i>FY 2026 Plans:</i> - Finish development of the SCEPTER technologies for use in demonstrations and transition it to U.S. Government stakeholders. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects program completion.		-	3.400
Accomplishments/Planned Programs Subtotals		-	30.904
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES				Project (Number/Name) AET-02 / ADVANCED DISRUPTIVE MICROT TECHNOLOGIES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
AET-02: ADVANCED DISRUPTIVE MICROT TECHNOLOGIES	0.000	0.000	212.361	212.483	-	212.483	229.016	237.140	238.502	238.502	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Disruptive Microtechnologies project supports activities to enable and accelerate the transition of advanced microsystems advancement, including those developed under the Disruptive Microtechnologies Sciences (EMR-02) and Disruptive Microtechnologies (MSL-02) projects. Funding under this project will include quantum circuits for computation and sensing; integrated photonic-electronic components for timing and data transfer; and developing new technologies and capabilities that intelligently leverage commercial scaling while still preserving advanced capabilities for the warfighter.

Prior to FY 2026, efforts in this Project were funded in PE 0603739E, Projects MT-15 and MT-16.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Next Generation Microelectronics Manufacturing (NGMM)	-	175.000	175.000
Description: The Next Generation Microelectronics Manufacturing (NGMM) program is creating a domestic capability for research, development, and production of three-dimensional heterogeneously-integrated (3DHI) microelectronics, with the goal of sustaining U.S. leadership and innovation in microelectronics manufacturing. The NGMM program aims to advance the state of the art in 3DHI microelectronics through the formation of a domestic open-access prototyping and pilot line center accessible to users in academia, government, and industry. The center will offer equipment resources driving common manufacturing processes for R&D prototyping and a common three-dimensional assembly design kit (3D-ADK), 3DHI electronic design automation (EDA) workflows, simulation / emulation methodologies using digital twins, and test standards to simplify design and performance estimation of 3DHI prototypes. The center will foster a collaborative infrastructure between the foundry, packaging engineers, and EDA engineers to support 3D-ADK development and updates that track with foundry process design kits (PDKs). There will be shared, co-located fabrication, assembly, metrology, and test equipment with interface and process standardization to reduce prototyping cycle time aided by automation of 3DHI fabrication, assembly, and test processes. The center will also create roadmaps to define current and expected future capabilities and business practices.			
The end goal of the program is to establish a self-sustaining 3DHI manufacturing center at an existing facility that is owned and operated by a non-federal entity. Success will be measured by the ability to support the design, fabrication, assembly, and test of a wide range of high-performance 3DHI microsystems at reasonable cost with cycle times supporting fast-paced innovative research. Applied research associated with this effort is funded within PE 0602025E, Project MSL-02.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TE CHNOLOGIES	Project (Number/Name) AET-02 / ADVANCED DISRUPTIVE MICROTECHNOLOGIES		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Prior to FY 2026, this program was funded in PE 0603739E, Project MT-16. FY 2026 Plans: - Develop baseline plan for user access model that addresses membership structure and intellectual property management. - Complete design review of 3DHI process and test vehicles. - Develop and optimize 3DHI fabrication process modules, including thermocompression bonding, hybrid bonding, and metal patterning. - Demonstrate baseline 3DHI test vehicles, including a power amplifier, low noise amplifier, and interposer. - Update design for test, design for manufacturing/yield, and design for reliability standards. FY 2027 Plans: - Update self-sufficiency plans based on projections from members and customer outreach. - Update roadmap with revised assessment of technical and business capabilities. - Improve yield of 3DHI fabrication process modules, including thermocompression bonding, hybrid bonding, and metal patterning. - Establish capability for processing non-silicon material substrates on standard silicon processing equipment. - Develop advanced thermal management solutions for integration in 3DHI assemblies. - Demonstrate user portal for external users to access 3D-ADK and purchase design and prototyping services.				
Title: Continuous-correctness On Opaque Processors (COOP) Description: The Continuous-correctness On Opaque Processors (COOP) program will validate that continuous correctness of software enables adoption of the latest processors with low overhead. Instead of creating new threat-specific signatures to detect the threats, COOP detects the physical manifestations of software errors and continuously corrects the errors with mathematical guarantees. COOP intends to transition to commercial companies and DoW system designers. Prior to FY 2026, this program was funded in PE 0603739E, Project MT-16. FY 2026 Plans: - Develop proof-of-concept that errors detected can be corrected within a relevant timeframe. - Develop techniques to minimize overhead during error detection. - Validate proof-of-concept solutions to correlate signatures to software errors within a relevant timeframe. FY 2027 Plans: - Demonstrate proof-of-concept solutions in a real-world scenario.		-	13.801	7.382

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-02 / <i>ADVANCED DISRUPTIVE MICROT TECHNOLOGIES</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
- Generalize the techniques to multiple different processors. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects the shift from researching a wide array of hardware and software solutions to developing and validating a small number of proof-of-concept solutions. <i>Title:</i> Modular Efficient Laser Technology (MELT) <i>Description:</i> The Modular Efficient Laser Technology (MELT) program will demonstrate the first compact, high-power laser tile as a building block to enable the next generation of scalable high energy laser (HEL) sources for laser weapon systems (LWS). Today's LWS use fiber laser array HEL sources, complex optical benches, and beam directors. These systems are large and heavy, contain large numbers of individual components, and require skilled labor to fabricate and integrate. This makes current LWS difficult and costly to manufacture, limiting their deployment and application. MELT will leverage recent advances in coherent beam combining and photonic integrated circuits (PICs) fabrication techniques to develop tiled arrays integrated with semiconductor-based optical systems, low-loss waveguides, optical interconnects, and application-specific integrated circuits (ASIC) into a compact laser tile that can be integrated with a supporting backplane to provide scalable HEL sources. This will provide the LWS developer a scalable HEL architecture that maintains excellent beam quality and allows LWS deployment on size, weight, and power (SWaP)-constrained platforms. MELT will leverage a mature industrial base for semiconductor manufacturing, as well as recent advances in photonic integrated circuits, coherent beam combining algorithms, semiconductor cooling techniques, and optical lithography to achieve its program goals. Technologies from this program are intended for transition to Army, Air Force, and Navy. Prior to FY 2026, this program was funded in PE 0603739E, Project MT-15. <i>FY 2026 Plans:</i> - Fabricate full laser tile array of semiconductor amplifiers with good electrical-to-optical efficiency. - Integrate full laser tiles with semiconductor amplifiers, phase sensing and control, and power and thermal management. - Demonstrate a fully integrated laser tile, including a full array beam combination test to show traceability to integration in a panelized array with minimal degradation in beam quality. <i>FY 2027 Plans:</i> - Fabricate integrated laser tiles with optical amplifiers, phase control, and thermal management. - Integrate laser tiles into panelized high energy laser (HEL) array with high power and high efficiency. - Demonstrate a fully integrated HEL array. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i>		-	12.952
			7.397

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-02 / <i>ADVANCED DISRUPTIVE MICROTECHNOLOGIES</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects a shift from finalizing fabrication and assembly to demonstration of fully integrated laser tile.			FY 2027
Title: Advanced Disruptive Microtechnologies Studies and Concepts Description: The Advanced Disruptive Microtechnologies Studies and Concepts examines potential opportunities to enable and accelerate the transition of advanced microsystems. Potential topics include quantum circuits for computation and sensing; integrated photonic-electronic components for timing and data transfer; and developing new technologies and capabilities that intelligently leverage commercial scaling while still preserving advanced capabilities for the warfighter. FY 2026 Plans: - Evaluate potential opportunities for accelerating the transition of quantum technologies. - Perform studies on potential insertion use cases of photonic-electronic components for timing and data transfer. - Investigate candidate technology areas for intelligent leverage of commercial scaling to provide advanced capabilities for the warfighter. FY 2027 Plans: - Perform relevant activities related to accelerating the transition of quantum technologies. - Initiate design of photonic-electronic components for timing and data transfer for relevant use cases. - Initiate activities related to intelligently leveraging commercial scaling to provide advanced capabilities for the warfighter. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to accelerate the development and transition of quantum circuits for computation and sensing, to support the design and development of photonic-electronic components for timing and data transfer, and to intelligently leverage commercial scaling.		-	10.608
			22.704
Accomplishments/Planned Programs Subtotals		-	212.483
C. Other Program Funding Summary (\$ in Millions) N/A			
Remarks			
D. Acquisition Strategy N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES				Project (Number/Name) AET-03 / ADVANCED CYBER SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
AET-03: ADVANCED CYBER SYSTEMS	0.000	0.000	55.456	61.314	-	61.314	173.602	273.471	278.958	300.471	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Cyber Systems project develops, implements, and demonstrates techniques, tools, and frameworks for the full range of cyber operations. Cyber is now ubiquitous to warfighting. For non-kinetic operations in advance of lethal conflict, cyber can be a powerful enabler of operations that limit adversary options and deter adversary actions. For kinetic operations during lethal conflict, cyber can be a force multiplier and provide an asymmetric advantage. The Advanced Cyber Systems project aims to create operational prototypes based on the cyber technology developed in applied research programs in the private sector and in academia. The utility of the operational prototypes that are developed in this project will be assessed, and improvements will be made, based on demonstrations and evaluations conducted in collaboration with warfighters, acquisition programs, and combatant commands.

Prior to FY 2026, efforts in this Project were funded in PE 0603760E, Project CCC-05.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Constellation	-	19.970	25.000
Description: The Constellation program is comprised of multiple projects that are developing technologies, capabilities, and prototype systems to enable full spectrum military cyberspace operations to defend the U.S. and deter, disrupt, and defeat cyber adversaries. Technologies of interest include, but are not limited to, artificial intelligence, machine learning, and data science; resilient software, networking, and computing systems; formal methods and program analysis; data and information assurance; and cyber threat intelligence. High relevance is achieved through close coordination with U.S. cyber operators and the use of development, security, and operations (DevSecOps) and other collaborative development processes. High velocity is achieved through streamlined acquisition, assessment, approval, and deployment processes. Constellation development and deployment pipelines enable the rapid and continuous delivery of cyber technologies, capabilities, and prototype systems into operational use for the Department of War (DoW). The Constellation program is also funded in PE 0602025E, Project MSL-04 to facilitate rapid transition of cyber technologies and laboratory prototypes from applied research to operational prototypes. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-05, and PE 0602303E, Project IT-03. FY 2026 Plans: - Refine and integrate cyber operational prototypes with established DoW programs and systems to create a cohesive, robust, and effective cyber warfighting architecture.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES	Project (Number/Name) AET-03 / ADVANCED CYBER SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
- Develop, test, evaluate, and operationalize cyber technologies, prototypes, and capabilities in collaboration with DoW cyber stakeholders. - Coordinate with U.S. cyber operators to formulate cyber operations challenges as problems that research programs can address through new projects/prototypes. - Coordinate with cyber acquisition offices to expand pipelined development of cyber operational prototypes using DevSecOps and other collaborative development processes and through streamlined acquisition, assessment, approval, and deployment processes. FY 2027 Plans: - Refine and integrate cyber operational prototypes with established DoW programs and systems to create a cohesive, robust, and effective cyber warfighting architecture. - Develop, test, evaluate, and operationalize cyber technologies, prototypes, and capabilities in collaboration with DoW cyber stakeholders. - Coordinate with U.S. cyber operators to formulate cyber operations challenges as problems that research programs can address through new projects/prototypes. - Coordinate with cyber acquisition offices to expand pipelined development of cyber operational prototypes using DevSecOps and other collaborative development processes and through streamlined acquisition, assessment, approval, and deployment processes. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects the expansion of prototyping activities.				
Title: Advanced Cyber Security Studies and Concepts Description: Efforts conducted under this thrust study, refine, prototype, demonstrate, and evaluate technologies for cyber security and resilience of U.S. software systems. Security and resilience of the software in weapons, combat, and information systems is essential for warfighters who rely on the correct functioning of this software for success on the battlefield. Thrust emphasis is on dramatically enhancing the security and resilience of both new and legacy software systems that may be subject to cyber attack by adversaries. Focus areas include enabling the confident creation of software systems that are secure by design against exploits; the rapid remediation of software defects once discovered; and the cost-effective improvement of the legacy software on which critical missions often depend. Technical approaches of interest include, but are not limited to, automated vulnerability discovery and remediation, binary analysis, reverse engineering, formal methods, artificial intelligence (AI)-based approaches, and other advanced software generation, analysis, and understanding technologies. Tools and prototypes are created to explore new software security and resilience concepts, and evaluation and refinement of the concepts, tools, and prototypes are performed in collaboration with software developers and cyber red teams. This thrust creates the new concepts		-	3.149	16.114

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-03 / <i>ADVANCED CYBER SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
and prototypes required for the U.S. to develop and sustain software systems that are secure and resilient in the face of cyber attack by adversaries.			
FY 2026 Plans: - Explore AI technology for stack security.			
FY 2027 Plans: - Use formal analysis to prove the fidelity of scanned electronic devices. - Develop scalable software analysis, sandboxing, and surveillance techniques to secure software development. - Mitigate vulnerabilities to secure Department of War infrastructure. - Use models of emergent communications as the basis for networks with strong privacy guarantees. - Develop techniques for rapid reconstitution following any large-scale internet disruption/outage. - Develop rigorous engineering approaches for assuring the use of foundation models in safety-critical systems. - Develop physics-informed multimodal foundation models to enable rapid generation of highly accurate digital twins. - Develop methods and tools to provide real-time ephemeral tactical data protection.			
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to support secure software development through scalable software analysis, sandboxing, and surveillance techniques, to enable rapid reconstitution following any large-scale internet disruption/outage and to support development of physics-informed multimodal foundation models to enable rapid generation of highly accurate digital twins.			
Title: Advanced Cyber Operations Studies and Concepts Description: Efforts conducted under this thrust study, refine, prototype, demonstrate, and evaluate technologies to achieve decisive and enduring operational advantages in the cyber domain. The thrust has two focus areas: Offensive Cyberspace Operations (OCO) that aim to project power and deliver effects through actions in the cyber domain and Defensive Cyberspace Operations (DCO) that aim to defend U.S. cyberspace. OCO interests span strategic, which includes attacks on adversary military infrastructure, to tactical, which includes attacks on adversary weapons, combat, and information systems. In addition to overt denial/disruption/destruction of adversary systems, OCO offers the potential for action that can undermine adversary confidence in their systems and thereby deter adversary aggression. DCO focuses on defeating adversary malicious cyber activity by preventing adversary access to U.S. networks and systems and by promptly discovering and remediating any compromise of U.S. systems. For both OCO and DCO, prototypes are created to explore new cyber operations concepts and the evaluation and refinement of these concepts is performed in collaboration with cyber operators. This thrust creates the new concepts and prototypes required for the U.S. to achieve and maintain superior cyber warfighting capabilities. This effort also includes funding for potential classified program initiations.		-	4.875
			14.825

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-03 / <i>ADVANCED CYBER SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
FY 2026 Plans: - Discover logic flaws in security protocol implementations. - Develop algorithms that identify latent vulnerabilities in complex systems. FY 2027 Plans: - Develop techniques for reasoning about the circuitry inside devices. - Develop automated operational cyber reasoning to enable decisive cyber maneuver warfare. - Develop an automated exploit development platform. - Develop cyber weapons with provable performance guarantees. - Develop techniques to cause hallucinations in adversary AI agents. - Develop persona systems for use in highly engaged virtual environments. - Develop a synthetic red team that probes adversary cognitive and behavioral vulnerabilities. - Model adversary decision rules to craft paradoxical situations that induce logical paralysis. - Develop high-fidelity emulations of adversary individual and group psychological processes. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 increase reflects studies and concepts initiations including, but not limited to, the advancement of new technologies to support development of automated operational cyber reasoning to enable decisive cyber maneuver warfare, to model adversary decision rules to craft paradoxical situations that induce logical paralysis, and to support development of cyber weapons with provable performance guarantees and techniques to cause hallucinations in adversary AI agents.			
Title: Resilient Software Systems Capstones (RSS Capstones) Description: The Resilient Software Systems Capstones (RSS Capstones) program is demonstrating formal methods-based software system development techniques that result in significant and measurable reductions in relevant cyber vulnerabilities on actual Department of War (DoW) systems and will document the techniques in best practices guides that facilitate the adoption of formal methods by the DoW and Defense Industrial Base (DIB). Experience in commercial industry has shown that formal methods provide the basis for high assurance software development, sustainment, modification, improvement, and authority to operate (ATO) capabilities for new and legacy software systems. For new software systems, proof artifacts created using formal methods facilitate the development of high assurance software systems free of high consequence vulnerabilities. For legacy software systems, formal methods enable a granular ATO process and the rapid insertion of formally attested patches and other modifications and improvements. To operationalize these capabilities, the RSS Capstones program hardens and systematizes formal methods-based tools and demonstrates their utility on actual DoW platforms across operational domains to include air, land, sea, and space. Cybersecurity sustainment is a particular focus for the program, including capabilities to produce targeted patches for known security flaws along with proofs that the patches preserve the original baseline functionality and quality attributes of the system - these proofs are the actual enablers of granular ATO and rapid patch deployment. The RSS Capstones		-	15.775
			5.375

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-03 / <i>ADVANCED CYBER SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
program will enable formal methods-based development and sustainment of mission-critical software systems that provide higher levels of assurance than the approaches currently in use by the DoW and DIB. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-05. FY 2026 Plans: - Demonstrate the use of formal methods-based proofs to enable granular ATO and quantify the cost and time of granular sustainment for comparison with the cost and time of ATO with block updates. - Collaborate with DoW stakeholders to make the formal methods-based tools widely available and to institutionalize their use across the DoW and DIB. - Demonstrate code repair and integration capabilities and collect performance metrics, such as development and ATO time and cost for high assurance military software systems. FY 2027 Plans: - Document formal methods-based development and sustainment techniques in best practices guides that facilitate adoption by the DoW and DIB. FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the shift from formal methods-based development to creation of DoW and DIB best practices guides.			
Title: Carcosa Description: The Carcosa program is developing and demonstrating cyber technologies for use by warfighters during tactical operations. Carcosa cyber technology aims to provide warfighters in the field with enhanced situational awareness of their immediate battlespace. Carcosa technologies are being integrated in prototype tools suitable for use by warfighters with a range of cyber knowledge and skills, including both cyber novices and advanced cyber practitioners. Prior to FY 2026, this program was funded in PE 0603760E, Project CCC-05. FY 2026 Plans: - Improve durability of form factor to ensure survivability in real operational contexts. - Evaluate and demonstrate improved technologies to military stakeholders and potential transition partners. FY 2026 to FY 2027 Increase/Decrease Statement:		-	11.687

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603469E / <i>ADVANCED ENABLING TECHNOLOGIES</i>	Project (Number/Name) AET-03 / <i>ADVANCED CYBER SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
The FY 2027 decrease reflects program completion.			
Accomplishments/Planned Programs Subtotals		-	61.314
C. Other Program Funding Summary (\$ in Millions) N/A			
Remarks			
D. Acquisition Strategy N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603469E / ADVANCED ENABLING TECHNOLOGIES				Project (Number/Name) AET-07 / ADVANCED ENABLING TECHNOLOGIES SUPPORT			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
AET-07: ADVANCED ENABLING TECHNOLOGIES SUPPORT	0.000	0.000	18.739	24.665	-	24.665	40.002	46.605	41.346	41.346	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Advanced Enabling Technologies Support project contains management costs in support of DARPA functions and activities across the entire Advanced Enabling Technologies PE. These costs include: DARPA classified and unclassified network support and equipment; contractor support; classified program security; building security; commercial transition services that increase the likelihood that DARPA-funded technologies remain in the U.S. and provide new capabilities for national defense; DARPA outreach to universities and industry; external contracting, financial and support fees; Program Manager Intragovernmental Personnel Act (IPA) Funding; Program Managers from other Government Agencies; and similar operating expenses. Agency support is allocated on a pro-rata basis across the Agency's BA1, BA2 and BA3 PEs and, therefore, fluctuates per PE by fiscal year based on the total Agency budget in that fiscal year.

Prior to FY 2026, support requirements in this Project were funded in PE 0603760E, Projects CCC-02 and CCC-05, and PE 0603739E, Projects MT-15 and MT-16.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide I BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603739E / <i>ADVANCED ELECTRONICS TECHNOLOGIES</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	148.382	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MT-15: <i>MIXED TECHNOLOGY INTEGRATION</i>	0.000	53.427	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
MT-16: <i>BEYOND SCALING ADVANCED TECHNOLOGIES</i>	0.000	94.955	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Advanced Electronics Technologies Program that seeks to design and demonstrate state-of-the-art manufacturing and processing technologies for the production of various electronics and microelectronic devices, sensor systems, integrated photonic-electronic components that have military applications and potential commercial utility. Introduction of advanced product design capability and flexible, scalable manufacturing techniques will enable the commercial sector to rapidly and cost-effectively satisfy military requirements.

The Mixed Technology Integration project funds the advanced development and demonstration of selected basic and applied electronics research programs. Examples of technologies with funded development and demonstration activities include, but are not limited to: reducing the size, weight, and power (SWaP) of components for laser weapon systems that will protect airborne platforms from emerging surface-to-air missiles; integrated photonic-electronic components for positioning, navigation and timing in GPS-denied environments; flexible, software-defined cameras that enable real-time image analysis of complex scenes to provide more actionable information; and optical communications systems that rely on no moving parts enabling their use on SWaP-restricted platforms. Funding under this project is intended to advance transitioning novel technologies to use, providing advanced components compatible with mid-term and other future warfighting requirements.

The Beyond Scaling Advanced Technologies Project supports activities to enable and accelerate the transition of disruptive microelectronics advancement, including those developed under the Beyond Scaling Sciences (ES-02) and Beyond Scaling Technology (ELT-02) projects. Funding under this project will include developing new technologies and capabilities in commercial settings, establishing access to these new processes and to commercial state-of-the-art foundries, enabling prototyping, developing manufacturable processes for three-dimensional heterogeneous integration (including integrated photonics), advancing new architectures and integration technologies for advanced field programmable gate arrays (FPGAs), and innovating back end of line technologies for wide bandgap semiconductors.

Beginning in FY 2026, efforts in this PE will be funded in PE 0603469E, Advanced Enabling Technologies.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603739E / <i>ADVANCED ELECTRONICS TECHNOLOGIES</i>
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B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	157.844	0.000	0.000	-	0.000
Current President's Budget	148.382	0.000	0.000	-	0.000
Total Adjustments	-9.462	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	-4.350	0.000			
• SBIR/STTR Transfer	-5.112	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer and reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603739E / ADVANCED ELECTRONI CS TECHNOLOGIES				Project (Number/Name) MT-15 / MIXED TECHNOLOGY INTEGRATION			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MT-15: MIXED TECHNOLOGY INTEGRATION	0.000	53.427	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Mixed Technology Integration project funds the advanced development and demonstration of selected basic and applied electronics research programs. Examples of technologies with funded development and demonstration activities include, but are not limited to: reducing the size, weight, and power (SWaP) of components for laser weapon systems that will protect airborne platforms from emerging surface-to-air missiles; integrated photonic-electronic components for positioning, navigation and timing in GPS-denied environments; flexible, software-defined cameras that enable real-time image analysis of complex scenes to provide more actionable information; and optical communications systems that rely on no moving parts enabling their use on SWaP-restricted platforms. Funding under this project is intended to advance transitioning novel technologies to use, providing advanced components compatible with mid-term and other future warfighting requirements.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603469E, Project AET-02 and PE 0603467E, Project DAT-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Modular Efficient Laser Technology (MELT)	17.427	-	-
Description: The Modular Efficient Laser Technology (MELT) program will demonstrate the first compact, high-power laser tile as a building block to enable the next generation of scalable high energy laser (HEL) sources for laser weapon systems (LWS). Today's LWS use fiber laser array HEL sources, complex optical benches, and beam directors. These systems are large and heavy, contain large numbers of individual components, and require skilled labor to fabricate and integrate. This makes current LWS difficult and costly to manufacture, limiting their deployment and application. MELT will leverage recent advances in coherent beam combining and photonic integrated circuits (PICs) fabrication techniques to develop tiled arrays integrated with semiconductor-based optical systems, low-loss waveguides, optical interconnects, and application-specific integrated circuits (ASIC) into a compact laser tile that can be integrated with a supporting backplane to provide scalable HEL sources. This will provide the LWS developer a scalable HEL architecture that maintains excellent beam quality and allows LWS deployment on size, weight, and power (SWaP)-constrained platforms. MELT will leverage a mature industrial base for semiconductor manufacturing, as well as recent advances in photonic integrated circuits, coherent beam combining algorithms, semiconductor cooling techniques, and optical lithography to achieve its program goals. Technologies from this program are intended for transition to Army, Air Force, and Navy.			
Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-02.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603739E / <i>ADVANCED ELECTRONICS TECHNOLOGIES</i>	Project (Number/Name) MT-15 / <i>MIXED TECHNOLOGY INTEGRATION</i>	

B. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Box Kick Description: The Box Kick program is developing a capability to engage and defeat raids of adversary subsonic cruise missiles and unmanned aerial systems at low cost while preserving more capable and expensive weapons for higher-end threats. The program will integrate this capability into existing platforms with little-to-no platform modifications required. Technology developed under this program is planned for transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.	36.000	-	-
Accomplishments/Planned Programs Subtotals	53.427	-	-

C. Other Program Funding Summary (\$ in Millions) N/A	Remarks
D. Acquisition Strategy N/A	

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603739E / ADVANCED ELECTRONI CS TECHNOLOGIES				Project (Number/Name) MT-16 / BEYOND SCALING ADVANCED TECHNOLOGIES			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
MT-16: BEYOND SCALING ADVANCED TECHNOLOGIES	0.000	94.955	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Beyond Scaling Advanced Technologies Project supports activities to enable and accelerate the transition of disruptive microelectronics advancement, including those developed under the Beyond Scaling Sciences (ES-02) and Beyond Scaling Technology (ELT-02) projects. Funding under this project will include developing new technologies and capabilities in commercial settings, establishing access to these new processes and to commercial state-of-the-art foundries, enabling prototyping, developing manufacturable processes for three-dimensional heterogeneous integration (including integrated photonics), advancing new architectures and integration technologies for advanced field programmable gate arrays (FPGAs), and innovating back end of line technologies for wide bandgap semiconductors.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603469E, Project AET-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Next Generation Microelectronics Manufacturing (NGMM)	82.500	-	-
Description: The Next Generation Microelectronics Manufacturing (NGMM) program is creating a domestic capability for research, development, and production of three-dimensional heterogeneously-integrated (3DHI) microelectronics, with the goal of sustaining U.S. leadership and innovation in microelectronics manufacturing. The NGMM program aims to advance the state of the art in 3DHI microelectronics through the formation of a domestic open-access prototyping and pilot line center accessible to users in academia, government, and industry. The center will offer equipment resources driving common manufacturing processes for R&D prototyping and a common three-dimensional assembly design kit (3D-ADK), 3DHI electronic design automation (EDA) workflows, simulation / emulation methodologies using digital twins, and test standards to simplify design and performance estimation of 3DHI prototypes. The center will foster a collaborative infrastructure between the foundry, packaging engineers, and EDA engineers to support 3D-ADK development and updates that track with foundry process design kits (PDKs). There will be shared, co-located fabrication, assembly, metrology, and test equipment with interface and process standardization to reduce prototyping cycle time aided by automation of 3DHI fabrication, assembly, and test processes. The center will also create roadmaps to define current and expected future capabilities and business practices. The end goal of the program is to establish a self-sustaining 3DHI manufacturing center at an existing facility that is owned and operated by a non-federal entity. Success will be measured by the ability to support the design, fabrication, assembly, and test of a wide range of high-performance 3DHI microsystems at reasonable cost with cycle times supporting fast-paced innovative research. Applied research associated with this effort is funded within PE 0602716E, Project ELT-02.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603739E / <i>ADVANCED ELECTRONICS TECHNOLOGIES</i>	Project (Number/Name) MT-16 / <i>BEYOND SCALING ADVANCED TECHNOLOGIES</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-02.			
Title: Continuous-correctness On Opaque Processors (COOP) Description: The Continuous-correctness On Opaque Processors (COOP) program will validate that continuous correctness of software enables adoption of the latest processors with low overhead. Instead of creating new threat-specific signatures to detect the threats, COOP detects the physical manifestations of software errors and continuously corrects the errors with mathematical guarantees. COOP intends to transition to commercial companies and DoW system designers. Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-02.		12.455	-
Accomplishments/Planned Programs Subtotals		94.955	-
C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A			

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide I BA 3: Advanced Technology Development (ATD)</i>					R-1 Program Element (Number/Name) PE 0603760E / <i>COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS</i>							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	591.042	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
CCC-02: <i>INFORMATION INTEGRATION SYSTEMS</i>	0.000	51.624	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
CCC-05: <i>CYBER SYSTEMS</i>	0.000	82.758	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
CCC-06: <i>COMMAND, CONTROL AND COMMUNICATION SYSTEMS</i>	0.000	456.660	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Command, Control and Communications Systems Program focused on demonstrating and evaluating advanced information systems research and development concepts.

The Information Integration Systems project develops and demonstrates technologies that will provide effective communications to U.S. forces. The success of military operations depends on timely, reliable, secure, and synchronized dissemination of command and control and relevant situational awareness information to every military echelon. While wired communications and networks are fairly well developed, providing assured high-bandwidth mobile wireless capabilities that match or exceed commercial wired infrastructure is needed to meet the demands of military users. Approaches to this goal include developing technologies in these areas:

- High-Capacity Links technologies - enables greater back-haul capability.
- Advanced Networking technologies - supports resilience, adaptability, scalability, and composable systems to enable adaptive effects webs.
- Low Probability of Detection and Anti-Jam (LPD/AJ) technologies - provides assured communications in very high-threat environments.
- Novel Radio Frequency and Spectral Sensing (RF/SS) - supports efficient spectrum management in congested environments and detection of electromagnetic threats.

The Cyber Systems project develops, implements, and demonstrates techniques, tools, and frameworks for the full range of cyber operations. Cyber is now ubiquitous to warfighting. For non-kinetic operations in advance of lethal conflict, cyber can be a powerful enabler of operations that limit adversary options and deter adversary aggression. For kinetic operations during lethal conflict, cyber can be a force multiplier and provide an asymmetric advantage. The Cyber Systems project aims to create operational prototypes based on the cyber technology developed in applied research programs, in the private sector, and in academia. The utility of the operational prototypes that are developed in this project will be assessed, and improvements made, based on demonstrations and evaluations conducted in collaboration with warfighters, acquisition programs, and combatant commands.

Beginning in FY 2026, efforts in this PE will be funded in PE 0603467E, DARPA Advanced Technology Development, and PE 0603469E, Advanced Enabling Technologies.

PE 0603760E: *COMMAND, CONTROL AND COMMUNICATIONS SYST...*

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603760E / <i>COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS</i>
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B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	336.542	0.000	0.000	-	0.000
Current President's Budget	591.042	0.000	0.000	-	0.000
Total Adjustments	254.500	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	263.298	0.000			
• Reprogrammings	2.101	0.000			
• SBIR/STTR Transfer	-10.899	0.000			

Change Summary Explanation

FY 2025: Increase reflects a Congressionally approved Above Threshold Reprogramming (ATR) for classified program and other reprogrammings offset by SBIR/STTR transfer.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency **Date:** April 2026

Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603760E / COMMAND, CONTROL A ND COMMUNICATIONS SYSTEMS				Project (Number/Name) CCC-02 / INFORMATION INTEGRATION SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
CCC-02: INFORMATION INTEGRATION SYSTEMS	0.000	51.624	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Information Integration Systems project develops and demonstrates technologies that will provide effective communications to U.S. forces. The success of military operations depends on timely, reliable, secure, and synchronized dissemination of command and control and relevant situational awareness information to every military echelon. While wired communications and networks are fairly well developed, providing assured high-bandwidth mobile wireless capabilities that match or exceed commercial wired infrastructure is needed to meet the demands of military users. Approaches to this goal include developing technologies in these areas:

- High-Capacity Links technologies - enables greater back-haul capability.
- Advanced Networking technologies - supports resilience, adaptability, scalability, and composable systems to enable adaptive effects webs.
- Low Probability of Detection and Anti-Jam (LPD/AJ) technologies - provides assured communications in very high-threat environments.
- Novel Radio Frequency and Spectral Sensing (RF/SS) - supports efficient spectrum management in congested environments and detection of electromagnetic threats.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603469E, Project AET-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Space domain Wide Area Tracking & Characterization (Space-WATCH) Description: The Space domain Wide Area Tracking & Characterization (Space-WATCH) program will enable real-time persistent tracking of objects in low earth orbit (LEO) and provide actionable intelligence on tactical timescales. Space-WATCH will enable detection and tracking of objects orbiting the Earth on much faster timescales than current ground-based sensors are capable of by combining proliferated, on-orbit sensors with automated data fusion. By working with commercial companies operating in LEO to host low-cost sensors on their space platforms, Space-WATCH will employ thousands of sensors on orbit to continuously gather data. Space-WATCH will utilize automated algorithms to process and fuse all the collected data for anomaly detection and false alarm reduction, making the data useful and actionable to ground-based operators. This comprehensive data set of objects in LEO and real-time information on anomalies will greatly increase the accuracy of the Department of War's (DoW) space situational awareness, as well as enable appropriate responses to anomalies, such as maneuvering space assets out of the way of orbital debris. Technology developed under this program will transition to the U.S. Space Force and Space Development Agency (SDA). Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-01.	17.983	-	-

PE 0603760E: COMMAND, CONTROL AND COMMUNICATIONS
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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603760E / COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS	Project (Number/Name) CCC-02 / INFORMATION INTEGRATION SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Title: Generating Communications Channels to Operate (GeCCO) Description: The Generating Communications Channels to Operate (GeCCO) program will enable secure communications for military operations in contested environments by creating communications paths that assure privacy and availability. This effort will develop advanced and flexible communication architectures that employ new virtual network services. GeCCO will enable communications by leveraging commercial networks. Future distributed operations across the globe will require a small logistical footprint and the flexibility to adapt to the available communication environments (commercial and military). GeCCO will address the secure use of already widespread advanced cellular networks to preserve privacy of communications by preventing pattern-of-life analysis. Technology developed under this program will transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-01.		15.821	-	-
Title: Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency (SCEPTER) Description: The Strategic Chaos Engine for Planning, Tactics, Experimentation and Resiliency (SCEPTER) program will develop machine-generated strategies for strategic planning. SCEPTER will discover novel and surprising courses of action (CoAs) by exploring the high complexity state-action space of military engagements at high machine speeds. High CoA exploration speed will be enabled by tailorable abstraction of trusted, expert-informed models. A few of the highest performing CoAs will be validated in higher fidelity simulators along with a thorough human review. Initially, SCEPTER will generate synthetic CoAs to identify challenges in human- generated plans. In later stages of the program, SCEPTER will apply them to develop novel plans. Ultimately, SCEPTER will continually evaluate iterations in planning and plans as changes in theater occur (such as evolutions in force laydowns and capabilities) and will identify new challenges and opportunities that will help prevent surprise from competitors. Technology developed under this program will transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-01.		9.237	-	-
Title: Space-Based Adaptive Communications Node (Space-BACN) Description: The Space-Based Adaptive Communications Node (Space-BACN) program created a reconfigurable intersatellite optical communications terminal that has low size, weight, power, and cost (SWaP-C) and easily integrates onto small satellites, as well as a methodology for cross-constellation command and control (C2). Space-BACN enabled on-orbit communications and data relay between heterogeneous satellite constellations that operate on different optical intersatellite link (OISL) specifications. Without Space-BACN, government and commercial OISL-equipped satellites faced difficulty in communicating seamlessly with each other due to use of single-waveform terminals and a lack of standardization for waveform specifications. Space-BACN overcame this challenge by developing a modular, reconfigurable optical terminal that is standard-agnostic and able to support most current and future OISL protocols. Space-BACN also developed a C2 system that controls access and		5.700	-	-

PE 0603760E: *COMMAND, CONTROL AND COMMUNICATIONS*
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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603760E / <i>COMMAND, CONTROL AND COMMUNICATIONS SYSTEMS</i>	Project (Number/Name) CCC-02 / <i>INFORMATION INTEGRATION SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
configures connectivity between constellations based on availability and mission requirements, increasing satellite constellation interoperability, interchangeability, and resilience. Technology developed under this program transitioned to the Services and the Space Development Agency (SDA).			
Title: Mission Integrated Network Control (MINC) Description: The goal of the Mission Integrated Network Control (MINC) program was to develop networking resource management technology to enable agile, self-healing, heterogeneous communications that adapt autonomously to battlefield situations and information needs. Technology developed by MINC translated warfighter information needs and mission applications into requests for communication services and autonomously discovered and configured communications nodes and pathways to form and execute adaptive effects chains and move information where it is needed the most. MINC supported applications that provided up-to-date information to support warfighter situational awareness, a customized common operating picture, and adaptive effects chains across joint all-domain operations in a highly contested environment. Technology from this program transitioned to the Services.		2.883	-
Accomplishments/Planned Programs Subtotals		51.624	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			
Remarks			
D. Acquisition Strategy			
N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603760E / COMMAND, CONTROL A ND COMMUNICATIONS SYSTEMS				Project (Number/Name) CCC-05 / CYBER SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
CCC-05: CYBER SYSTEMS	0.000	82.758	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Cyber Systems project develops, implements, and demonstrates techniques, tools, and frameworks for the full range of cyber operations. Cyber is now ubiquitous to warfighting. For non-kinetic operations in advance of lethal conflict, cyber can be a powerful enabler of operations that limit adversary options and deter adversary aggression. For kinetic operations during lethal conflict, cyber can be a force multiplier and provide an asymmetric advantage. The Cyber Systems project aims to create operational prototypes based on the cyber technology developed in applied research programs, in the private sector, and in academia. The utility of the operational prototypes that are developed in this project will be assessed, and improvements made, based on demonstrations and evaluations conducted in collaboration with warfighters, acquisition programs, and combatant commands.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603469E, Project AET-03.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Carcosa	40.000	-	-
Description: The Carcosa program is developing and demonstrating cyber technologies for use by warfighters during tactical operations. Carcosa cyber technology aims to provide warfighters in the field with enhanced situational awareness of their immediate battlespace. Carcosa technologies are being integrated into prototype tools suitable for use by warfighters with a range of cyber knowledge and skills, including both cyber novices and advanced cyber practitioners.			
Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-03.			
Title: Constellation	28.000	-	-
Description: The Constellation program is comprised of multiple projects that are developing technologies, capabilities, and prototype systems to enable full spectrum military cyberspace operations to defend the U.S. and deter, disrupt, and defeat cyber adversaries. Technologies of interest include, but are not limited to, artificial intelligence, machine learning, and data science; resilient software, networking, and computing systems; formal methods and program analysis; data and information assurance; and cyber threat intelligence. High relevance is achieved through close coordination with U.S. cyber operators and the use of development, security, and operations (DevSecOps) and other collaborative development processes. High velocity is achieved through streamlined acquisition, assessment, approval, and deployment processes. Constellation development and deployment pipelines enable the rapid and continuous delivery of cyber technologies, capabilities, and prototype systems into operational			

PE 0603760E: *COMMAND, CONTROL AND COMMUNICATIONS SYST...*

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603760E / <i>COMMAND, CONTROL A ND COMMUNICATIONS SYSTEMS</i>	Project (Number/Name) CCC-05 / <i>CYBER SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
use for the Department of War (DoW). The Constellation program is also funded in PE 0602303E, Project IT-03 to facilitate rapid transition of cyber technologies and laboratory prototypes from applied research to operational prototypes.			
Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-03 and PE 0602025E, Project MSL-04.			
Title: Resilient Software Systems Capstones (RSS Capstones) Description: The Resilient Software Systems Capstones (RSS Capstones) program is demonstrating formal methods-based software system development techniques that result in significant and measurable reductions in relevant cyber vulnerabilities on actual Department of War (DoW) systems and documents the techniques in best practices guides that facilitate the adoption of formal methods by the DoW and Defense Industrial Base (DIB). Experience in commercial industry has shown that formal methods provide the basis for high assurance software development, sustainment, upgrade, and authority to operate (ATO) capabilities for new and legacy software systems. For new software systems, proof artifacts created using formal methods facilitate the development of high assurance software systems free of high consequence vulnerabilities. For legacy software systems, formal methods enable a granular ATO process and the rapid insertion of formally attested patches and other modifications and improvements. To operationalize these capabilities, the RSS Capstones program hardens and systematizes formal methods-based tools and demonstrates their utility on actual DoW platforms across operational domains to include air, land, sea, and space. Cybersecurity sustainment is a particular focus for the program, including capabilities to produce targeted patches for known security flaws along with proofs that the patches preserve the original baseline functionality and quality attributes of the system - these proofs are the actual enablers of granular ATO and rapid patch deployment. The RSS Capstones program will enable formal methods-based development and sustainment of mission-critical software systems that provide higher levels of assurance than the approaches currently in use by the DoW and DIB. Beginning in FY 2026, this program will be funded in PE 0603469E, Project AET-03.		10.000	-
Title: Mechanical Reckoning Description: The Mechanical Reckoning program developed capabilities to conduct digital site surveys for force protection. The survey included a combination of appropriately sourced publicly and commercially available data. The program enabled service personnel to conduct their own surveys while improving digital survey coverage, automating analytics, and validating kit performance and usability by service personnel.		4.758	-
Accomplishments/Planned Programs Subtotals		82.758	-
C. Other Program Funding Summary (\$ in Millions)			
N/A			

PE 0603760E: *COMMAND, CONTROL AND COMMUNICATIONS SYST...*

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603760E / <i>COMMAND, CONTROL A ND COMMUNICATIONS SYSTEMS</i>	Project (Number/Name) CCC-05 / <i>CYBER SYSTEMS</i>
C. Other Program Funding Summary (\$ in Millions)		
Remarks		
D. Acquisition Strategy		
N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603760E / COMMAND, CONTROL A ND COMMUNICATIONS SYSTEMS				Project (Number/Name) CCC-06 / COMMAND, CONTROL AND COMMUNICATION SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
CCC-06: COMMAND, CONTROL AND COMMUNICATION SYSTEMS	0.000	456.660	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

This project funds classified DARPA programs that are reported in accordance with the SAP Annual Report to Congress.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-06.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)					R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WARFARE TECHNOLOGY							
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	838.984	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
NET-01: JOINT WARFARE SYSTEMS	0.000	77.026	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
NET-02: MARITIME SYSTEMS	0.000	95.664	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
NET-06: NETWORK-CENTRIC WARFARE TECHNOLOGY	0.000	666.294	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Network-Centric Warfare Technology Program that addresses high payoff opportunities to develop and rapidly mature advanced technologies and systems required for today's network-centric warfare concepts. It is imperative for the future of the U.S. forces to operate flawlessly with each other, regardless of which services and systems are involved in any particular mission. The overarching goal of this PE is to enable technologies at all levels, regardless of service component, to operate as one system.

The objective of the Joint Warfare Systems project is to create enabling technologies for seamless joint operations, from strategic planning to tactical and urban operations. Joint Warfare Systems leverage current and emerging network, robotic, and information technology and provide next generation U.S. forces with greatly increased capability, lethality, and rapid responsiveness. Critical issues facing this project are: (1) U.S. opponents using systems that are flexible, robust, and difficult to neutralize; and (2) U.S. doctrine that limits the use of firepower to lessen the impact of operations on noncombatants. These problems are magnified in urban and semi-urban areas where combatants and civilians are often co-located and in peacekeeping operations where combatants and civilians are often indistinguishable. Meeting these challenges places a heavy burden on joint war planning. Understanding opponent networks is essential so that creative options can be developed to counter their strategies. Synchronization of air and ground operations to apply force only where needed and with specific effects is required. This project supports all levels of the force structure including: (1) the strategic/operational level by generating targeting options against opponents' centers of gravity that have complex networked relationships; (2) the tactical/operational level by managing highly automated forces with tight coupling between air and ground platforms; and (3) the focused tactical level by developing platforms and tools, which acquire targets of opportunity and cue network-based analysis of likely enemy operations thus maximizing the effectiveness of ground forces in stability and support operations.

The Maritime Systems project is identifying, developing and rapidly maturing critical advanced technologies and system concepts for the naval forces' role in today's network-centric warfare concept. Improvements in communications between and among submarines, surface ships and naval aircraft have allowed these forces to operate seamlessly with each other and with other Services' network-centric systems across domains. Naval forces will play an ever-increasing role in network-centric warfare because of their forward deployed nature, their unique capability to operate simultaneously in the air, on the sea and under the sea, and their versatile ability to provide both rapid strike and project sustained force. The technologies developed under this project will capitalize on these attributes, improve them and enable them to operate with other network-centric forces.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency			Date: April 2026		
Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)		R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WARFARE TECHNOLOGY			
Beginning in FY 2026, efforts in this PE will be funded in PE 0603467E, DARPA Advanced Technology Development.					
B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	866.533	0.000	0.000	-	0.000
Current President's Budget	838.984	0.000	0.000	-	0.000
Total Adjustments	-27.549	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.515	0.000			
• SBIR/STTR Transfer	-28.064	0.000			
Change Summary Explanation					
FY 2025: Decrease reflects SBIR/STTR transfer.					
FY 2026: N/A					
FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.					

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WARFARE TECHNOLOGY				Project (Number/Name) NET-01 / JOINT WARFARE SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
NET-01: JOINT WARFARE SYSTEMS	0.000	77.026	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The objective of the Joint Warfare Systems project is to create enabling technologies for seamless joint operations, from strategic planning to tactical and urban operations. Joint Warfare Systems leverage current and emerging network, robotic, and information technology and provide next generation U.S. forces with greatly increased capability, lethality, and rapid responsiveness. Critical issues facing this project are: (1) U.S. opponents using systems that are flexible, robust, and difficult to neutralize; and (2) U.S. doctrine that limits the use of firepower to lessen the impact of operations on noncombatants. These problems are magnified in urban and semi-urban areas where combatants and civilians are often co-located and in peacekeeping operations where combatants and civilians are often indistinguishable. Meeting these challenges places a heavy burden on joint war planning. Understanding opponent networks is essential so that creative options can be developed to counter their strategies. Synchronization of air and ground operations to apply force only where needed and with specific effects is required. This project supports all levels of the force structure including: (1) the strategic/operational level by generating targeting options against opponents' centers of gravity that have complex networked relationships; (2) the tactical/operational level by managing highly automated forces with tight coupling between air and ground platforms; and (3) the focused tactical level by developing platforms and tools, which acquire targets of opportunity and cue network-based analysis of likely enemy operations thus maximizing the effectiveness of ground forces in stability and support operations.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Time-definite Capability Delivery	42.999	-	-
Description: As an outgrowth from the success of Awareness in Joint Warfighting Technology, the Time-definite Capability Delivery effort will plan, research, and execute activities designed to field capabilities based on proven technologies on relevant timelines and at sufficient quantities. The effort will achieve this objective by planning and coordinating virtual, live, constructive modelling and simulation in conjunction with in-theater demonstrations and experimentation to help orient programs on acute warfighting problems. Additionally, the effort will establish industry and policy-oriented working groups, host technical exchanges and exercises, conduct research, and develop analytic tools that will explore systemic options to speed time to field and ramp production to relevant volumes to support national security objectives. The effort will probe the boundaries of the current institutional processes through rapid experimentation, novel transition frameworks, and technology analysis to provide proof points that rationally challenge existing process boundaries and identify ways to complement, adapt, or replace institutional processes in order to deliver operationally relevant capabilities on strategically relevant timescales. Time-definite Capability Delivery will create frameworks to evaluate technology investment, policy changes, and other government levers to generate competitive advantage.			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WA RFARE TECHNOLOGY	Project (Number/Name) NET-01 / JOINT WARFARE SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.			
Title: Autonomy Standards and Ideals with Military Operational Values (ASIMOV) Description: The Autonomy Standards and Ideals with Military Operational Values (ASIMOV) program developed an initial autonomy benchmarking framework to objectively and quantitatively measure the readiness of future autonomous systems and the operational complexity and difficulty of proposed use-cases in support of Military Operational Values (MOV) (e.g., international humanitarian law, rules of engagement, commander's intent, etc.) in increasingly complex and changing scenarios. In order to accelerate the development and eventual use of autonomous systems, an implementable measurement and benchmarking framework of military autonomy must be developed. ASIMOV's benchmark will enable future autonomous systems that undergo intensive testing to be evaluated and scored with autonomy readiness levels (ARL) much like how technology readiness levels (TRL) and manufacturing readiness levels (MRL) are used to describe the maturity of technology and manufacturing processes, respectively. ASIMOV decomposed the five U.S. Responsible Artificial Intelligence (AI) Principles (Responsibility, Equitability, Reliability, Traceability, and Governability) and other MOVs in a structured, observable, and independently verifiable manner to measure the readiness of specific autonomous systems to perform within complex and dynamic scenarios. Technology developed under ASIMOV will support future autonomous systems research and development.		22.789	-
Title: FLexible networking Using Intelligent Dialecting (FLUID) Description: FLexible networking Using Intelligent Dialecting (FLUID) will enable Command, Control, Computing, Communications, Cyber, Intelligence, Surveillance, Reconnaissance, and Targeting (C5ISRT) under extremely constrained internet protocol (IP) network environments, including those with link capacities in the range from 10s to single digit Kilobits per second. FLUID will research technologies that reduce network and user communications with semantically correct but condensed data, application, and/or protocol information. FLUID will apply intelligent mechanisms to reduce the link capacity required for C5ISRT systems to operate at target scale. Research will focus on extracting and communicating compressed but semantically similar bits from user data, application messages, and network protocol messages to fit within the network capacity constraints; thus, creating the potential for greatly increased efficiency at and across the tactical edge. Software developed under the program will transition to the Services.		11.238	-
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.			
Accomplishments/Planned Programs Subtotals		77.026	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WARFARE TECHNOLOGY	Project (Number/Name) NET-01 / JOINT WARFARE SYSTEMS
C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WA RFARE TECHNOLOGY				Project (Number/Name) NET-02 / MARITIME SYSTEMS			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
NET-02: MARITIME SYSTEMS	0.000	95.664	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Maritime Systems project is identifying, developing and rapidly maturing critical advanced technologies and system concepts for the naval forces' role in today's network-centric warfare concept. Improvements in communications between and among submarines, surface ships and naval aircraft have allowed these forces to operate seamlessly with each other and with other Services' network-centric systems across domains. Naval forces will play an ever-increasing role in network centric warfare because of their forward deployed nature, their unique capability to operate simultaneously in the air, on the sea and under the sea, and their versatile ability to provide both rapid strike and project sustained force. The technologies developed under this project will capitalize on these attributes, improve them and enable them to operate with other network-centric forces.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Projects DAT-01.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Advanced Propulsor, Experimental (APEX)	33.855	-	-
Description: Current submarine propulsor and propeller designs compose restraints on ship layouts and maneuvering capabilities. The Advanced Propulsor, Experimental (APEX) program is developing and demonstrating a new generation of submarine propulsor designs enabling revolutionary improvements in submarine design, maneuverability, speed, and quieting that will transform future submarine designs. The anticipated transition is to the Navy.			
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.			
Title: Pulling Guard	10.404	-	-
Description: The Pulling Guard program will revolutionize Freedom of Navigation escort operations. The program will create semi-autonomous, modular, point-defense overwatch/escort systems, and an ecosystem that commercially delivers these escort services in partnership with the Services. Pulling Guard will dramatically reduce risk and insurance costs for logistics platforms in peacetime (piracy and terror) and wartime (blockades), while increasing large combatant availability for other critical missions.			
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.			
Title: Scorpionfish	11.927	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WA RFARE TECHNOLOGY	Project (Number/Name) NET-02 / MARITIME SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
Description: The Scorpionfish program will design and test novel undersea weapon technologies. Successful execution of the program will be a revolutionary step in the execution of missions in the undersea domain. The anticipated transition is to the Navy. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.				
Title: Willow Description: The Willow program is developing innovative payloads to conduct Acoustic Warfare (AW) to counter sonars using a unique combination of acoustic hardware and waveforms provided by advanced sonar signal processing algorithms. Willow provides a robust capability to help the Navy respond to sonar threats. This is a novel method for challenging adversary sonars. Willow uses advanced hardware-in-the-loop simulations, Independent Verification and Validation (IV&V), and stressing at-sea testing to create this capability. Technology developed under this program will transition to the Navy. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.		7.022	-	-
Title: Goblin Description: The undersea domain has significant importance to national security and military operations, but as in every domain, manned missions have some limitations. The Goblin program will enhance U.S. autonomous capabilities in the challenging undersea domain by developing and demonstrating complex underwater systems able to search, locate, and execute mission objectives without the need for human control. Navigation approaches will focus on the use of commercial, low-cost navigation hardware combined with environmental feature-based algorithm approaches to eliminate reliance on the Global Positioning System (GPS) for long-duration missions. Key Goblin technical challenges include sensing techniques that provide high-resolution navigation without GPS, perception and effector strategies for objects with unknown parameters, long-duration autonomy approaches to support mission execution, and autonomy approaches that do not rely on human interaction. The anticipated transition is to the Navy. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.		7.030	-	-
Title: Trenton Description: The Trenton program is enabling the production of a wide range of structural objects and underwater repairs through the optimization of three-dimensional concrete printing (3DCP) formulations and fabrications at depth. Trenton is developing the first underwater 3DCP (U3DCP) formulations utilizing in-situ materials to support various marine engineering needs. Current technologies rely on land-based techniques that require forming on land then deployment by barge or ship. Trenton overcomes		8.201	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WA RFARE TECHNOLOGY	Project (Number/Name) NET-02 / MARITIME SYSTEMS	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
this challenge by developing the formulations and fabrication technologies necessary for U3DCP. Technology developed under this program will transition to the Services and commercial partners.			
Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.			
Title: No Manning Required Ship (NOMARS) Description: No Manning Required Ship (NOMARS) is developing small, low-cost, disaggregated naval platforms to demonstrate the ability to perform persistent power projection and force application combat missions currently conducted from large, high value capital ships. The NOMARS program entails designing, building, and sailing a ship that can operate autonomously for long durations at sea, enabling a ship design process that eliminates considerations associated with crew. NOMARS focuses on exploring novel approaches to the design of the sea frame (the ship without mission systems) while accommodating representative payload size, weight, and power. The goal of the program is to demonstrate the feasibility of Unmanned Surface Vessels (USVs) that operate autonomously for months to years without human intervention, in large numbers, with only periodic, depot-based maintenance. This capability will enable disaggregated persistent USVs, allowing the surface fleet to credibly threaten peer adversaries and negate their investments in high-cost weapon systems designed to counter large naval targets such as aircraft carriers. A successful NOMARS program will prove feasibility of a small, unmanned ship with significantly improved reliability and functional performance over current USVs providing a pathway to fully realize the greatest potential of distributed lethality concepts: small ships, in large numbers, each of which is individually low-cost and low value, but in aggregate presents a significant deterrent and warfighting advantage. The transition partner is the Navy. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-01.		11.899	-
Title: Manta Ray Description: The Manta Ray program developed and demonstrated a new class of long-duration, long-range, unmanned underwater vehicle (UUV) at an acquisition and lifecycle cost significantly less than previous payload-capable UUVs. This new class of UUV gives the combatant commander an amplification of capacity without disrupting current operations by remaining independent of manned vessels and ports once deployed. The primary goal of the Manta Ray program was to open a design space for future UUVs capable of both long-duration missions and large payload capacity. A secondary goal of the program was to advance key technologies benefiting other naval designs such as low lifecycle cost UUV operations, energy management technologies to enable long-duration operations, biofouling reduction technologies, and long-duration navigational enablers. The transition partner was the Navy.		5.326	-
Accomplishments/Planned Programs Subtotals		95.664	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WA RFARE TECHNOLOGY	Project (Number/Name) NET-02 / MARITIME SYSTEMS
C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603766E / NETWORK-CENTRIC WARFARE TECHNOLOGY				Project (Number/Name) NET-06 / NETWORK-CENTRIC WARFARE TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
NET-06: NETWORK-CENTRIC WARFARE TECHNOLOGY	0.000	666.294	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

This project funds classified DARPA programs that are reported in accordance with the SAP Annual Report to Congress.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-06.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity	R-1 Program Element (Number/Name)											
0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	PE 0603767E / <i>SENSOR TECHNOLOGY</i>											
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	259.283	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
SEN-01: <i>SURVEILLANCE AND COUNTERMEASURES TECHNOLOGY</i>	0.000	37.043	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
SEN-02: <i>SENSORS AND PROCESSING SYSTEMS</i>	0.000	72.976	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
SEN-06: <i>SENSOR TECHNOLOGY</i>	0.000	149.264	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Advanced Technology Development associated with the Sensor Technology Program focused on sensor efforts that will improve the accuracy and timeliness of our surveillance and targeting systems for improved battlefield awareness, strike capability and battle damage assessment.

The Surveillance and Countermeasures Technology project funds sensor efforts that will improve the accuracy and timeliness of our surveillance and targeting systems for improved battlefield awareness, strike capability, and battle damage assessment. Timely surveillance of enemy territory under all weather conditions is critical to providing our forces with the tactical information needed to succeed in future wars. This operational surveillance capability must continue to perform during enemy efforts to deny and deceive the sensor systems, and operate, at times, in a clandestine manner. This project will exploit recent advances in multispectral target phenomenology, signal processing, low-power high-performance computing, and low-cost microelectronics to develop advanced surveillance and targeting systems. In addition, this project encompasses several advanced technologies related to the development of techniques to counter advanced battlefield threats.

The Sensors and Processing Systems project develops and demonstrates the advanced sensor and processing technologies and systems necessary for Intelligence, Surveillance, and Reconnaissance (ISR) missions. Future battlefields will continue to be populated with targets that use mobility and concealment as key survival tactics, and high-value targets will range from specific individual insurgents and vehicles to groups of individuals and large platforms such as mobile missile launchers and artillery. The Sensors and Processing Systems project is primarily driven by four needs: (a) providing day-night ISR capabilities against the entire range of potential targets; (b) countering camouflage, concealment, and deception of mobile ground targets; (c) detecting and identifying objects of interest/targets across wide geographic areas in near-real-time; and (d) enabling reliable identification, precision fire control tracking, timely engagement, and accurate battle damage assessment of ground targets. The Sensors and Processing Systems project develops and demonstrates technologies and system concepts that combine novel approaches to sensing with emerging sensor technologies and advanced sensor and image processing algorithms, software, and hardware to enable comprehensive knowledge of the battlespace and detection, identification, tracking, engagement, and battle damage assessment for high-value targets in all weather conditions and combat environments.

Beginning in FY 2026, efforts in this PE will be funded in PE 0603467E, DARPA Advanced Technology Development.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 3: Advanced Technology Development (ATD)</i>	R-1 Program Element (Number/Name) PE 0603767E / <i>SENSOR TECHNOLOGY</i>
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B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	267.961	0.000	0.000	-	0.000
Current President's Budget	259.283	0.000	0.000	-	0.000
Total Adjustments	-8.678	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	-8.678	0.000			

Change Summary Explanation

FY 2025: Decrease reflects SBIR/STTR transfer.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603767E / SENSOR TECHNOLOGY				Project (Number/Name) SEN-01 / SURVEILLANCE AND COUNTERMEASURES TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
SEN-01: SURVEILLANCE AND COUNTERMEASURES TECHNOLOGY	0.000	37.043	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Surveillance and Countermeasures Technology project funds sensor efforts that will improve the accuracy and timeliness of our surveillance and targeting systems for improved battlefield awareness, strike capability, and battle damage assessment. Timely surveillance of enemy territory under all weather conditions is critical to providing our forces with the tactical information needed to succeed in future wars. This operational surveillance capability must continue to perform during enemy efforts to deny and deceive the sensor systems, and operate, at times, in a clandestine manner. This project will exploit recent advances in multispectral target phenomenology, signal processing, low-power high-performance computing, and low-cost microelectronics to develop advanced surveillance and targeting systems. In addition, this project encompasses several advanced technologies related to the development of techniques to counter advanced battlefield threats.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Dynamic Optimization for Defense of Ground bases with Electromagnetic warfare (DODGEball) Description: The Dynamic Optimization for Defense of Ground bases with Electromagnetic warfare (DODGEball) program will develop algorithms for optimization of non-kinetic countermeasures for efficient and effective resource management in extended campaign warfare. DODGEball will optimize heterogeneous applications of electromagnetic warfare for the defense of surface forces and infrastructure for long duration campaigns. Technology developed under this program will transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-02.	25.210	-	-
Title: Ouija Description: The Ouija program quantified the High Frequency (HF) noise environment in space and improved the characterization of the ionosphere in support of warfighter capabilities. Ouija successfully created ionospheric measurements of unprecedented granularity using ground equipment and satellites in very low earth orbit (VLEO) to improve ionospheric models and better predict long-range HF propagation. Ouija technology improved performance and characterization of radars and communication systems operating in the HF band. Technology developed under this program transitioned to the Services.	11.833	-	-
Accomplishments/Planned Programs Subtotals	37.043	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603767E / <i>SENSOR TECHNOLOGY</i>	Project (Number/Name) SEN-01 / <i>SURVEILLANCE AND COUNTERMEASURES TECHNOLOGY</i>
C. Other Program Funding Summary (\$ in Millions) N/A		
Remarks		
D. Acquisition Strategy N/A		

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603767E / <i>SENSOR TECHNOLOGY</i>				Project (Number/Name) SEN-02 / <i>SENSORS AND PROCESSING SYSTEMS</i>			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
SEN-02: <i>SENSORS AND PROCESSING SYSTEMS</i>	0.000	72.976	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Sensors and Processing Systems project develops and demonstrates the advanced sensor and processing technologies and systems necessary for Intelligence, Surveillance, and Reconnaissance (ISR) missions. Future battlefields will continue to be populated with targets that use mobility and concealment as key survival tactics, and high-value targets will range from specific individual insurgents and vehicles to groups of individuals and large platforms such as mobile missile launchers and artillery. The Sensors and Processing Systems project is primarily driven by four needs: (a) providing day-night ISR capabilities against the entire range of potential targets; (b) countering camouflage, concealment, and deception of mobile ground targets; (c) detecting and identifying objects of interest/targets across wide geographic areas in near-real-time; and (d) enabling reliable identification, precision fire control tracking, timely engagement, and accurate battle damage assessment of ground targets. The Sensors and Processing Systems project develops and demonstrates technologies and system concepts that combine novel approaches to sensing with emerging sensor technologies and advanced sensor and image processing algorithms, software, and hardware to enable comprehensive knowledge of the battlespace and detection, identification, tracking, engagement, and battle damage assessment for high-value targets in all weather conditions and combat environments.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-02.

B. Accomplishments/Planned Programs (\$ in Millions)

	FY 2025	FY 2026	FY 2027
Title: Cancun Description: The Cancun program will create distributable nodes to measure the radio high frequency (HF) environment for improved warfighter situational awareness. Cancun will enable cost-effective wide-area deployment of low size, weight, power, and cost (SWaP-C) nodes. Cancun will also develop the command and control (C2) network and planning tools required to address the challenge of coordinating large numbers of Cancun nodes deployed over distances of well over 1000 kilometers. The Cancun nodes will measure the state of the ionosphere using a sounding function, as well as record and relay portions of the HF radio band for analysis. The mission planning tool will be developed with warfighter input to optimize functionality. Technologies developed under the Cancun program will transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-02.	24.341	-	-
Title: Large Area Photocathode (LAP) Description: Photocathodes can be found in devices such as photomultiplier tubes, image intensifiers, high-power electronics, and even free electron lasers. Photocathodes typically use low-work function metals or semiconductors and are applied in sub-	19.718	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603767E / SENSOR TECHNOLOGY	Project (Number/Name) SEN-02 / SENSORS AND PROCESSING SYSTEMS		
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
micron thicknesses on planar surfaces with millimeter to centimeter-scale areas. In addition to size and geometry limitations, current materials are fragile and require special handling in ultra-high vacuum conditions. The Large Area Photocathode (LAP) program seeks to overcome these limitations by developing manufacturable photocathode material systems with high quantum efficiency that can be produced in large areas and curved geometries with substantial lifetimes. New materials and assembly methods will be refined and tested through the application of large area photocathodes to a high current, fast electron device prototype. The prototype will demonstrate material feasibility and enable new classes of fast electron devices with relevant applications. Technology developed under this program will transition to the Services. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-02.				
Title: Track at Big Distances with Track-Before-Detect (TBD2) Description: The Track at Big Distances with Track Before Detect (TBD2) program will enable continuous space-based detection and tracking of objects in cislunar space on relevant timelines, thereby increasing the safety of cislunar commercial and civilian traffic for the benefit of all nations and enabling a sustainable space ecosystem. Existing space situational awareness capabilities are primarily focused on objects in geosynchronous orbit (or closer) and are not optimized to scan the much greater volume of space or detect objects at the greater distances required for cislunar space. The TBD2 program will develop signal processing algorithms and payload designs necessary to enable detection from space of faint objects in space. Technology developed under this program will be transitioned to the U.S. Air Force or U.S. Space Force. Beginning in FY 2026, this program will be funded in PE 0603467E, Project DAT-02.		4.500	-	-
Title: X-Ray Extreme-range Non-imaging Analysis (XENA) Description: The X-ray Extreme-range Non-imaging Analysis (XENA) program intended to discover new methods for long-standoff X-ray characterization through the development of data processing algorithms. XENA enabled the processing of imperfect radiography datasets to gain useful insights. Algorithm enhanced radiography datasets are relevant to numerous fields ranging from regulatory industries to quality control. The specific enhancements XENA focused on included improvements to overcome low contrast and motion blur to mimic data collected in real-world environments. Technology developed under this program transitioned to the Services and other U.S. Government organizations.		10.275	-	-
Title: Distributed Radar Image Formation Technology (DRIFT) Description: Based on developments in small synthetic aperture radar (SAR) satellites in commercial industry, there were new opportunities to experiment with novel SAR-related concepts. The Distributed Radar Image Formation Technology (DRIFT) program demonstrated advanced capabilities enabled by a cluster of SAR satellites flown in a coordinated way. DRIFT acquired		8.663	-	-

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026	
Appropriation/Budget Activity 0400 / 3	R-1 Program Element (Number/Name) PE 0603767E / <i>SENSOR TECHNOLOGY</i>	Project (Number/Name) SEN-02 / <i>SENSORS AND PROCESSING SYSTEMS</i>	
B. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026
data from SAR satellites flown in such ways and demonstrated novel processing algorithms on this data. This expanded the utility of small SAR satellites, including commercial satellites, for military applications. Technology developed under this program transitioned to the U.S. Government.			
Title: Painter Description: The Painter program sought to create revolutionary advancements in laser technologies for future active optical systems. Painter translated efficiency benefits from critical laser components into compact optical sources. The objective of Painter was to simultaneously increase the power and decrease the size of laser sources compared to state of the art. Aggressive packaging objectives were met by overcoming thermal management challenges. Painter development was guided and constrained by spectral properties required to support multiple mission applications. Technologies from Painter were transitioned to the Services.		5.479	-
Accomplishments/Planned Programs Subtotals		72.976	-
C. Other Program Funding Summary (\$ in Millions) N/A Remarks D. Acquisition Strategy N/A			

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Exhibit R-2A, RDT&E Project Justification: PB 2027 Defense Advanced Research Projects Agency										Date: April 2026		
Appropriation/Budget Activity 0400 / 3					R-1 Program Element (Number/Name) PE 0603767E / SENSOR TECHNOLOGY				Project (Number/Name) SEN-06 / SENSOR TECHNOLOGY			
COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
SEN-06: SENSOR TECHNOLOGY	0.000	149.264	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

This project funds classified DARPA programs that are reported in accordance with the SAP Annual Report to Congress.

Beginning in FY 2026, efforts in this Project will be funded in PE 0603467E, Project DAT-06.

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 6:</i> <i>RDT&E Management Support</i>	R-1 Program Element (Number/Name) PE 0605001E / <i>MISSION SUPPORT</i>
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	122.998	115.673	108.101	-	108.101	111.203	114.381	117.134	119.655	-	-
MST-01: <i>MISSION SUPPORT</i>	0.000	122.998	115.673	108.101	-	108.101	111.203	114.381	117.134	119.655	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Mission Support Program Element provides funding for the costs of mission support activities for the Defense Advanced Research Projects Agency. The funds provide personnel compensation for mission support civilians as well as costs for building rent, physical security, travel, supplies and equipment, communications, printing and reproduction.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	113.007	115.673	0.000	-	0.000
Current President's Budget	122.998	115.673	108.101	-	108.101
Total Adjustments	9.991	0.000	108.101	-	108.101
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	9.991	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	108.101	-	108.101

Change Summary Explanation

FY 2025: Increase reflects reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

C. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Mission Support Civilian Personnel Salaries	80.611	79.443	76.367
Description: Mission Support Civilian Personnel Salaries			

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 6: RDT&E Management Support</i>		R-1 Program Element (Number/Name) PE 0605001E / <i>MISSION SUPPORT</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
<i>FY 2026 Plans:</i> - Continue to fund Mission Support civilian salaries and benefits, including additional technical and support civilian personnel costs for increased mission requirements and administrative support costs. <i>FY 2027 Plans:</i> - Continue to fund Mission Support civilian salaries and benefits to meet mission requirements and administrative support costs. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects civilian pay repricing.				
<i>Title:</i> Facilities and Infrastructure Costs <i>Description:</i> Facilities and Infrastructure Costs <i>FY 2026 Plans:</i> - Continue to fund building rent, General Services Administration lease overhead costs, building maintenance and repairs, building utilities and cleaning services; Pentagon Force Protection Agency security guards; Defense Information Systems Agency telecommunication service; printing and reproduction. <i>FY 2027 Plans:</i> - Continue to fund building rent, General Services Administration lease overhead costs, building maintenance and repairs, building utilities and cleaning services; Pentagon Force Protection Agency security guards; Defense Information Systems Agency telecommunication service; printing and reproduction. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i> The FY 2027 decrease reflects a reduction in planned facilities projects.		28.429	32.332	28.630
<i>Title:</i> Mission Support Travel <i>Description:</i> Mission Support Travel <i>FY 2026 Plans:</i> - Continue to fund Mission Support travel requirements. <i>FY 2027 Plans:</i> - Continue to fund Mission Support travel requirements. <i>FY 2026 to FY 2027 Increase/Decrease Statement:</i>		6.098	3.898	3.104

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide</i> / BA 6: <i>RDT&E Management Support</i>		R-1 Program Element (Number/Name) PE 0605001E / <i>MISSION SUPPORT</i>		
C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
The FY 2027 decrease reflects estimated Mission Support travel requirements.				
Title: Accounting and Audit		7.860	-	-
Description: Accounting and Audit				
Accomplishments/Planned Programs Subtotals		122.998	115.673	108.101
D. Other Program Funding Summary (\$ in Millions) N/A				
Remarks				
E. Acquisition Strategy N/A				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 6: RDT&E Management Support	R-1 Program Element (Number/Name) PE 0605502E / SMALL BUSINESS INNOVATION RESEARCH
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	130.133	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
SB-01: SMALL BUSINESS INNOVATION RESEARCH	0.000	130.133	0.000	0.000	-	0.000	0.000	0.000	0.000	0.000	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

In accordance with Public Law No: 116-92 (National Defense Authorization Act 2020) and the Small Business Act (15 U.S.C. 638), the DARPA Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs are designed to provide small, high-tech businesses and academic institutions the opportunity to propose radical, innovative, high-risk approaches to address existing and emerging national security threats, thereby supporting DARPA's overall strategy to enable fundamental discoveries and technological breakthroughs that provide new military capabilities.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	0.000	0.000	0.000	-	0.000
Current President's Budget	130.133	0.000	0.000	-	0.000
Total Adjustments	130.133	0.000	0.000	-	0.000
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	0.000	0.000			
• SBIR/STTR Transfer	130.133	0.000			

Change Summary Explanation

FY 2025: Increase reflects SBIR/STTR transfer.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

C. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Small Business Innovation Research	130.133	0.000	0.000
Description: The Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs are			

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 6: RDT&E Management Support</i>		R-1 Program Element (Number/Name) PE 0605502E / <i>SMALL BUSINESS INNOVATION RESEARCH</i>		
C. Accomplishments/Planned Programs (\$ in Millions) designed to provide small, high-tech businesses and academic institutions the opportunity to propose radical, innovative, high-risk approaches to address existing and emerging national security threats; thereby supporting DARPA's overall strategy to enable fundamental discoveries and technological breakthroughs that provide new military capabilities.		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Once reauthorized, Small Business Program Office (SBPO) will implement any new legislative changes and processes outlined within the new SBIR/STTR Policy Directive. - Once reauthorized, SBPO will continue to utilize DoW's yearlong BAA to release SBIR/STTR topics on a Just-in-Time basis via DoW's monthly release schedule. - Once reauthorized, and if still applicable, SBPO will continue to release an SBIR and/or STTR Open topic in accordance with the 2022 SBIR/STTR Reauthorization Act. Lessons learned from the FY 2025 release will be identified and implemented for FY 2026. - Once reauthorized, and if still applicable, SBPO will continue and/or adjust in accordance with new legislation its Due Diligence Program Business Assessment Program, taking lessons learned from FY 2025 and improving and streamlining them in FY 2026. - Once reauthorized, will continue to leverage DARPA SBIR/STTR topics in support of larger DARPA Programs to the highest extent possible to ensure successful transition of SBIR/STTR technologies. - Once reauthorized, SBPO will continue to utilize various funding pathways available to the SBIR/STTR programs. This includes, Phase I, Phase II, Direct to Phase II, co-funds, cross agency awards, Phase II Enhancements, and SBIR XL Pilot. - Once reauthorized, SBPO will continue to use DARPA's SBIR XL pilot, which aims to increase opportunities for DARPA funded technology by reimagining SBIRs to transform ideas into successful small businesses that scale. The goals of SBIR XL include: (1) increase relevance of SBIR Program for Technology Development in DARPA; (2) emphasize transition and commercialization as part of evaluation process including establishment of concrete commercialization milestones; (3) raise award ceilings to support efforts for operation scale deployment, increasing the probability of technology transition and commercialization; (4) decrease award timelines. - Once reauthorized, SBPO will continue to link wherever possible to the OUSD(R&E) Critical Technology Areas which, for FY 2026, include: (1) Applied Artificial Intelligence; (2) Biomanufacturing; (3) Contested Logistics Technologies; (4) Quantum and Battlefield Information Dominance; (5) Scaled Hypersonics; and (6) Scaled Directed Energy. - DARPA SBIR/STTR contracting will continue streamlining the acquisition process for Phase II awards and standardizing the contracting templates for all acquisition vehicles. The team plans to focus on leveraging Other Transactions (OT) for suitable SBIR/STTR actions.				

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency		Date: April 2026		
Appropriation/Budget Activity 0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 6: RDT&E Management Support</i>		R-1 Program Element (Number/Name) PE 0605502E / <i>SMALL BUSINESS INNOVATION RESEARCH</i>		
C. Accomplishments/Planned Programs (\$ in Millions) - SBPO will continue to support the transition of SBIR/STTR efforts using tools and programs available to them including: OSW SBIR/STTR Transitions SBIR/STTR Technologies (OTST), Accelerate the Procurement and Fielding of Innovative Technologies (APFIT), and Prime Accelerator and Challenge events, etc. FY 2027 Plans: - Will continue to utilize DoW's yearlong BAA to release SBIR/STTR topics on a Just-in-Time basis via DoW's monthly release schedule. - If still applicable, will continue to release an SBIR and/or STTR Open topic in accordance with the 2022 SBIR/STTR Reauthorization Act. Lessons learned from the FY 2026 release will be identified and implemented for FY 2027. - As applicable, will continue its Due Diligence Program Business Assessment Program, taking lessons learned from FY 2026 and improving and streamlining them in FY 2027. - Will continue to leverage DARPA SBIR/STTR topics in support of larger DARPA Programs to the highest extent possible to ensure successful transition of SBIR/STTR technologies. - Will continue to utilize various funding pathways available to the SBIR/STTR programs. This includes, Phase I, Phase II, Direct to Phase II, co-funds, cross agency awards, Phase II Enhancements, and SBIR XL Pilot. - Will continue to use DARPA's SBIR XL pilot, which aims to increase opportunities for DARPA funded technology by reimagining SBIRs to transform ideas into successful small businesses that scale. The goals of SBIR XL include: (1) increase relevance of SBIR Program for Technology Development in DARPA; (2) emphasize transition and commercialization as part of evaluation process including establishment of concrete commercialization milestones; (3) raise award ceilings to support efforts for operation scale deployment, increasing the probability of technology transition and commercialization; (4) decrease award timelines. - Will continue to link wherever possible to the OUSD(R&E) Critical Technology Areas which currently include: (1) Applied Artificial Intelligence; (2) Biomanufacturing; (3) Contested Logistics Technologies; (4) Quantum and Battlefield Information Dominance; (5) Scaled Hypersonics; and (6) Scaled Directed Energy. - DARPA SBIR/STTR contracting will continue streamlining the acquisition process for Phase II awards and standardize the contracting templates for all acquisition vehicles. The team will implement processes and procedures to decrease time to award SBIR/STTR efforts. - SBPO will continue to support the transition of SBIR/STTR efforts using tools and programs available to them including: OSW SBIR/STTR Transitions SBIR/STTR Technologies (OTST), Accelerate the Procurement and Fielding of Innovative Technologies (APFIT), and Prime Accelerator and Challenge events, etc.		FY 2025	FY 2026	FY 2027
Accomplishments/Planned Programs Subtotals		130.133	0.000	0.000
D. Other Program Funding Summary (\$ in Millions) N/A				

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D. Other Program Funding Summary (\$ in Millions)		
Remarks		
E. Acquisition Strategy N/A		

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Exhibit R-2, RDT&E Budget Item Justification: PB 2027 Defense Advanced Research Projects Agency	Date: April 2026
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Appropriation/Budget Activity 0400: Research, Development, Test & Evaluation, Defense-Wide / BA 6: RDT&E Management Support	R-1 Program Element (Number/Name) PE 0605898E / MANAGEMENT HQ - R&D
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COST (\$ in Millions)	Prior Years	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total	FY 2028	FY 2029	FY 2030	FY 2031	Cost To Complete	Total Cost
Total Program Element	0.000	15.587	13.623	5.383	-	5.383	5.507	5.644	5.738	5.846	-	-
MH-01: MANAGEMENT HQ - R&D	0.000	15.587	13.623	5.383	-	5.383	5.507	5.644	5.738	5.846	-	-
Quantity of RDT&E Articles	-	-	-	-	-	-	-	-	-	-		

A. Mission Description and Budget Item Justification

The Management HQ - R&D Program Element provides funding for the administrative support costs of the Defense Advanced Research Projects Agency. This project provides funding for DARPA Management Headquarters Activities (MHA). The funds provide personnel compensation for management headquarters civilians as well as associated travel and support contract costs.

B. Program Change Summary (\$ in Millions)	FY 2025	FY 2026	FY 2027 Base	FY 2027 OOC	FY 2027 Total
Previous President's Budget	14.577	13.623	0.000	-	0.000
Current President's Budget	15.587	13.623	5.383	-	5.383
Total Adjustments	1.010	0.000	5.383	-	5.383
• Congressional General Reductions	0.000	0.000			
• Congressional Directed Reductions	0.000	0.000			
• Congressional Rescissions	0.000	0.000			
• Congressional Adds	0.000	0.000			
• Congressional Directed Transfers	0.000	0.000			
• Reprogrammings	1.010	0.000			
• SBIR/STTR Transfer	0.000	0.000			
• TotalOtherAdjustments	-	-	5.383	-	5.383

Change Summary Explanation

FY 2025: Increase reflects reprogrammings.

FY 2026: N/A

FY 2027: The FY 2026 President's Budget submission did not include FY 2027 funding projections. Therefore, no comparison between budget positions is available for FY 2027.

C. Accomplishments/Planned Programs (\$ in Millions)	FY 2025	FY 2026	FY 2027
Title: Management Headquarters Civilian Personnel Salaries	5.566	5.037	4.931
Description: Management Headquarters Civilian Personnel Salaries			

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C. Accomplishments/Planned Programs (\$ in Millions)		FY 2025	FY 2026	FY 2027
FY 2026 Plans: - Continue to fund management headquarters civilian salaries and benefits.				
FY 2027 Plans: - Continue to fund management headquarters civilian salaries and benefits.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects civilian pay repricing.				
Title: Management Headquarters Travel Description: Management Headquarters Travel		0.521	0.565	0.452
FY 2026 Plans: - Continue to fund Management Headquarters travel requirements.				
FY 2027 Plans: - Continue to fund Management Headquarters travel requirements.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects estimated Management Headquarters travel requirements.				
Title: Management Headquarters Strategic Operational Support Description: Management Headquarters Strategic Operational Support		9.500	8.021	-
FY 2026 Plans: - Continue to fund operational support which includes Public Affairs, Legislative Affairs, General Council, International Cooperation, and Human Resources Support.				
FY 2026 to FY 2027 Increase/Decrease Statement: The FY 2027 decrease reflects the Management Headquarters Strategic Operational Support costs being realigned to PE 0603467E, Project DAT-07.				
Accomplishments/Planned Programs Subtotals		15.587	13.623	5.383
D. Other Program Funding Summary (\$ in Millions)				
N/A				
Remarks				

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E. Acquisition Strategy N/A		

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